Administering and Configuring Amazon Connect



Student Guide

sunset LEARNING INSTITUTE



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What You Will Learn

Participants going through this course will lean how to:

- Describe the components and features that comprise the Amazon Connect service
- Identify the AWS ecosystem of services that commonly integrate with Amazon Connect
- Create an (or use an existing) AWS account and launch an instance of Amazon Connect
- Utilize the Amazon Connect Console to build a basic configuration
- Design and implement Contact Flows to define the customer experience for chat and voice contacts
- Manage Contact Attributes to populate data for use in Contact Flows
- Demonstrate the ability to use Polly and SSML in Contact Flows
- Configure and use a Lex Bot for voice and chat in Contact Flows
- Invoke Lambda functions in Contact Flows
- Construct custom queue logic for contacts
- Utilize Agent Workspace by enabling Customer Profiles, Cases, and Views
- Implement a Queued Callback strategy in a Contact Center environment
- Demonstrate the ability to use and customize the Agent interface
- Describe Recording and Monitoring features available in Amazon Connect
- Utilize and customize Dashboards, monitor Metrics, and generate Reports
- Troubleshoot Amazon Connect issues

Course Description

The Administering and Configuring Amazon Connect course is a 4-day, hands-on, instructor-led training intended for administrators, systems engineers, and AWS partners. This lab-intensive course enables learners to implement Amazon Connect as a cloud-native Contact Center.

The lab environment emulates a typical deployment and provides each learner with a dedicated instance of Amazon Connect. Learners initially create a reproduceable lab environment by performing the instantiation of Amazon Connect instance and then apply a configuration using the Amazon Connect Console.

Course topics include setting up accounts, navigating the Amazon Connect Dashboard, configuring basic Contact Flows to establish the customer experience, and performing day-to-day operational tasks. The course also includes advanced features that focus on creating custom call queues, defining contact attributes, using Polly and SSML in Contact Flows, and creating a typical Call Center Agent operating environment. Configuring more complex Contact Flows utilizing Lex Bots and Lambda provides experience integrating with other AWS services. Functional testing and problem isolation are featured in the Monitoring, Recording, and Reporting module. Sunset Learning developed this course to provide Amazon Connect training for those who have basic on-premise Contact Center experience or are considering a virtual Contact Center environment. Knowledge of AWS and associated services is helpful, although not required. This is not a certification course.

Pre-requisites

- Have dedicated browser access, a telephony device with local call access and a PC/laptop with audio/video capability
- Complete or have equivalent knowledge of AWS-CLD-ESS or AWS-ESS
- Knowledge of AWS and associated services is helpful, although not required
- Have a basic understanding of inbound call handling in a Call Center
- Have a familiarity with applications and services available in the Cloud

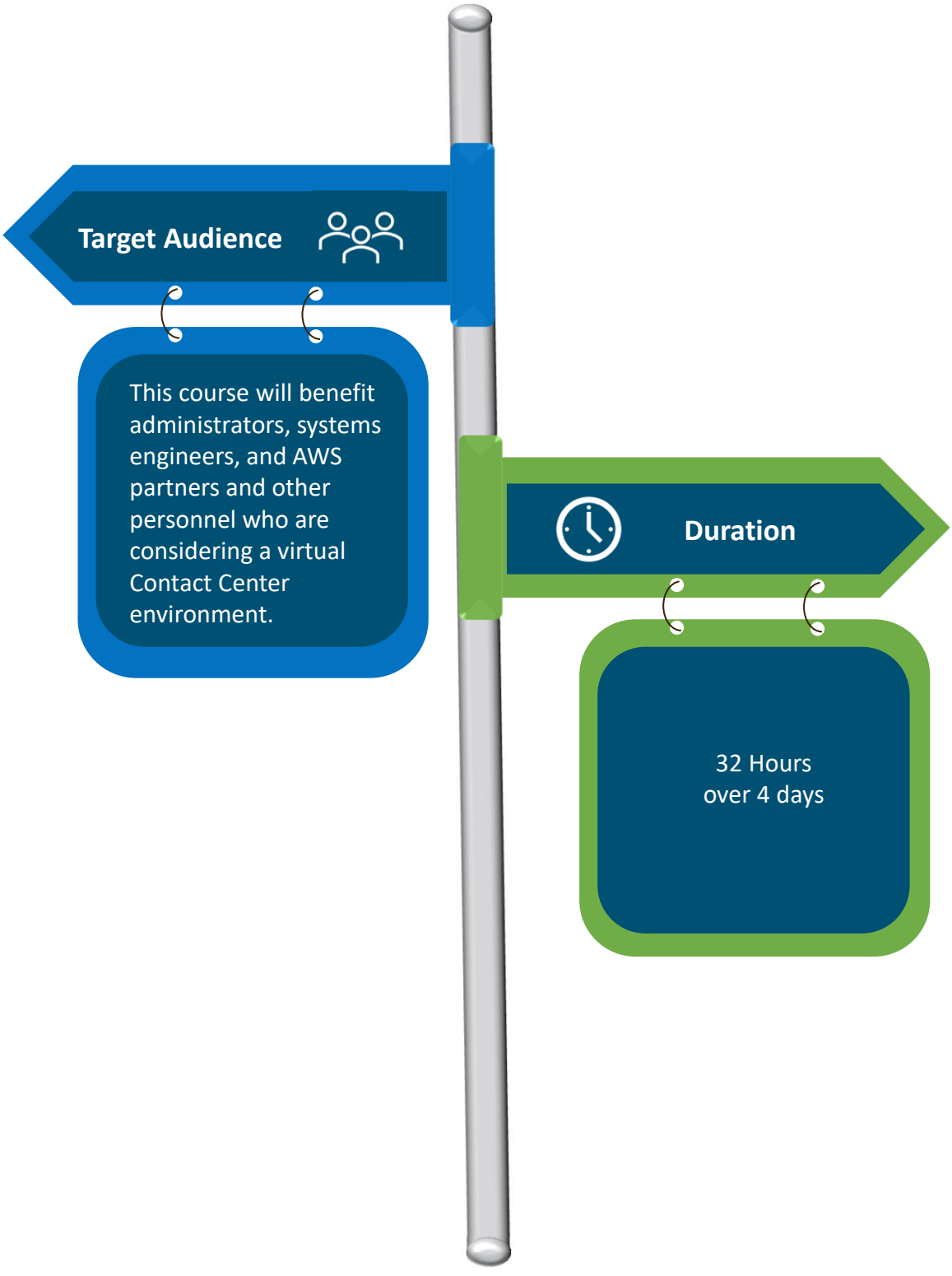

Target Audience
This course will benefit administrators, systems engineers, and AWS partners and other personnel who are considering a virtual Contact Center environment.
Duration
32 Hours
over 4 days

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Module 1

Introduction to Amazon Connect and Related Amazon Web Services (AWS)

Overview

This module provides you with the knowledge and skills to identify and understand the base components of Amazon Connect. Amazon Connect enables you to create an omnichannel Contact Center that provides a unified experience across multiple channels, such as voice, chats, and tasks. For example, you can dynamically offer chat and/or voice contact based on such factors as customer preference, estimated wait times, and agent skill. In this module, you will learn about the power of AWS and Amazon Connect, browsers, and features supported by Amazon Connect.

Objectives

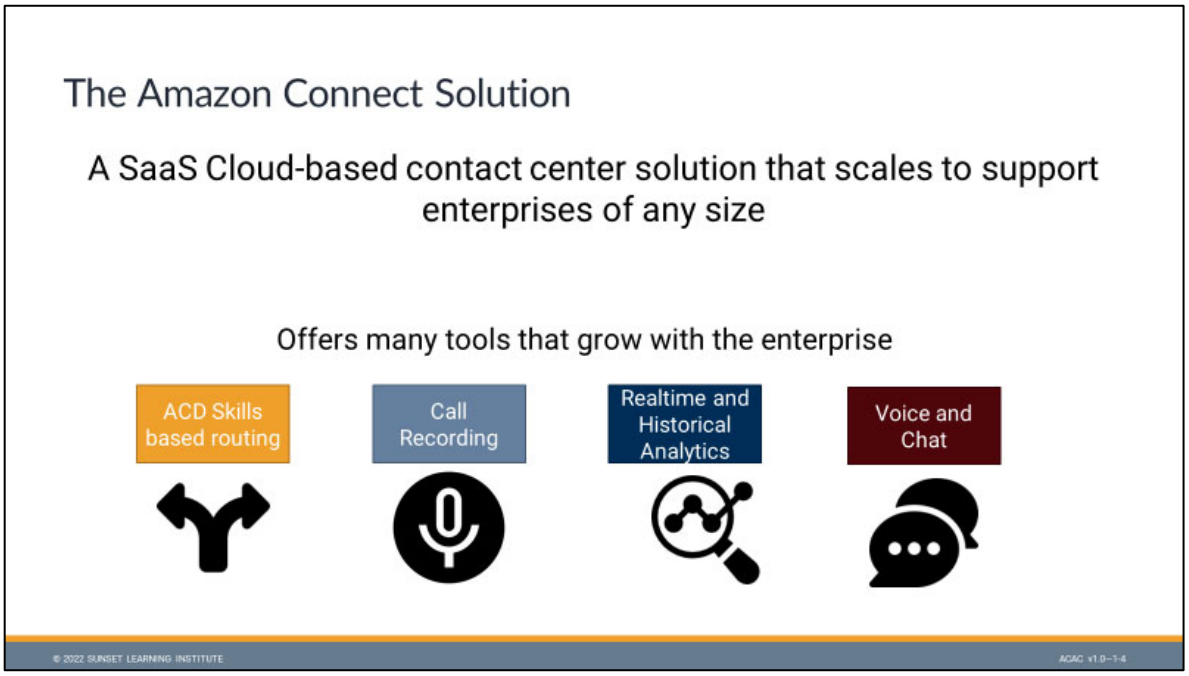
Upon completing this module, you will show an understanding of Amazon Connect. This ability includes being able to meet this objective:

- Describe features and benefits of Amazon Connect

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Introducing Amazon Connect

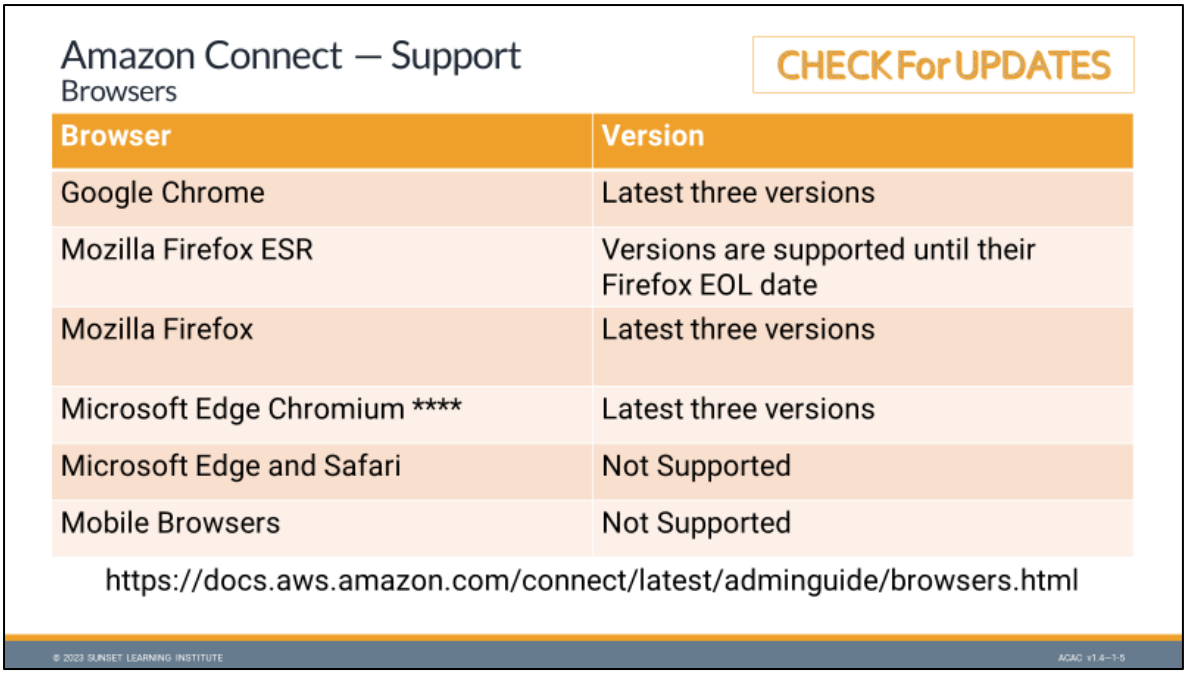
This section describes AWS, Amazon Connect, browsers, and features supported by Amazon Connect.



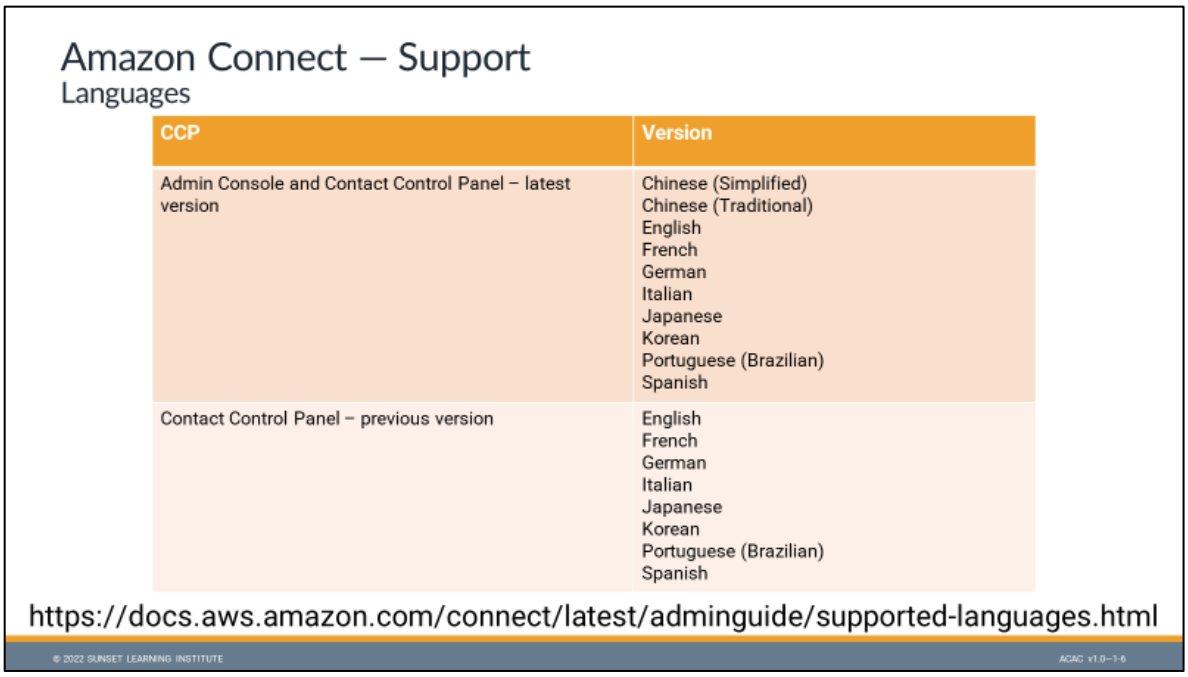
Amazon Connect is an omnichannel Cloud Contact Center. You can **set up a Contact Center** in a few steps, add agents who are located anywhere, and start engaging with your customers.

You can create personalized experiences for your customers using omnichannel communications. For example, you can **dynamically offer chat and voice contact**, based on such factors as customer preference and estimated wait times. Agents, meanwhile, conveniently handle all customers from just **one interface**. For example, they can chat with customers, and create or respond to tasks as they are routed to them.

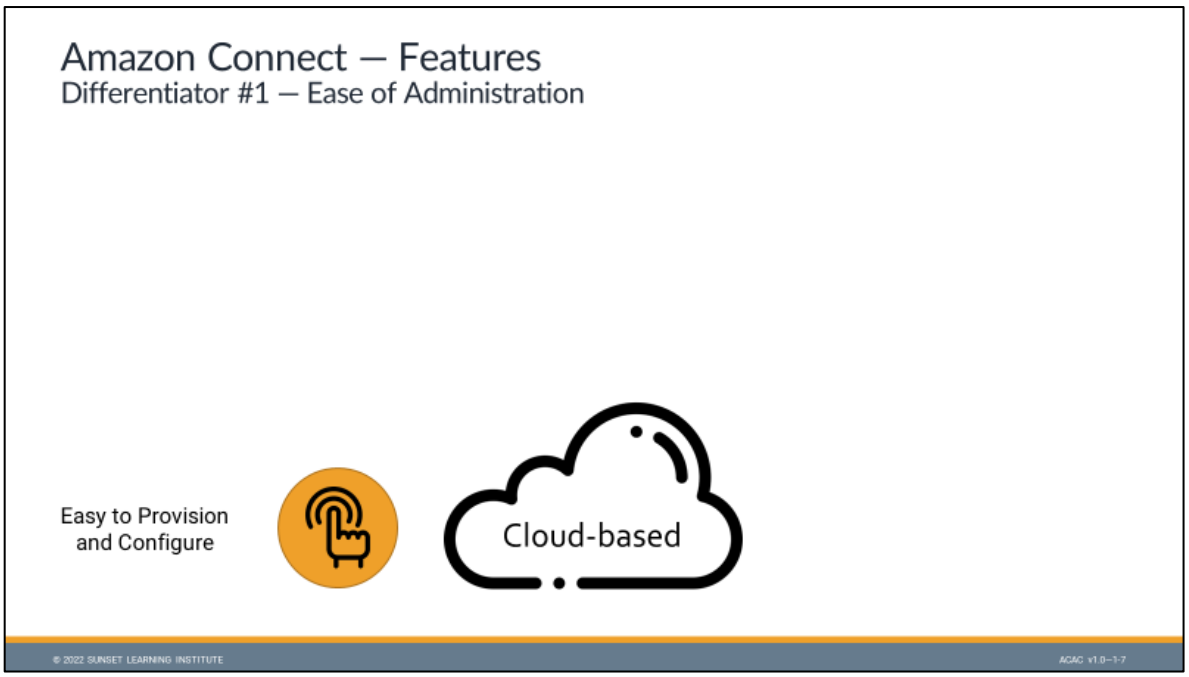
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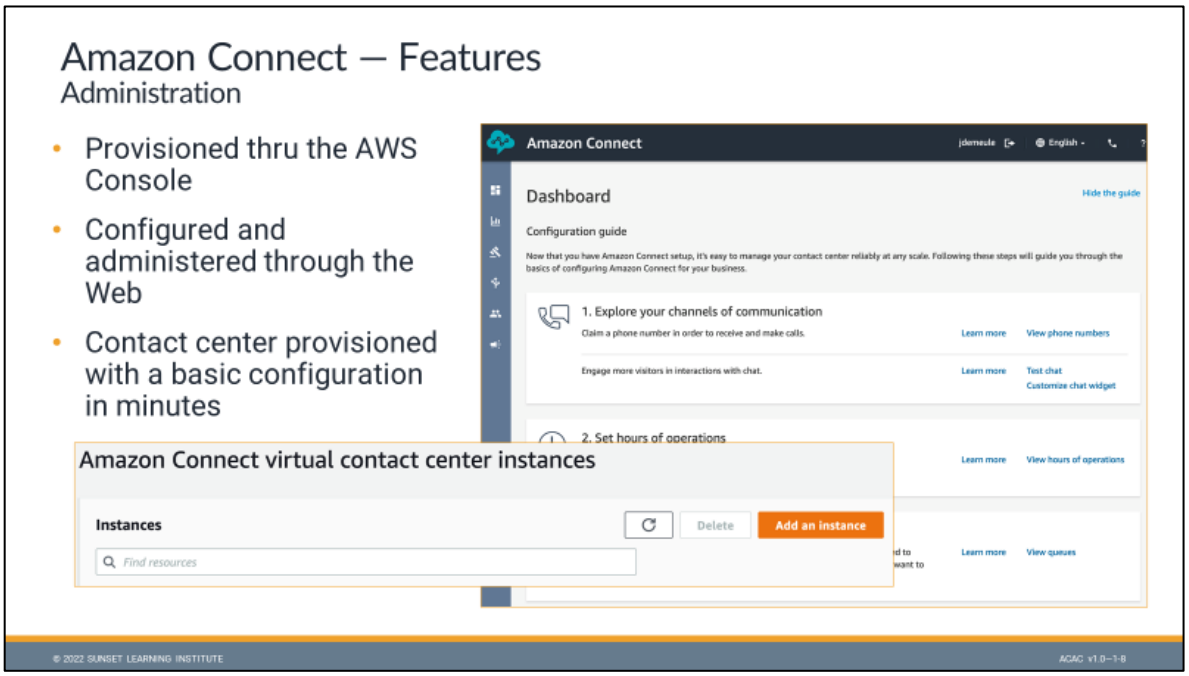
For the most up to date guidance, refer to https://docs.aws.amazon.com/connect/latest/adminguide/browsers.html.



For the most up to date guidance, refer to https://docs.aws.amazon.com/connect/latest/adminguide/supported-languages.html.

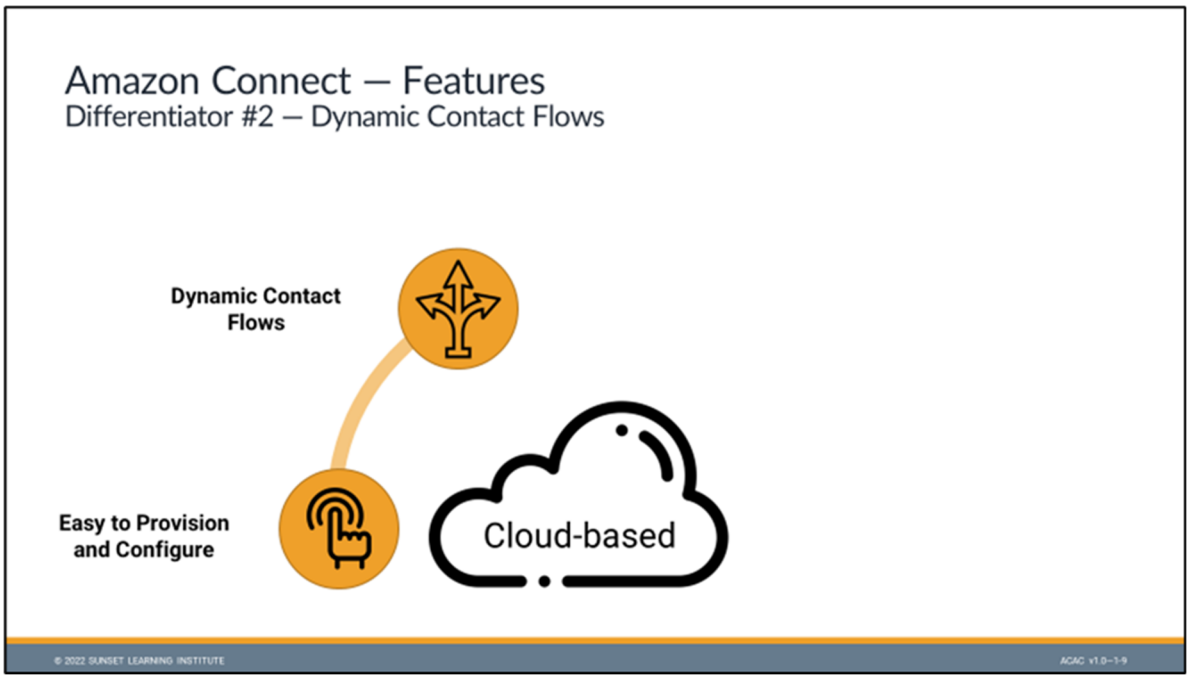


Amazon Connect is simple to set up and use; with only a few clicks you can set up a virtual Contact Center in minutes. **Remote agents can be trained in less than 20 minutes**. All they need is an Internet connection and a headset to provide the best possible customer service.



Use the AWS console to provision and instance of Amazon Connect which takes just a few minutes. Then use the Connect Admin Console to configure and administer the instance. An initial basic configuration takes just a few minutes. You start with nothing and can be taking a call in less than 10 minutes.

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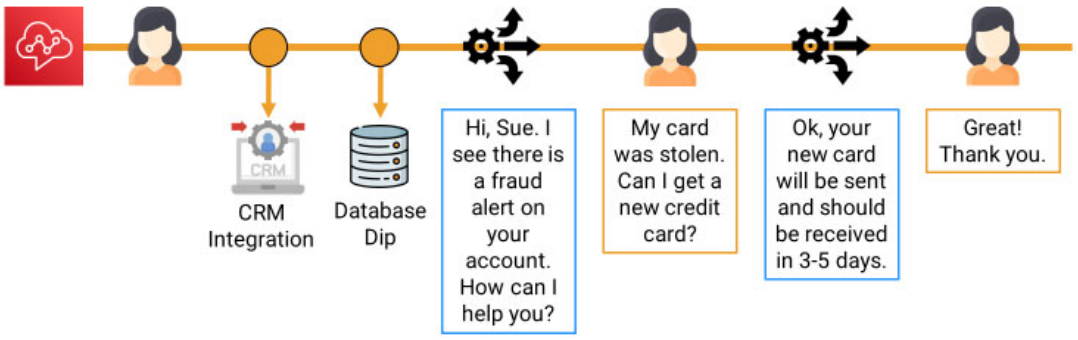
Amazon Connect Contact Flow Editor allows you to configure IVR settings easily through a graphical interface. Dynamic configuration comes in handy for more complex use cases. That way, it's possible to manage the setup through a central place, usually a database.

A good example of that is international contact flows. Let's say you want exactly the same customer experience in five or more different languages. You can leverage a single Amazon Connect Contact Flow for that and load the configuration dynamically based on language preference.

Currently, you can configure the following resources dynamically in a Contact Flow:

- Text-to-speech prompt
- Pre-recorded audio prompt
- Menu options
- Queue
- Contact flow transfer
- Hold flow
- Whisper flow
- Contact attribute
- Outbound caller ID

Amazon Connect — Features
Dynamic Contact Flows



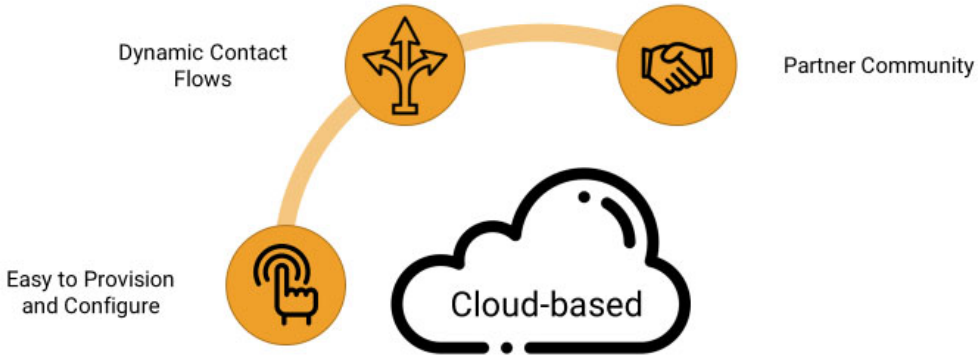
Dynamic
Anticipate customers' needs or questions before they ask

Personal
Contact flows can adapt on a per customer basis

Natural
Lex Chatbots use the same technology as Alexa

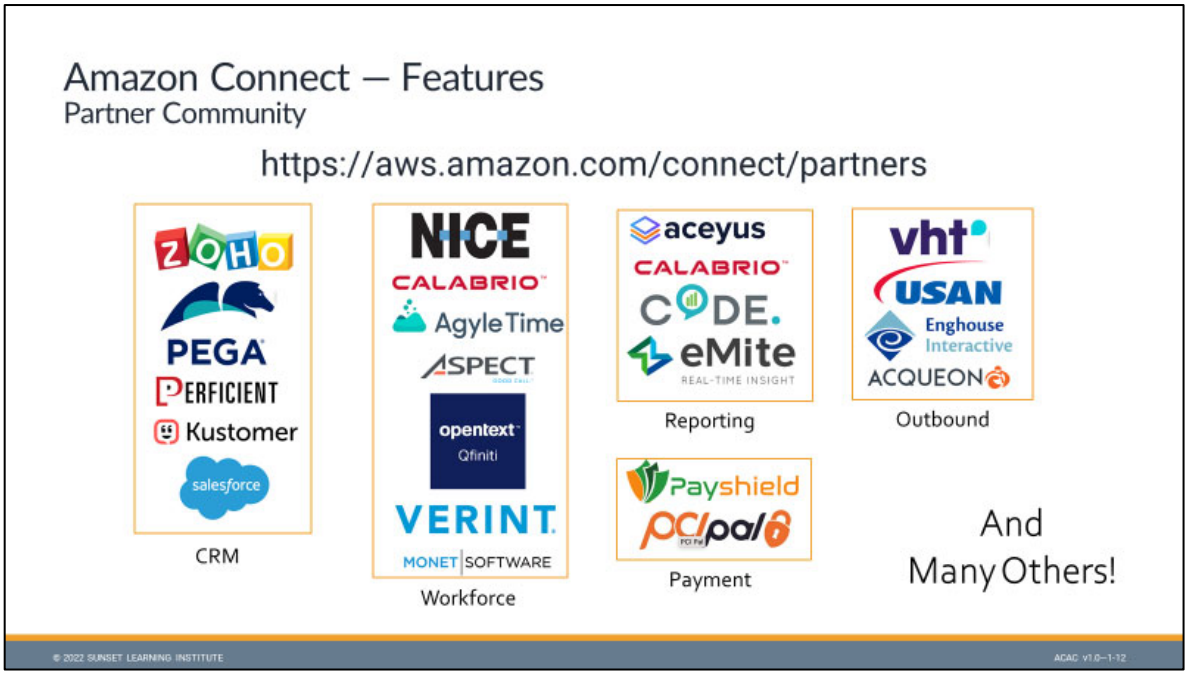
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Amazon Connect — Features
Differentiator #3 — Access to the AWS Partner Community



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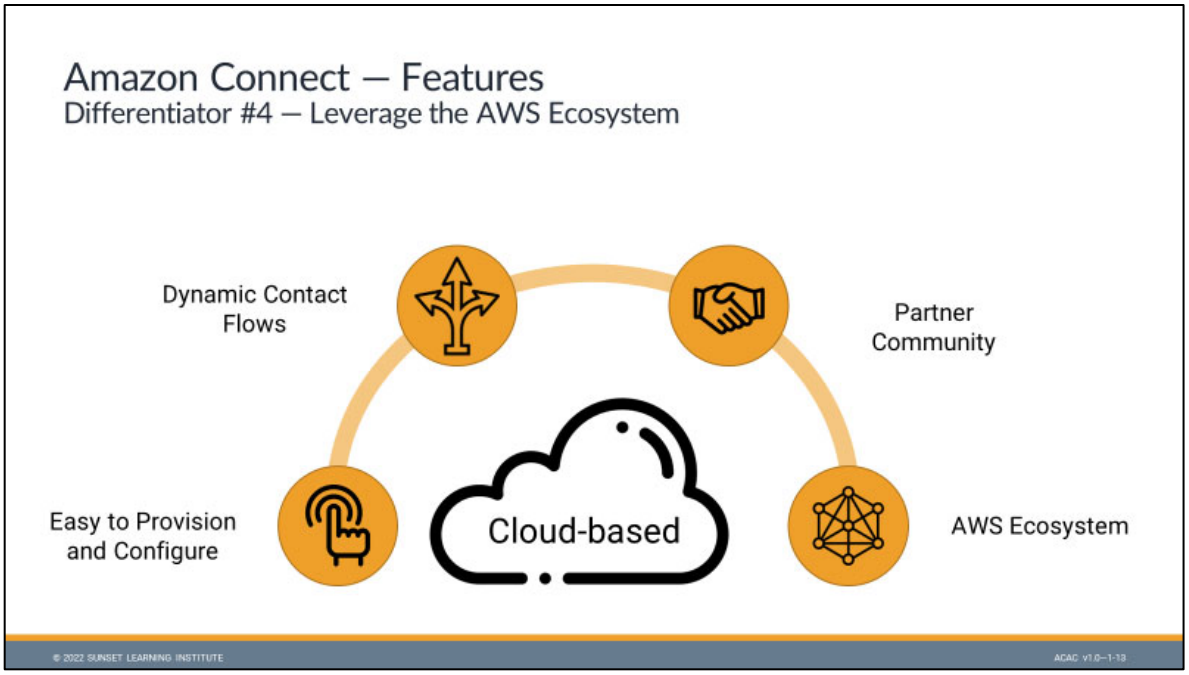
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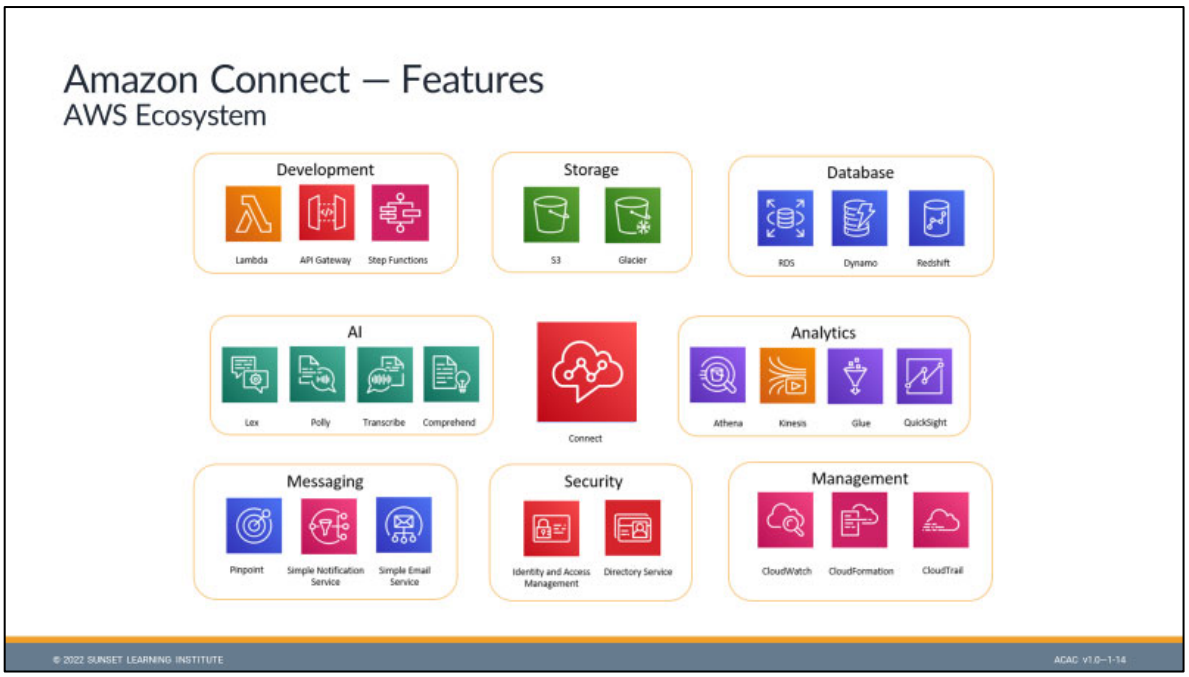


Amazon Connect Delivery Partners are AWS Partners who help companies build Cloud Contact Centers with Amazon Connect. By leveraging Amazon Connect, these AWS Partners can help you to improve customer experiences and outcomes at a lower cost.

For more information, refer to https://aws.amazon.com/connect/partners.

The **AWS Service Delivery Program** enables AWS customers to identify AWS Partners with experience and a deep understanding of specific AWS services. These AWS Partners follow best practices for AWS services and have proven success delivering AWS services to customers. You administer your own system and have the freedom to work with any of our partners to innovate your customer experience.





Development

You can use AWS Lambda functions to either look up or post data to sources outside of Amazon Connect. For example, you can look up an inbound caller on Salesforce based on the customer's phone number. The function may return such results as the customer's name, membership level (for example, frequent flyer), last order, and order status. Then based on that information, the call can be routed to an Amazon Lex bot or an agent.

You can also use Lambda with AWS databases like DynamoDB to create dynamic routing abilities. For example, you can retrieve a prompt in a specific language, based on input from the customer.

API Gateway and Step Functions further enhance the abilities of Lambda.

Storage

Amazon Connect uses Amazon Simple Storage Service (Amazon S3) to store recorded conversations and exported reports. When you set up Amazon Connect, it creates default buckets for these requirements, or you can point it to existing Amazon S3 infrastructure.

Database

You can use AWS databases with Amazon Connect for a variety of reasons. For example, with DynamoDB, you can create quick tables of data.

You can also create tables of dynamic information for call routing. For example, a Lambda function can write inbound calls to a DynamoDB table and then query the table to see if there are other matches for the phone number. If so, a decision can be made to send the caller to the same queue as before, or to flag that caller as a repeat caller.

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Analytics

Amazon Connect tracks all interactions using **contact records**. Contact records are used for real-time and historical metrics reports. You can also use Amazon Kinesis to stream them to an AWS database like Amazon Redshift or Amazon Athena for BI analysis (Amazon QuickSight, or a third party such as Tableau). AWS CloudFormation templates are available to set up this functionality for Amazon Redshift and Athena.

To perform analysis on your contact flow logs, you can set up an Amazon Kinesis stream to stream your contact flow log data from CloudWatch to a data warehouse service, such as Amazon Redshift. You can combine the contact flow log data with other Amazon Connect data in your warehouse, or run queries to identify trends or common issues with a contact flow.

Machine Learning (ML) and Artificial Intelligence (AI)

Amazon Connect uses the following services for ML/AI:

- **Amazon Lex**: Lets you create a chatbot to use as Interactive Voice Response (IVR). For more information, see *Add an Amazon Lex bot*.
- **Amazon Polly**: Provides text-to-speech in all contact flows. For more information, see *Add text-to-speech to prompts* and *SSML tags supported by Amazon Connect*.
- **Amazon Transcribe**: Grabs conversation recordings from Amazon S3 and transcribes them to text so you can review them.
- **Amazon Comprehend**: Takes the transcription of recordings, and applies speech analytics machine learning to the call to identify sentiment, keywords, adherence to company policies, and more.

Messaging

Amazon Connect uses the following services for messaging:

- **Amazon Pinpoint**: Use as an outbound messaging trigger for events; for example, bulk messaging (such as outbound marketing campaigns).
- **Amazon Simple Notification Service (Amazon SNS)**: Use to send and receive SMS and other channel notifications. Amazon SNS is particularly useful for sending alerts and validations.
- **Amazon Simple Email Service (Amazon SES)**: Use to send validation e-mails, such as a password reset bot sending a confirmation of the transaction.

Security

Amazon Connect uses the following services for added security:

- **AWS Identity and Access Management (IAM)**: Use to manage permissions for users. Amazon Connect users require permission for services.
- **AWS Directory Service**: Amazon Connect supports user federation through the internal directory (created in the Amazon Connect instance), using Active Directory integration (MAD, ADFS) or SAML 2.0.

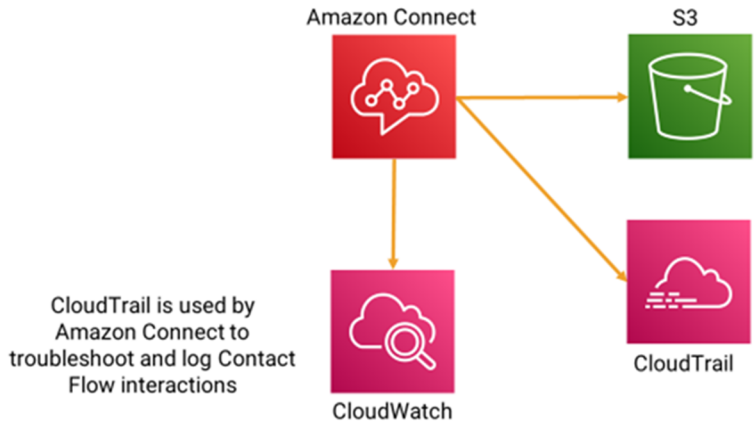
Management

Amazon Connect uses the following services for monitoring usage:

- **Amazon CloudWatch**: Collects logs, service metrics, and performance metrics for Amazon Connect.
- **AWS CloudFormation**: Amazon Connect does not support this directly for creating instances. However, it does support AWS CloudFormation templates for associated services, like integrations, database export, and so on.
- **AWS CloudTrail**: Provides a record of Amazon Connect API calls. This is especially useful for tracking who accessed recorded conversations.

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Amazon Connect — Features
AWS Ecosystem Example 1



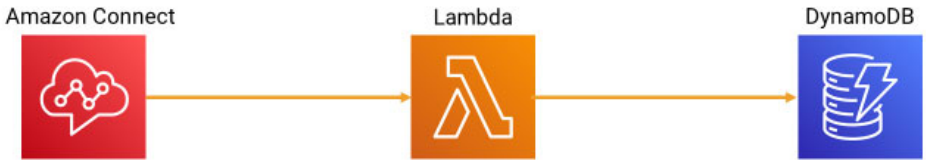
S3 is used by Amazon Connect to store voice recordings and chat transcripts

CloudTrail is used by Amazon Connect to track API calls and recording access

CloudTrail is used by Amazon Connect to troubleshoot and log Contact Flow interactions

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Amazon Connect — Features
AWS Ecosystem Example 2

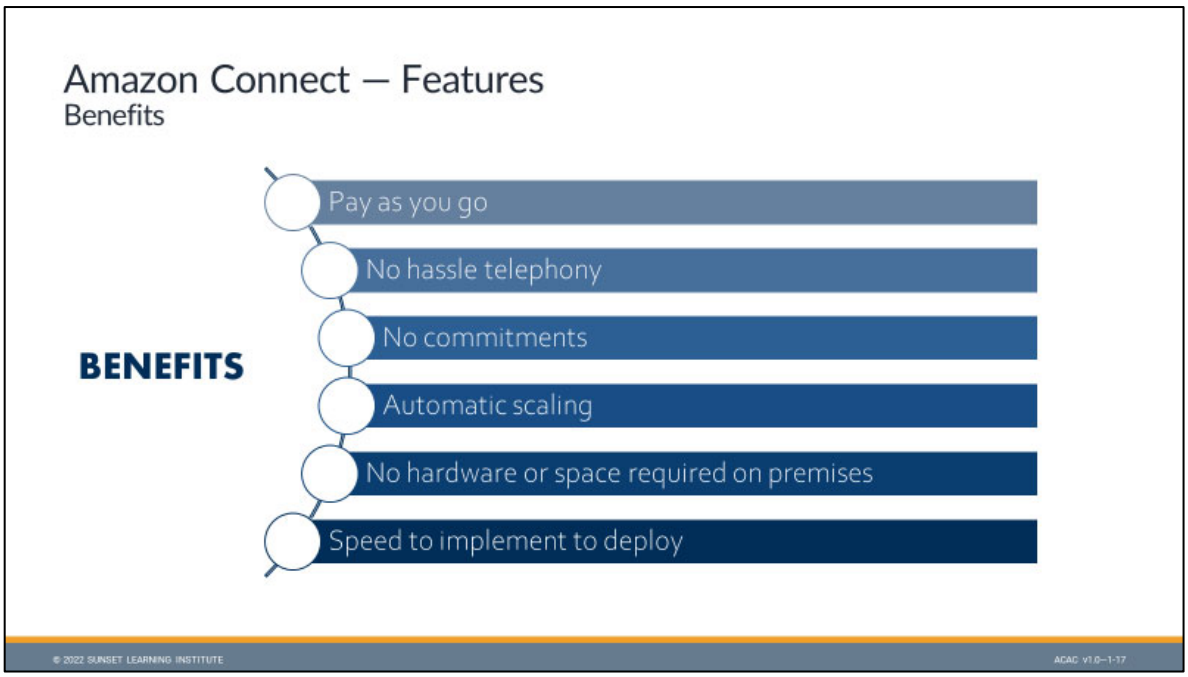


Amazon Connect may wish to write or query some data to a database so it sends a trigger to a Lambda function

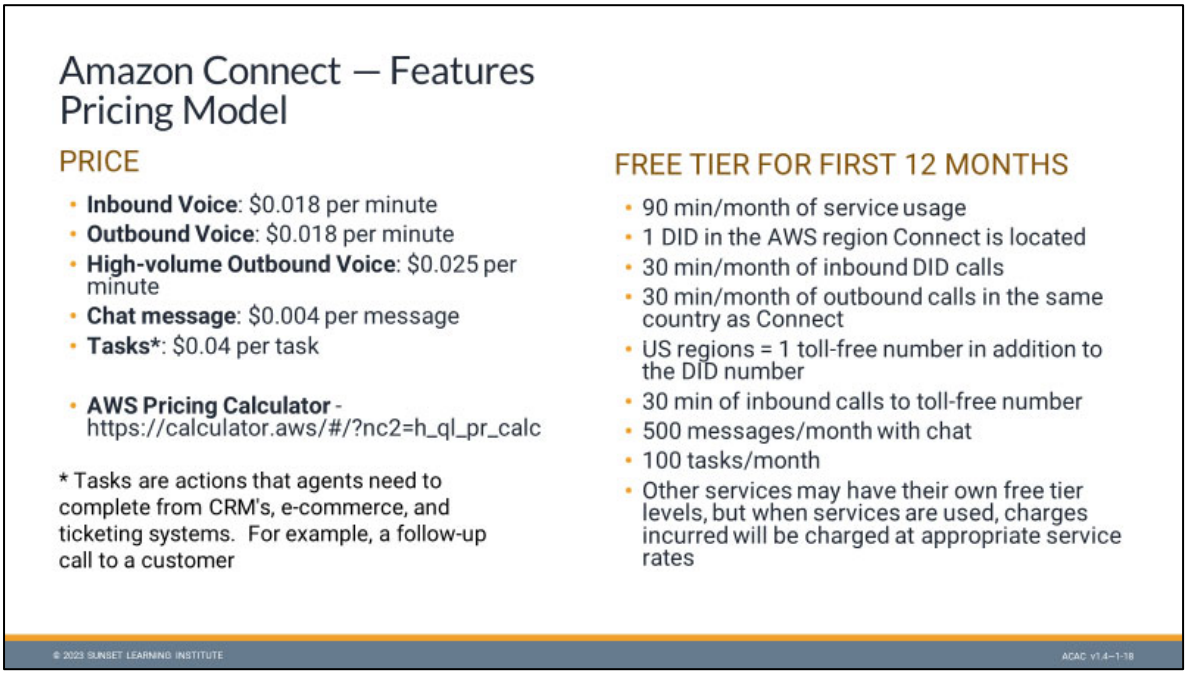
The Lambda function runs some code that will connect to the DynamoDB and perform the required query or write action. A response will then be delivered to the Amazon Connect containing a confirmation or data

The DynamoDB is queried or written to and sends a confirmation or data to the Lambda function

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Voice

- **Inbound Voice usage**: $0.018 per minute
- **Outbound Voice usage**: $0.018 per minute
- **High-volume Outbound Voice usage**: $0.025 per minute

AWS provides tools to assist with budget planning. The AWS Pricing Calculator, **https://calculator.aws/#/?nc2=h_ql_pr_calc**, can be used to evaluate different scenarios to determine the cost to use various services. Within the budget planning tools, administrators can create alerts based on various usage and cost levels to provide notifications before a budget is exceeded.

For **Amazon Connect Voice**, there are two charges associated with usage: voice service charges and telephony charges.

- **Voice service usage**: You are billed on a per-second basis (minimum 10s). Amazon Connect Voice service usage for inbound calls as well as manual outbound calls dialed from the contact control panel (CCP) is determined by the seconds your end-customer is connected to the service. For outbound voice usage initiated by invoking the startOutboundVoiceContact API or high-volume outbound voice usage, you are billed on a per-second basis (minimum 10s) from the time the contact attempt is initiated by Connect.
- **Telephony charges**: Amazon Connect offers direct inward dial (DID) and toll-free phone numbers in more than 20 countries worldwide, which are charged on a per-day rate. Inbound calls are charged per-second (minimum 60s). Also 200+ outbound calling destinations are available that charge per-second (minimum 60s) for each interaction. To check for country-specific pricing, refer to the tables below. Taxes, surcharges, and fees may apply.

When you use an uploaded contact list to schedule and manage high-volume outbound communications, you pay a Monthly Targeted Audience (MTA) charge for processing a phone number, known as an **endpoint**, in an attempt to reach a customer and associated costs for usage of the channel. (Just uploading a

contact address does not incur any charges.) MTA is defined as the number of unique phone numbers that were processed in a month †.

Inbound Pricing Example

An end customer call is using an Amazon Connect German direct inward dial (DID) number in the EU (Frankfurt) region, answered by an agent on the Amazon Connect softphone. The call lasts seven minutes.

There are three separate charges that apply for this call:

- Amazon Connect service usage charge, based on end-customer call duration. At $0.018 per minute, 7 minutes = $0.126.
- The day charge for use of the German DID number. At $0.10 per day, 1 day = $0.10.
- An inbound call per minute charge for German DID numbers. At $0.0040 per minute, 7 minutes = $0.0280.

So, the total for this call is $0.25 (plus applicable taxes, fees, and surcharges).

High-Volume Outbound Pricing Example

An Amazon Connect Contact Center in US East (N. Virginia) wants to notify all its customers about the change in its terms of use. This notification is a pre-recorded message which will be played irrespective of whether a call is answered by a live person or an answering machine. 25% of all contact attempts connect to a live customer or an answering machine, and each contact lasts 30 seconds. The bill for this communication using high-volume outbound communications would be calculated at the end of the month in US East (N. Virginia) as follows:

- † Monthly Targeted Audience (**MTA**) Charges:
  - 10,000 endpoints targeted in the month would cost $1.20 per 1,000 endpoints after the first 5,000 endpoints. Total = $6.00.
- Voice Service Charges:
  - Number of contact attempts made = 10,000
  - At a connection rate of 25%, number of calls that result in the recorded message being played = 2,500.
  - With a single call duration of 30 seconds, the minutes of voice service usage for connected calls = 30 seconds. 30 seconds ✳ 2,500 messages = 1,250 minutes
  - Total billable voice service usage for connected calls less the 90 minutes of Amazon Connect free tier usage per month = 1,250 minutes. 1,250 minutes – 90 minutes = 1,160 minutes
  - Since each contact attempt is charged for a minimum of 10 seconds, minutes of voice service usage for unsuccessful contact attempts = 7,500. 7,500 ✳ 10/60 = 1,250 minutes
  - Total voice service charges for the notification campaign = (1,160 minutes + 1,250 minutes) ✳ $0.025 per minute. **Total** = $55.43
- Telephony Charges:
  - Assume that you have claimed a US toll-free number to make these calls, and all your calls were completed within a single day.
  - Day charge for the use of the US toll-free number = $0 for free tier usage in the first 12 months. $0.06 per day after that.
  - Total minutes of connected calls = 2,500 calls that connect to a voicemail or live customer ✳ 30 seconds = 1,250 minutes

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- Outbound call charges for calls that connected, less the 30 minutes of free tier outbound telephony usage per month = $0.0048 outbound call charges per minute ✳ (1,250 minutes of connected calls – 30 minutes of free tier). **Total** = $5.86

For the month, your bill will be $67.29, a total that includes $6.00 for Pinpoint MTA charge to process your endpoints through Pinpoint list orchestration, $55.43 for Connect voice service usage, $5.86 for Connect telephony charges, and the complete usage of your free tier.

Chat

Chat usage: $0.004 per message

Pricing example
An end-customer sends a message to begin a chat through your website. The chat conversation occurs over multiple hours with two different agents, since the end-customer stopped responding and then later resumed the chat when the first agent was no longer online. During the chat, the customer sent nine messages and the agents sent 15 messages. There is one charge for this chat:

- Amazon Connect Chat charge, based on the total messages sent during the chat (including initial message when chat was started). At $0.004 per message ✳ (9 customer messages + 15 agent messages) = $0.096
- So, the total for this chat is $0.096 (plus applicable taxes, fees, and surcharges).

AWS Free Tier

As part of the **AWS Free Tier**, you can get started with Amazon Connect for free.

For the first 12 months of your initial Amazon Connect Cloud Contact Center deployment in any region, you receive:

- 90 minutes per month of Amazon Connect service usage
- A direct inward dial (DID) number from the country in which the AWS region is located
- 30 minutes per month of inbound DID calls
- 30 minutes per month of outbound calls made to numbers in the country in which the AWS region is located

For US regions, you also receive:

- One US toll-free number
- 30 minutes per month of inbound toll-free calls made from the US

For example: If you create your instance in the Asia Pacific (Sydney) region, you get one DID number from Australia, receive 30 minutes of inbound DID calls to that number, and can use Amazon Connect to place 30 minutes of outbound calls to numbers in Australia.

Other Amazon Connect capabilities that are also part of the Free Tier during your first year include:

- 500 messages per month with Chat
- 100 tasks per month with Tasks
- 1,000 profiles per month with Customer Profiles
- 90 minutes per month for audio calls with Contact Lens
- 180 transactions per month with Voice ID

For more information on Amazon Connect pricing refer to https://aws.amazon.com/connect/pricing/. Additionally, the AWS Pricing Calculator can assist with pricing questions for other AWS services and can be found at https://calculator.aws/#/?nc2=h_ql_pr_calc.

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Module 2

Instantiating and Configuring the Amazon Connect

Overview

This module provides an overview of the process and key tasks for successfully creating and configuring an AWS account, an IAM user, and an Amazon Connect Instance using AWS management console.

Objectives

Upon completing this module, you will show an understanding of the basic premise of Amazon Connect. This ability includes being able to meet this objective:

- Identify the essential requirements and steps to successfully create an Amazon Connect Instance and configure basic settings using the Amazon Connect dashboard.

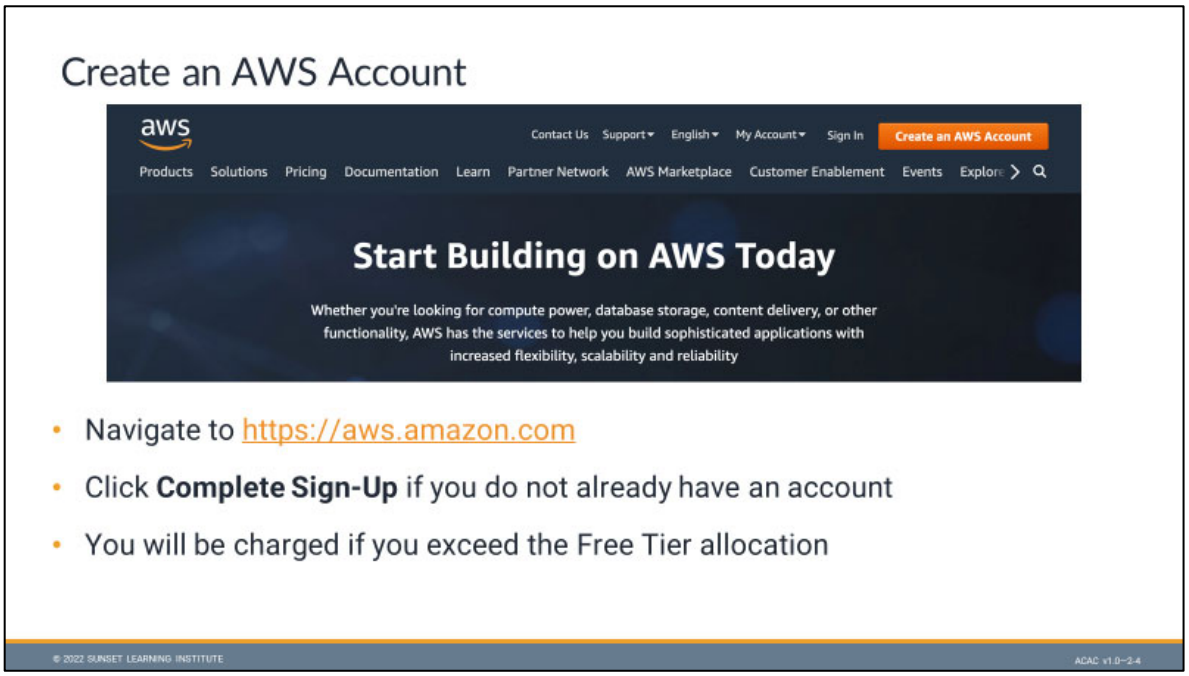
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IAM and Console Access

In order to instantiate and Amazon Connect Instance, you need to create an AWS account and an IAM User.

Create an AWS Account

When setting up a new AWS account, you need to create the root user and create additional AWS IAM users to log into the account. The root user is a special account with full access to the account and can perform all actions, including the ability to change the payment methods or close the account. This section covers creating an account for the root user.



Sign up using your email address:

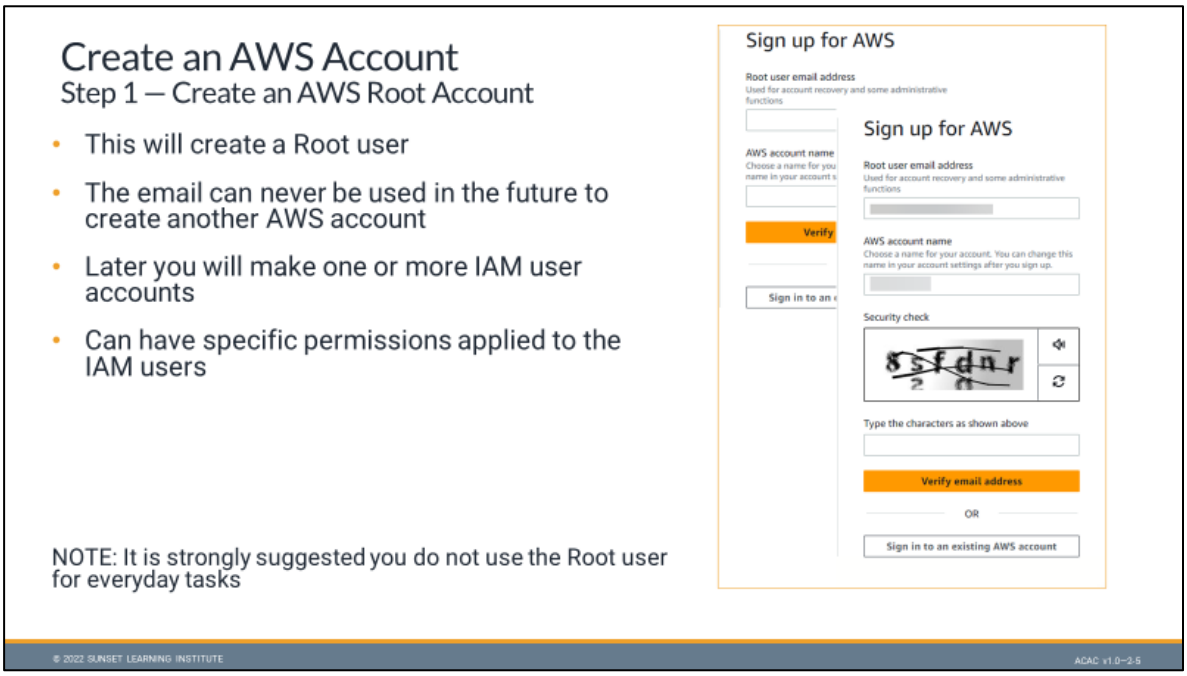
Step 1 Open the Amazon Web Services (AWS) home page.

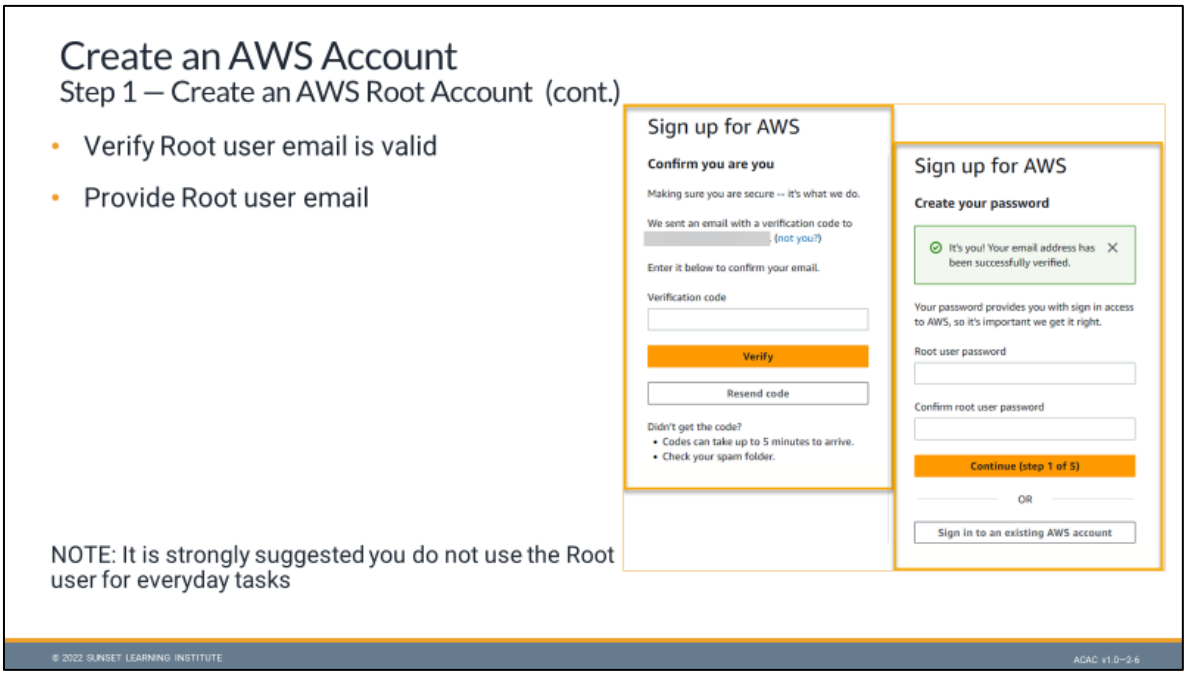
Step 2 Choose **Create an AWS Account**.

Note If you signed in to AWS recently, choose **Sign in to the Console**. If **Create a new AWS account** isn't visible, first choose **Sign in to a different account**, and then choose **Create a new AWS account**.

Step 3 Enter your email address, password, AWS account name, and then choose **Continue**. Be sure that you enter your account information correctly, especially your email address. If you enter your email address incorrectly, you can't access your account.

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There are two different types of users in AWS. You are either the account owner (root user), or you are an AWS Identity and Access Management (IAM) user. The root user is created when the AWS account is created, and IAM users are created by the root user or an IAM Administrator for the account. All AWS users have security credentials.

Tasks that require root user credentials:

We recommend that you use an IAM user with appropriate permissions to perform tasks and access AWS resources. However, you can perform the tasks listed below only when you sign in as the root user of an account.

Tasks

Step 1 **Change your account settings.** This includes the account name, email address, root user password, and root user access keys. Other account settings, such as contact information, payment currency preference, and Regions, do not require root user credentials.

Step 2 **Restore IAM user permissions.** If the only IAM Administrator accidentally revokes their own permissions, you can sign in as the root user to edit policies and restore those permissions.

Step 3 **Activate IAM access to the Billing and Cost Management console.**

Step 4 **View certain tax invoices.** An IAM user with the aws-portal:ViewBillingpermission can view and download VAT invoices from AWS Europe, but not AWS Inc or Amazon Internet Services Pvt. Ltd (AISPL).

Step 5 **Close your AWS account.**

Step 6 **Change** your AWS Support plan **or Cancel** your AWS Support plan. For more information, see **IAM for AWS Support**.

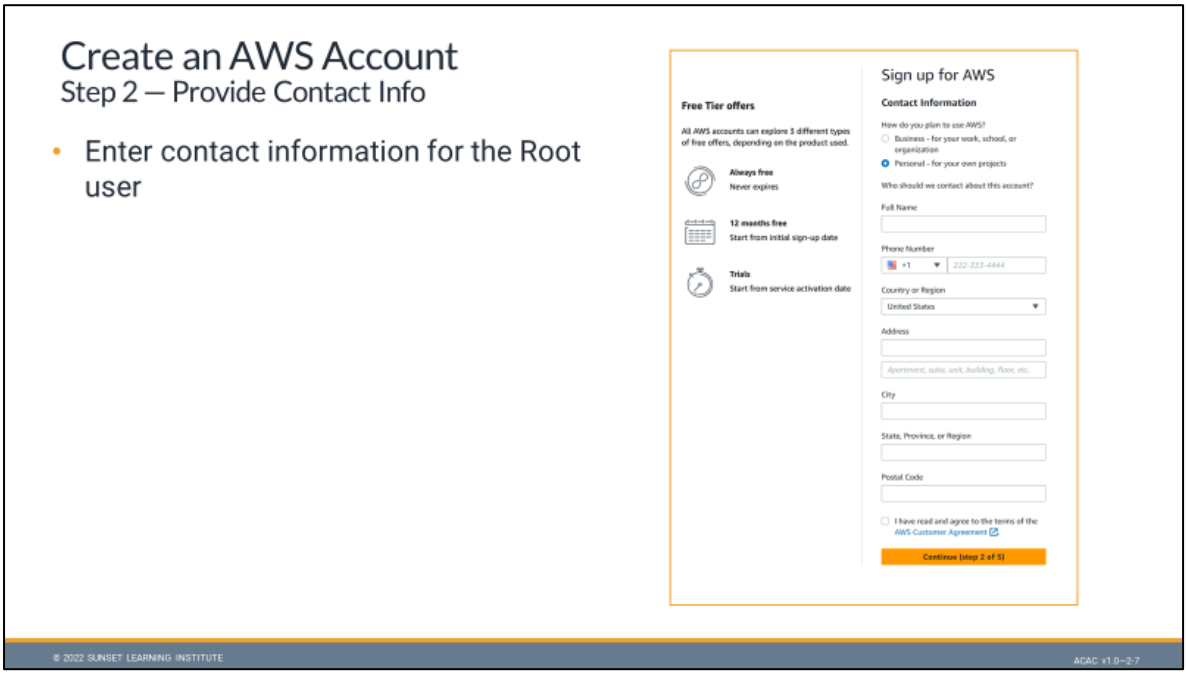
Step 7 **Register as a seller** in the Reserved Instance Marketplace.

Step 8 **Configure** an Amazon S3 bucket to enable **MFA** (multi-factor authentication) **Delete**.

Step 9 **Edit** or **delete** an Amazon S3 bucket policy that includes an invalid VPC ID or VPC endpoint ID.

Step 10 **Sign up for GovCloud**.

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Add your contact information:

Step 1 Select Business or Personal.

Note	Personal accounts and business accounts have the same features and functions.

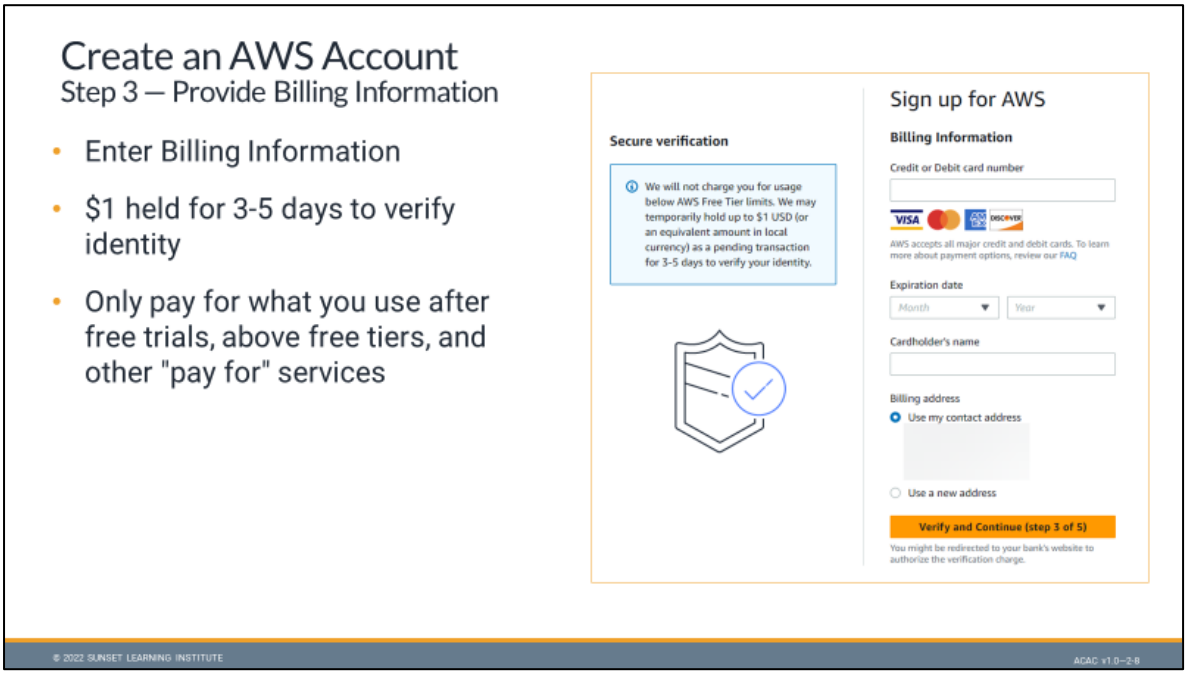
Step 2 Enter your company or personal information.

IMPORTANT: For business AWS accounts, it's a best practice to enter the company phone number rather than a personal cell phone. Configuring a root account with an individual email address or a personal phone number can make your account insecure.

Step 3 Read and **accept** the AWS Customer Agreement.

Step 4 Choose **Continue**.

Step 5 You will receive an email to confirm that your account has been created. You can sign in to your new account using the email address and password you registered with. However, you can't use AWS services until you finish activating your account.



Add a payment method:

Step 1 On the **Billing information** page, enter the information about your payment method.

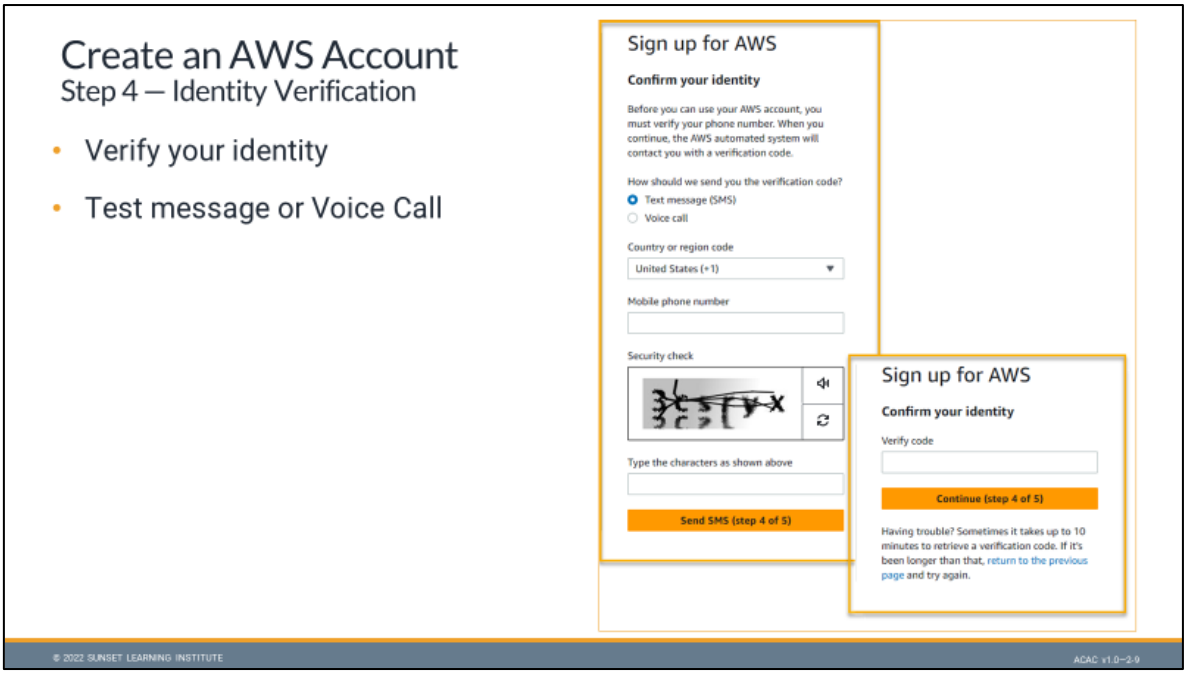
If you are signing up in India for an Amazon Internet Services Private Limited (AISPL) account, you must provide your CVV as part of the verification process. You might also have to enter a one-time password, depending on your bank. AISPL charges your Payment Method 2 Indian Rupees (INR) as part of the verification process. AISPL refunds the 2 INR after the verification is completed.

Step 2 If you want to use a different billing address for your AWS billing information, choose **Use a new address**.

Step 3 Then choose **Verify and Continue**.

Important: You can't proceed with the sign-up process until you add a valid payment method.

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Verify your phone number:

Step 1 On the **Confirm your identity** page, select a contact method to receive a verification code.

Step 2 Select your phone number, country, or region code from the list.

Step 3 Enter a mobile phone number where you can be reached in the next few minutes.

Step 4 If presented with a CAPTCHA, enter the displayed code and then submit.

Step 5 In a few moments, an automated system communicates with you.

Step 6 Enter the PIN you receive, and then choose **Continue**.

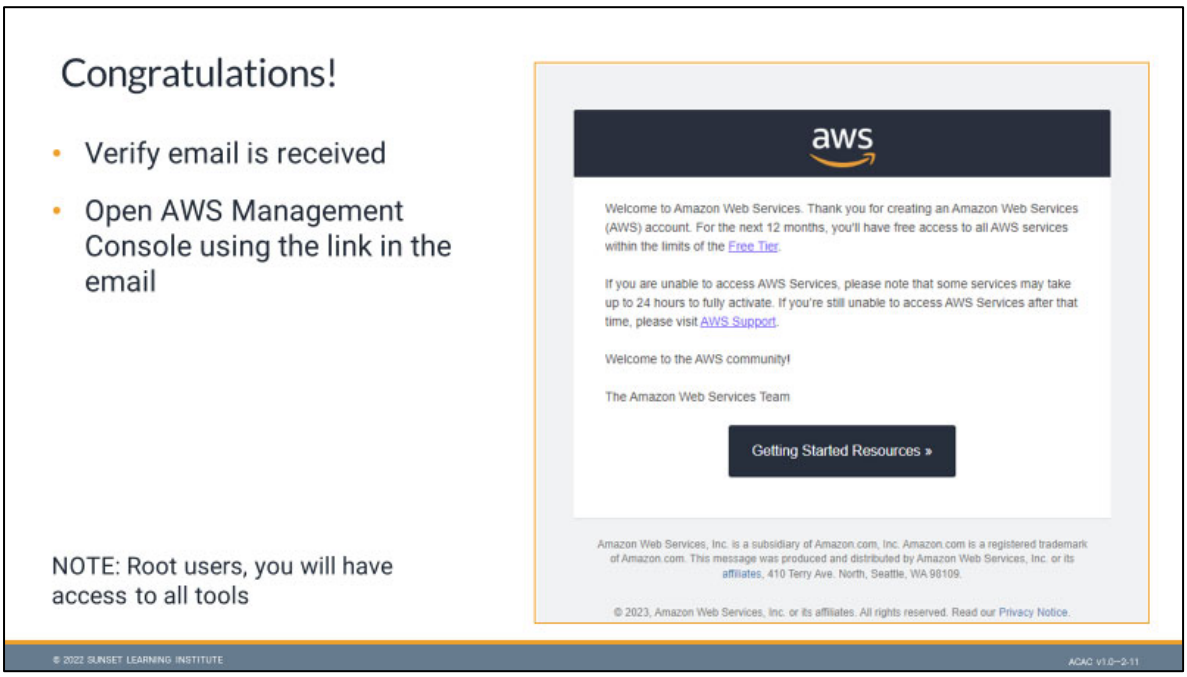


Choose an AWS Support plan:

Step 1 On the **Select a support plan** page, choose one of the available Support plans. For a description of the available Support plans and their benefits, see **Compare AWS Support plans**.

Step 2 Choose **Complete sign up**.

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Wait for account activation:

Step 1 After you choose a Support plan, a confirmation page indicates that your account is being activated. Accounts are usually activated within a few minutes, but the process might take up to 24 hours.

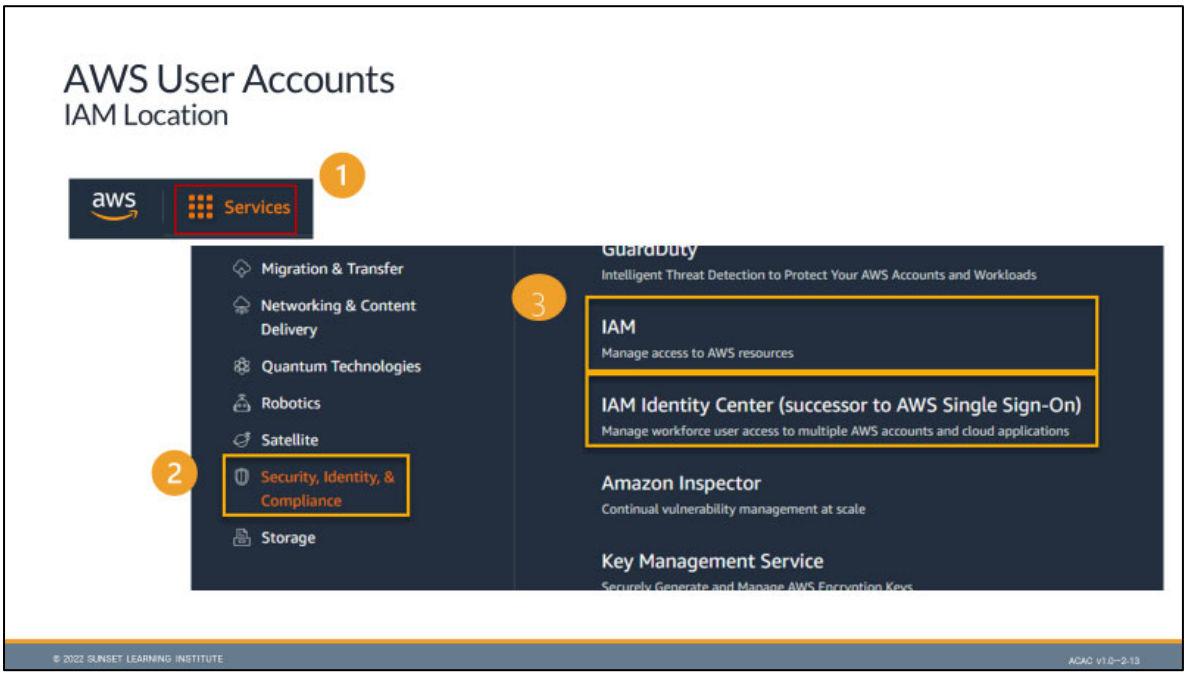
Step 2 You can sign in to your AWS account during this time. The AWS home page might display a **Complete Sign-Up** button during this time, even if you've completed all the steps in the sign-up process.

Step 3 When your account is fully activated, you receive a confirmation email. Check your email and spam folder for the confirmation email. After you receive this email, you have full access to all AWS services.

Create an IAM User

It is considered a security best practice not to use your root account for day-to-day use but instead, create separate users for specific roles and functions. Keep in mind that at this point, our focus is accounts to access and manage the AWS account, not the Amazon Connect instance.

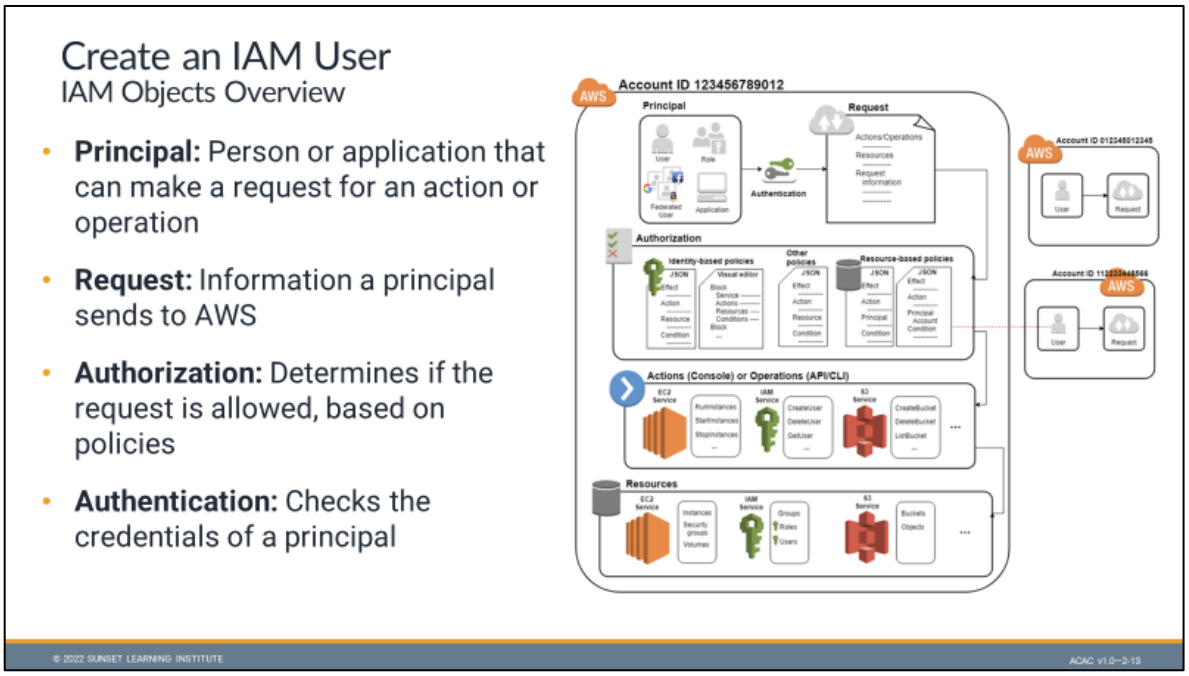
There are two methods to manage AWS accounts, IAM and IAM Identity Center. While IAM Identity Center is the recommended tool and will facilitate integration with your organizations Identity Provider (IdP). However, for class purposes combined with the fact that this is not for production use, our focus will be to use IAM.



As part of setting up an IAM User, you will first create an IAM User group. This will have a set of permissions that applies to any user that is part of the group. Lastly, you will create an IAM user. This section covers creating an IAM user.

Use the AWS Management Console to find and open the **Identity & Access Management** (IAM) **tool**. The AWS Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources. You use IAM to control who is authenticated (signed in) and authorized (has permissions) to use resources.

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First, to understand IAM Users, it is helpful to understand that IAM Users are referred to as a Principal. Principals may submit requests via the AWS Management Console, CLI, SDKs, or APIs.

Principal

A **principal** is a person or application that can make a request for an action or operation on an AWS resource. The principal is authenticated as the AWS account root user or an IAM entity to make requests to AWS.

As a best practice, do not use your root user credentials for your daily work. Instead, create IAM entities (users and roles). You can also support federated users or programmatic access to allow an application to access your AWS account.

Request

When a principal tries to use the AWS Management Console, the AWS API, or the AWS CLI, that principal sends a *request* to AWS. The request includes the following information:

- **Actions or operations:** The actions or operations that the principal wants to perform. This can be an action in the AWS Management Console, or an operation in the AWS CLI or AWS API.
- **Resources:** The AWS resource object upon which the actions or operations are performed.
- **Principal:** The person or application that used an entity (user or role) to send the request. Information about the principal includes the policies that are associated with the entity that the principal used to sign in.
- **Environment data:** Information about the IP address, user agent, SSL enabled status, or the time of day.
- **Resource data** – Data related to the resource that is being requested. This can include information such as a DynamoDB table name or a tag on an Amazon EC2 instance.

AWS gathers the request information into a *request context*, which is used to evaluate and authorize the request.

Authorization

You must also be authorized (allowed) to complete your request. During authorization, AWS uses values from the request context to check for policies that apply to the request. It then uses the policies to determine whether to allow or deny the request.

Most policies are stored in AWS as JSON documents and specify the permissions for principal entities. There are several types of policies that can affect whether a request is authorized.

To provide your users with permissions to access the AWS resources in their own account, you need only identity-based policies. Resource-based policies are popular for granting cross-account access. The other policy types are advanced features and should be used carefully.

AWS checks each policy that applies to the context of your request. If a single permissions policy includes a denied action, AWS denies the entire request and stops evaluating. This is called an **explicit deny**. Because requests are **denied by default**, AWS authorizes your request only if every part of your request is allowed by the applicable permissions policies.

The evaluation logic for a request within a single account follows these general rules:

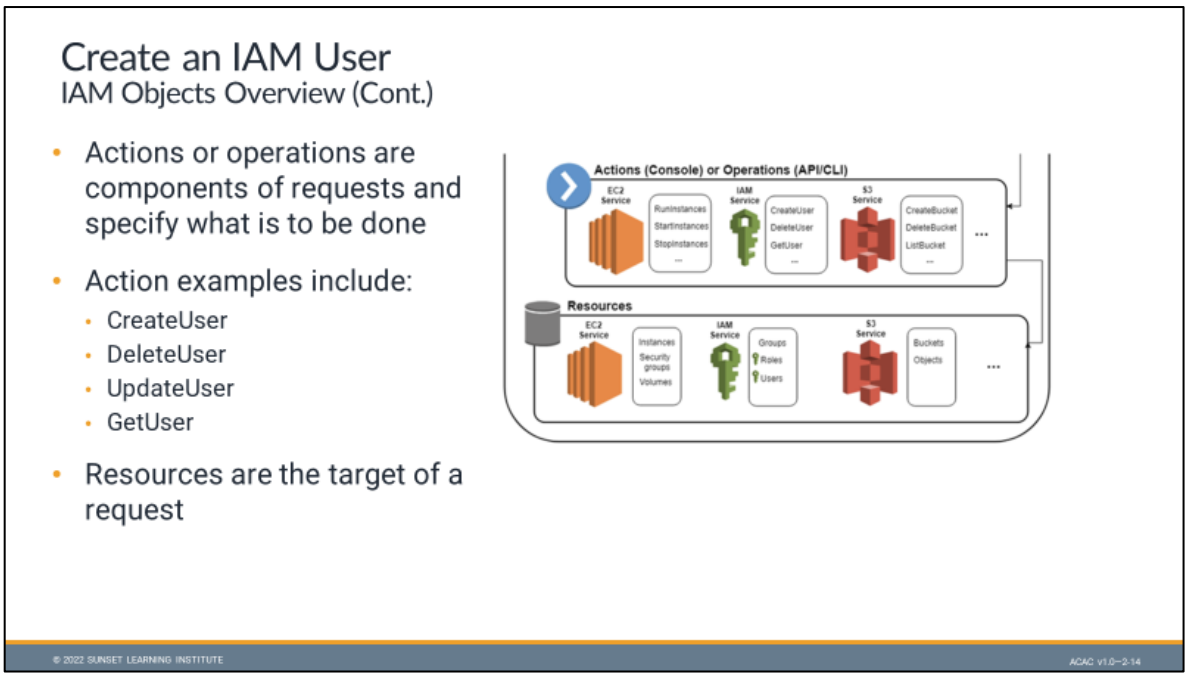
- By default, all requests are denied. (In general, requests made using the AWS account root user credentials for resources in the account are always allowed.)
- An explicit allow in any permissions policy (identity-based or resource-based) overrides this default.
- The existence of an Organizations SCP, IAM permissions boundary, or a session policy overrides the allow. If one or more of these policy types exists, they must all allow the request. Otherwise, it is implicitly denied.
- An **explicit deny** in any policy overrides any allows.
- If you need to make a request in a different account, a policy in the other account must allow you to access the resource, and the IAM entity that you use to make the request must have an identity-based policy that allows the request.

Authentication

A principal must be authenticated (signed in to AWS) using the credentials to send a request to AWS. Some services, such as Amazon S3 and AWS STS, allow a few requests from anonymous users. However, they are the exception to the rule.

- To authenticate from the console as a root user, you must sign in with your email address and password.
- As an IAM user, provide your account ID or alias and then your username and password.
- To authenticate from the API or AWS CLI, you must provide your access key and secret key.
- You might also be required to provide additional security information. For example, AWS recommends that you use multi-factor authentication (MFA) to increase the security of your account.

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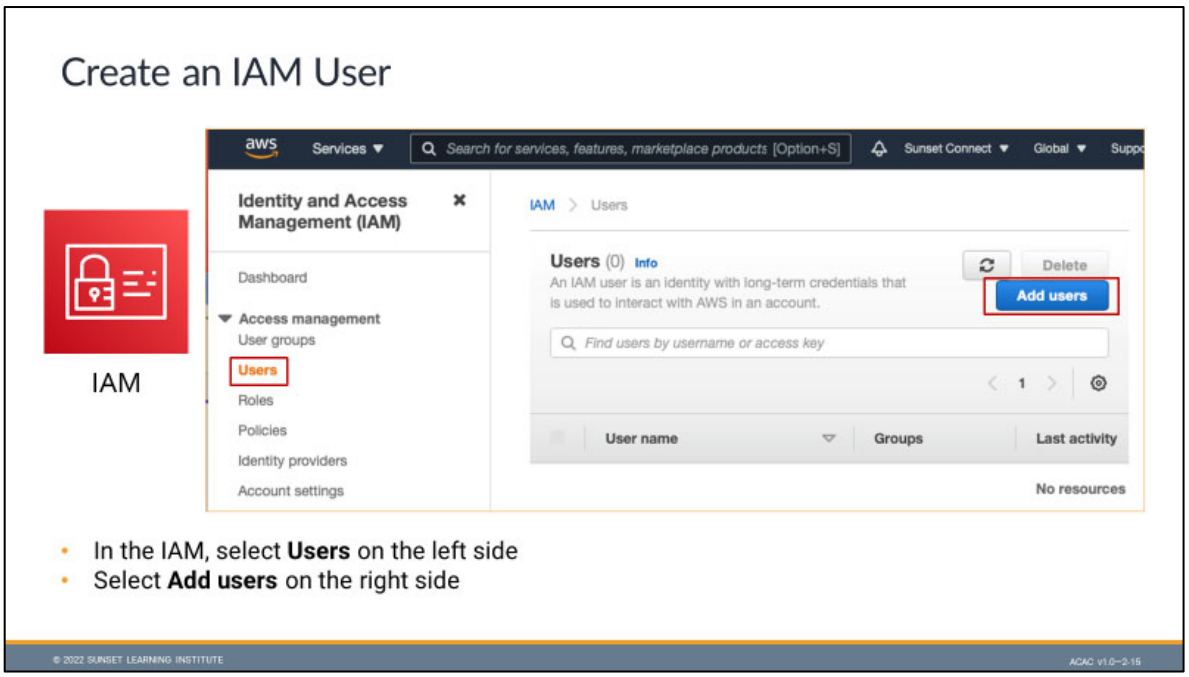
Actions or operations

After your request has been authenticated and authorized, AWS approves the actions or operations in your request.

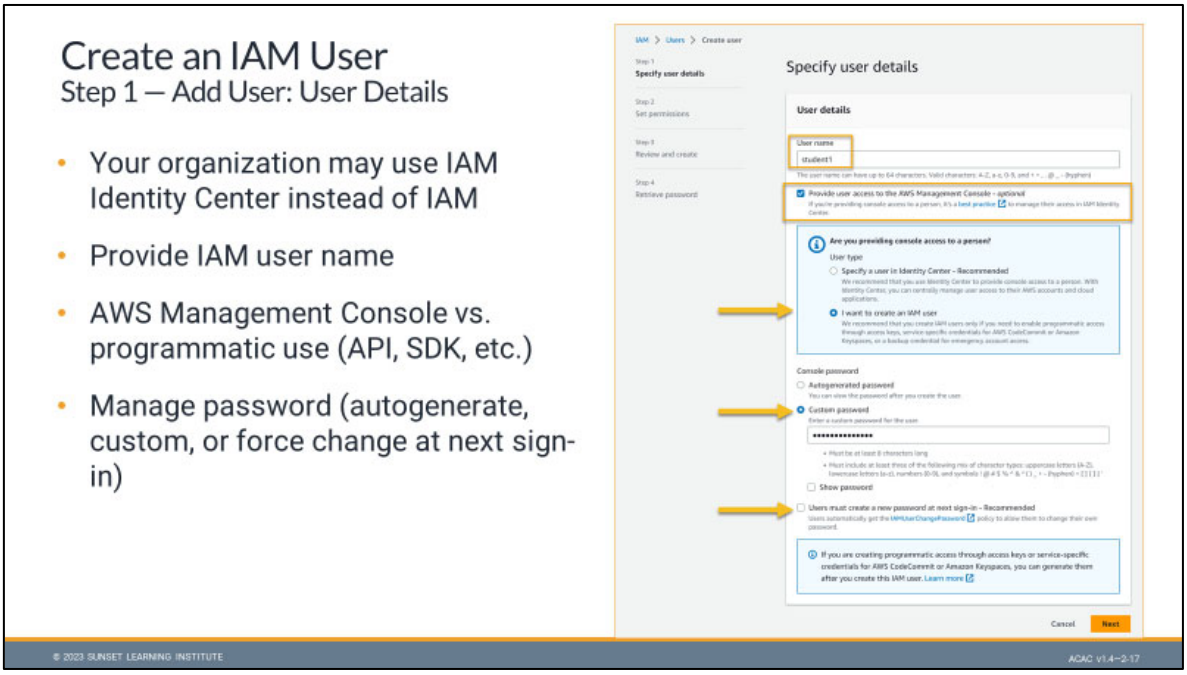
Operations are defined by a service and include things that you can do to a resource, such as viewing, creating, editing, and deleting that resource. For example, IAM supports approximately 40 actions for a user resource, including the following actions:

- CreateUser
- DeleteUser
- GetUser
- UpdateUser

To allow a principal to perform an operation, you must include the necessary actions in a policy that applies to the principal or the affected resource.



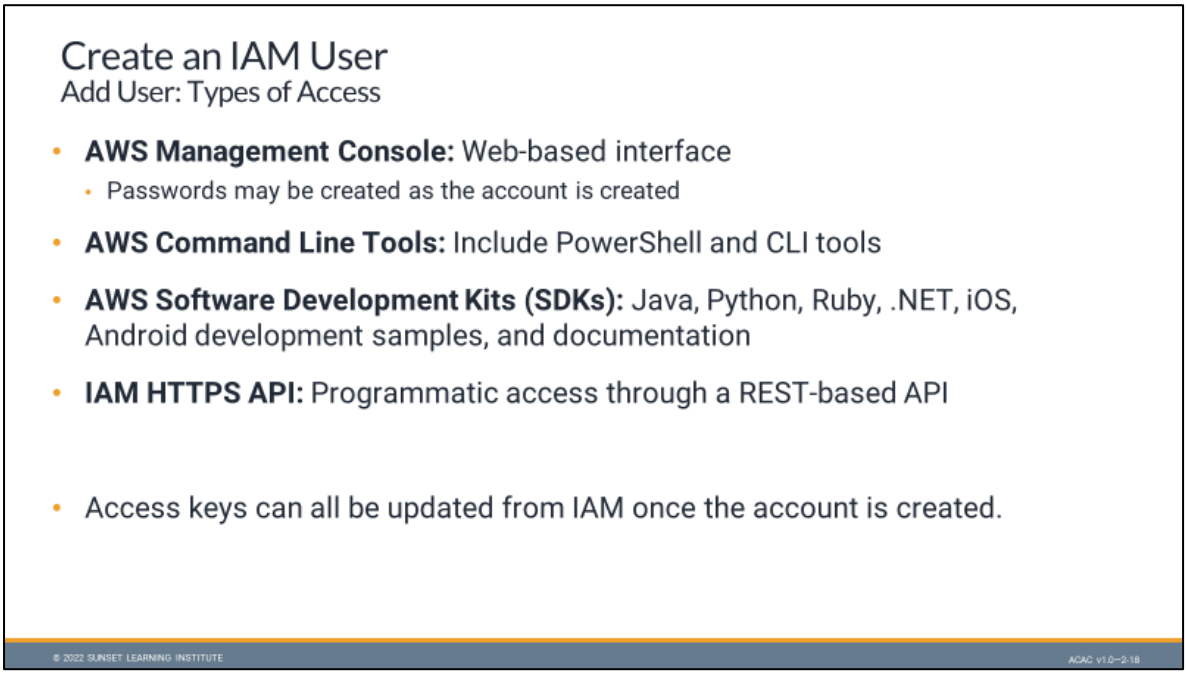
In the navigation pane, choose **Users** and then choose **Add users**.



Provide an IAM user name. Note which characters are valid. Select **Provide user access to the AWS Management Console**. Since we are not creating this account in IAM Identity Center, select **I want to create an IAM user**. Define the options for password, whether Autogenerated, Custom, or update at next sign-in).

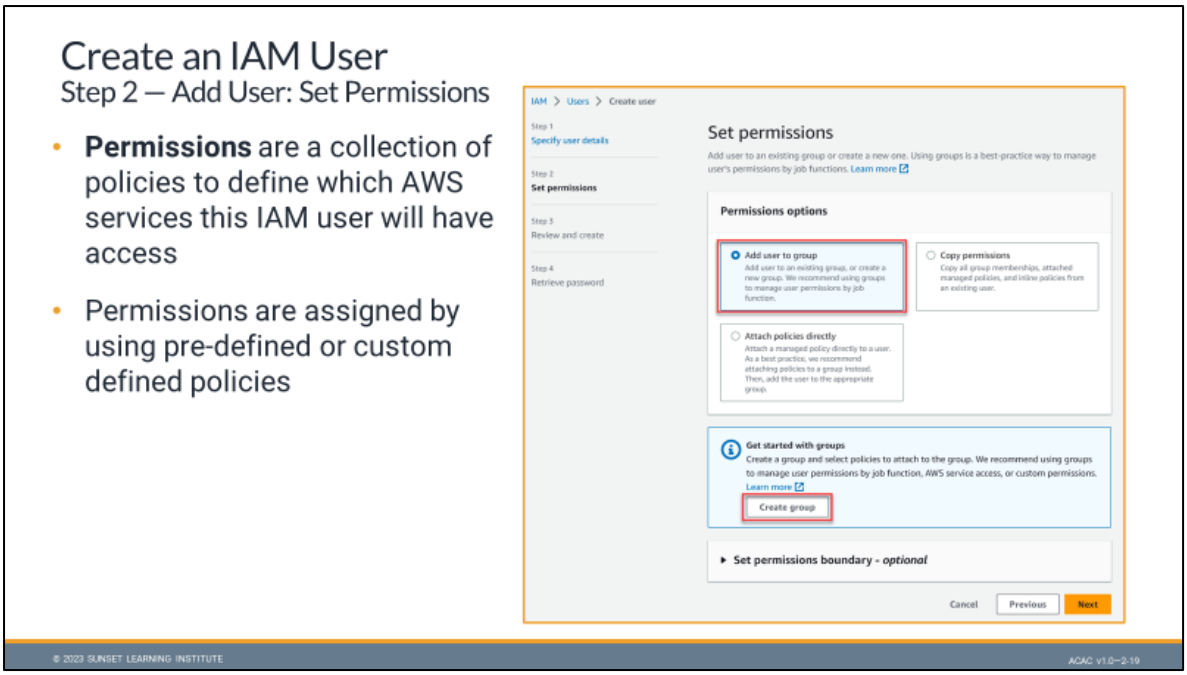
By default, AWS forces the new user to create a new password when first signing in. You can optionally clear the check box next to **User must create a new password at next sign-in** to allow the new user to reset the password after signing in.

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You can work with AWS Identity and Access Management in any of the following ways:

- **AWS Management Console:** The console is a browser-based interface to manage IAM and AWS resources. For more information about accessing IAM through the console, see **Signing in to the AWS Management Console as an IAM user or root user**. For a tutorial that guides you through using the console, see **Creating your first IAM admin user and user group**.
- **AWS Command Line Tools:** You can use the AWS command line tools to issue commands at your system's command line to perform IAM and AWS tasks. Using the command line can be faster and more convenient than the console. The command line tools are also useful if you want to build scripts that perform AWS tasks.
- AWS provides **two sets of command line tools:** the **AWS Command Line Interface** (AWS CLI) and the **AWS Tools for Windows PowerShell**. For information about installing and using the AWS CLI, see the **AWS Command Line Interface User Guide**. For information about installing and using the Tools for Windows PowerShell, see the **AWS Tools for Windows PowerShell User Guide**.
- **AWS Software Development Kits (SDKs):** AWS provides SDKs that consist of libraries and sample code for various programming languages and platforms (Java, Python, Ruby, .NET, iOS, Android, etc.). The SDKs provide a convenient way to create programmatic access to IAM and AWS. For example, the SDKs take care of tasks, such as cryptographically signing requests, managing errors, and retrying requests automatically. For information about the AWS SDKs, including how to download and install them, see the **Tools for Amazon Web Services** page.
- **IAM HTTPS API:** You can access IAM and AWS programmatically by using the IAM HTTPS API, which lets you issue HTTPS requests directly to the service. When you use the HTTPS API, you must include code to digitally sign requests using your credentials. For more information, see **Calling the IAM API using HTTP query requests** and the **IAM API Reference**.



On the **Set Permissions** page, do the following to perform the following to create an Administrator group with full administration permissions:

Step 1 Choose **Create group**.

Step 2 In the **Create group** dialog box, for Group name type **Administrators**.

Step 3 Select the **check box** for the **AdministratorAccess** policy.

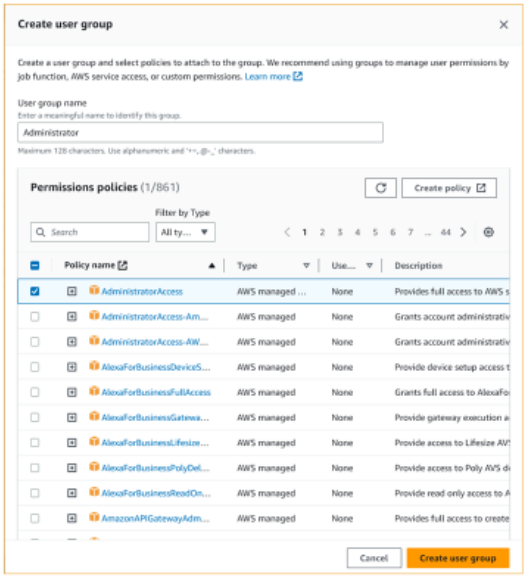
Step 4 Choose **Create user group**.

Step 5 On the page with the list of user groups, select the **check box** for your New user group. Choose **Refresh** if you don't see the new user group in the list.

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Create an IAM User
Step 2 — Add User: Create New Group

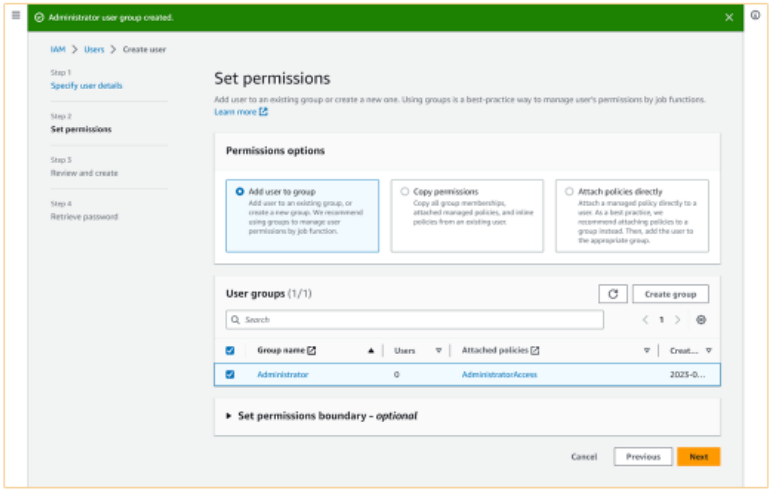
- Create a new Group for Administrators
- Other options: Directly assign permissions to the user or copy permissions from another existing user



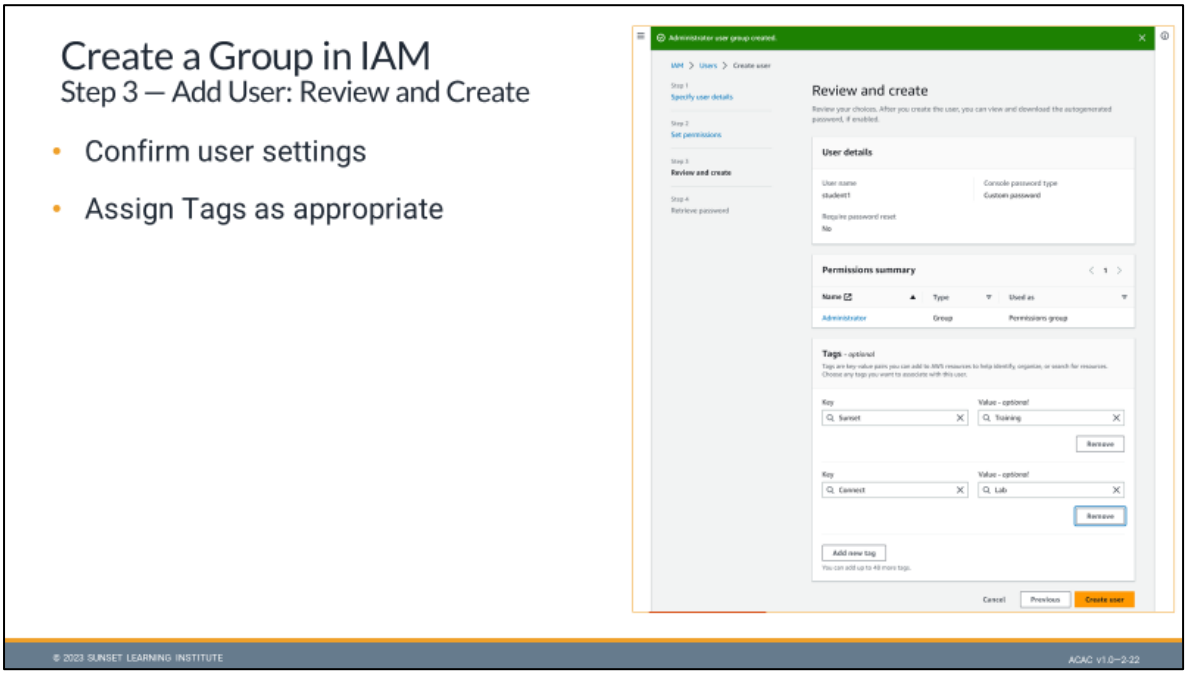
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Create an IAM User
Step 2 — Add User: Set Permissions (cont.)

- Wizard returns to IAM interface
- Make sure the appropriate permissions are assigned



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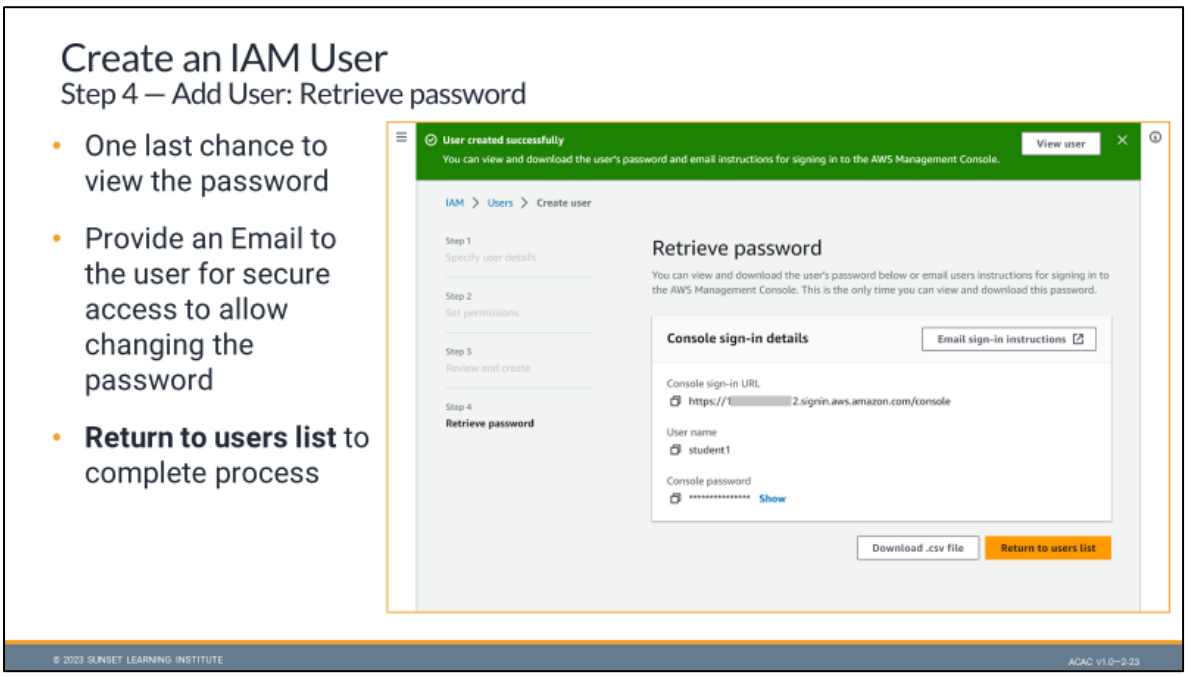
A **tag** is a custom attribute label that you can assign to an AWS resource. Each tag has two parts:

- A **tag key** (for example, CostCenter, Environment, Project, or Purpose). Tag keys are case sensitive.
- An optional field known as a **tag value** (for example, 111122223333, Production, or a team name). Omitting the tag value is the same as using an empty string. Like tag keys, tag values are case sensitive.

Together these are known as **key-value pairs**.

Tags help you identify and organize your AWS resources. Many AWS services support tagging, so you can assign the same tag to resources from different services to indicate that the resources are related. For example, you can assign the same tag to an IAM role that you assign to an Amazon S3 bucket.

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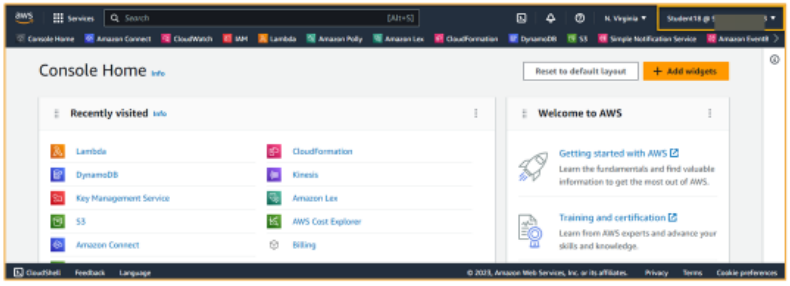


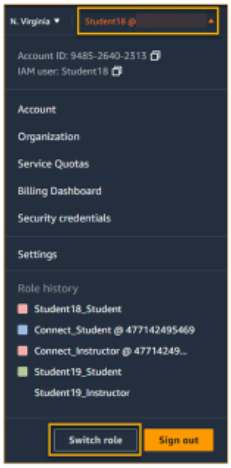
As the last step, you will have one last chance to view the password as well as an option to provide an Email to the user for secure access to allow changing the password. You will also be provided with an option for a csv file of this information. Select **Return to users list** to complete the process.



To access the labs while in class, , use the URL https://<parent_account>.signin.aws.amazon.com/console. We will use the **IAM user name** and **password** as assigned by your instructor.

Access for Lab Environment (cont.)
Part 2 — Switch Role into your student account



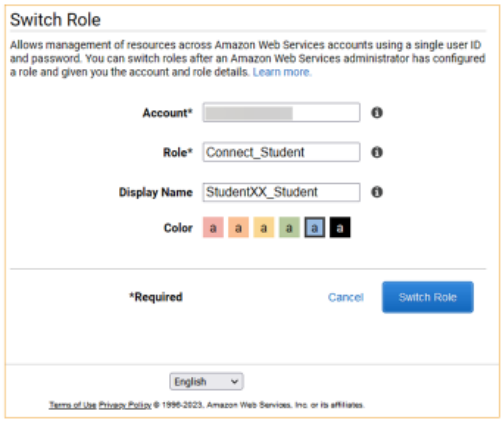


- Select the user login in the upper right corner
- Select Switch role

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Access for Lab Environment (cont.)
Part 3 — Switch Role into your student account

- Use the 12-digit Account number as provided
- Enter the role: **Connect_Student**
- Enter a Display Name of your choice
- Select a display color of your choice



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Switch role and change the account to use for class.

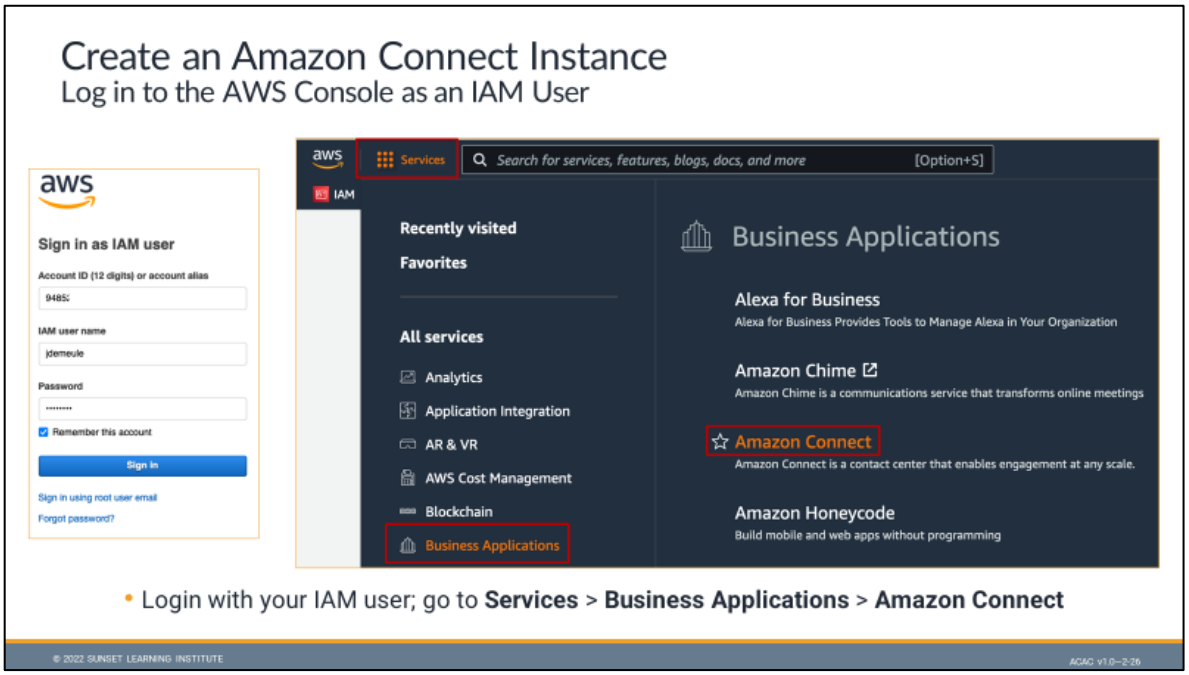
- Use the 12-digit Account number as provided
- Enter the role: **Connect_Student**
- Enter a Display Name and color of your choice
- Select Switch Role

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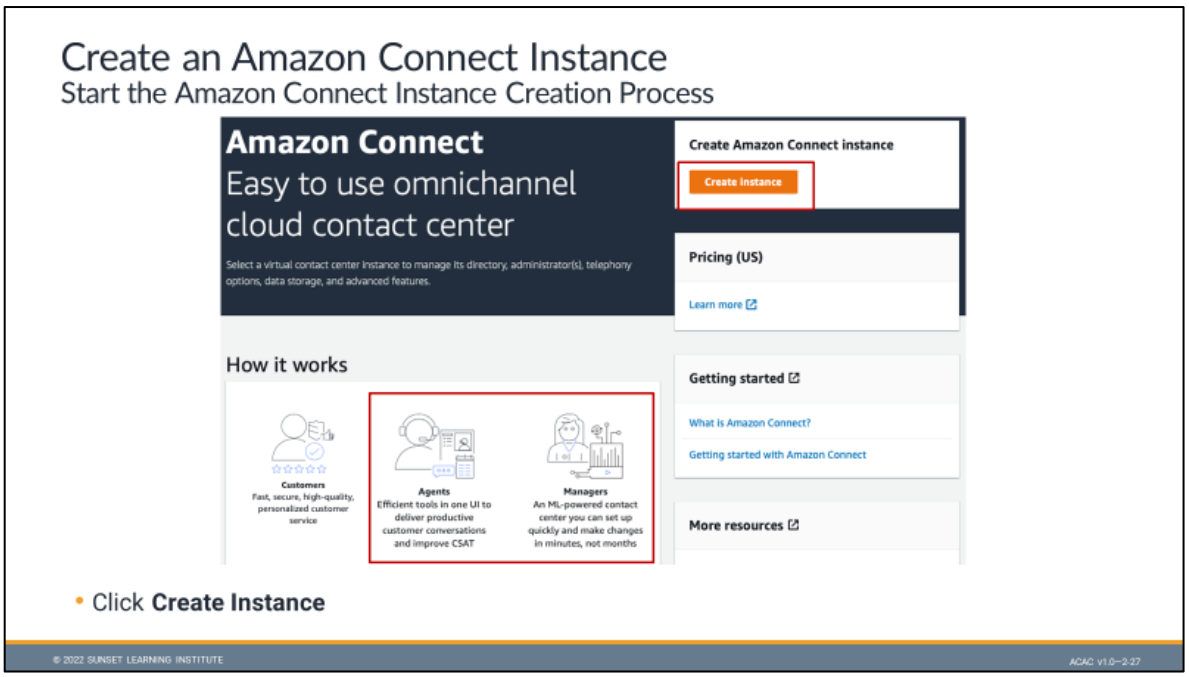

Lab 1: Accessing your AWS Account:
perform Task 1a OR 1b (not both)

Create an Amazon Connect Instance

To create an Amazon Connect Instance, you must log in to the AWS Management Console as an IAM user. This section covers creating an Amazon Connect Instance.



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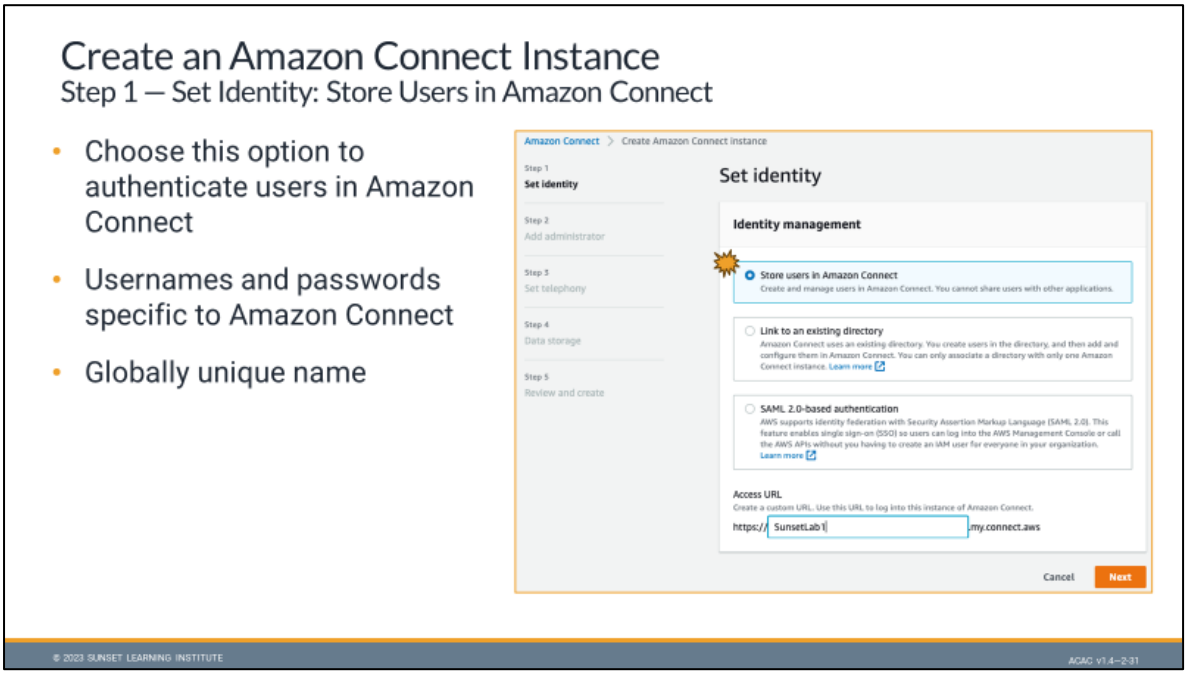
Things to know before you begin

When you sign up for Amazon Web Services (AWS), your AWS account is automatically signed up for all services in AWS, including Amazon Connect. You are charged only for the services that you use.

To allow an IAM user to create an instance, ensure that the user has the permissions granted by the AmazonConnect_FullAccess policy.

Amazon Connect is not available to customers in India using Amazon Web Services through Amazon Internet Services Pvt. Ltd (AISPL). You will receive an error message if you try to create an instance in Amazon Connect.

When you create an instance, you must decide how you want to manage users. You can't change the Identity Management option after you create the instance.

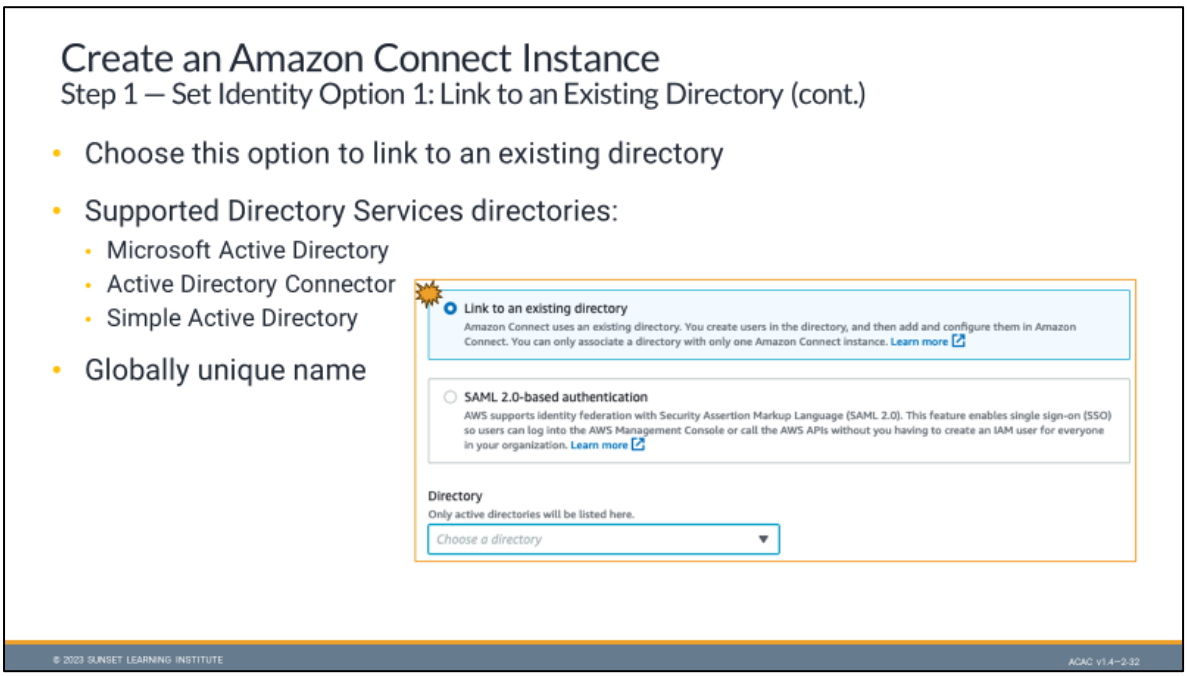


Store users with Amazon Connect:

Choose this option if you want to create and manage user accounts within Amazon Connect. An AWS Directory Service instance will be created in your AWS account.

When you manage users in Amazon Connect, the username and password for each user is specific to Amazon Connect. Users must remember a separate username and password to log in to Amazon Connect.

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Use an existing directory for Identity Management

If you are already using an AWS Directory Service directory to manage users, you can use the same directory to manage user accounts in Amazon Connect.

You can also create a new directory in AWS Directory Service to use for Amazon Connect. The directory you choose must be associated with your AWS account and must be active in the AWS Region in which you create your instance.

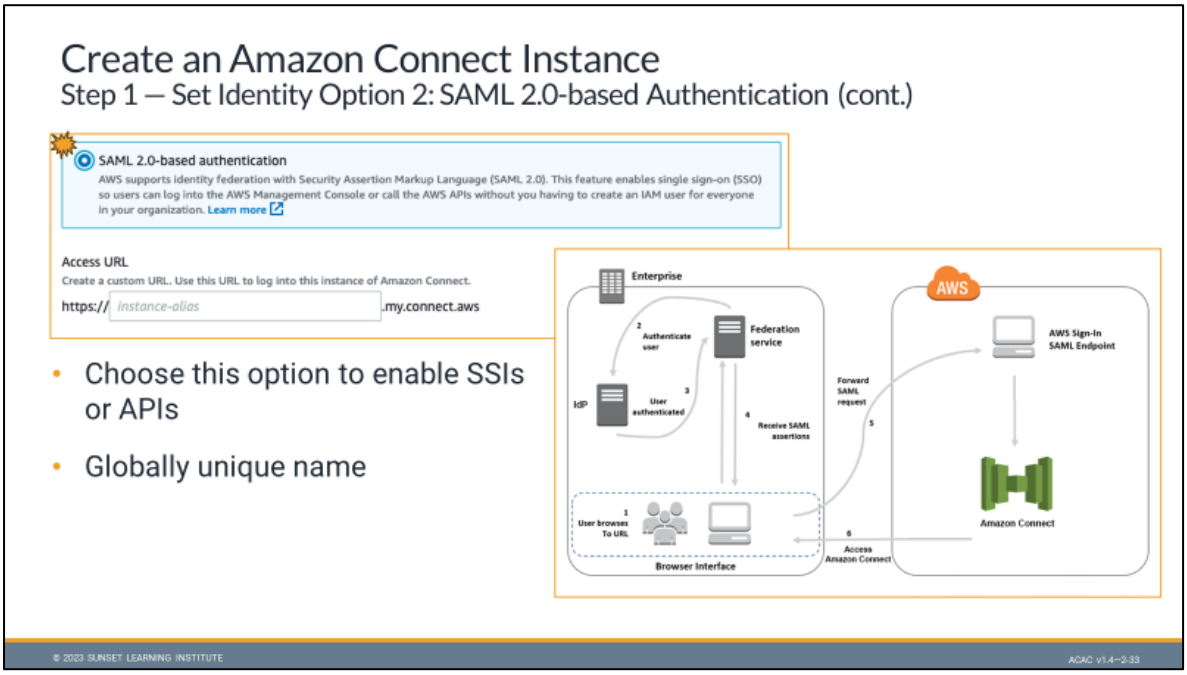
You can associate an AWS Directory Service directory with only one Amazon Connect instance at a time. To use the directory with a different instance, you must delete the instance with which it is already associated.

The following AWS Directory Service directories are supported in Amazon Connect:

- **Microsoft Active Directory**: AWS Directory Service lets you run Microsoft Active Directory as a managed service.
- **Active Directory Connector:** AD Connector is a directory gateway you can use to redirect directory requests to your on-premises Microsoft Active Directory.
- **Simple Active Directory**: Simple AD is a stand-alone managed directory that is powered by a Samba 4 Active Directory compatible server.

You cannot change the option you select after you create the instance. If you decide to change the directory you selected, you can delete the instance and create a new one. When you delete an instance, you lose all configuration settings and metrics data for it.

There is no additional charge for using an existing or a proprietary directory in Amazon Connect. For information about the costs associated with using AWS Directory Service, see **AWS Directory Service Pricing Overview**.



Configure SAML with IAM for Amazon Connect

Amazon Connect supports Identity Federation by configuring Security Assertion Markup Language (SAML) 2.0 with AWS IAM to enable web-based single sign-on (SSO) from your organization to your Amazon Connect instance. This allows your users to sign in to a portal in your organization hosted by a SAML 2.0 compatible identity provider (IdP) and log in to an Amazon Connect instance with a single sign-on experience without having to provide separate credentials for Amazon Connect.

Important notes

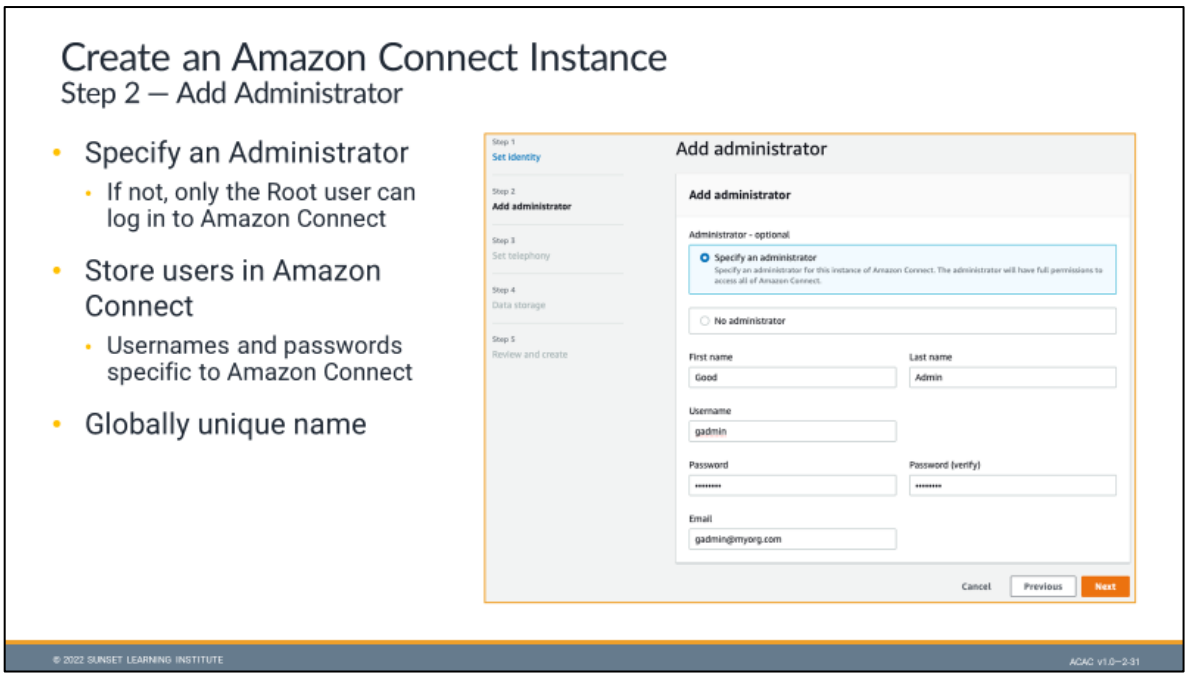
Before you begin, note the following:

- Choosing SAML 2.0-based authentication as the Identity Management method for your Amazon Connect instance requires the configuration of **AWS Identity and Access Management Federation**.
- The username in Amazon Connect must match the RoleSessionName SAML attribute specified in the SAML response returned by the identity provider.
- An Amazon Connect user can only be associated with a single AWS IAM Role. Changing the AWS IAM Role used for Federation will cause previously federated users to fail on login.

The following limitations apply to all new directories created using AWS Directory Service:

- Directories can only have alphanumeric names. Only the '.' character can be used.
- Directories cannot be unbound from an Amazon Connect instance after they have been associated.
- Only one directory can be added to an Amazon Connect instance.
- Directories cannot be shared across multiple Amazon Connect instances.

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Add Administrator

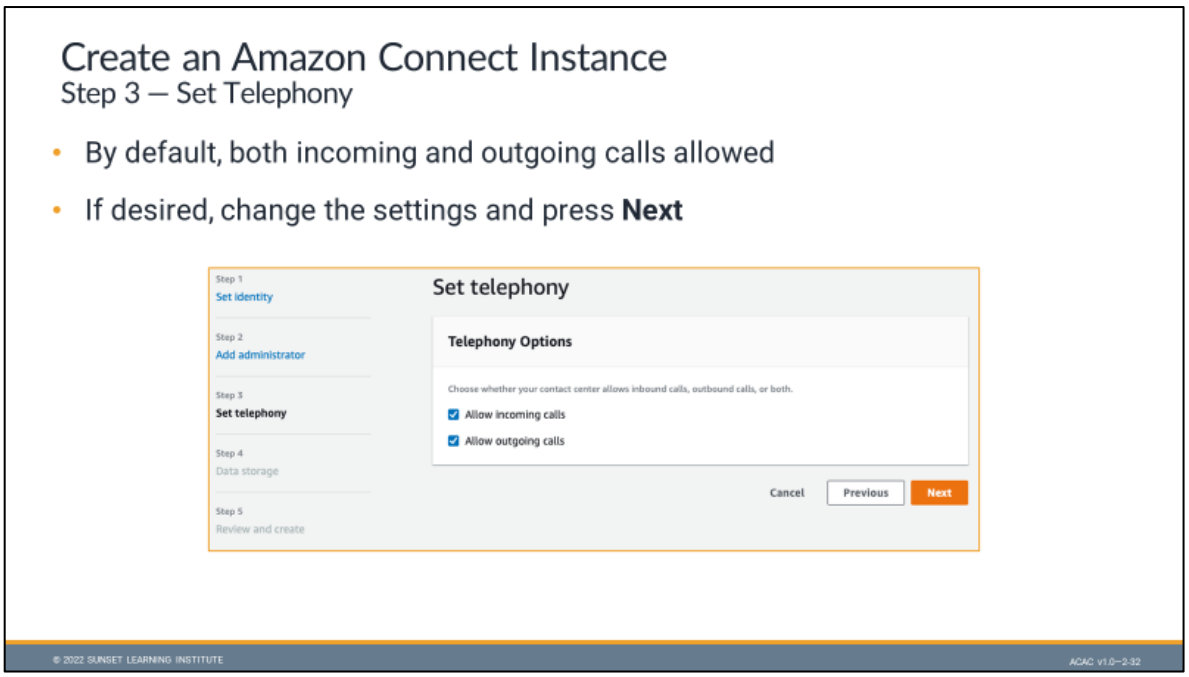
After you specify the username of the Administrator for the Amazon Connect instance, a user account is created in Amazon Connect and the user is assigned the Admin security profile.

To specify the Administrator for your instance

Do one of the following, based on the option that you chose in the previous step:

- If you chose **Store users within Amazon Connect**, select **Specify an Administrator**, and provide a name, password, and email address for the user account in Amazon Connect.
- If you chose **Link to an existing directory** for **Username,** type the name of an existing user in the AWS Directory Service directory. The password for this user is managed through the directory.
- If you chose **SAML 2.0-based authentication**, select **Add a new admin** and provide a name for the user account in Amazon Connect. The password for this user is managed through the IdP.

Choose **Next**.



Set telephony

Use the options in this section to choose whether you want your agents to receive calls from customers, make outbound calls, or hear early media audio.

Early media

When early media audio is enabled, for outbound calls your agents can hear pre-connection audio, such as busy signals, failure-to-connect errors, or other informational messages provided by telephony providers.

By default, early media is enabled for you. Note the following exception:

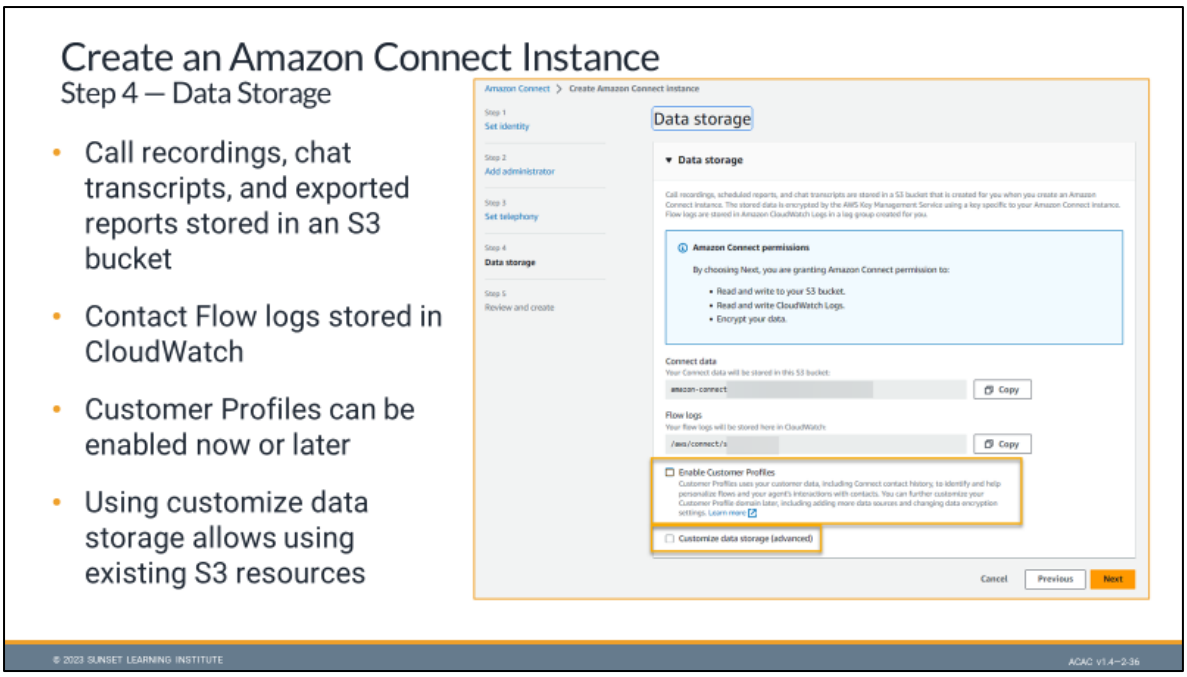
- Your instance was created before April 17, 2020, and you weren't enrolled in the preview program. You need to enable early media audio.

To configure telephony options for your instance

- To allow inbound calls to your Contact Center, choose **Allow incoming calls**.
- To enable outbound calling from your Contact Center, choose **Allow outgoing calls**.

Choose **Next**.

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Data Storage

When you create an instance, by default we create an Amazon S3 bucket. Data, such as reports and recordings of conversations, is encrypted using AWS Key Management Service and then stored in the Amazon S3 bucket.

This bucket and key are used for both recordings of conversations and exported reports. Alternatively, you can specify separate buckets and keys for recordings of conversations and exported reports.

By default, Amazon Connect creates S3 buckets for storing call recordings, chat transcripts, exported reports, and contact flow logs.

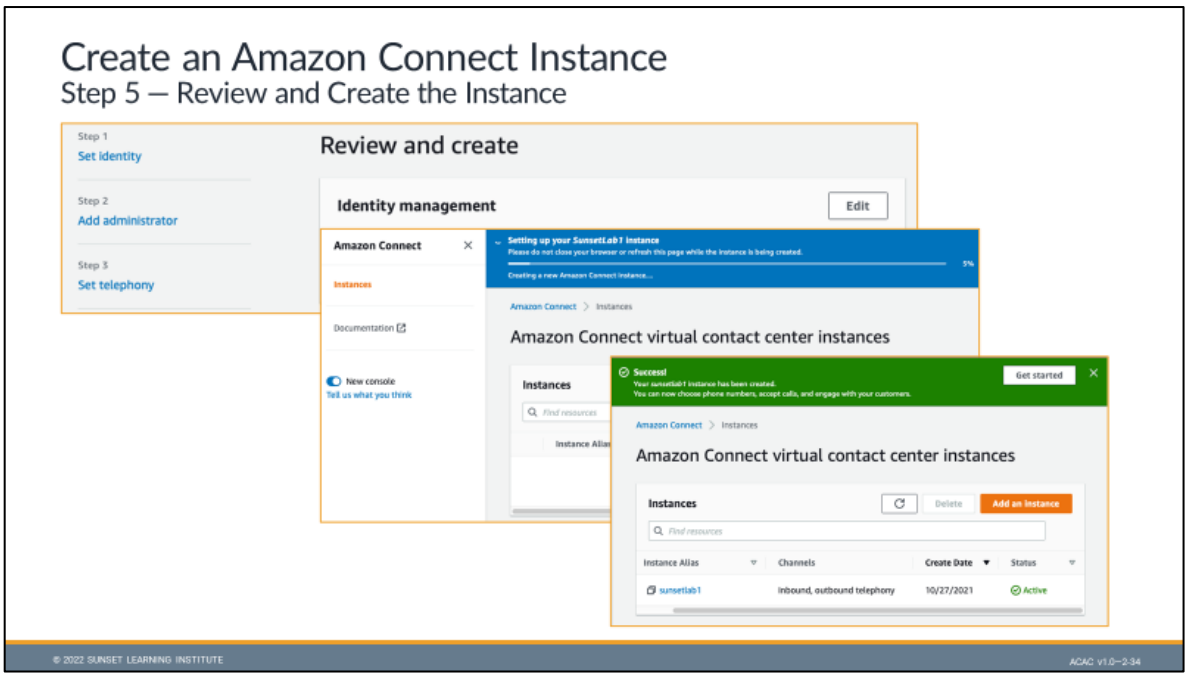
- When a bucket is created to store call recordings, Call Recording is enabled at the instance level.
- When a bucket is created to store chat transcripts, Chat Transcription is enabled at the instance level. Now all chat transcripts will be stored. Only if you want to monitor chat conversations do you need to set up recording behavior in a contact flow.
- Live media streaming is not enabled by default.

Customer Profiles may be enabled at this time or can be enabled later. For the purposes of this course, we will NOT enable Profiles at this time.

Review and copy the location of the S3 bucket and contact flow logs

- If desired, copy the location of the S3 bucket where your data encryption is stored and the location of the contact flow logs into CloudWatch.

Choose **Next**.

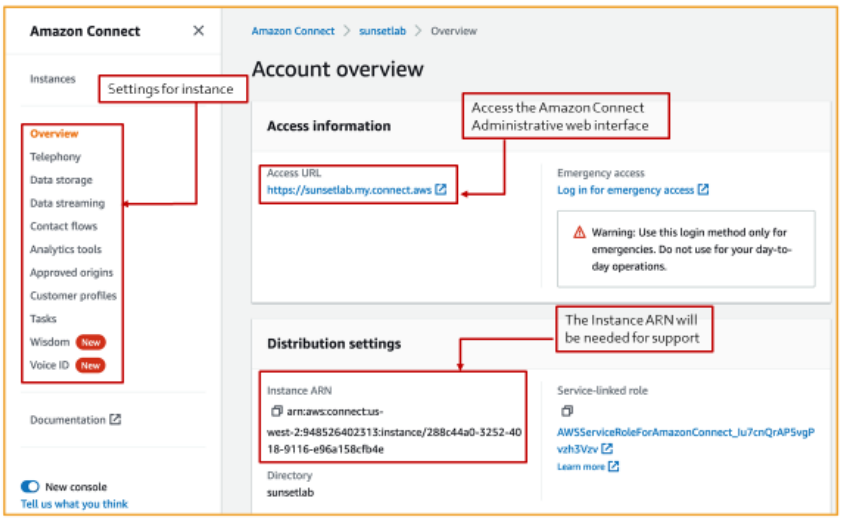


To create your instance:

- Review the configuration choices. Remember that you cannot change the Identity Management options after you create the instance.
- (Optional) To change any of the configuration options, choose **Edit**.
- Choose **Create instance**.
- (Optional) To continue configuring your instance, choose **Get started** and then choose **Let's go**. If you prefer, you can access your instance and configure it later. If you chose to manage your users directly within Amazon Connect or through an AWS Directory Service directory, you can access the instance using its access URL. If you chose to manage your users through SAML-based authentication, you can access the instance using the IdP.

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Create an Amazon Connect Instance
View the Amazon Connect Instance in the AWS Console



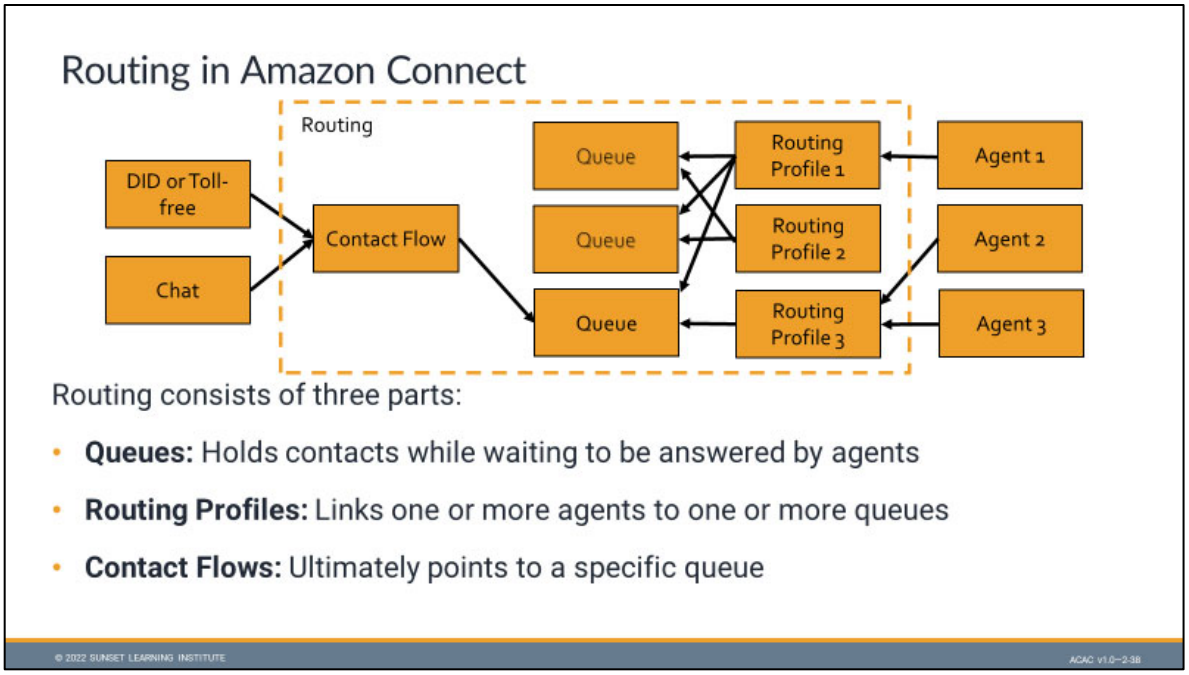
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Configuration Components of Amazon Connect

Amazon Connect routing includes three parts: queues, routing profiles, and contact flows.



Routing: In Amazon Connect, routing consists of three parts: queues, routing profiles, and contact flows.

- **Queue**: A queue holds contacts waiting to be answered by agents. You can use a single queue to handle all incoming contacts, or you can set up multiple queues.

- **Routing Profiles:** Queues are linked to agents through a routing profile. When you create a routing profile, you specify the following items:
  - Which queues will be in the profile
  - Whether one queue should be prioritized over another
  - What channels agents will handle in the Contact Control Panel (CCP)
  - How many contacts agents can handle simultaneously for each channel
  - Whether individual queues are for all channels or specific ones

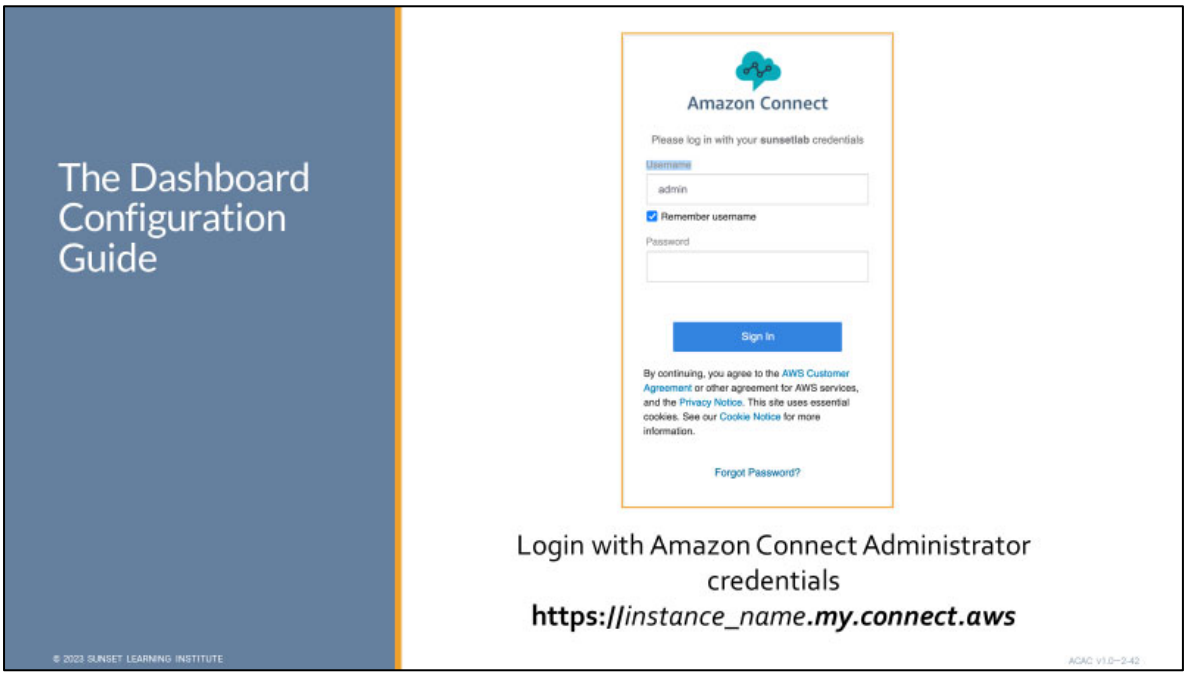
Each agent is assigned to one routing profile.

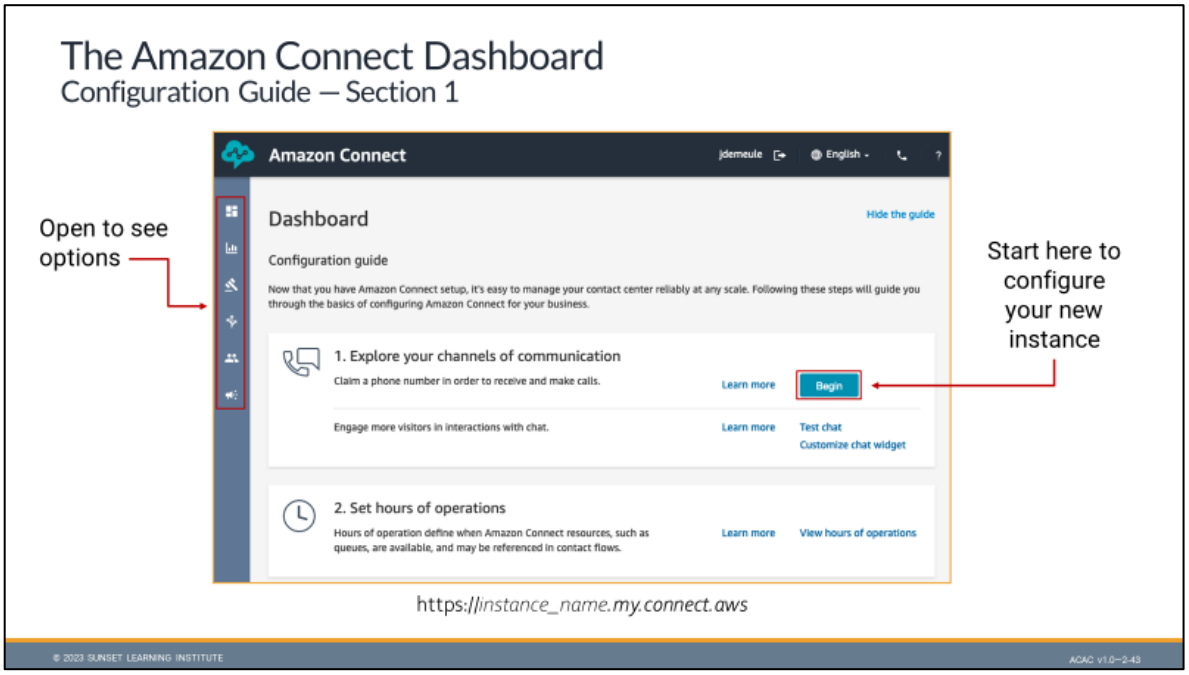
- **Contact Flows:** A contact flow defines the customer experience with your Contact Center from start to finish. Amazon Connect includes a set of default contact flows so you can quickly set up and run a Contact Center.

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The Dashboard Configuration Guide

Amazon Connect provides a variety of choices to enable companies to make and receive telephone calls. This section explains Amazon Connect's options for toll-free numbers, telephony architecture, direct-in-dial (DID) numbers. In addition, you will learn how to use the Amazon Connect dashboard to claim and test your phone number, test chat, set hours of operation, create queues, create prompts, create contact flows, and create routing profiles.





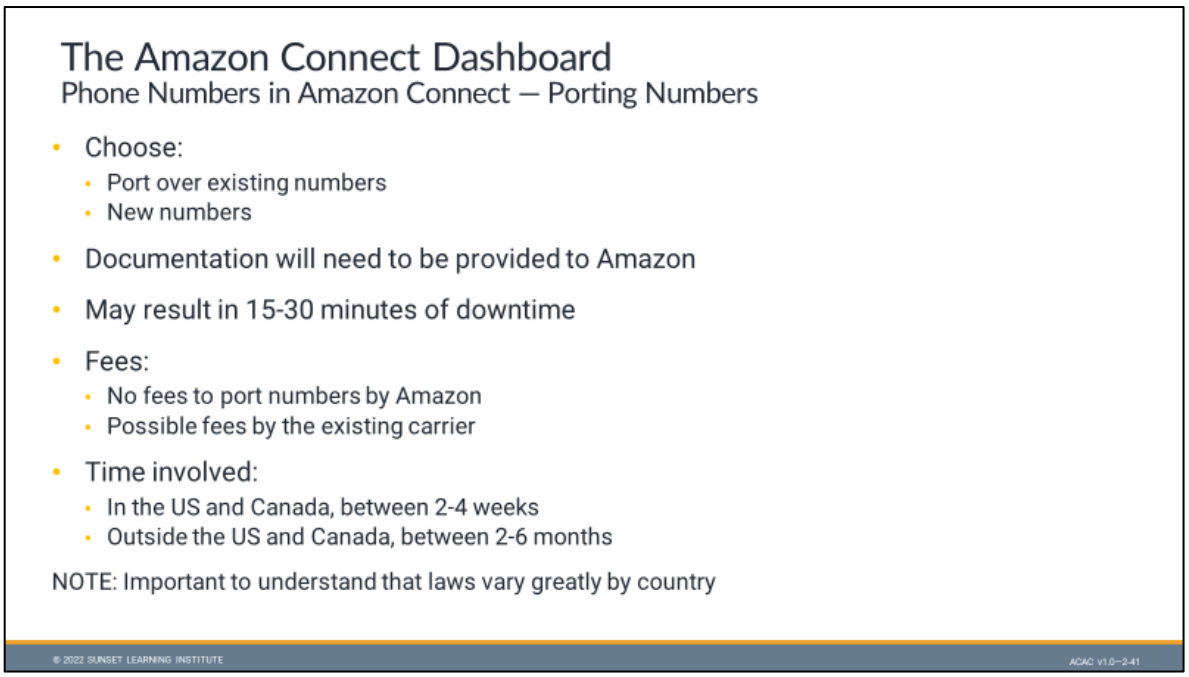
Toll-free numbers

Toll-free numbers are telephone numbers with distinct prefix codes that can be dialed with no charge to the person placing the call. Such numbers allow callers to reach businesses and/or individuals out of the area without being charged a long-distance fee for the call.

In the United States, the FCC provides rules for obtaining and using toll-free numbers. In other countries, similar governing bodies ensure that toll-free numbers are managed and distributed in accordance with local laws.

AWS manages toll-free numbers as a Responsible Organization (RespOrg). When you claim or port a number into Amazon Connect, we register that number. Once the number is registered, we can select multiple carriers to provide BOTH route and carrier redundancy. This provides the highest level of availability, ensuring the number will remain available even in the event of a complete carrier outage. This level of service does come at an additional cost, as toll-free numbers are a higher price than direct dial, but this the most attractive option because of the service reliability and the overall customer experience.

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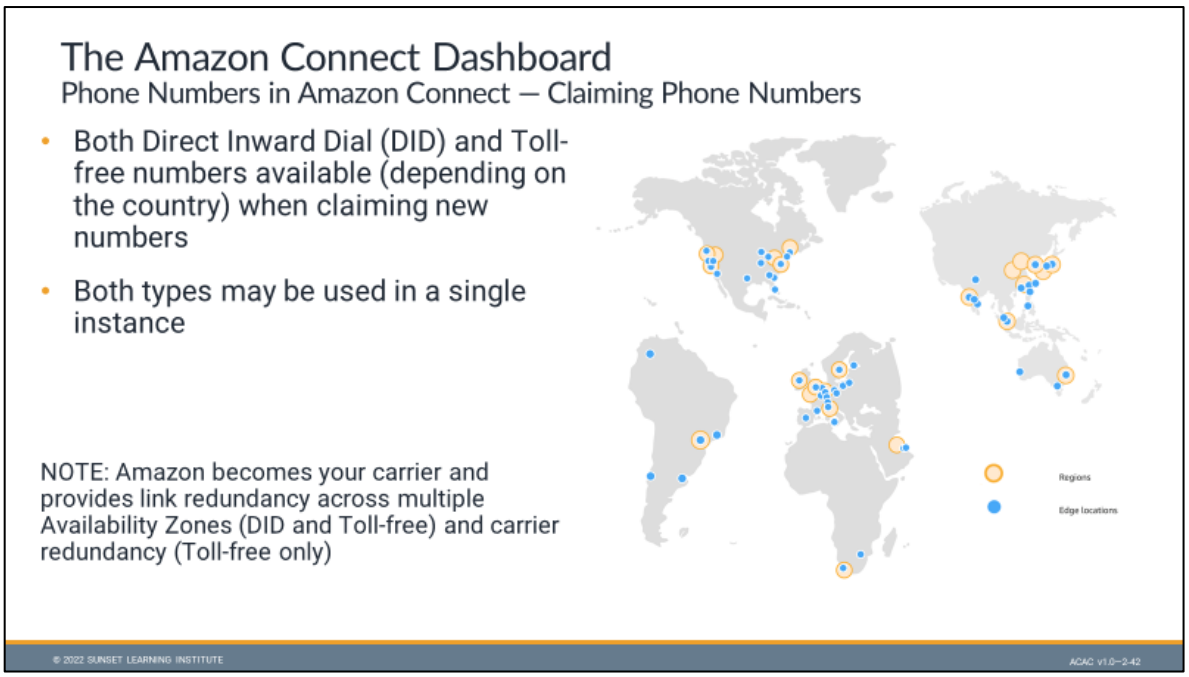


Telephony architecture

Amazon Connect provides capabilities to host both toll-free and direct dial numbers (DID) in all AWS Regions supported by Amazon Connect. You can use both types of numbers in a single instance.

AWS manages the connectivity to our network of carriers providing diverse connections to multiple carriers in each region supported by Amazon Connect. When Amazon Connect is deployed in a Region, we take advantage of the built-in redundancy of the AWS Availability Zone design to provide multiple carrier interfaces into multiple data centers.

In addition to the Amazon Connect service being spread across multiple Availability Zones, AWS also has multiple telephony providers. These providers have multiple links into the data centers in those Availability Zones. This ensures that if a single or even multiple links fail from a carrier, there are alternate routes available to ensure the service remains available.



Direct-in-dial (DID) numbers

Direct inward dialing (DID), also called direct dial-in (DDI) in Europe, is a telecommunication service offered by telephone companies to subscribers. DID numbers provide a locally formatted telephone number that can match the dialing pattern of a local subscriber. For example, in Seattle, Washington, USA, the local dialing pattern is +1(206)-NXX-XXXX. The provider of the DID number would provide numbers with the +1(206) pattern to match local dialing.

In the United States, DID numbers are regulated by State Public Utilities commissions. DID numbers are managed by a single carrier. While they are portable, they can't be load balanced or managed across multiple carriers. This makes them less reliable than toll-free numbers.

DID numbers offer you the ability to present a local calling line identification when placing outbound calls and a local presence to inbound callers. This can be very useful to increase the likelihood outbound and queued callback calls get answered by your customers. It can also show a customer that you are local to their area and provide a cheaper inbound route than a long-distance call if you don't publish a toll-free number.

Because DID numbers are threaded to a single carrier, Amazon Connect doesn't offer carrier redundancy for DID numbers. We do offer link redundancy across multiple Availability Zones, so in the event of a link failure, that carrier still has facilities available in another location to deliver calls. DID numbers also have a capacity limitation on how many calls a single number can accommodate, and this number does vary by Region. It is important to work with your AWS account team to ensure you are properly enabled with the right type of DID numbers if you plan on using DID numbers as your primary inbound channel and have an expectation of over 100 concurrent calls per number.

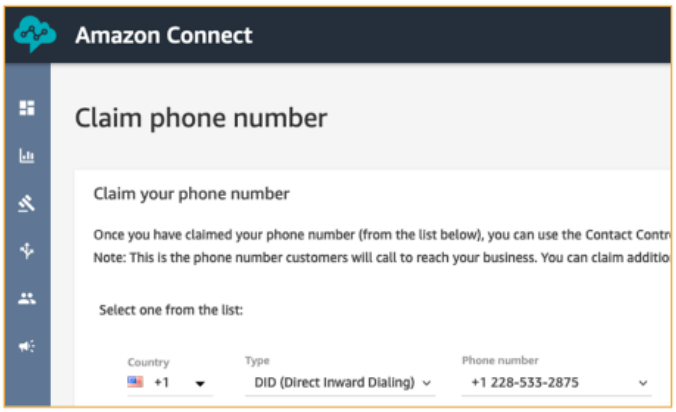
DID numbers are less expensive than toll-free numbers but don't have the redundancy and broad geographical coverage of a toll-free number. The ability to localize numbers may be an attractive option for your business.

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The Amazon Connect Dashboard

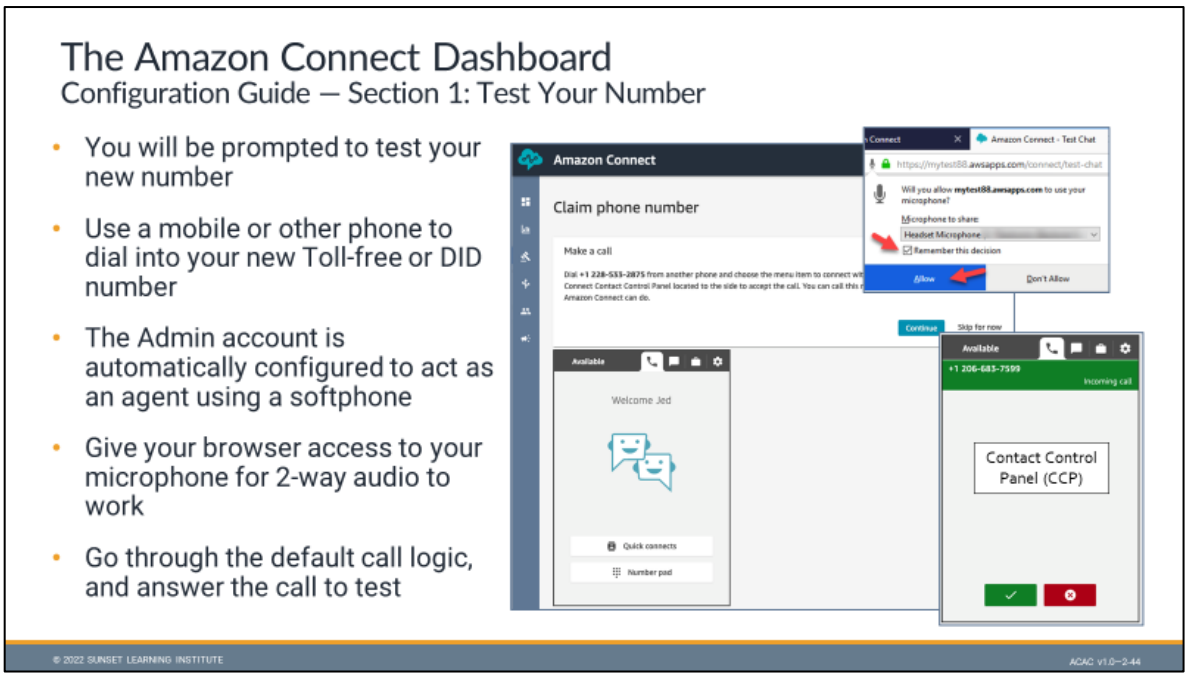
Configuration Guide — Section 1: Claim Your Phone Number

- Select the Country
- Choose the type of new number:
  - DID
  - Toll-free
- Five numbers presented; choose your preference



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To better understand what the voice and chat experiences are like for your agents and customers, you can test them without doing any development.

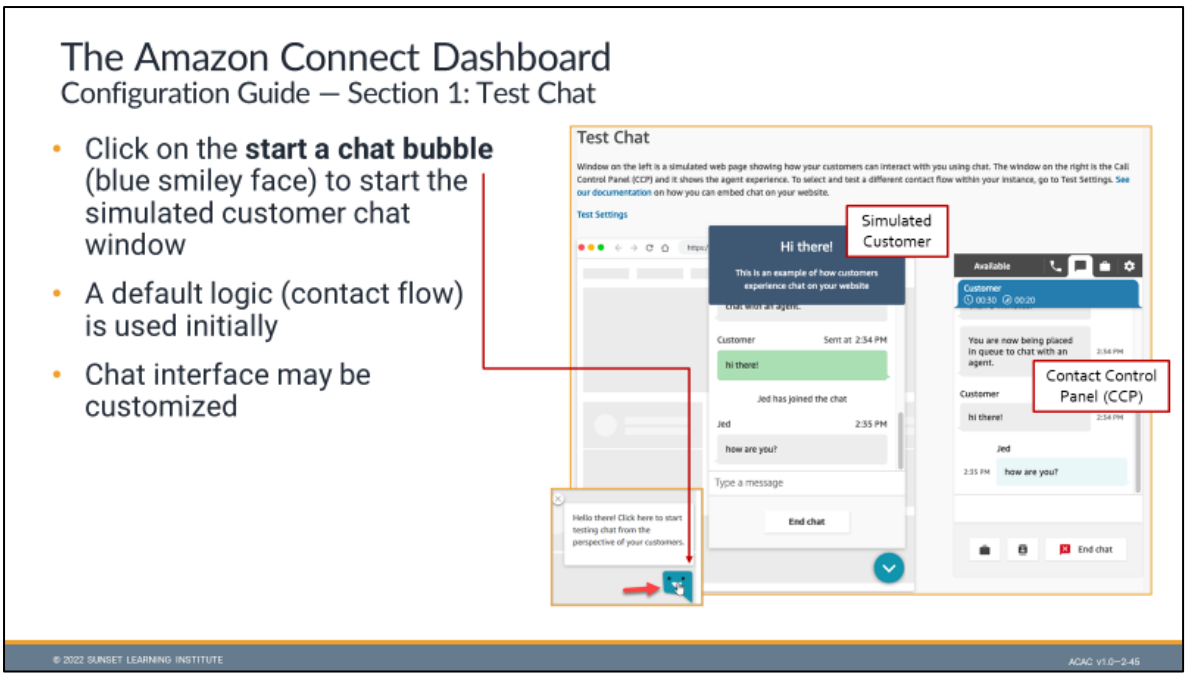
Step 1 Use your mobile phone to call the phone number that you claimed earlier. If you didn't write down the number, you can find it by going to **Routing** > **Phone numbers**.

Step 2 When your call is joined to Amazon Connect, you'll hear “Press 1 to be put in queue for an agent, 2 to ...”. This is the **Sample inbound flow** that Amazon Connect runs by default.

Step 3 You can play around with the different options in the Sample inbound contact flow. To connect to an agent, press **1**, **1**, **1**.

Step 4 In the Contact Control Panel (CCP), choose **Accept call**.

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You may also test the chat experience using the Contact Control Panel (CCP).

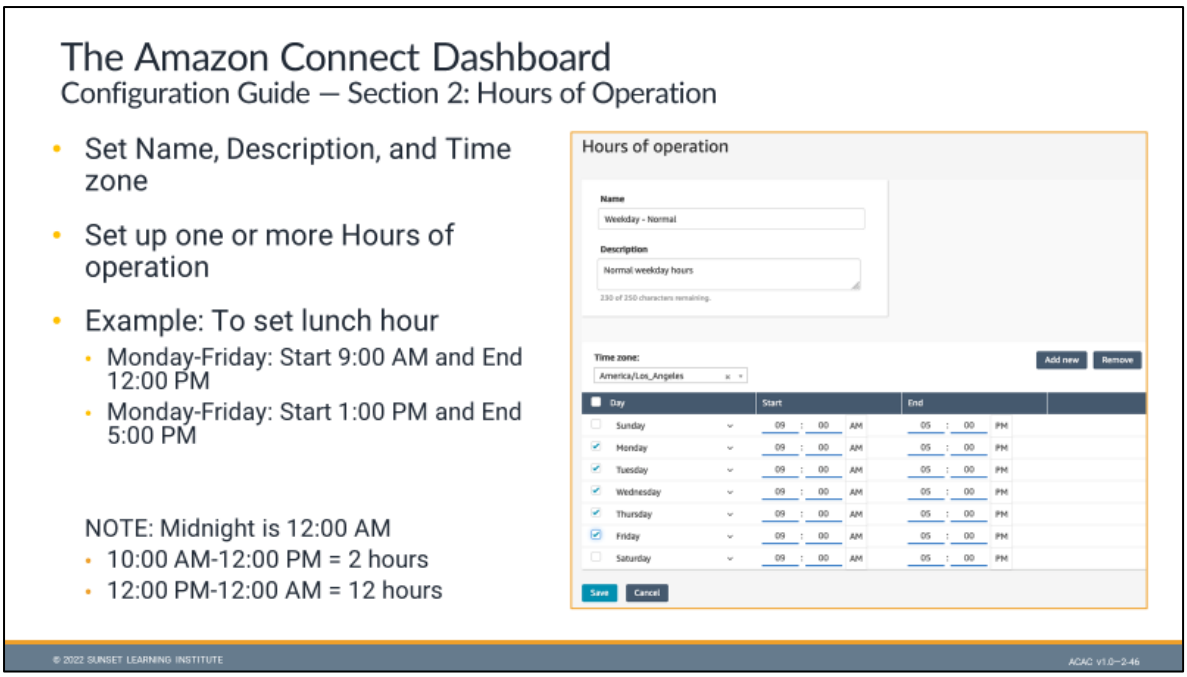
Step 1 Choose the chat bubble to start a chat.

Step 2 The Sample inbound flow automatically transfers to you a queue. However, you can type a message as the customer and the agent receives it. For example, *I need help resetting my password*.

Step 3 In the CCP, accept the incoming chat.

Step 4 Use the CCP to send chat messages to the customer.

Step 5 When you're done chatting, choose **End chat**. Then in the CCP, choose **Close contact**.



The first thing you need to do when you set up a queue is to specify the hours of operation and time zone. The hours may be referenced in contact flows. For example, when routing contacts to agents, you might check the hours first, and then route the contact to the appropriate queue.

To set the hours of operation and time zone for a queue:

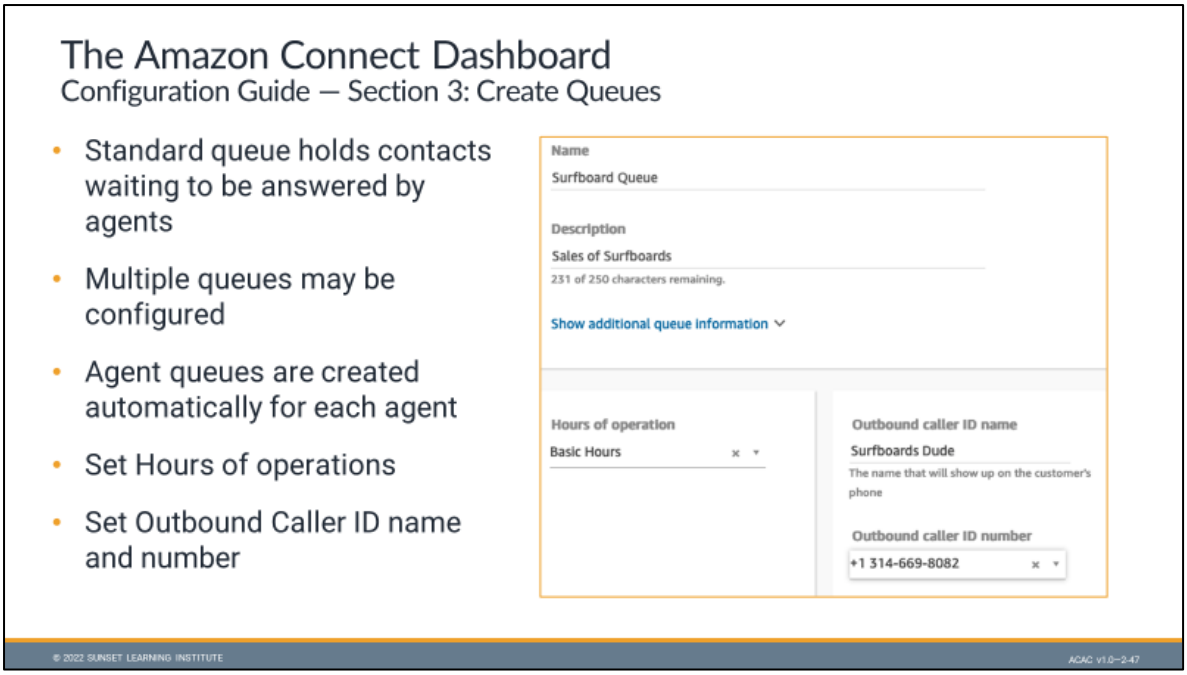
- On the navigation menu, choose **Routing**, **Hours of operation**.
- To create a template, choose **Add new hours**, and enter a name and a description.
- For **Time zone**, select a value.
- For **Add new**, set new hours.
- Choose **Save**.

Now you can specify these the hours of operation when you create a queue and check them in the Amazon Connect routing logic.

What happens during daylight savings time:

Amazon Connect uses the time zone to determine whether daylight savings time is in effect for the queues and adjusts automatically. When a contact comes in, Amazon Connect looks at the hours and time zone to determine whether the contact can be routed to the given queue.

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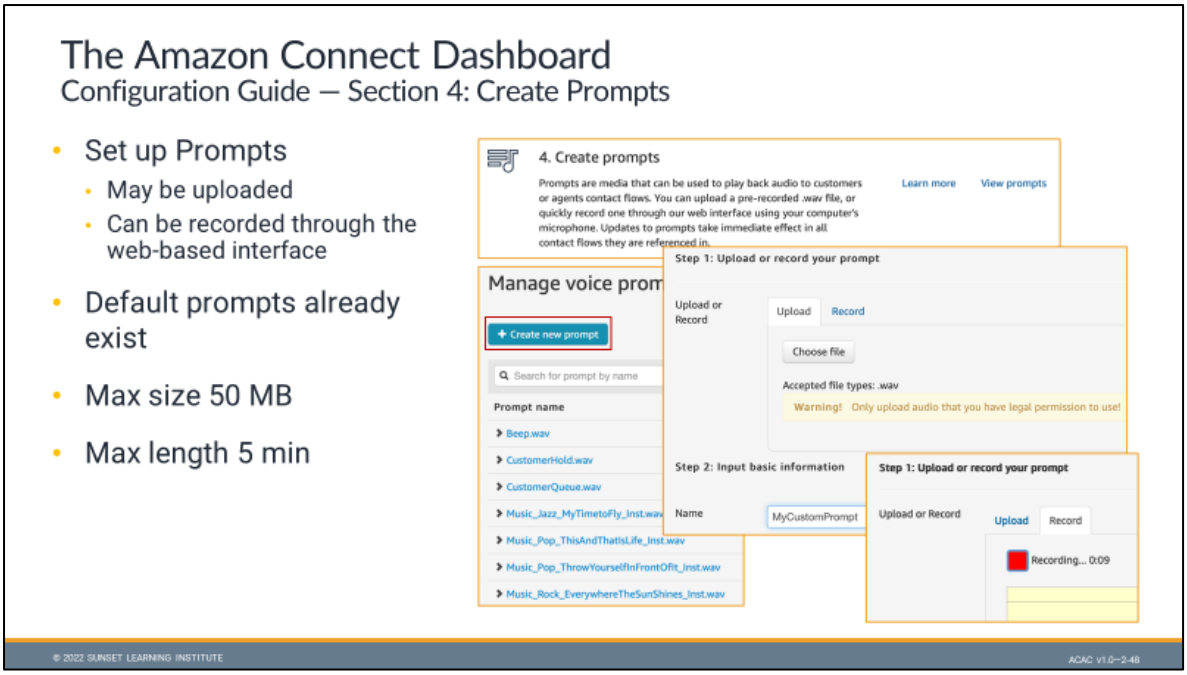
There are two types of queues:

- **Standard queues**: This is where contacts wait before they are routed to, and accepted by, agents.
- **Agent queues**: These queues are created automatically when you add an agent to your Contact Center.

Contacts are only routed to agent queues when explicitly sent there as part of a contact flow. For example, you might route contacts to a specific agent who's responsible for certain customer issues, such as billing or premium support. Or you might use agent queues to route to an agent's voice-mail.

Contacts waiting in agent queues have a higher priority than contacts waiting in standard queues. Contacts in agent queues have the highest priority and zero delay:

- **Highest priority**: If there's another contact in the basic queue, Amazon Connect chooses to give the agent the contact from the agent queue first.
- **Zero delay**: If the agent is available, the contact immediately gets routed to them.



Prompts are audio files played in call flows. For example, hold music is a prompt. Amazon Connect comes with a set of prompts that you can add to your contact flows. Or, you can add your own recordings.

We recommend that you align your prompts and routing policies with each other to ensure a smooth call flow for customers.

To create a prompt:

Step 1 In the navigation pane, choose **Routing**, **Prompts**.

Step 2 On the **Manage voice prompts** page, choose **Create new prompt**.

Step 3 Choose the following actions:

- **Upload**: Select the file to upload.
- **Record**: Select the red circle to begin recording. Use the red square to stop. You can choose **Crop** to cut the recorded prompt or **Discard** to record a new prompt.

Step 4 For **Step 2: Input basic information**, enter the name of the file and then choose **Create**.

Supported file types:

You can upload a pre-recorded .wav file to use for your prompt or record one in the web application.

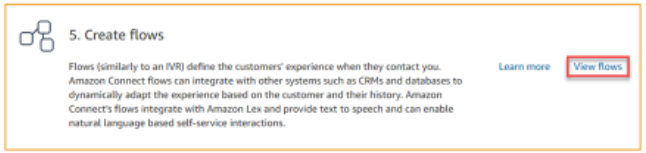
Amazon recommends using 8 KHz .wav files that are less than 50 MB and less than 5 minutes long. If you use higher rated audio libraries, such as 16 KHz or 16-bit files, Amazon Connect will have to down sample them into 8 KHz samples due to PSTN limitations. This down sampling may result in low quality audio.

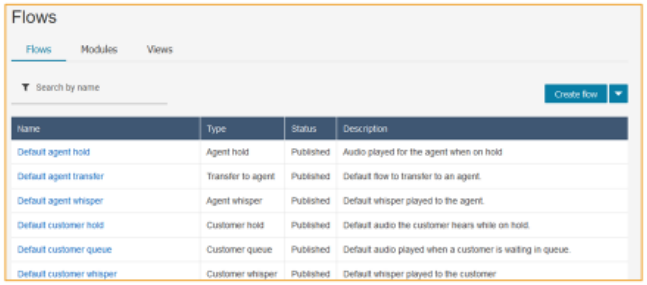
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The Amazon Connect Dashboard

Configuration Guide — Section 5: Create Flows

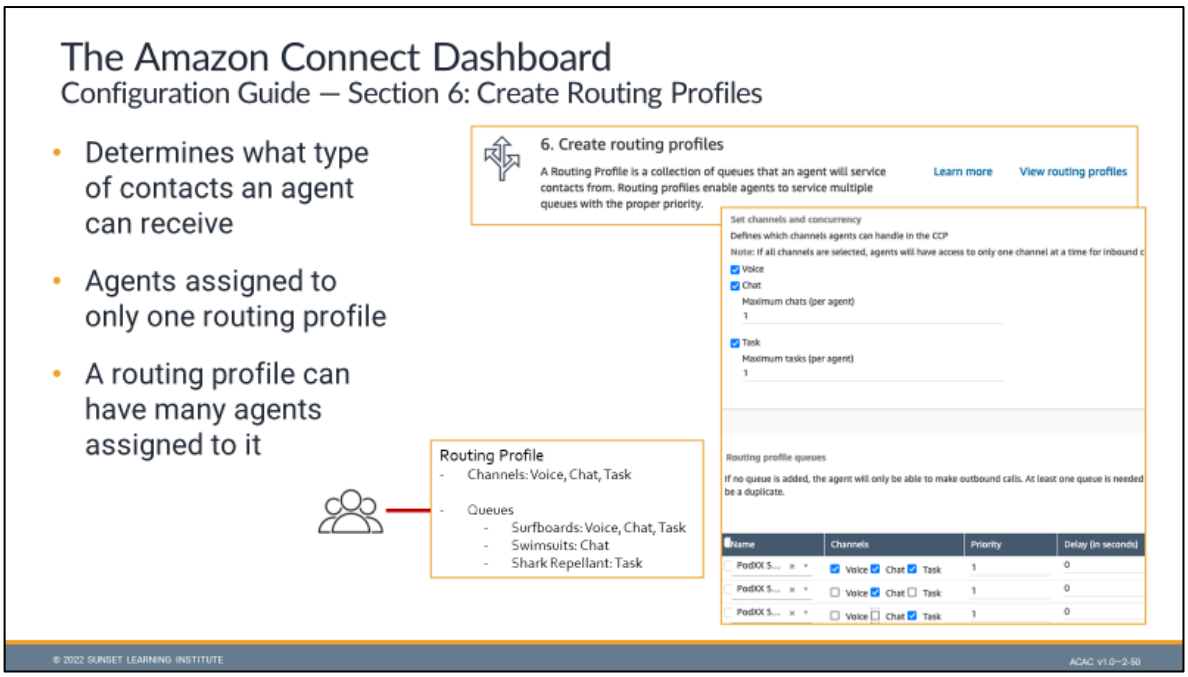
- Defines the customer experience with your contact center from start to finish
- Requires Admin or CallCenterManager security profile
- Default and Sample Flows available
- Flow Designer
  - A drag-and-drop visual interface
  - Use to create or modify flows





Flow Designer will be covered in more detail later in the course

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Routing Profiles

A routing profile determines what types of contacts an agent can receive and the routing priority.

- Each agent is assigned to one routing profile.
- A routing profile can have multiple agents assigned to it.

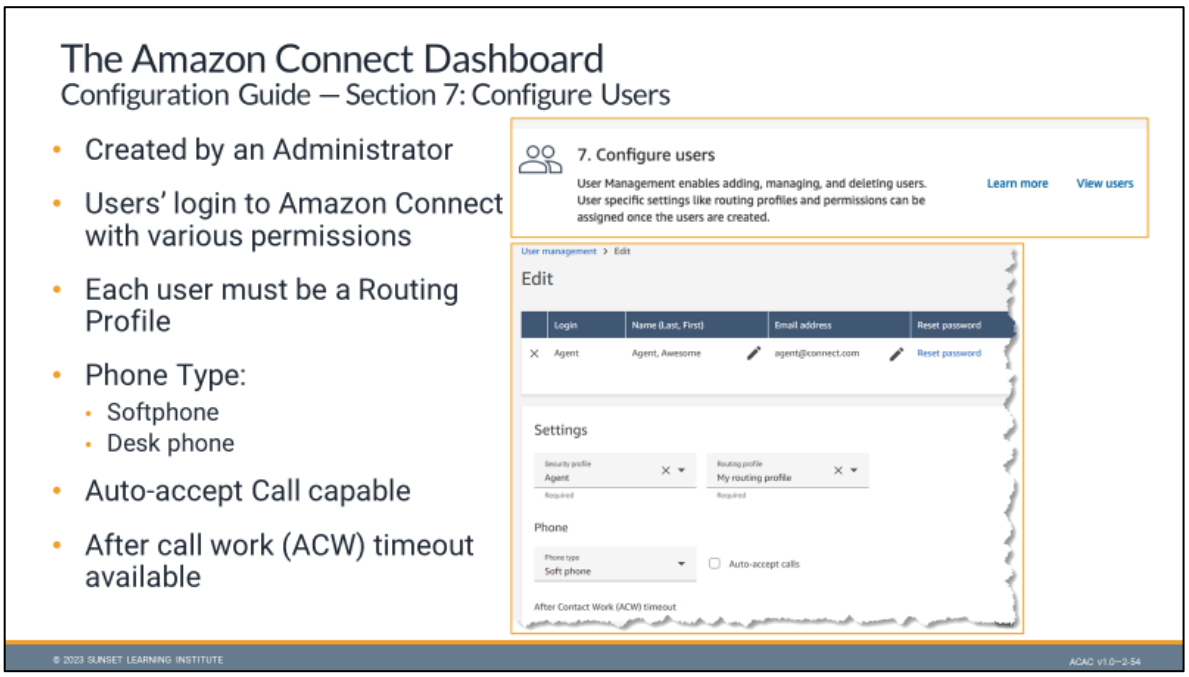
Amazon Connect uses routing profiles to allow you to manage your Contact Center at scale. To quickly change what a group of agents does, you only need to make an update in one place: the routing profile.

Routing Profiles Link Queues and Agents

When you create a routing profile, you specify:

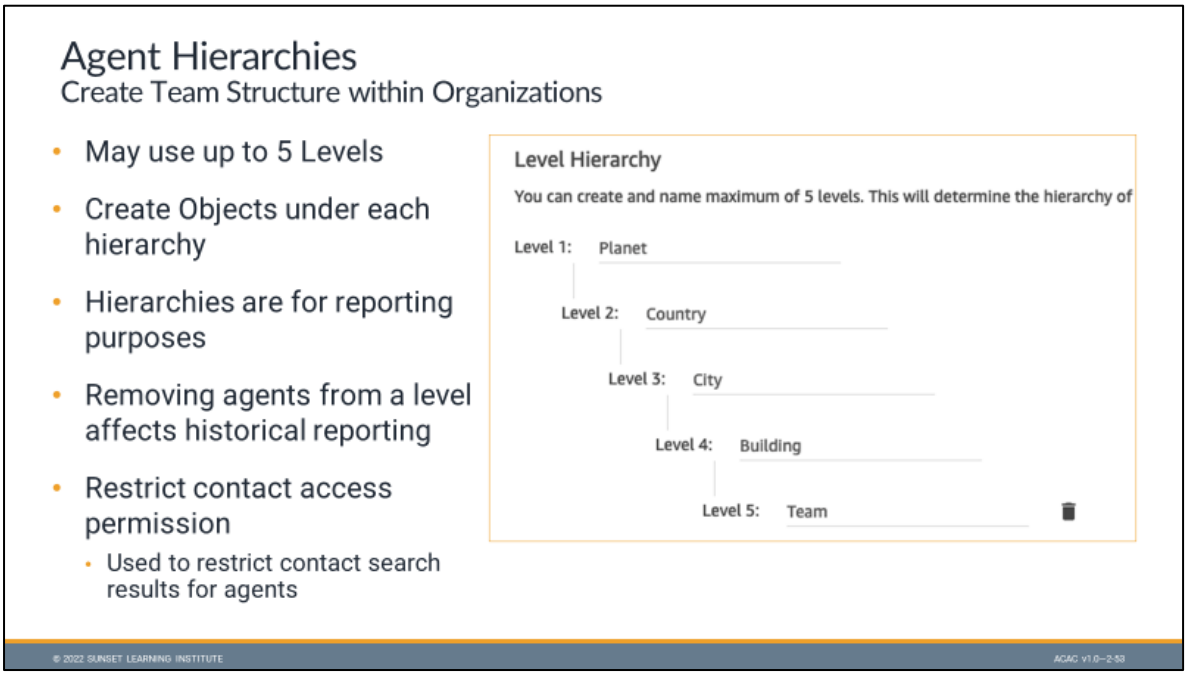
- The channels the agents will support.
- The queues of customers that the agents will handle. You can use a single queue to handle all incoming contacts, or you can set up multiple queues. Queues are linked to agents through a routing profile.
- Priority and delay of the queues.

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As the admin, one of your key responsibilities is to manage users: add users to Amazon Connect, give them their credentials, and assign the appropriate permissions so they can access the features needed to do their job.

Agent Hierarchies

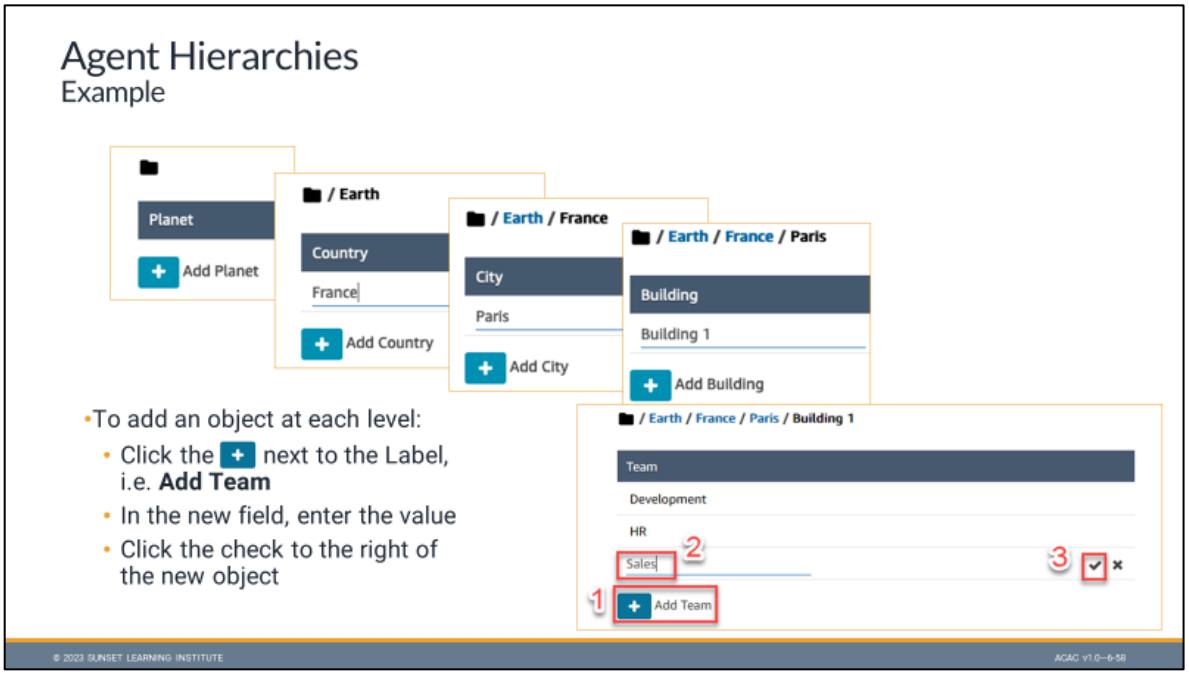


Agent hierarchies are a way for you to organize agents into teams and groups for reporting purposes. It's useful to organize them based on their location and their skill sets. For example, you might want to create large groups, such as all agents who work on a specific continent, or smaller groups such as all agents working in a specific department.

You can also configure hierarchies with up to five levels and segment agents or teams. Here are a couple of things to note about using hierarchies:

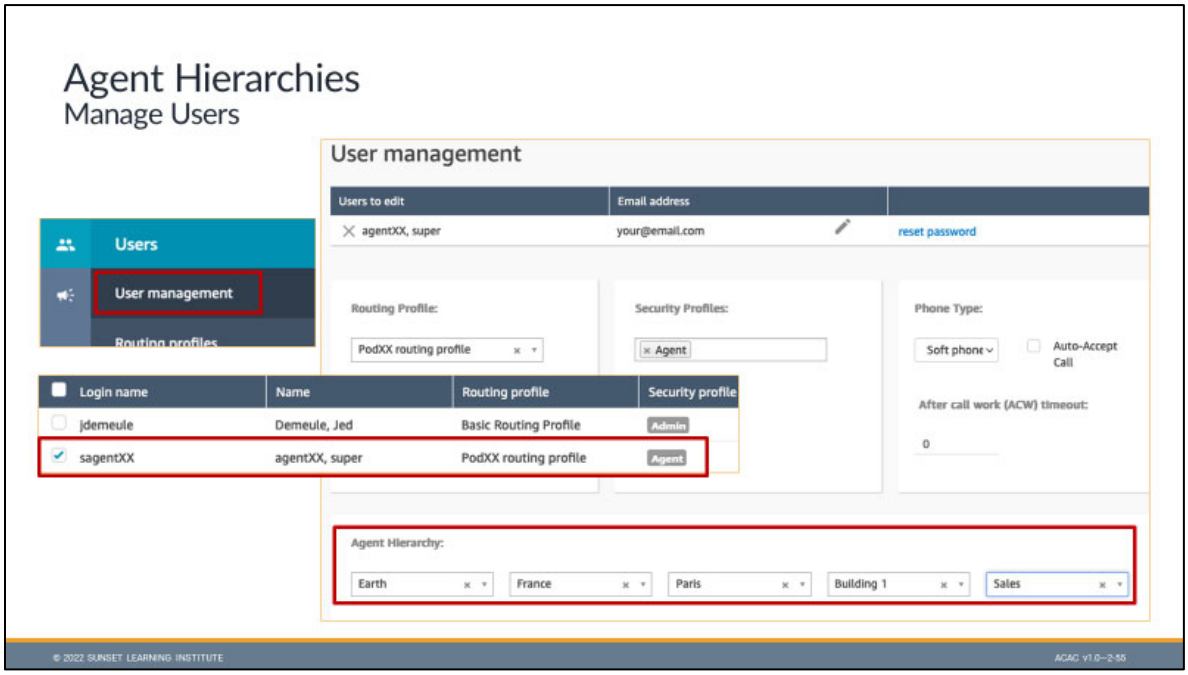
- Removing agents from a level affects historical reporting.
- When you use the **Restrict contact access** permission, you can restrict contact search results based on the agent's hierarchy.

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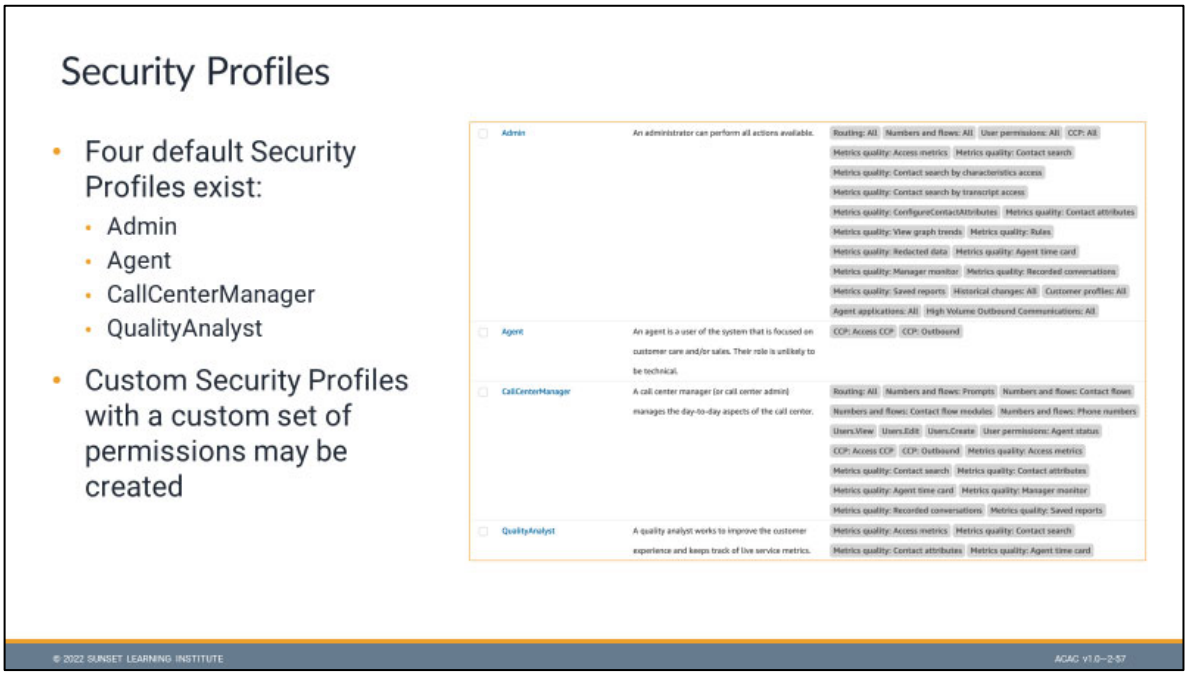
Once the team structure hierarchy has been defined, populate the hierarchies for your agents. At each level, add the objects by:

- Click the plus icon next to the label
- In the new field, enter the value for that level
- Click the check box to the right of the new object



Each user can be assigned to the appropriate hierarchy.

Security Profiles

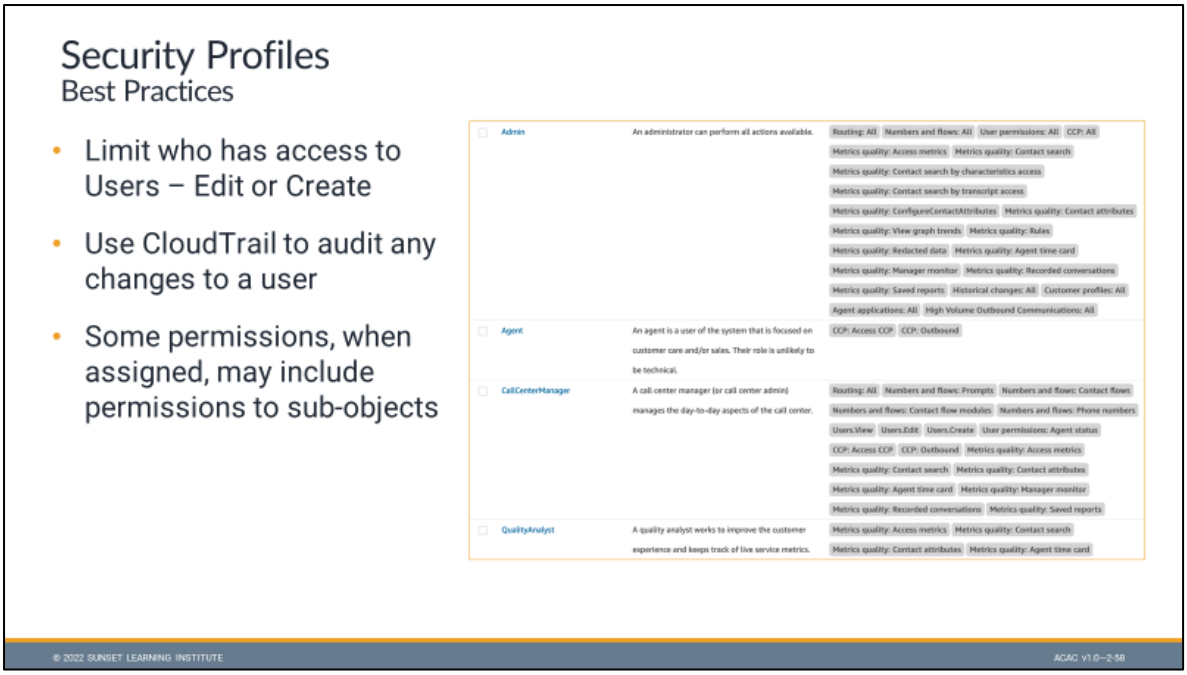


A security profile is a group of permissions that map to a common role in a Contact Center. For example, the Agent security profile contains permissions needed to access the Contact Control Panel (CCP).

Security profiles help you manage who can access the Amazon Connect dashboard and Contact Control Panel (CCP), and who can perform specific tasks.

Amazon Connect includes default security profiles for general roles. You can review the permissions granted by these profiles and use them if they align with the permissions that your users need. Otherwise, create a security profile that grants your users only the permissions they need.

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Best practices for security profiles:

- Limit who has Users - Edit or Create permissions
  - People with these permissions pose a risk to your Contact Center because they can do the following:
    - Reset passwords, including that of the administrator.
    - Grant other users' permission to the Admin security profile. People assigned to the Admin security profile have full access to your Contact Center.

Doing these things would enable someone to lock out those who need to access Amazon Connect and allow in others who can steal customer data and damage your business.

To reduce the risk, as a best practice we recommend limiting the number of people who have **Users - Edit or Create** permissions.

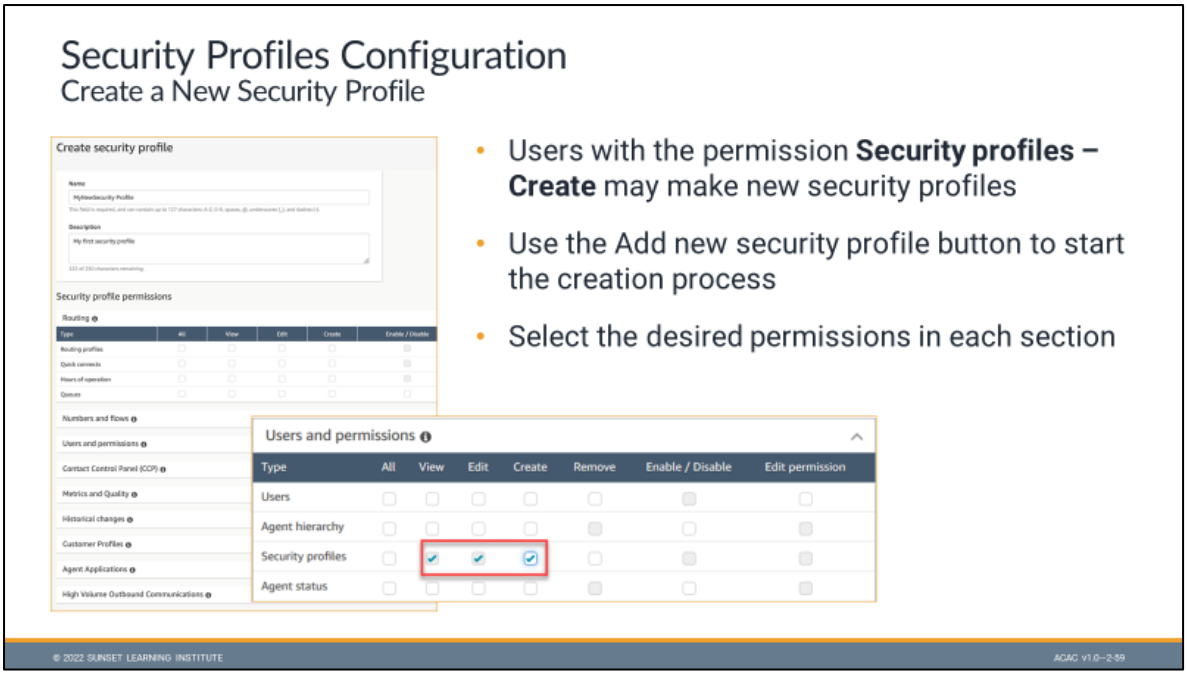
Use AWS Cloudtrail

Cloudtrail allows you to log the requests and responses of UpdateUserIdentityInfo. This enables you to track changes made to user information. Someone who has the ability to call the UpdateUserIdentityInfo API can change a user's email address to one owned by an attacker, and then reset the password through email.

Understand inherited permissions

Some security profiles included inherited permissions: when you assign dedicated permissions to one object, by default permissions are granted to sub-objects. For example, when you grant dedicated permission to edit users, you also grant them permission to list all security profiles for your Amazon Connect instance because the person has access to drop-down list of security profiles.

Before assigning security profiles, review the list of inherited permissions.



Create a security profile

Creating a security profile enables you to grant your users only the permissions that they need.

For each permission group, there is a set of resources and supported set of actions. For example, users are part of the **Users and permissions** group, which supports the following actions: view, edit, create, remove, enable/disable, and edit permission.

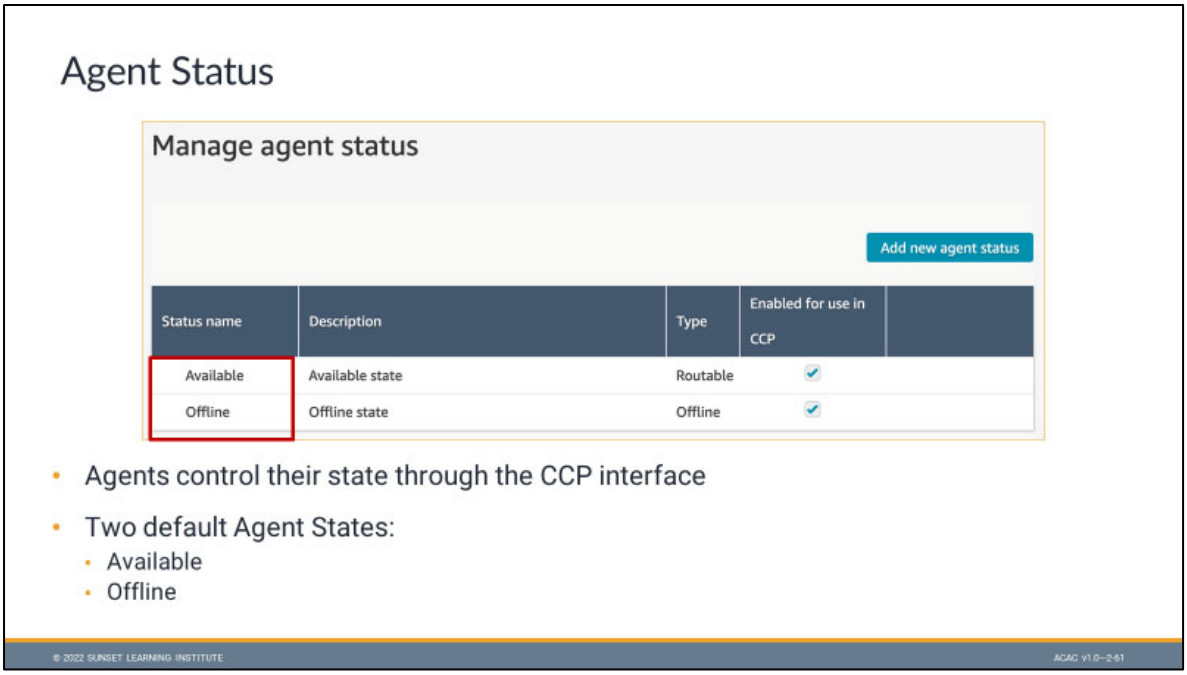
Some actions depend on other actions. When you choose an action that depends on another action, the dependent action is automatically chosen and must also be granted. For example, if you add permission to edit users, we also add permission to view users.

Required permissions to create security profiles

Before you can create a new security profile, you must be logged in with an Amazon Connect account that has the following permissions: **Security profiles - Create.**

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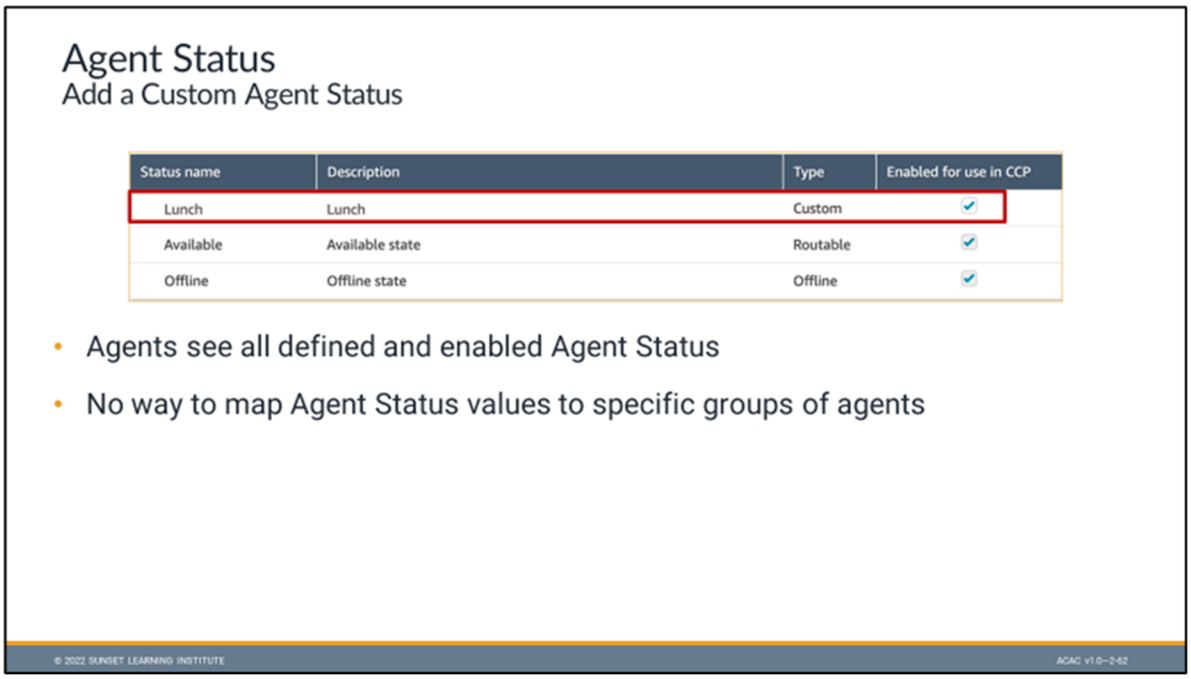
Agent Status and Settings



Agents are responsible for setting their status in the Contact Control Panel (CCP). In fact, the only time an agent's status changes is when they manually change it in the CCP, or when their supervisor changes it in a real-time metrics report.

Amazon Connect provides two default status values:

- Available
- Offline

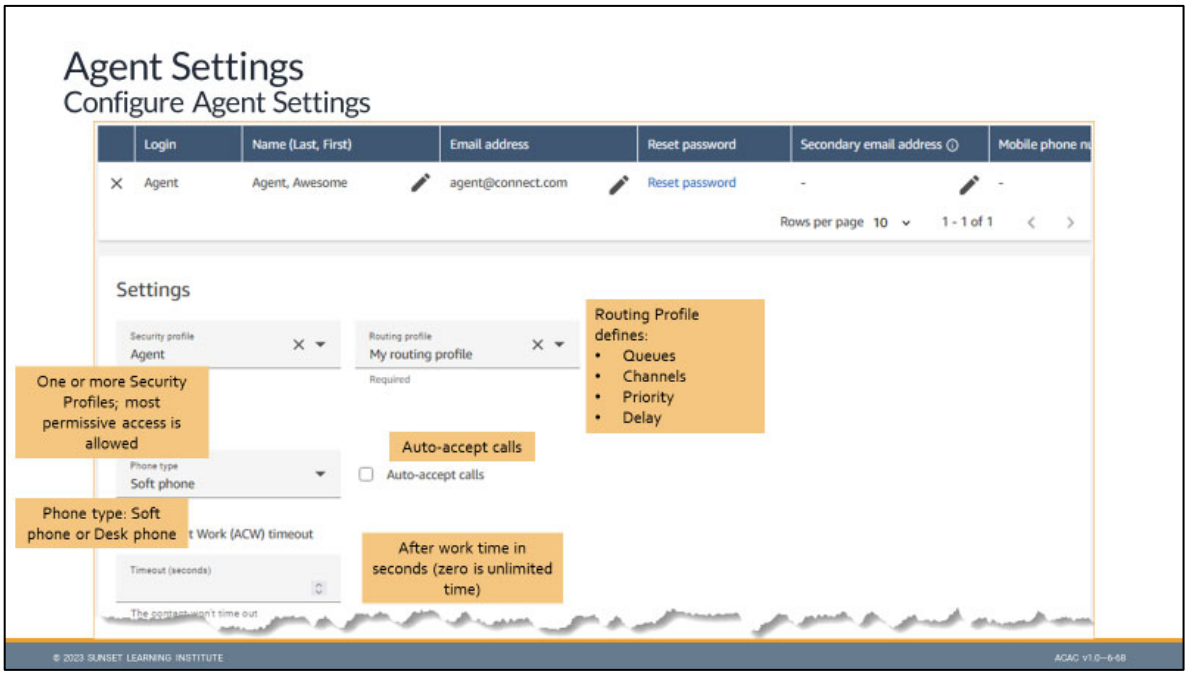


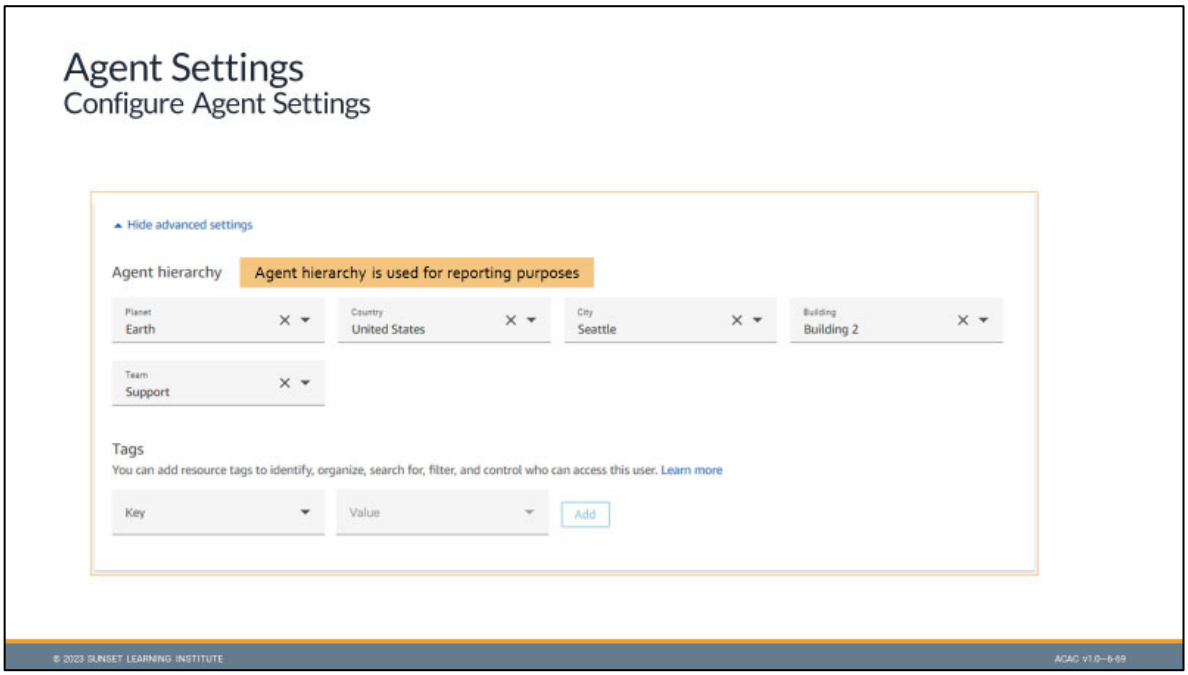
You can change the name of these values, and you can add new ones. For example, you might add a status for Lunch and another for Training. These and the default status values will be used for reporting, metrics, and resource management.

When you add a new status, it will always be **Custom**, not routable.

You can't delete a status value, but you can disable it, so it doesn't appear on the agent's CCP.

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Module 3

Contact Flows

Overview

A contact flow defines how a customer experiences your Contact Center from start to finish. Contact flows enable you to customize your IVR (interactive voice response) system at the most basic level. For example, you can give customers a set of menu options and route customers to Agents based on what they enter on their phones. With Amazon Connect, however, contact flows are significantly more powerful than that: you can create dynamic, personalized flows that interact with other AWS services.

When you create an instance and claim a number, you automatically have a working Contact Center in a few minutes. This is because Amazon Connect includes a set of default contact flows that have already been published. It uses them to power your Contact Center.

When you customize your Contact Center and create new flows, you replace the default contact flows with your own.

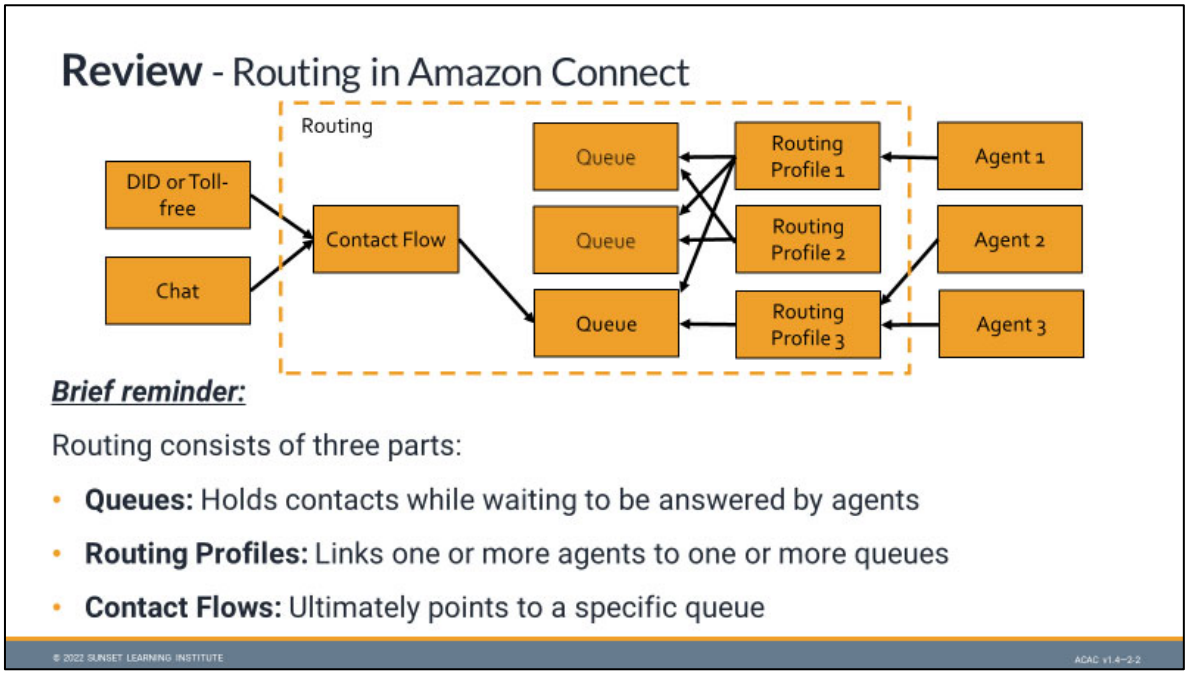
Objectives

Upon completing this module, you will show an understanding of how contact flows work in Amazon Connect. This ability includes being able to meet this objective:

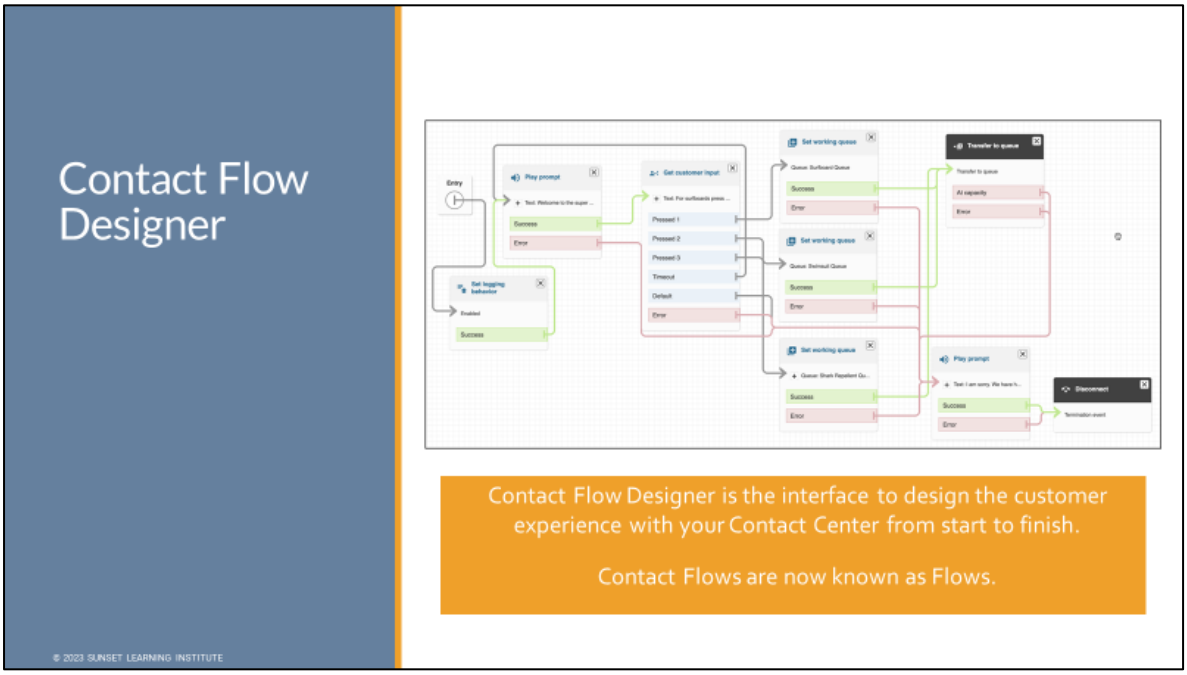
- Create a simple contact flow, create a custom queue logic, demonstrate how to use contact attributes using the Amazon Connect dashboard, and integrate Amazon Polly and SSML.

Contact Flow Designer

To customize your Contact Center, you use the Contact Flow designer. It is a drag-and-drop interface that allows you to customize your Contact Center without coding. This section describes the contact flow designer.

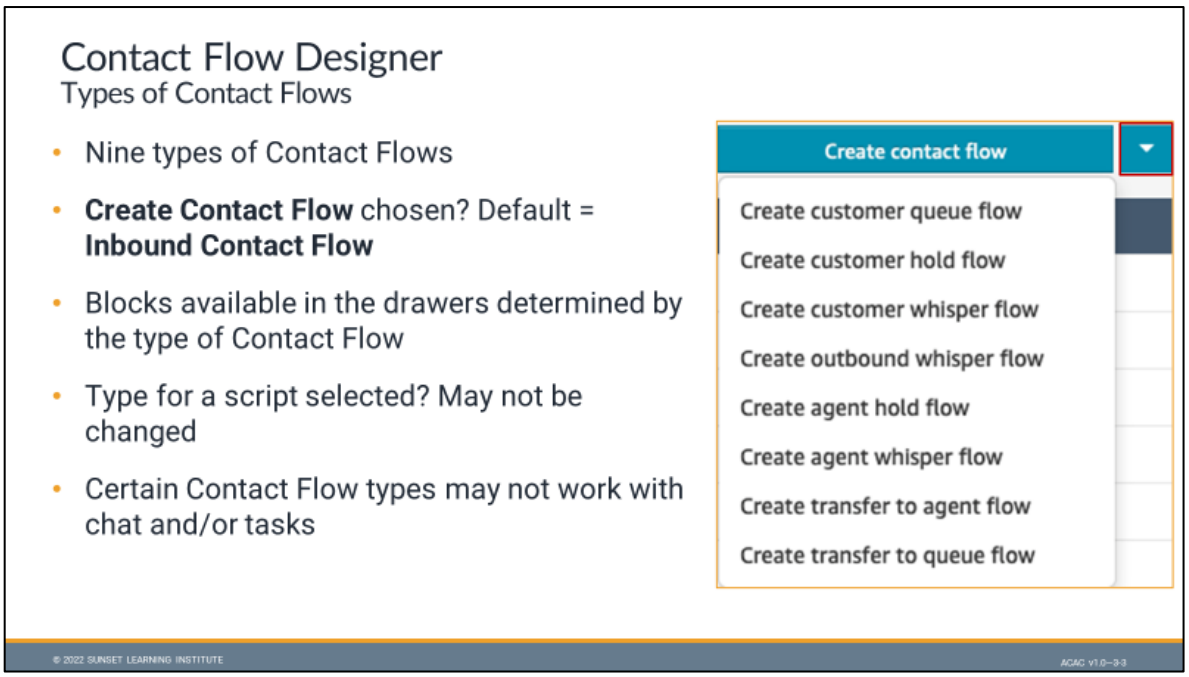


This is such an important concept, that is warrants review. In Amazon Connect, routing consists of three parts: queues, routing profiles, and contact flows.



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To customize your Contact Center, you use the Contact Flow designer. It's a drag-and-drop interface that allows you to customize your Contact Center without any coding.



Choose a Contact Flow type:

Amazon Connect includes a set of nine Contact Flow types. Each type has only those blocks for a specific scenario. For example, the Contact Flow type for transferring to a queue contains only the appropriate contact blocks for that type of flow.

The nine types of Contact Flows are:

- Inbound Contact Flow
- Customer queue flow
- Customer hold flow
- Customer whisper flow
- Outbound whisper flow
- Agent hold flow
- Agent whisper flow
- Transfer to Agent flow
- Transfer to queue flow

See https://docs.aws.amazon.com/connect/latest/adminguide/create-contact-flow.html for additional details.

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Contact Flow Designer
Types of Contact Flows (Cont.)

Type	When to use
Inbound Contact Flow	This is the generic Contact Flow type that's created when you choose the **Create Contact Flow** button, and don't select a type using the drop-down arrow. It creates an inbound Contact Flow. This Contact Flow works with voice, chat, and tasks.
Customer queue flow	Use to manage what the customer experiences while in queue, before being joined to an agent. Customer queue flows are interruptible and can include actions such as an audio clip apologizing for a delay and offering an option to receive a callback, leveraging the **Transfer to queue** block. This Contact Flow works with voice, chat, and tasks.
Customer hold flow	Use to manage what the customer experiences while the customer is on hold. With this flow, one or more audio prompts can be played to a customer using the **Loop prompts** block while waiting on hold. This Contact Flow works with **voice**.
Customer whisper flow	Use to manage what the customer experiences as part of an inbound call immediately before being joined with an agent. The agent and customer whispers are played to completion; then the two are joined. This Contact Flow works with voice and chat.

https://docs.aws.amazon.com/connect/latest/adminguide/create-contact-flow.html

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Contact Flow Designer
Types of Contact Flows (Cont.)

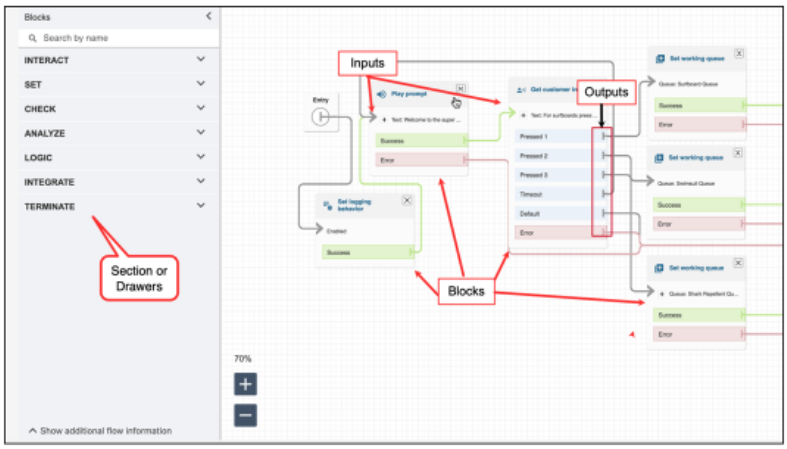
Type	When to use
Outbound whisper flow	Use to manage what the customer experiences as part of an outbound call before being connected with an agent. In this flow, the customer whisper is played to completion; then the two are joined. For example, this flow can be used to enable call recordings for outbound calls with the **Set recording behavior** block. This Contact Flow works with voice and chat.
Agent hold flow	Use to manage what the agent experiences when on hold with a customer. With this flow, one or more audio prompts can be played to an agent using the **Loop prompts** block while the customer is on hold. This Contact Flow works with **voice**.
Agent whisper flow	Use to manage what the agent experiences as part of an inbound call immediately before being joined with a customer. The agent and customer whispers are played to completion; then the two are joined. This Contact Flow works with voice, chat, and tasks.
Transfer to agent flow	Use to manage what the agent experiences when transferring to another agent. This type of flow is associated with transfer-to-agent quick connects, and often plays messaging, then completes the transfer using the **Transfer to agent** block. This Contact Flow works with voice, chat, and tasks.
Transfer to queue flow	Use to manage what the agent experiences when transferring to another queue. This type of flow is associated with transfer-to-queue quick connects, and often plays messaging, then completes the transfer using the **Transfer to queue** block. This Contact Flow works with voice, chat, and tasks

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Contact Flow Designer

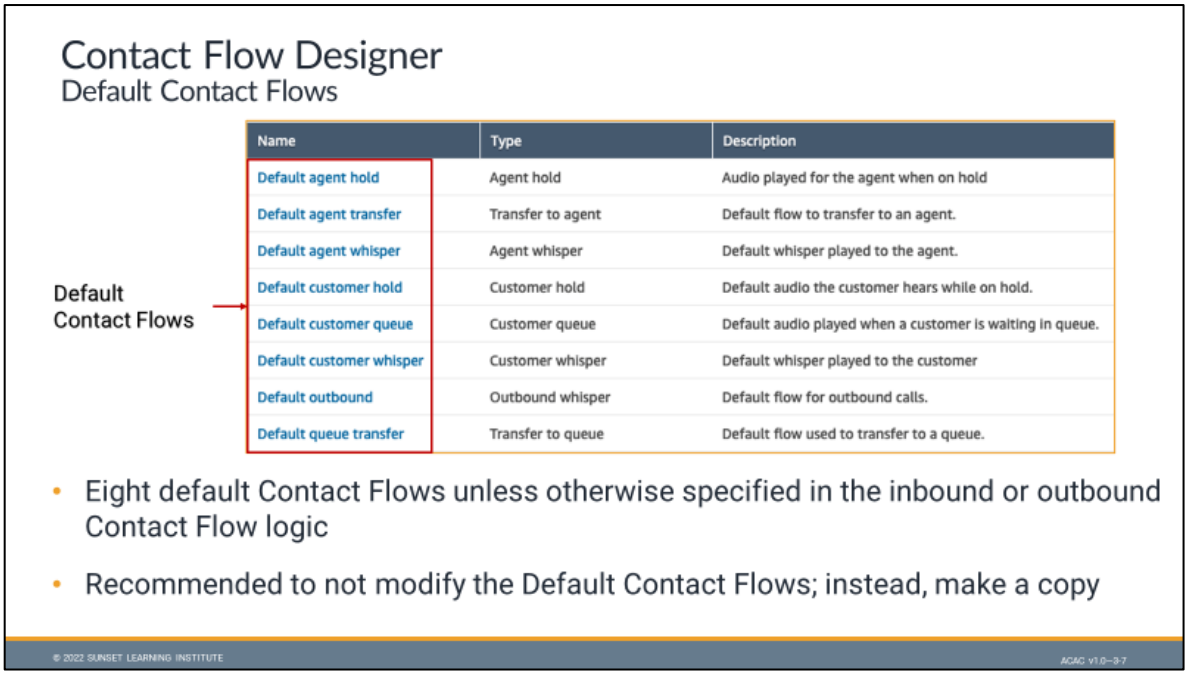
Sample Inbound Flow (First Contact Experience)

- Contact Flow
  - Used by default
  - Composed of blocks connected together
- Always an Entry point block
- Sections contain blocks to be dragged to the workspace
- All outputs must be connected somewhere



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Default Contact Flows

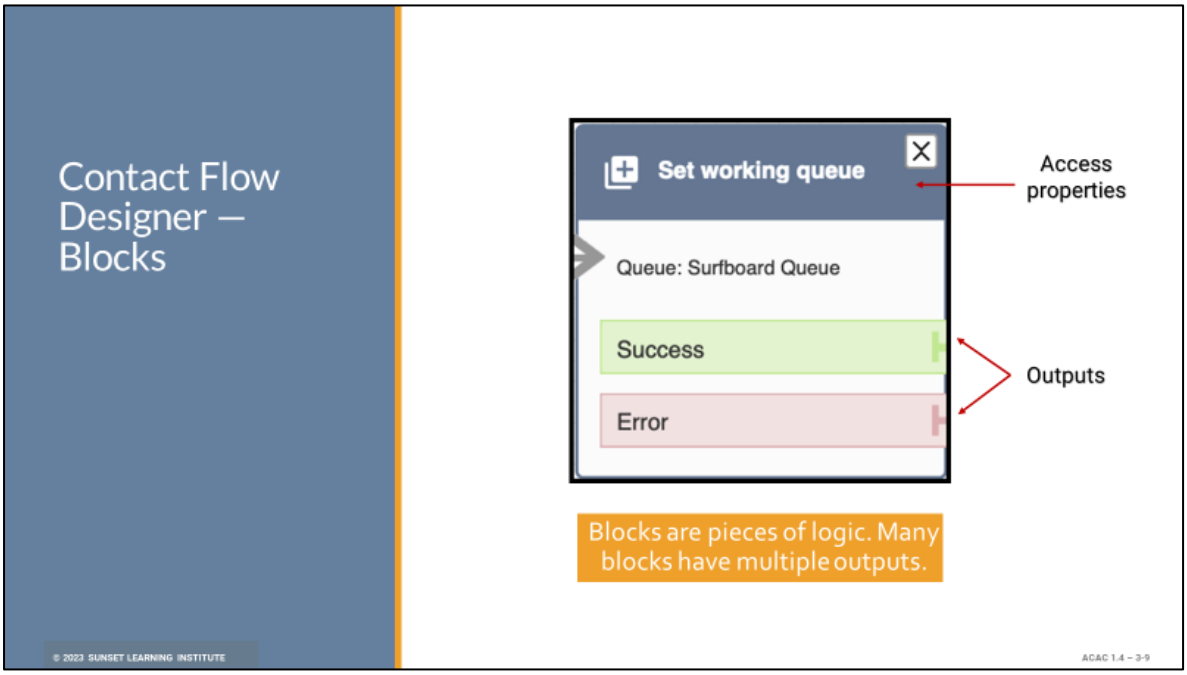
Amazon Connect includes a set of default Contact Flows that have already been published. It uses them to power your Contact Center.

For example, say you create a Contact Flow that includes putting the customer on hold, but you don't create a prompt for it. The default Contact Flow, Default Agent hold, will be played automatically. This is a way to help you get started with your call center quickly.

If you want to change the behavior of a default Contact Flow, we recommend creating a new customized flow based on the default. Then call the new flow intentionally in your Contact Flows rather than defaulting to it. This gives you better control over how your Contact Flows work.

Contact Flow Designer — Blocks

Contact blocks are the building blocks of your Contact Flows. Each block is designed for a specific function a business might want in a Contact Center. This section describes how to configure contact flow designer – blocks.



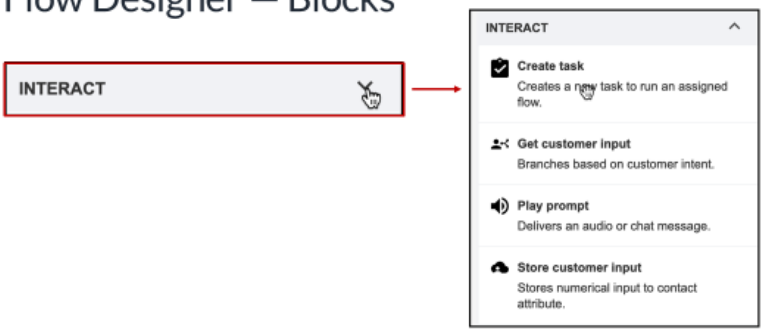
Contact blocks

Contact blocks are the building blocks of your Contact Flows. Each block is designed for a specific function a business might want in a Contact Center.

Contact blocks will have a variety of outputs in most cases. These outputs must always be connected somewhere.

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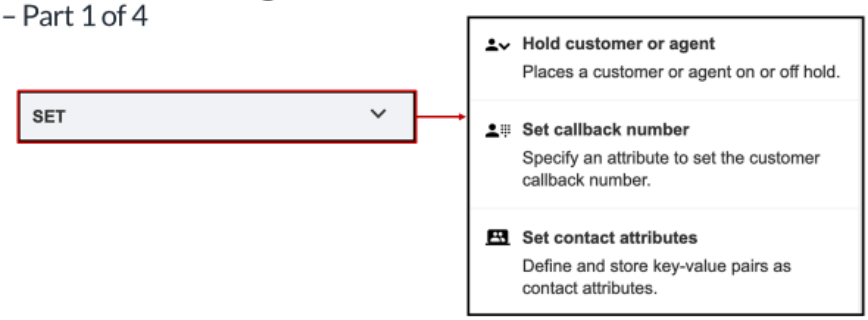
Contact Flow Designer — Blocks
Interact



- **Create Task:** Creates a new task to run an assigned flow
- **Play prompt:** Interruptible audio prompt, a text-to-speech message, or a chat response
- **Get customer input:** Branches based on customer intent
- **Store customer input:** Stores numerical input to a contact attribute

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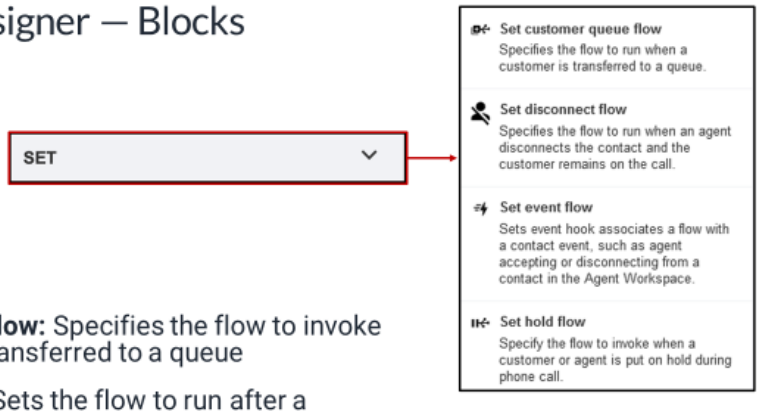
Contact Flow Designer — Blocks
Set – Part 1 of 4



- **Hold customer or agent:** Puts a person on or off hold
- **Set callback number:** Specifies an attribute to set the customer callback number
- **Set contact attributes:** Stores key-value pairs as contact attributes

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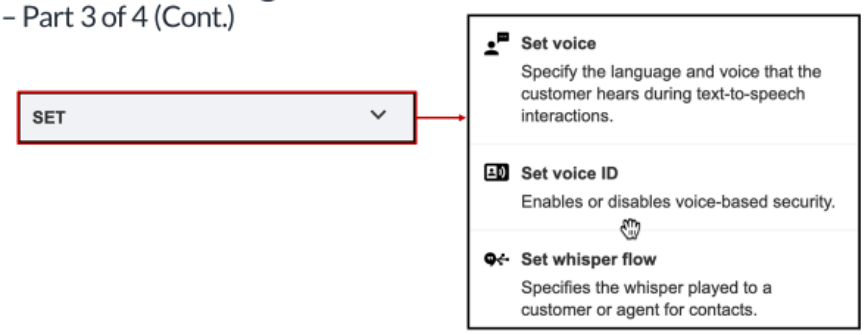
Contact Flow Designer — Blocks
Set – Part 2 of 4 (Cont.)



- **Set customer queue flow:** Specifies the flow to invoke when a customer is transferred to a queue
- **Set disconnect flow:** Sets the flow to run after a disconnect event
- **Set event flow:** Associates a flow with contact event
- **Set hold flow:** Links from one Contact Flow type to another

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Contact Flow Designer — Blocks
Set – Part 3 of 4 (Cont.)

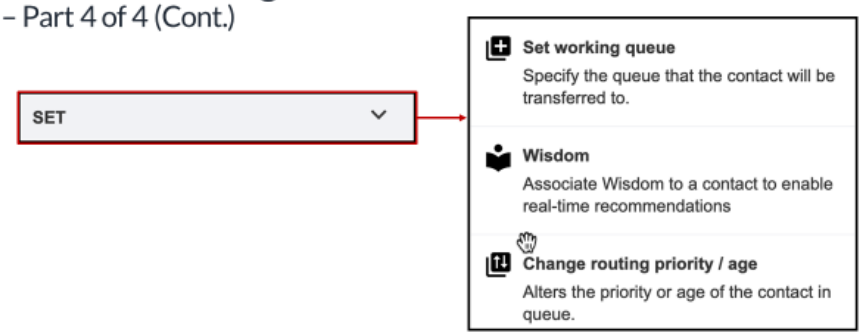


- **Set voice:** Specify the language and voice that the customer hears during text-to-speech interactions
- **Set voice ID:** Enables or disables voice-based security
- **Set whisper flow:** Overrides the default whisper by linking to a whisper flow

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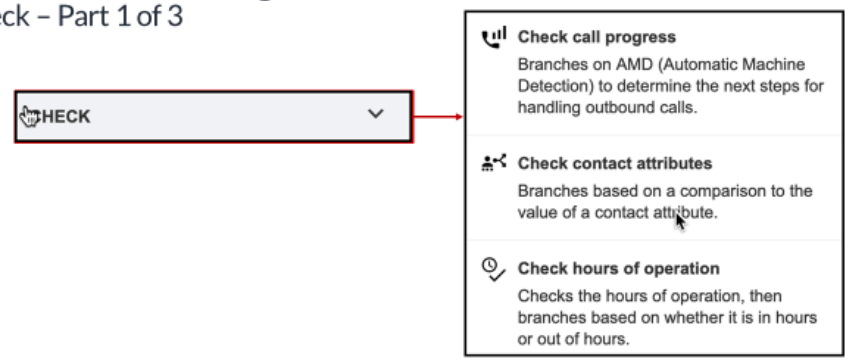
Contact Flow Designer — Blocks
Set – Part 4 of 4 (Cont.)



- **Set callback number:** Set a callback target number to the selected queue
- **Set contact attributes:** Stores key-value pairs as contact attributes
- **Set working queue:** Specifies the queue to be used when Transfer-to-queue is invoked

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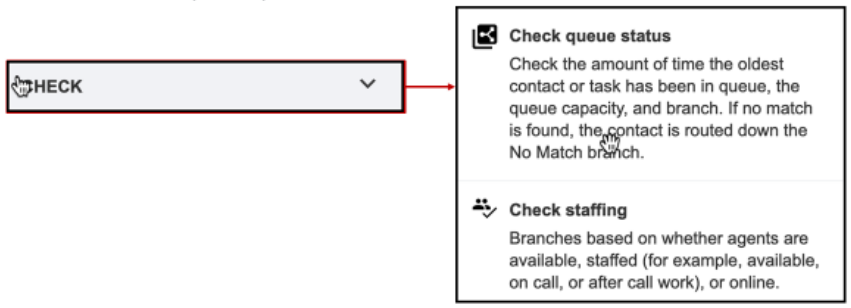
Contact Flow Designer — Blocks
Check – Part 1 of 3



- **Check call progress:** Branches on AMD to determine the next steps for handling outbound calls
- **Check contact attributes:** Checks the values of contact attributes
- **Check hours of operation:** Checks whether the contact is occurring within or outside of the hours of operation defined for the queue

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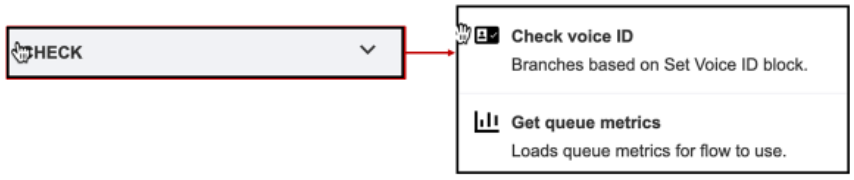
Contact Flow Designer — Blocks
Check — Part 2 of 3 (Cont.)



- **Check queue status:** Checks the status of the queue based on specified conditions
- **Check staffing:** Checks the current working queue or queue you specify in the block, for whether agents are available, staffed, or online

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Contact Flow Designer — Blocks
Check — Part 3 of 3 (Cont.)

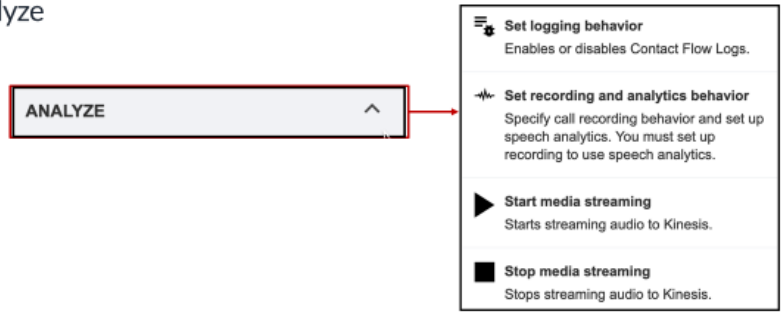


- **Check voice ID:** Voice ID leads to branches based on these items:
  - Enrollment status
  - Voice authentication status
  - Status of detection of fraudsters in a watchlist of the caller
- **Get queue metrics:** Retrieves real-time metrics about queues and agents, and returns them as attributes

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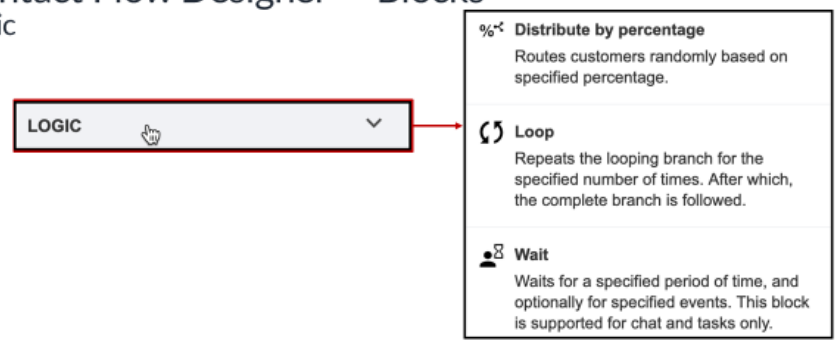
Contact Flow Designer — Blocks
Analyze



- **Set logging behavior:** Enables or disables Contact Flow Logs
- **Set recording and analytics behavior:** Specify call recording behavior and set up speech analytics. You must set up recording to use speech analytics
- **Start media streaming:** Starts capturing customer audio for a contact
- **Stop media streaming:** Stops capturing customer audio with a Start media streaming block

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Contact Flow Designer — Blocks
Logic



- **Distribute by percentage:** Routes customers randomly based on a percentage
- **Loop:** Loops through, or repeats, the Looping branch for the number of loops specified
- **Wait:** Pauses the Contact Flow

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Contact Flow Designer — Blocks
Integrate

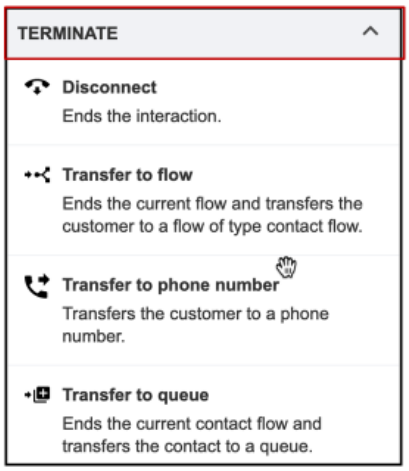
- **Case:** Get, update or create a case
- **Customer profile:** Get, update, or create a customer profile
- **Invoke AWS Lambda function:** Calls AWS Lambda, optionally returns key-value pairs
- **Invoke module:** Select a module to invoke in a contact flow
- **Show View:** Renders a View in a user application and waits for a response from the application



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Contact Flow Designer — Blocks
Terminate / Transfer

- **Disconnect:** Disconnects a contact
- **Transfer to flow:** Transfers the customer to another Contact Flow
- **Transfer to phone number:** Transfers the customer to a phone number external to your instance
- **Transfer to queue:**
  - In most Contact Flows, ends the current Contact Flow and places the customer in queue
  - In a customer queue flow, transfers a contact already in a queue to another queue

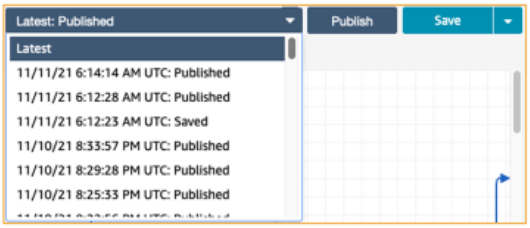


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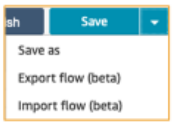
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Contact Flow Designer — Blocks
Saving and Publishing

- Saving a Contact Flow? Does not make it live
- Invalid script may be saved
- Must be Published to go live with a version of a Contact Flow



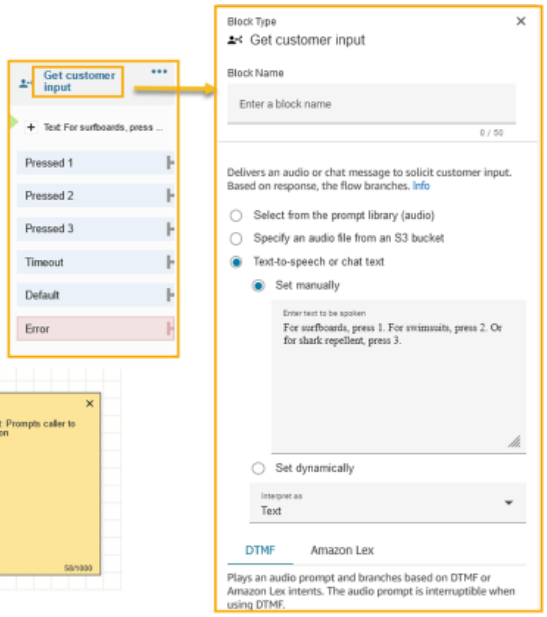
- Only valid Contact Flows may be Published
- Contact Flows possibilities:
  - Reverted to an historical version
  - Can be imported or exported (This feature is in Beta)



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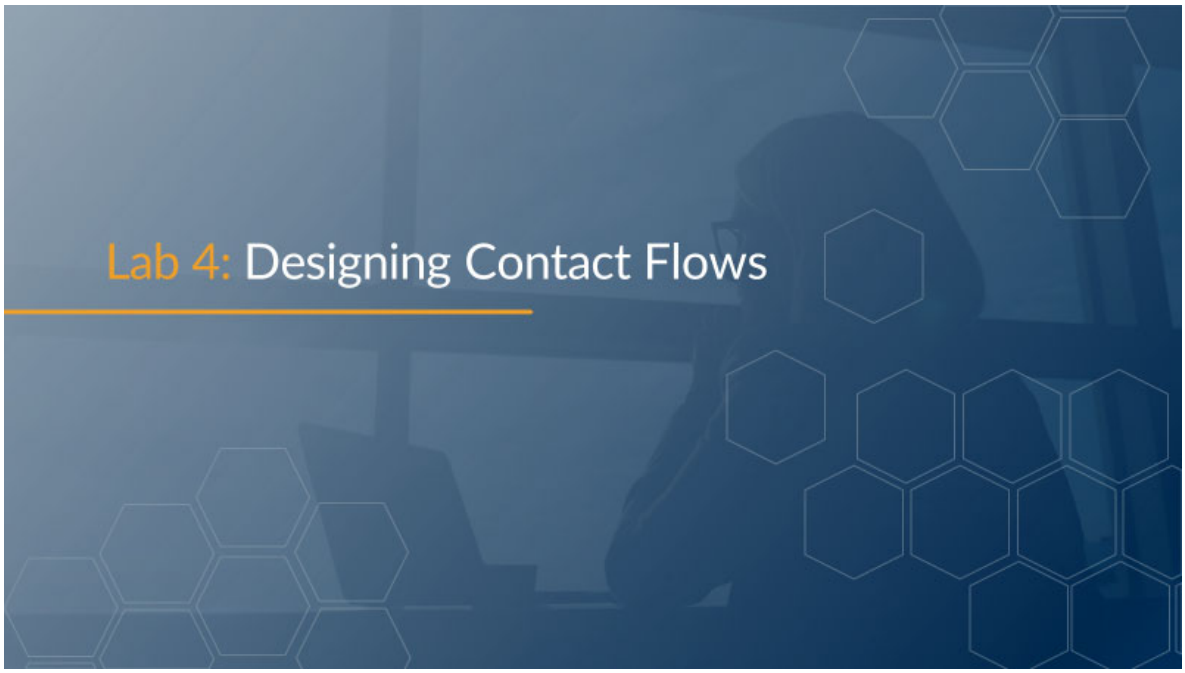
Contact Flow Designer — Blocks
Properties of a Block

- Select title of the block to access the properties of the block
- Differing settings are based on the block
- Recent update to Designer adds Block Name and Notes
- **Save** when done.



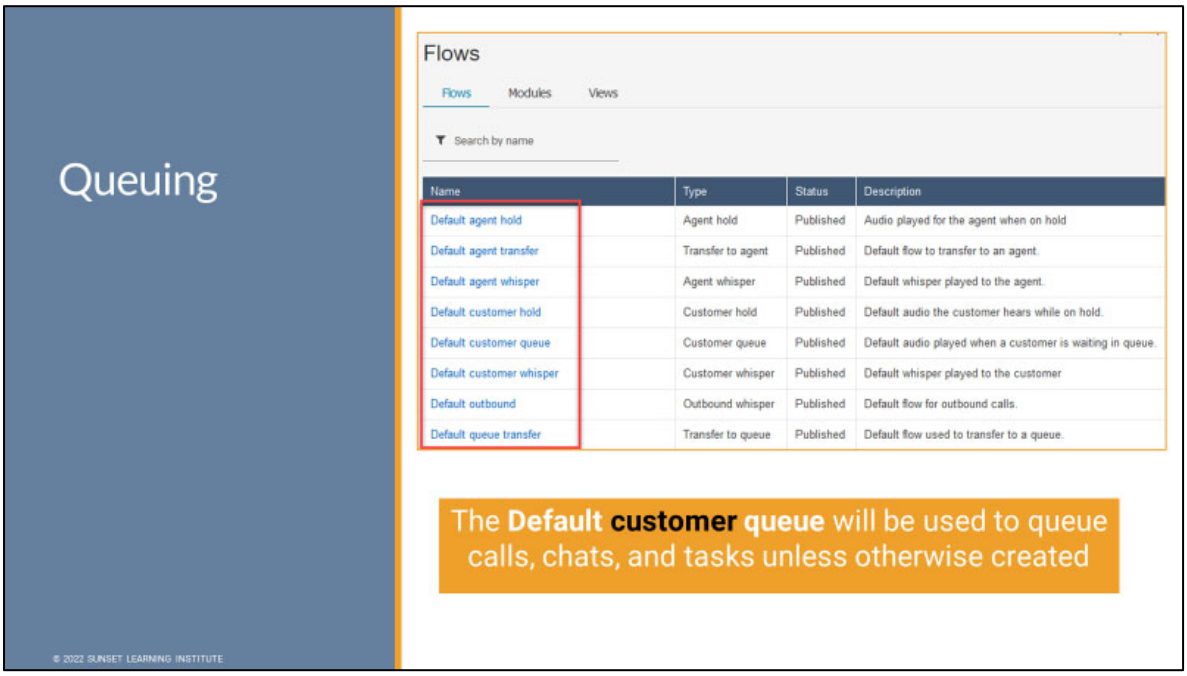
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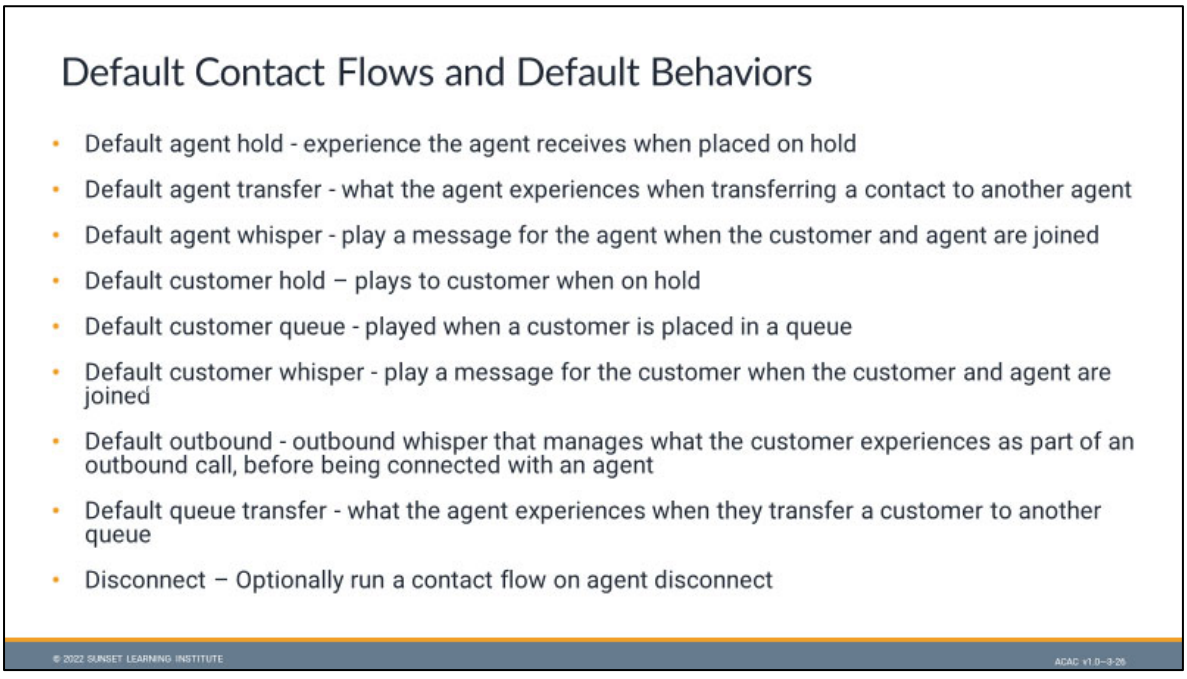




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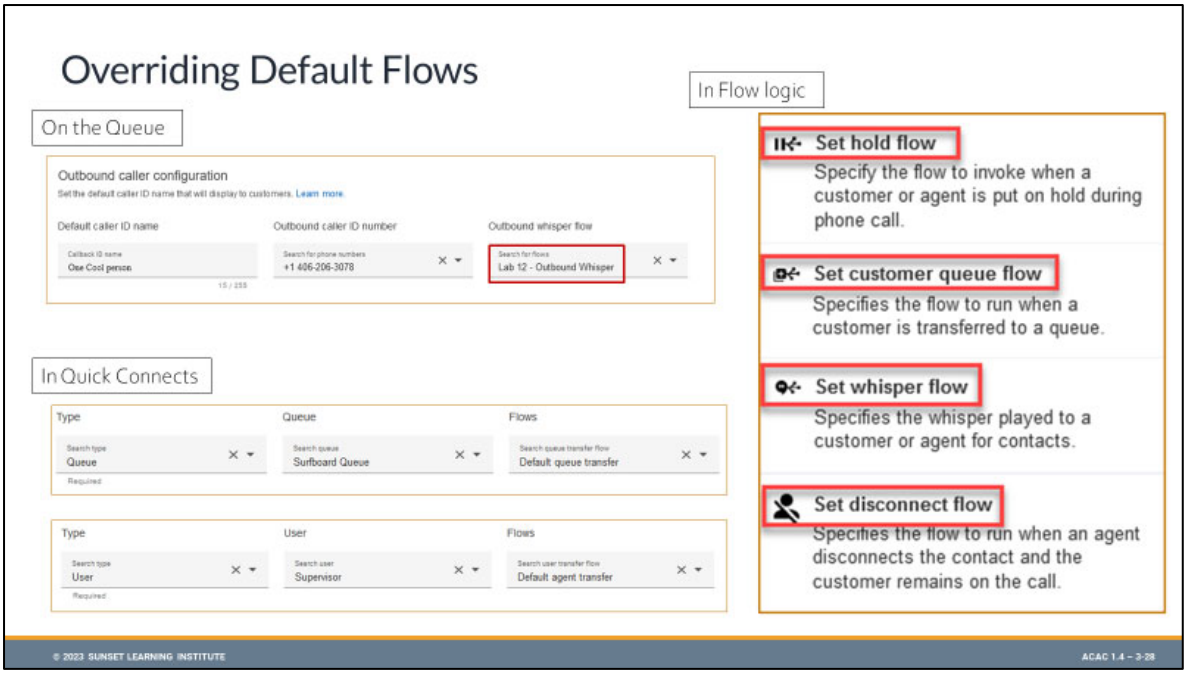
Default Contact Flows and Behaviors





You can override the way the default flows work by editing them directly.

Generally, it is recommended to create new flows based on the defaults rather than editing the default flow directly. You can make a copy of the default flow, assign a name that indicates it's a custom version, and then edit that one. This gives you more control over how your contact flows work.

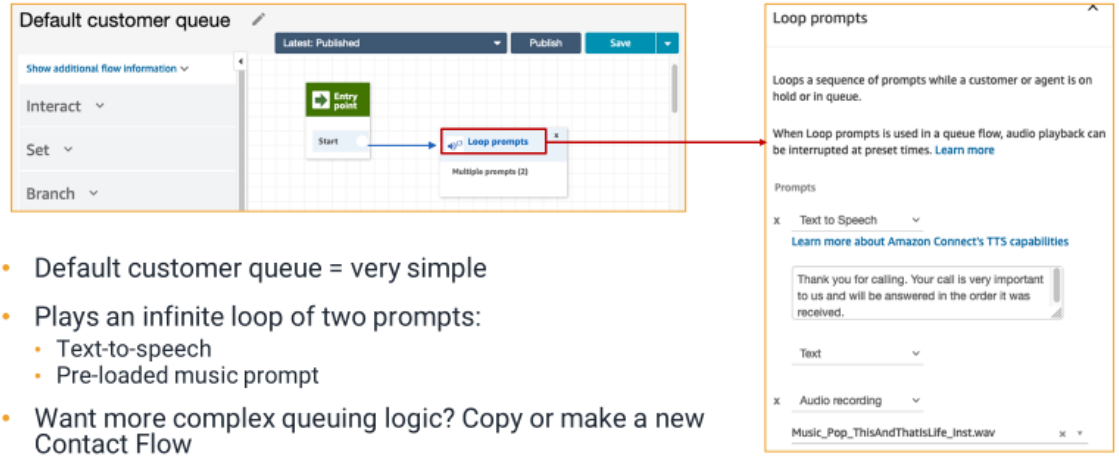


While using default flows provides a starting point for most contact centers, part of the power of Amazon Connect is to be able to customize the behavior to support the needs of the business. Once you have created alternate flows, you may apply them as part of the Queue for Outbound Flows and as part of the Quick Connect. Possibly more critically, as part of the development of individual Flows, you have the ability to customize the behavior for each Flow but also for different branches of the Flow.

For example, the **Set customer queue flow** block defines the customer's experience once the contact is put into queue. However, contacts in a Sales queue may hear a message regarding the latest promotion whereas a contact in a Support queue may a hear a message reminding customers of the current software version. Creating multiple customer queue flows can be referenced by different **Set customer queue flow** blocks from within a Flow.

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Queuing
Default Customer Queue Contact Flow

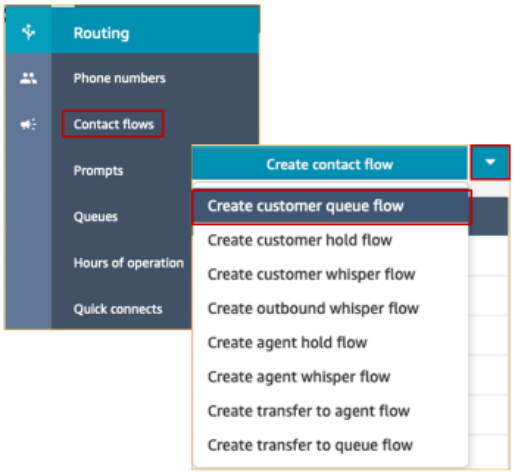


- Default customer queue = very simple
- Plays an infinite loop of two prompts:
  - Text-to-speech
  - Pre-loaded music prompt
- Want more complex queuing logic? Copy or make a new Contact Flow

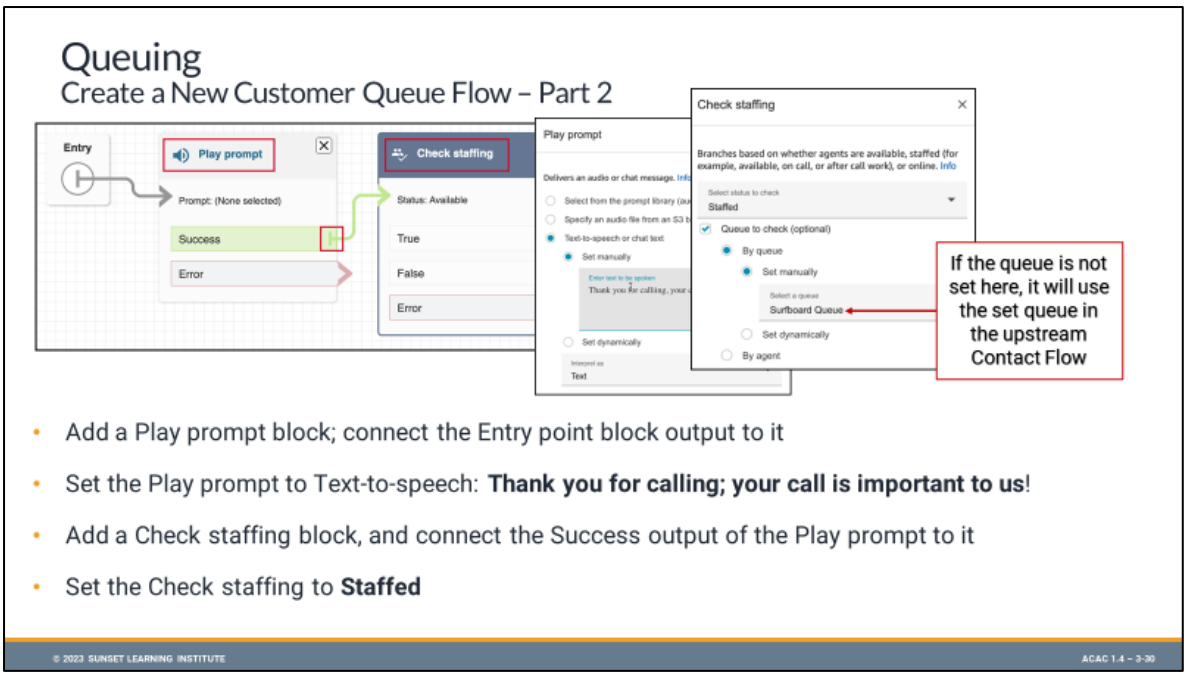
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Queuing
Create a New Customer Queue Flow – Part 1

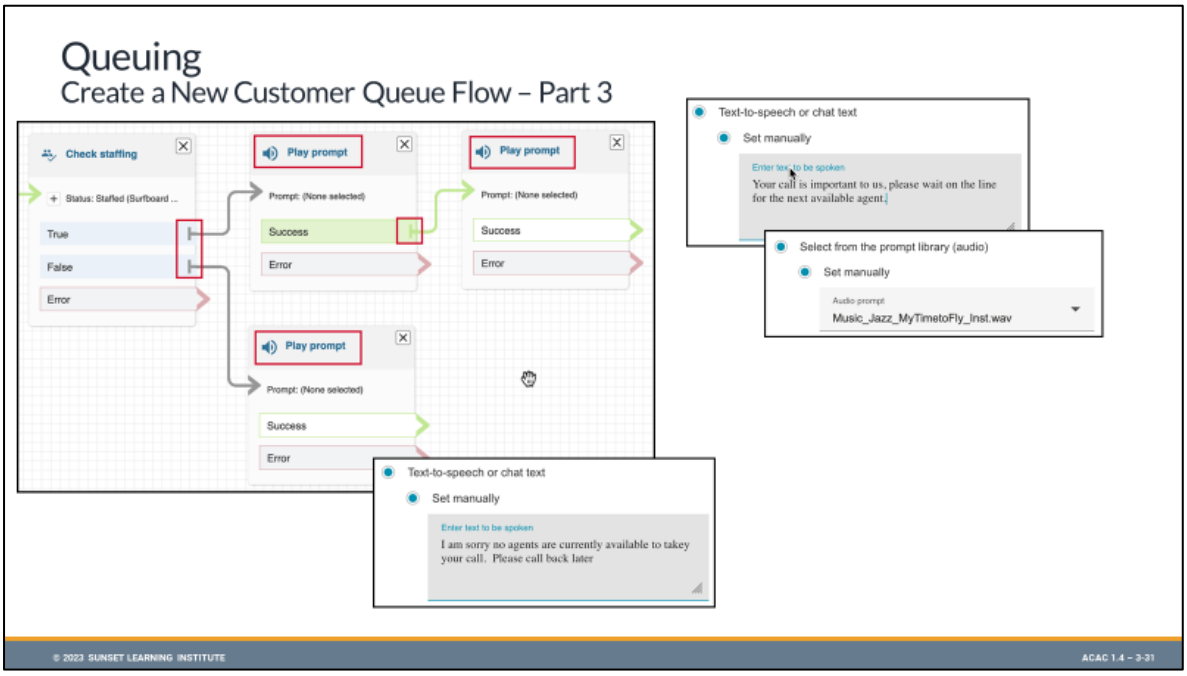
- Navigate to **Routing** > **Contact Flows**
- Use the drop-down to select **Create customer queue flow**
- Design the desired queue logic



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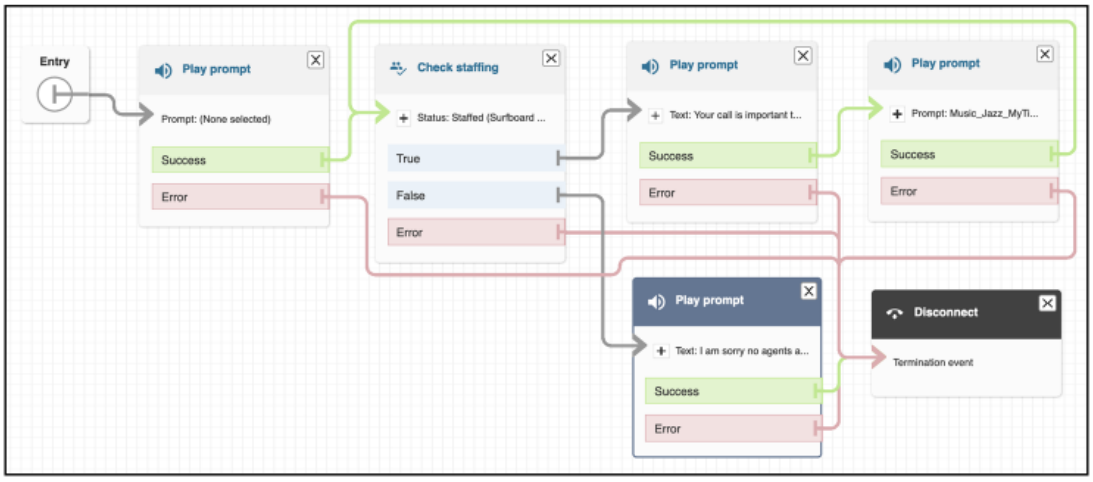


The **Check staffing** block offers three values: **Available**, **Staffed**, or **Online**. When set as **Available**, the **Check staffing** block returns true if the number of agents in the **Available** state is greater than or equal to 1. Having agents are **Available**, **On Call**, or in **After Contact** states are considered to be **Staffed**. The state **Online** applies if agents are in any state other than logged off.



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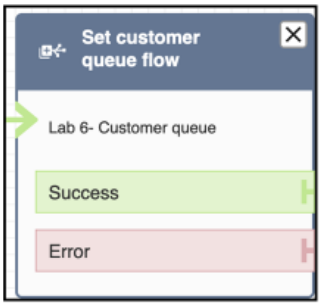
Queuing
Create a New Customer Queue Flow – Part 4

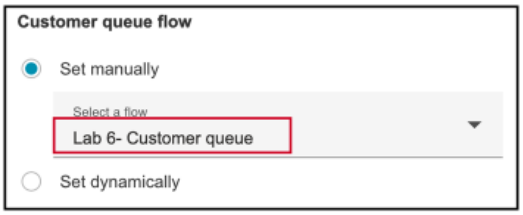


- See instructions on the following slide

Queuing
Invoke the New Customer Queue Flow

- To use something other than the **Default customer queue flow,** set it in the incoming Contact Flow
- Use the **Set customer queue flow** block and select the queue desired
- Customer queue flows may be set per customer, if desired

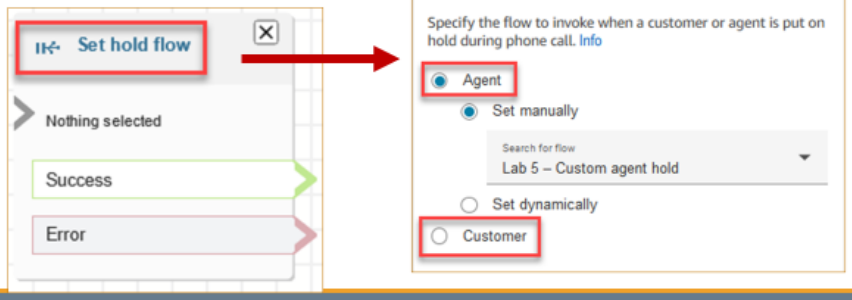




Hold

Default Customer Hold and Default Agent Hold Flows

- Default Hold treatment for Customer is simple music loop
- Default Hold treatment for Agent is simple prompt in a loop
- Want more complex hold logic? Copy or make a new Flow

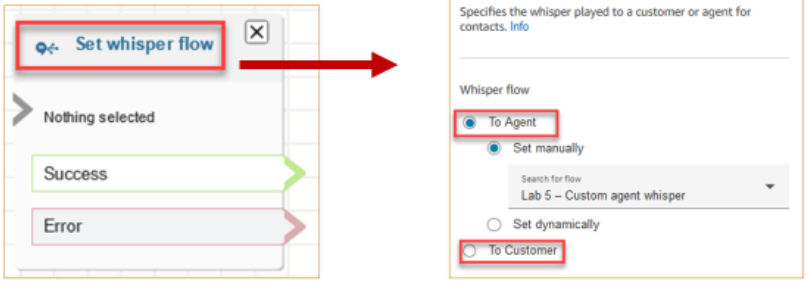


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Whisper

Default Customer Whisper and Default Agent Whisper Flows

- Default Hold treatment for Customer is a wave files that plays a “beep” sound
- Default Whisper treatment for Agent the name of the selected Queue
- Want more complex whisper logic? Copy or make a new Flow



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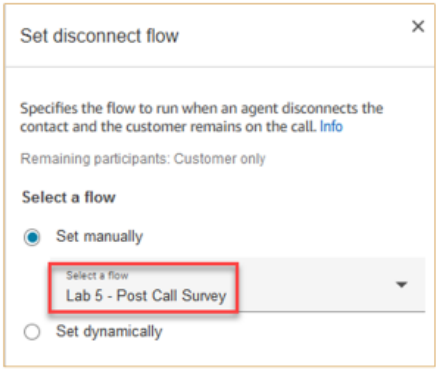
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Disconnect

Evoking a Disconnect Flows

- There is no Disconnect logic unless set in a contact flow for the contact
- Use the Set Disconnect Flow block to set a Disconnect contact flow when an agent disconnects
- If the customer disconnects not contact flow is run



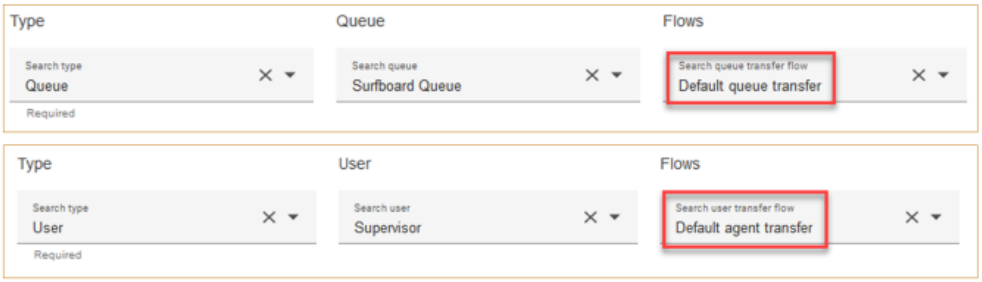


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Transfer

Default queue transfer and Default agent transfer

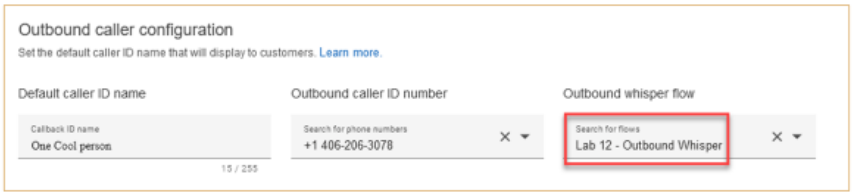
- Set in the Quick Connect
- Default queue transfer logic checks hours and staffing and then transfers to the queue
- Default agent transfer plays a transfer message and then transfers to a specified agent



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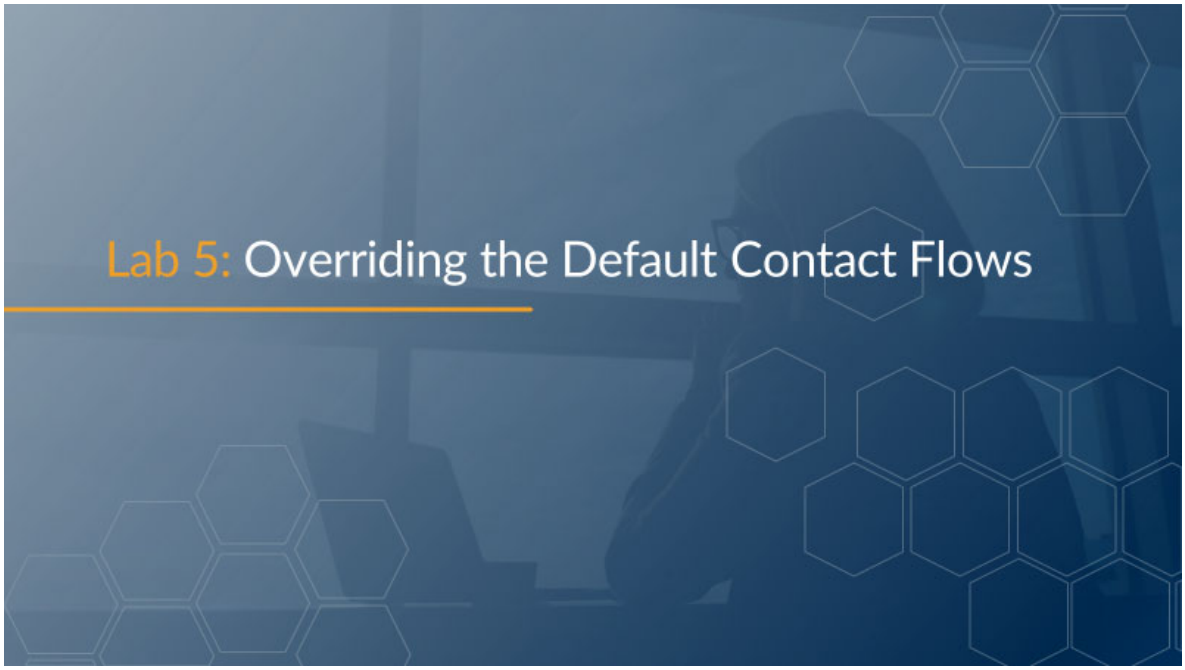
Outbound
Default Outbound Whisper

- When a customer is called back, this defines the treatment the caller will hear
- Default outbound whisper logic plays a prompt stating the call is not being recorded
- This can be overridden on the Queue settings



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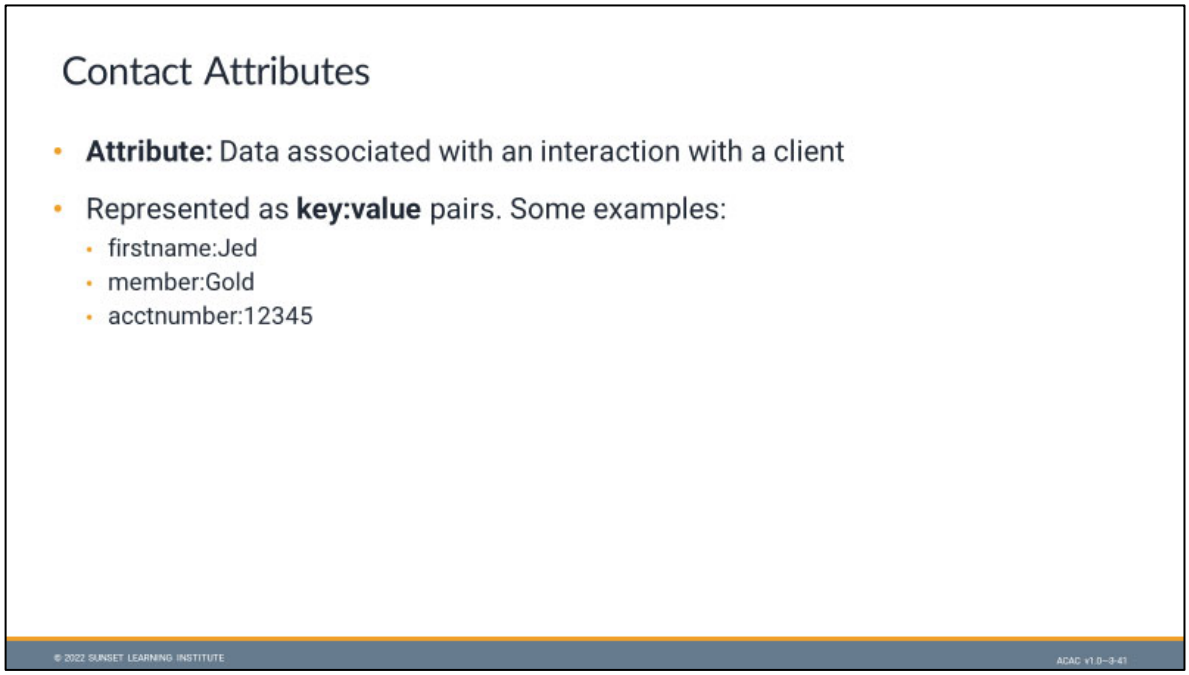
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Contact Attributes

One way to make your customers feel cared for is to create personalized experiences for them in your Contact Center. For example, you can deliver one welcome message for customers who are using a phone and another for customers using chat. To do this, you need a way to store information about the contact and then decide what to use based on the value. In Amazon Connect, each interaction with a customer is a contact. The interaction can be a phone call (voice), a chat, or an automated interaction using an Amazon Lex Bot. This section describes contact attributes and use cases for where they are used.



In Amazon Connect, each interaction with a customer is a **contact**. The interaction can be a phone call (voice), a chat, or an automated interaction using an Amazon Lex Bot.

Each contact can have some data that is specific to a particular interaction. This data can be accessed as a contact attribute. For example:

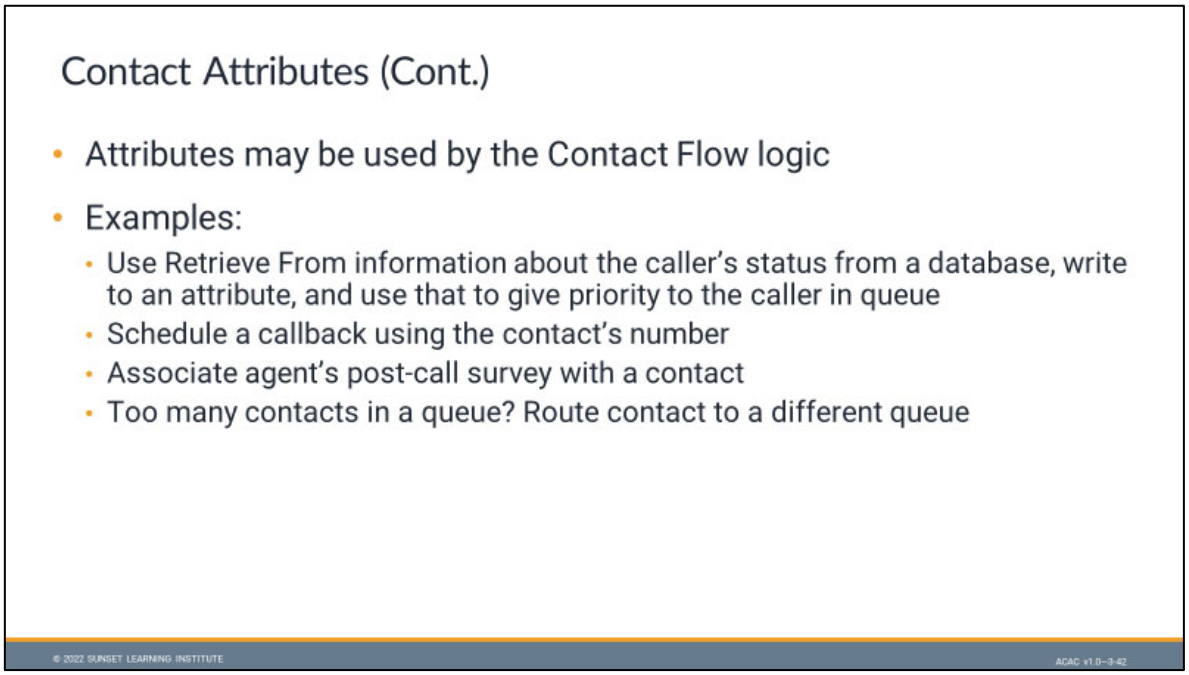
- The name of the customer
- The name of the Agent
- The channel used for the contact, such as phone or chat
- And more

A contact attribute represents this data as a **key-value** pair. You might think of it as a field name together with the data entered in that field.

The advantage of contact attributes is that they enable you to store temporary information about the contact so you can use it in the Contact Flow.

For example, in your welcome messages, you can say their name or thank them for being a member. To do this, you need a way of retrieving data about that specific customer and using it in a Contact Flow.

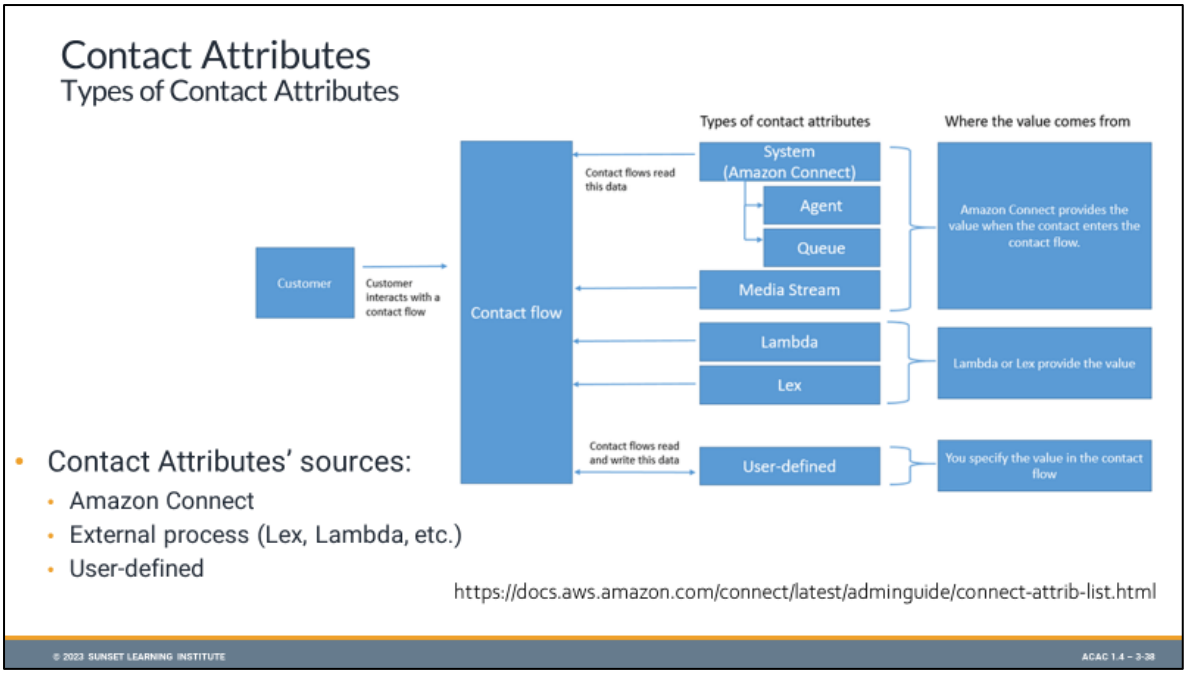
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Common use cases:

Here are some common use cases for where contact attributes are used:

- Use the customer phone number to schedule a queued callback.
- Identify which Agent is interacting with a customer so that a post call survey can be associated with a contact.
- Identify the number of contacts in a queue to decide if the contact should be routed to a different queue.
- Get the corresponding media streaming ARN to store in a database.
- Use the customer phone number to identify the status of a customer (for example, are they a member), or the status of their order (shipped, delayed, etc.) to route them to the appropriate queue.
- Based on a customer interaction with a Bot, identify the slot (for example, the type of flowers to order) to be used in a flow.



Types of contact attributes

To make it faster for you to find and choose the attributes you want to use, attributes are grouped into types. For each contact block, we only surface those types of attributes that work with it.

Another way to think about types of contact attributes is to categorize them based on where the value comes from. The values for contact attributes have three sources:

- **Amazon Connect:** Connect provides the value, such as the Agent's name, during the contact interaction. This is known as providing the value at runtime.
- **An external process:** A process, such as Amazon Lex or AWS Lambda, provides the value.
- **User-defined:** User is defined in the Contact Flow; you can specify the value for an attribute.

Contact attributes in the contact record

In contact records, contact attributes are shared across all contacts with the same InitialContactId.

For example, while carrying out transfers, a contact attribute updated in the transfer flow updates the attribute's value in the contact attributes of Both contact records, that is, the Inbound and Transfer contact attributes.

For the full list of contact attributes, refer to https://docs.aws.amazon.com/connect/latest/adminguide/connect-attrib-list.html.

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Contact Attributes
System Attributes

https://docs.aws.amazon.com/connect/latest/adminguide/connect-attrib-list.html

Attribute	Description	JSONPath Reference
Customer number	The customer's phone number	$.CustomerEndpoint.Address
Customer ID	The customer's identification number	$.CustomerId
Dialed number	The number the customer dialed to call your contact center	$.SystemEndpoint.Address
Customer callback number	The number that Amazon Connect uses to call back the customer	N/A
Stored customer input	An attribute created from the most recent invocation of a **Store customer input** block	N/A
Queue name	The name of the queue	$.Queue.Name
Queue ARN	The ARN of the queue	$.Queue.ARN
Queue outbound number	The Outbound caller ID number for the selected queue	
Text to speech voice	The name of the Amazon Polly voice to use for text-to-speech	$.TextToSpeechVoiceId

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Contact Attributes
System Attributes (Cont.)

Attribute	Description	JSONPath Reference
Contact id	The unique identifier of the contact	$.ContactId
Initial Contact id	The unique identifier for the contact associated with the first interaction between the customer and your contact center	$.InitialContactId
Task Contact id	The unique identifier for the task contact. Use the task contact ID to track tasks between Contact Flows	$.Task.ContactId
Previous Contact id	The unique identifier for the contact before it was transferred. Use the previous contact ID to trace contacts between Contact Flows	$.PreviousContactId
Channel	The method used to contact your contact center: VOICE, CHAT, TASK	$.Channel
Instance ARN	The ARN for your Amazon Connect instance	$.InstanceARN
Initiation method	How the contact was initiated. Valid values include: INBOUND, OUTBOUND, TRANSFER, CALLBACK, QUEUE_TRANSFER, DISCONNECT, and API.	$.InitiationMethod
Name	The name of the task	$.Name
Description	A description of the task	$.Description

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Contact Attributes
System Attributes (Cont.)

Attribute	Description	JSONPath Reference
References	Links to other documents that are related to a contact	s.References.ReferenceKey.Value and s.References.ReferenceKey.Type
Language	The language of content. Use the standard java.util.Locale. For example, en-US for United States English, jp-JP for Japanese, etc.	s.LanguageCode
System Endpoint Type	The type of the system endpoint. Valid value is TELEPHONE_NUMBER	s.SystemEndpoint.Type
Customer Endpoint Type	The type of the customer endpoint. Valid value is TELEPHONE_NUMBER	s.CustomerEndpoint.Type
Queue Outbound Caller ID number	The outbound caller ID number defined for the queue. This can be useful for reverting the caller ID after setting a custom caller ID	s.Queue.OutboundCallerId.Address
Queue Outbound Caller ID number type	The type of the outbound caller ID number. Valid value is TELEPHONE_NUMBER	s.Queue.OutboundCallerId.Type

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Contact Attributes
Agent Attributes

Attribute	Description	JSONPath Reference
Agent Username	The username an agent uses to log in to Amazon Connect	s.Agent.UserName
Agent First Name	The agent's first name as entered in the Amazon Connect user account	s.Agent.FirstName
Agent Last Name	The agent's last name as entered in the Amazon Connect user account	s.Agent.LastName
Agent ARN	The ARN of the agent	s.Agent.ARN

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Contact Attributes
Queue Attributes

Attribute	Description	JSONPath Reference
Queue name	The name of the queue for which metrics were retrieved	$.Metrics.Queue.Name
Queue ARN	The ARN of the queue for which metrics were retrieved	$.Metrics.Queue.ARN
Contacts in queue	The number of contacts currently in the queue	$.Metrics.Queue.Size
Oldest contact in queue	For the contact that has been in the queue the longest, the length of time that the contact has been in the queue, in seconds	$.Metrics.Queue.OldestContactAge
Agents online	The number of agents currently online, which means logged in and in any state other than offline	$.Metrics.Agents.Online.Count
Agents available	The number of agents whose state is set to Available	$.Metrics.Agents.Available.Count

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Contact Attributes
Queue Attributes (Cont.)

Attribute	Description	JSONPath Reference
Agents staffed	The number of agents currently staffed, which is agents logged in and in Available, ACW, or Busy states	$.Metrics.Agents.Staffed.Count
Agents in After contact work	The number of agents currently in the ACW state	$.Metrics.Agents.AfterContactWork.Count
Agents busy	The number of agents currently active on a contact	$.Metrics.Agents.Missed.Count
Agents missed count	The number of agents in the Missed state, which is the state an agent enters after a missed contact	$.Metrics.Agents.Missed.Count
Agents in non-productive state	The number of agents in a non-productive (NPT) state	$.Metrics.Agents.NonProductive.Count

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Contact Attributes
User-Defined Attributes

Attribute	Description	JSONPath Reference
Any name you choose	A user-defined attribute has two parts: • Destination key: This is any name you choose for the key. However, the **$** and . (period) characters are not allowed because they are both used in defining the attribute paths in JSONPath. • Value: This is can be any value you choose. You can enter several paragraphs worth of text if you want! (For the **Max size of the contact record attributes section**)	$.Attributes.*name_of_your_destination_key*

A user defined attribute of **accountNumber** would have a JSONPath reference of **$.Attributes.accountNumber**

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Contact Attributes
Lex Attributes and Lambda Attributes

Lex Attribute	Description	JSONPath Reference
Dialog state	The last dialog state returned from an Amazon Lex bot. The value is 'Fulfilled' if an intent was returned to the contact flow.	$.Lex.DialogState
Intent name	The user intent returned by Amazon Lex.	$.Lex.IntentName
Slots	Map of intent slots (key/value pairs) Amazon Lex detected from the user input during the interaction.	$.Lex.Slots.slotName
Session Attributes	Map of key-value pairs representing the session-specific context information.	$.Lex.SessionAttributes.attributeKey

Lambda Attribute	Description	JSONPath Reference
attributeName	Lambda attributes are returned as key-value pairs from the most recent invocation of an **Invoke AWS Lambda function** block. External attributes are overwritten with each invocation of the Lambda function	$.External.*attributeName*

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Contact Attributes
Lex Attributes and Lambda Attributes

Lex Attribute	Description	JSONPath Reference
Dialog state	The last dialog state returned from an Amazon Lex bot. The value is 'Fulfilled' if an intent was returned to the contact flow.	$.Lex.DialogState
Intent name	The user intent returned by Amazon Lex.	$.Lex.IntentName
Slots	Map of intent slots (key/value pairs) Amazon Lex detected from the user input during the interaction.	$.Lex.Slots.slotName
Session Attributes	Map of key-value pairs representing the session-specific context information.	$.Lex.SessionAttributes.attributeKey

Lambda Attribute	Description	JSONPath Reference
attributeName	Lambda attributes are returned as key-value pairs from the most recent invocation of an **Invoke AWS Lambda function** block. External attributes are overwritten with each invocation of the Lambda function	$.External.*attributeName*

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Contact Attributes
Other Attributes

- Other types of contact attributes exist and will not be covered in detail here. These include:
  - SIP metadata
  - Media stream
  - Apple Business Chat
  - Customer profile

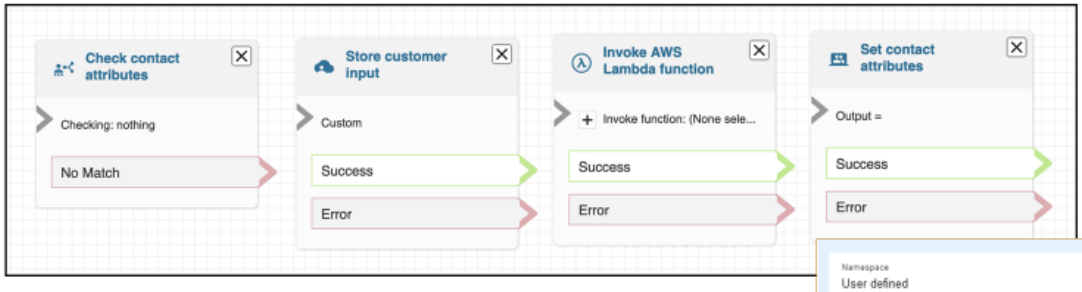
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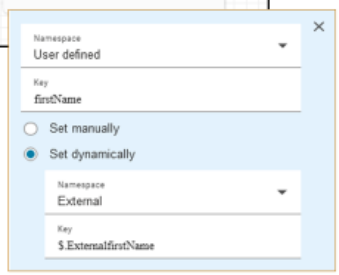
See the following link for more details:

https://docs.aws.amazon.com/connect/latest/adminguide/connect-attrib-list.html

Contact Attributes

Referencing a Contact Attribute in a Contact Flow





- Some blocks: Require a contact attribute be set, checked, stored, or returned
- Optional for other blocks (Ex. Play prompt block can play what is in an attribute)

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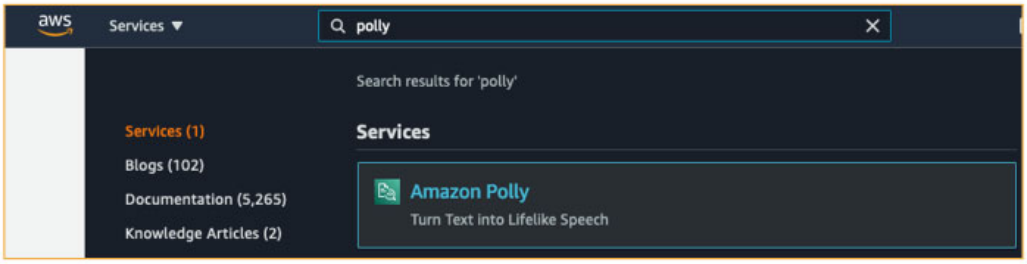


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Amazon Polly and SSML

Amazon Connect offers a native integration with Amazon Polly, allowing you to play dynamic and natural Text-to-Speech (TTS), use Speech Synthesis Markup Language (SSML), and take advantage of Neural Text-to-Speech (NTTS) to achieve the most natural and human-like text-to-speech voices possible. This section describes how to configure Amazon Polly.

Amazon Polly

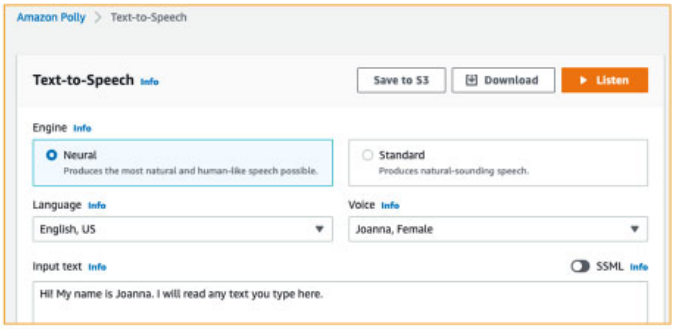


- Uses lifelike Text-to-Speech
- Supports multiple voices and multiple languages
- Includes Neural Text-to-Speech (NTTS) voices
  - Advanced improvements in speech quality through a new machine-learning approach
- Want custom voices for your organization? Amazon will do that

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Amazon Polly (Cont.)

- Select between Neural and Standard
- Select the language and voice
- Enter desired input text
- Save results to an S3 bucket or to your computer



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Amazon Polly (Cont.)

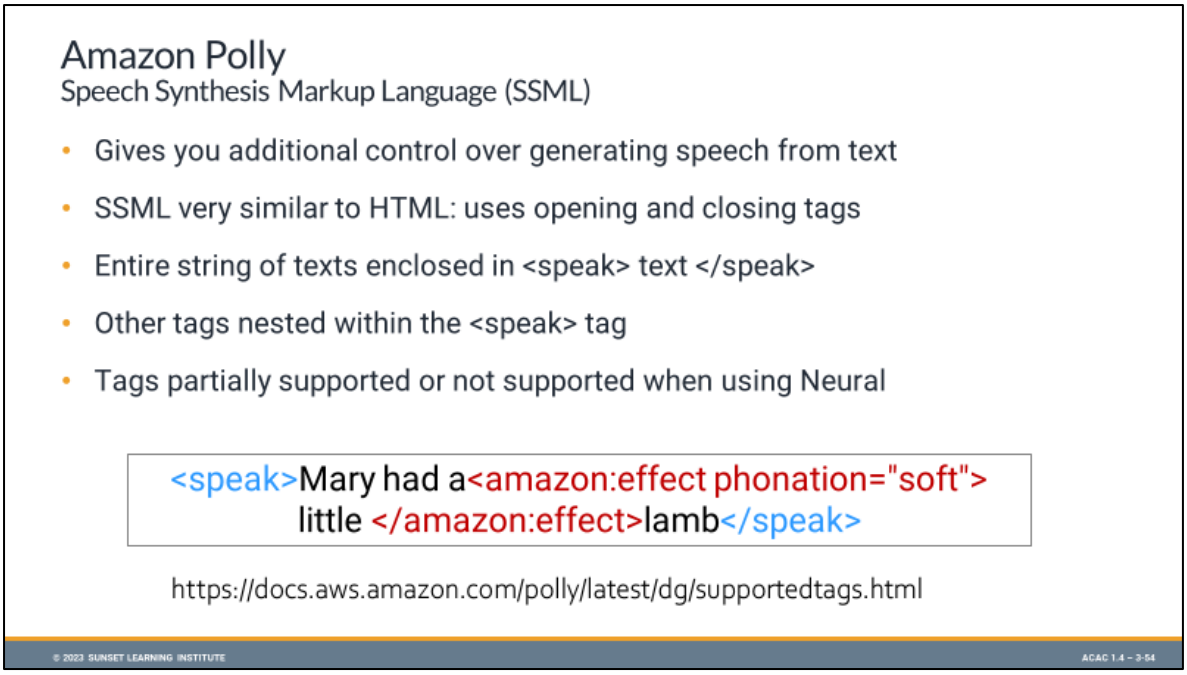
- Select format and sample rate of audio file
- Supports MP3, OGG, PCM, and Speech Marks



PCM, 8000Hz is recommended for Amazon Connect prompts

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You can use Amazon Polly to generate speech from either plain text or from documents marked up with Speech Synthesis Markup Language (SSML). Using SSML-enhanced text gives you additional control over how Amazon Polly generates speech from the text you provide.

For example, you can include a long pause within your text, or change the speech rate or pitch. Other options include:

- Emphasizing specific words or phrases
- Using phonetic pronunciation
- Including breathing sounds
- Whispering
- Using the Newscaster speaking style

For more information on SSML Tags go to:

https://docs.aws.amazon.com/polly/latest/dg/supportedtags.html

Amazon Polly
Speech Synthesis Markup Language (SSML) (Cont.)

Action	SSML Tag	Availability with Neural Voices
Adding a pause	<break>	Full availability
Emphasizing Words	<emphasis>	Not available, Standard only
Specifying Another Language for Specific Words	<lang>	Full availability
Placing a Custom Tag in Your Text	<mark>	Full availability
Adding a Pause Between Paragraphs	<p>	Full availability
Using Phonetic Pronunciation	<phoneme>	Full availability

https://docs.aws.amazon.com/polly/latest/dg/ssml.html

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Amazon Polly
Speech Synthesis Markup Language (SSML) (Cont.)

Action	SSML Tag	Availability with Neural Voices
Controlling Volume, Speaking Rate and Pitch	<prosody>	Partial availability, pitch not supported
Setting a Maximum Duration for Synthesized Speech	<prosody amazon:max-duration>	Not available, Standard only
Adding a Pause Between Sentences	<s>	Full availability
Controlling how Special Types of Words are Spoken	<say-as>	Partial availability, characters or spell-out is not supported
Identifying SSML-Enhanced Text	<speak>	Full availability
Pronouncing Acronyms and Abbreviations	<sub>	Full availability

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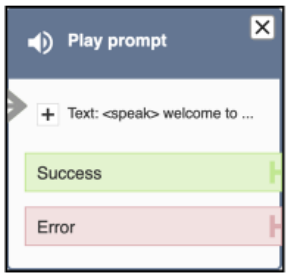
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Amazon Polly
Speech Synthesis Markup Language (SSML) (Cont.)

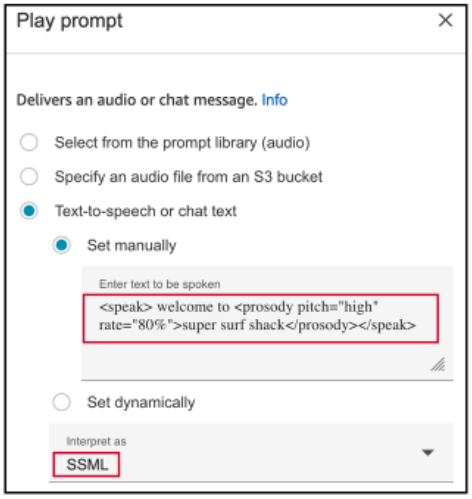
Action	SSML Tag	Availability with Neural Voices
Improving Pronunciation by Specifying Parts of Speech	<w>	Full availability
Adding the Sound of Breathing	<amazon:auto-breaths>	Not available
Newscaster speaking style	<amazon:domain name="news">	Select neural voices only
Adding Dynamic Range Compression	<amazon:effect name="drc">	Full availability
Speaking Softly	<amazon:effect phonation="soft">	Not available
Controlling Timbre	<amazon:effect vocal-tract-length>	Not available
Whispering	<amazon: effect name="whispered">	Not available

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Amazon Polly
Invoking Speech Synthesis Markup Language (SSML)



Any block that accepts Text-to-speech will support SSML



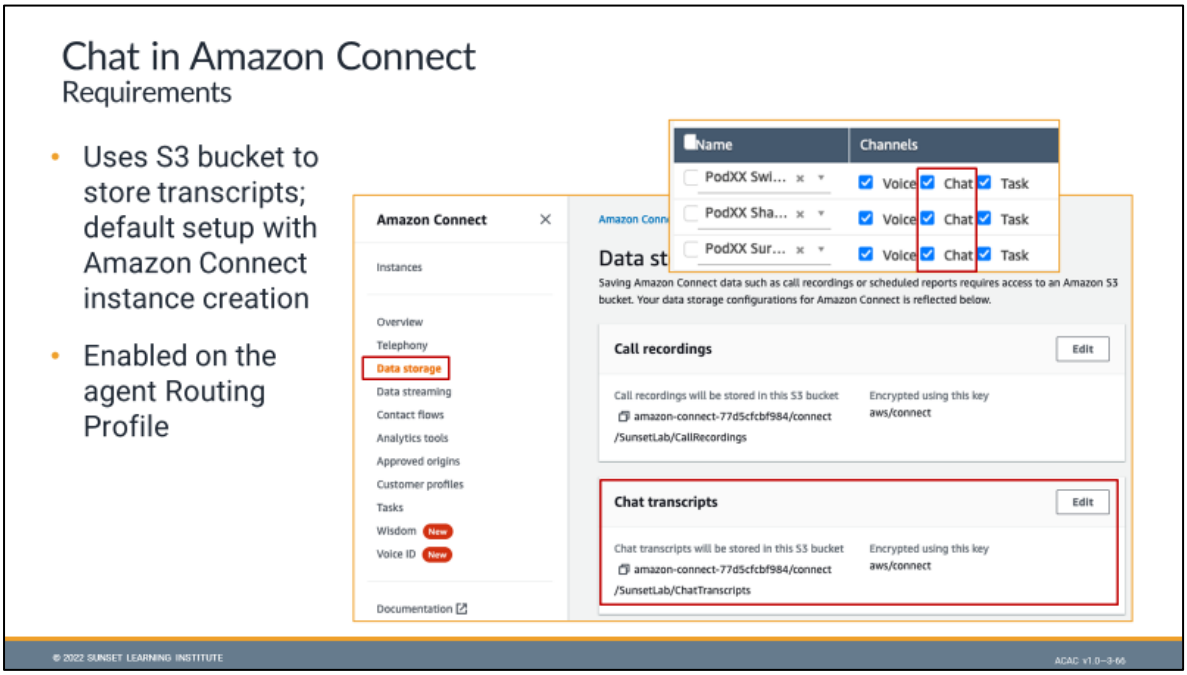
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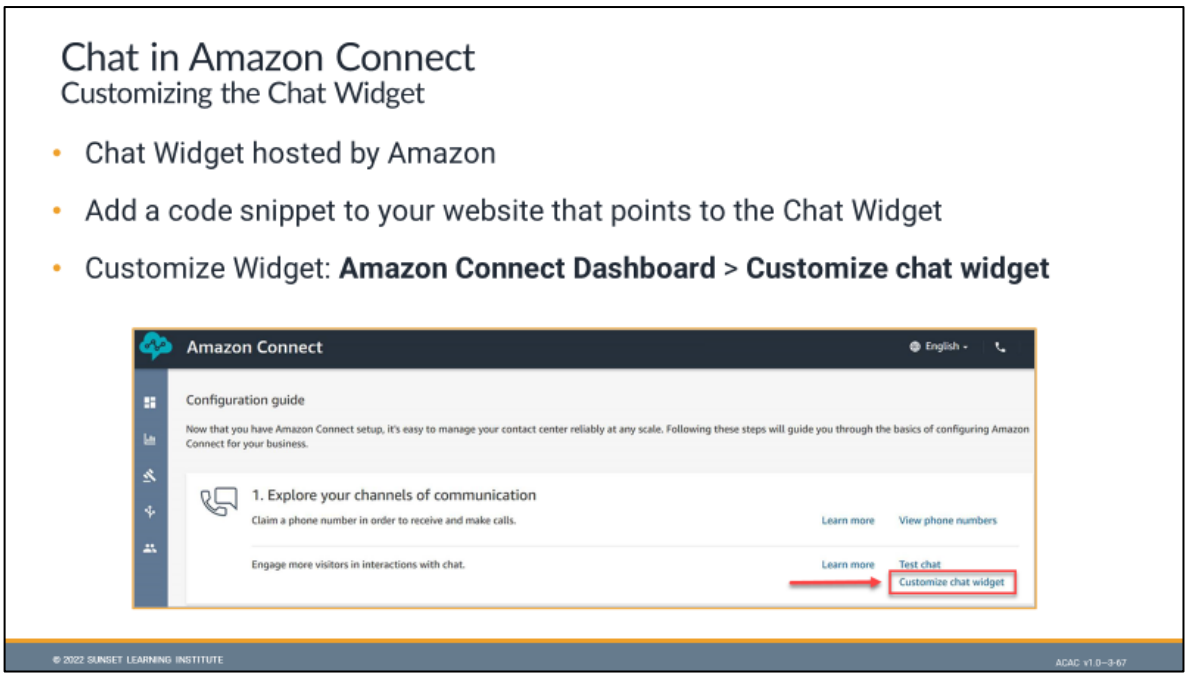
Chat and Contact Flows

Amazon Connect Chat enables your customers to start chatting with Contact Center agents from any of your business applications, web, or mobile. Interactions are asynchronous, allowing your customers to start a chat with an agent or Amazon Lex Bot, step away from it, and then resume the conversation again. This section describes how to add a chat widget using Amazon Connect.



Amazon Connect Chat enables your customers to start chatting with Contact Center Agents from any of your business applications: web or mobile. Interactions are asynchronous, enabling your customers to start a chat with an Agent or Amazon Lex Bot, step away from it, and then resume the conversation again. They can even switch devices and continue the chat.

Agents have a single user interface to help customers using Both voice and chat. This reduces the number of tools that Agents have to learn and the number of screens they have to interact with. Chat activities integrate into your existing Contact Center flows and the automation that you built for voice. You build your flows once and reuse them across multiple channels. Likewise, for metrics collection and the dashboards you built, they automatically benefit from the unified metrics across multiple channels.



To support your customers through chat, you can add a chat widget to your website that is hosted by Amazon Connect. You can configure the chat widget in the Amazon Connect console: customize the font and colors, and secure the widget so that it can be launched only from your website. As a result, you will have a short code snippet that you add to your website.

Because Amazon Connect hosts the widget, it ensures that the latest version is always live on your website.

Supported browsers

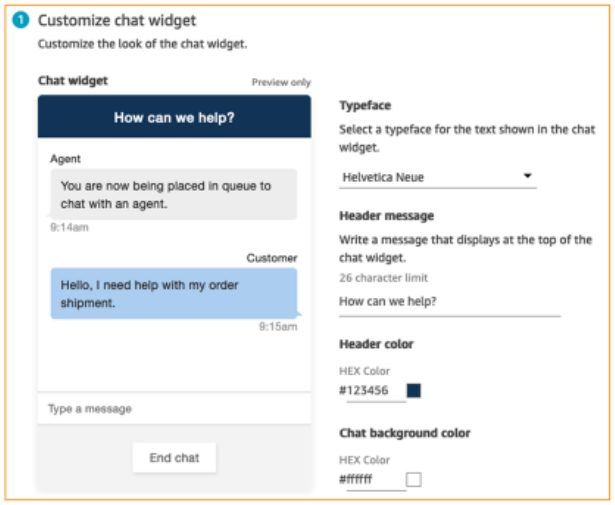
The pre-built chat widget supports the following browser versions and higher:

- Google Chrome 85.0
- Safari 13.1
- Microsoft Edge version 85
- Mozilla Firefox 81.0

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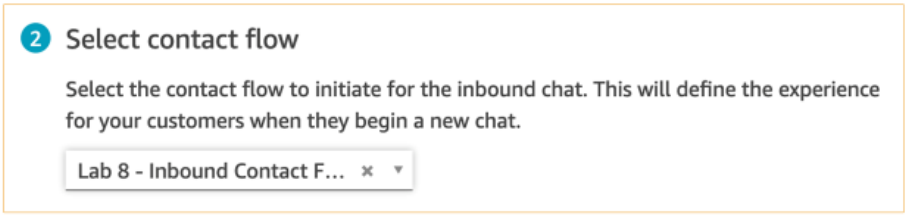
Chat in Amazon Connect
Customizing the Chat Interface

- Customize:
  - Fonts
  - Header Message
  - Header color
  - Chat background and Bubble color
  - Start Icon colors and background
  - Minimize Chat Icon colors and background



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Chat in Amazon Connect
Customizing the Chat Interface (Cont.)

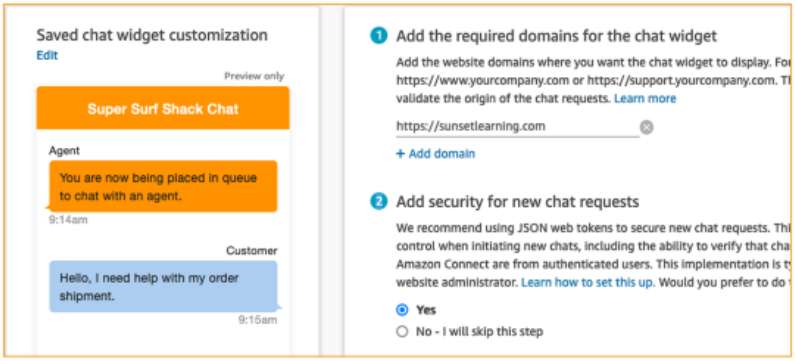


- Set the Contact Flow that will be used for inbound chats

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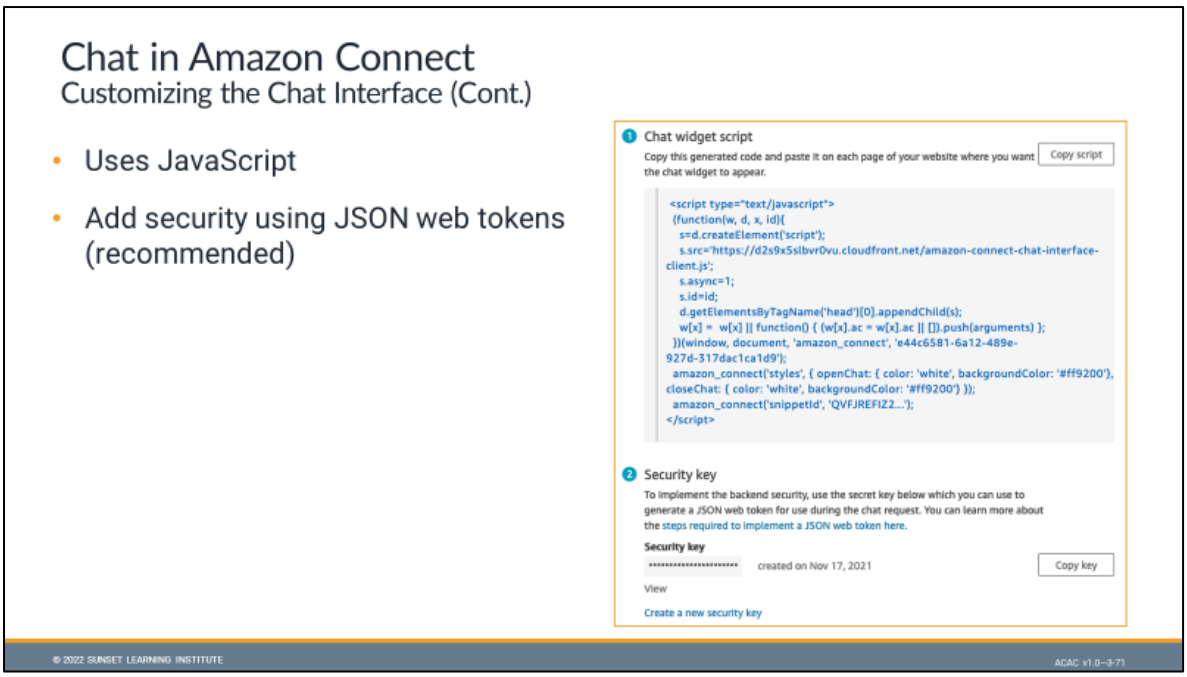
Chat in Amazon Connect
Customizing the Chat Interface (Cont.)



- Add up to five domains
- Add security using JSON web tokens (recommended)

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In this step, you confirm your selections and copy the code for the chat widget and embed it in your website.

Chat widget script

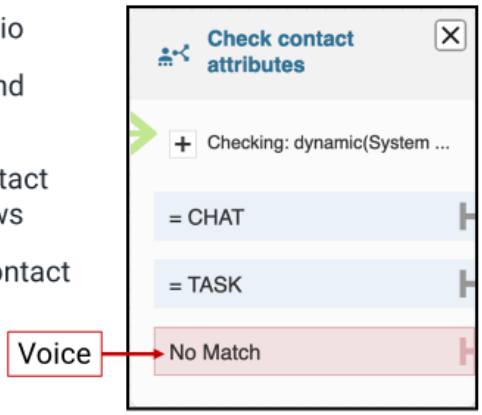
The above image shows an example of the JavaScript that you embed on the websites where you want customers to chat with Agents. This script displays the widget in the Bottom-right corner of your website.

Security key

Use this 44-character security key to generate JSON web tokens from your web server. You can also update or rotate keys if you need to change them. When you do this, Amazon Connect provides you with a new key and maintains the old key until you have a chance to replace it. After you have the new key deployed, you can come back to Amazon Connect and delete the old key.

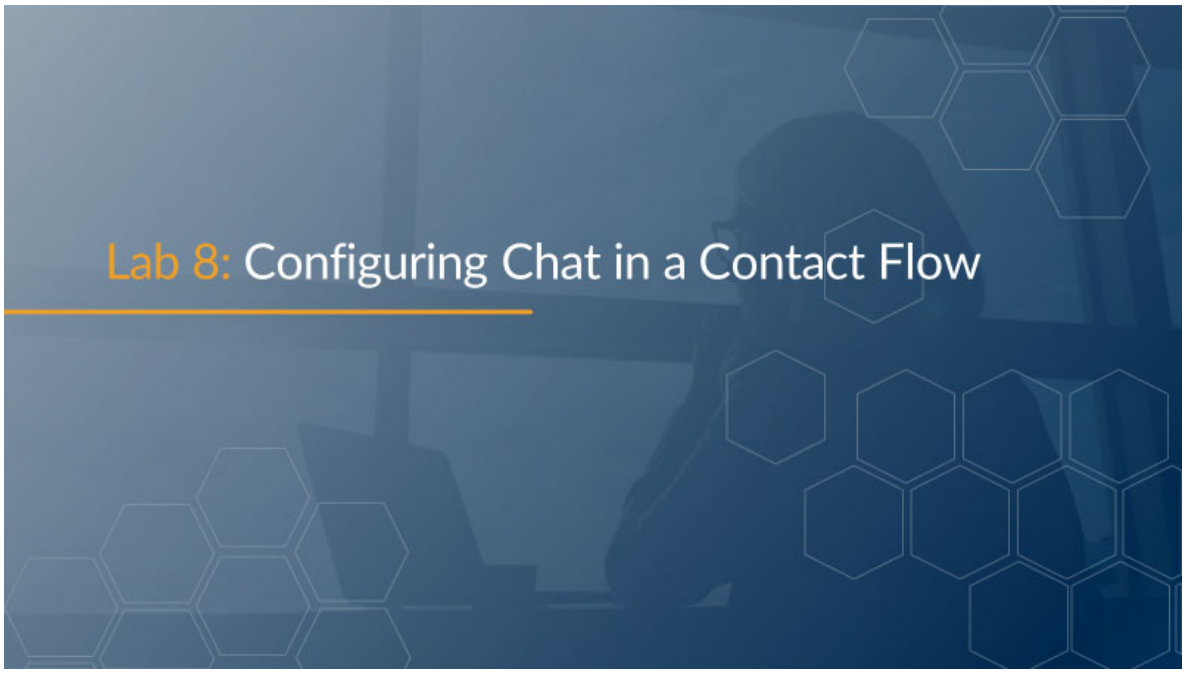
Chat in Amazon Connect
Contact Flows and Chat

- Blocks that use Text-to-speech = chat or audio
- Chat doesn't understand SSML and will try and pronounce the SSML code
- Chat and voice may be combined in one Contact Flow or separated into different Contact Flows
- Split your logic as necessary with a Check contact attributes block (shown to the right)



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Module 4

Advanced Contact Flows

Overview

You can optimize your Contact Center in many ways. One way is the use of AWS Lambda that uses your organization's code in Lambda functions. Lambda runs your function only when needed and scales automatically, from a few daily requests to thousands per second. Another way to optimize your Contact Center is the use of Amazon Lex to reduce the load on your Agents.

Objectives

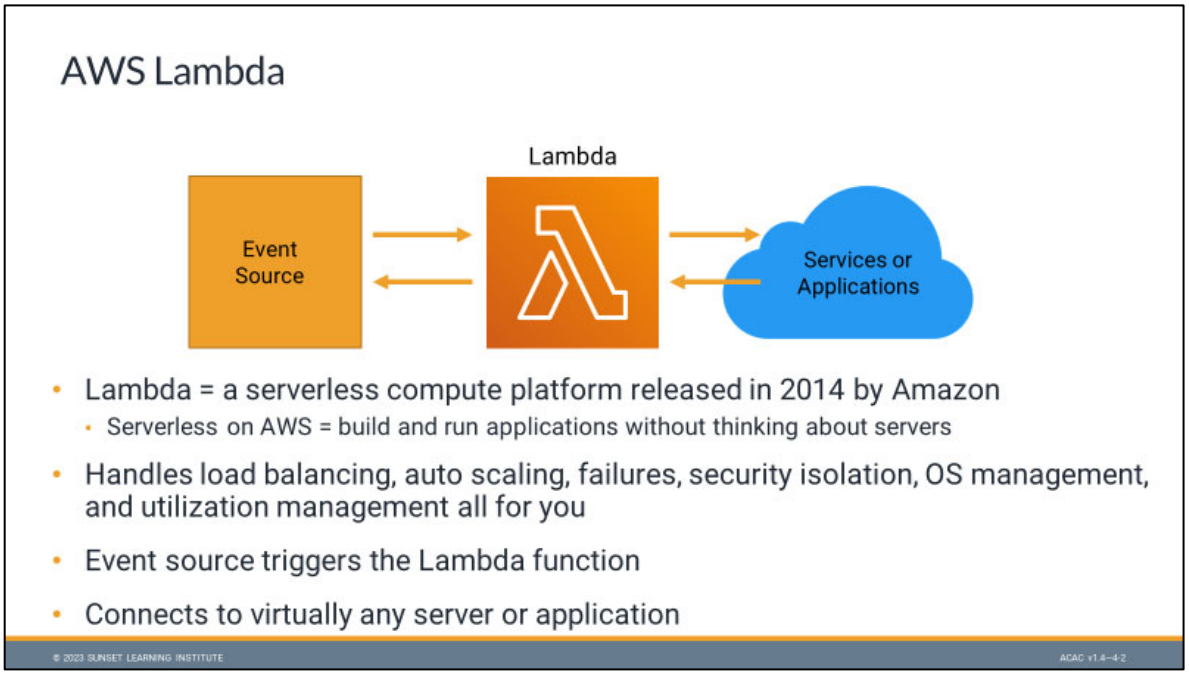
Upon completing this module, you will be able to use two methods to optimize your Contact Center. This ability includes being able to meet this objective:

- Invoke and use a Lambda function and construct an Amazon Lex Bot for chat and voice interactions.

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Calling on a Lambda Function

Lambda is a compute service that lets you run code without provisioning or managing servers. You can create AWS Lambda functions to use as code looks for your Amazon Lex Bot. This section describes AWS Lambda and how to create a Lambda function using Amazon Connect.



Serverless on AWS allows users to build and run applications without thinking about servers.

Lambda is a compute service that lets you run code without provisioning or managing servers. Lambda runs your code on a high-availability compute infrastructure and performs all of the administration of the compute resources, including server and operating system maintenance, capacity provisioning and automatic scaling, code monitoring and logging. With Lambda, you can run code for virtually any type of application or backend service.

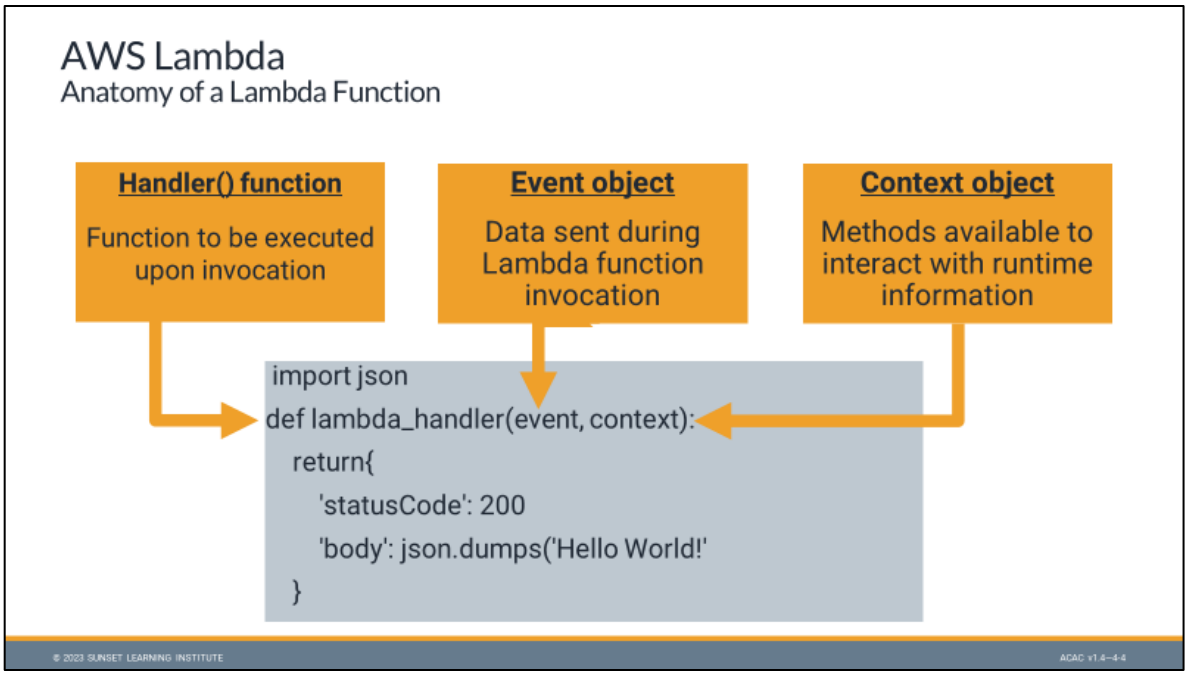
You organize your code into Lambda functions. Lambda runs your function only when needed and scales automatically, from a few requests per day to thousands per second. You pay only for the compute time that you consume: there is no charge when your code is not running.

You can invoke your Lambda functions using the Lambda API, or Lambda can run your functions in response to events from other AWS services. For example, you can use Lambda to perform these options:

- Build data-processing triggers for AWS services, such as Amazon Simple Storage Service (Amazon S3) and Amazon DynamoDB.
- Process streaming data stored in Amazon Kinesis.
- Create your own backend that operates at AWS scale, performance, and security.

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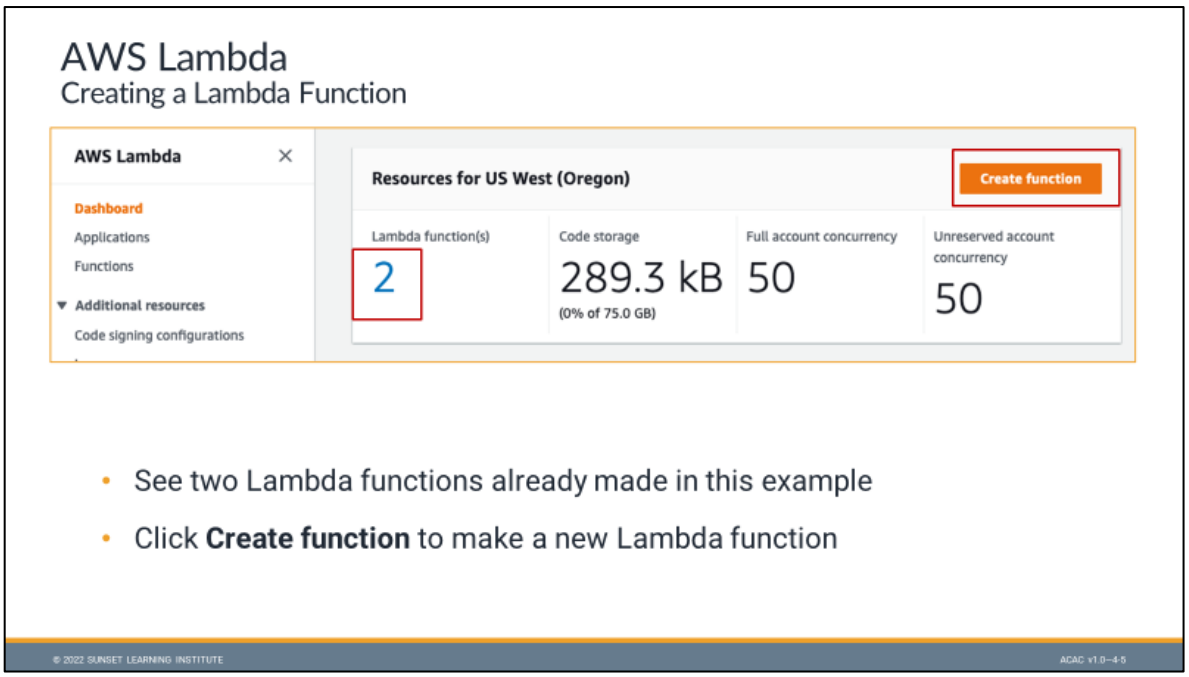




The code in the grey box is the code that will execute for the Lambda function.

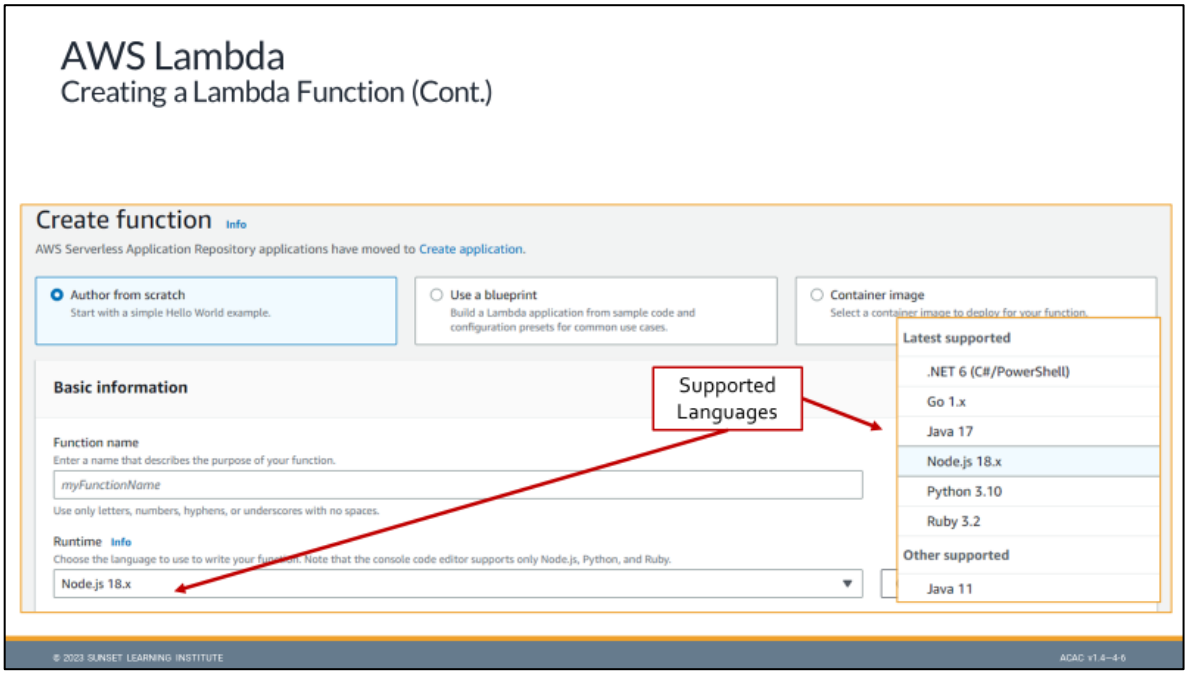
In a Lambda function, you must have a **Handler**: this is the insertion point for execution of the function from the Lambda platform. The Handler will pass in an Event object which contains any data that you want to pass to the Lambda function. In an Amazon Connect environment you might, for example, pass the calling phone number or an account number that was entered by the customer.

The Context object contains things, like configured timeout values or source platform specific values that may be needed in the function.



You create a Lambda function using the console. The function uses the default code that Lambda creates. The Lambda console provides a code editor for non-compiled languages that lets you modify and test code quickly.

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To create a Lambda function with the console:

Step 1 Open the **Amazon Web Services** (AWS) home page.

Step 2 Open the **Functions** page on the Lambda console.

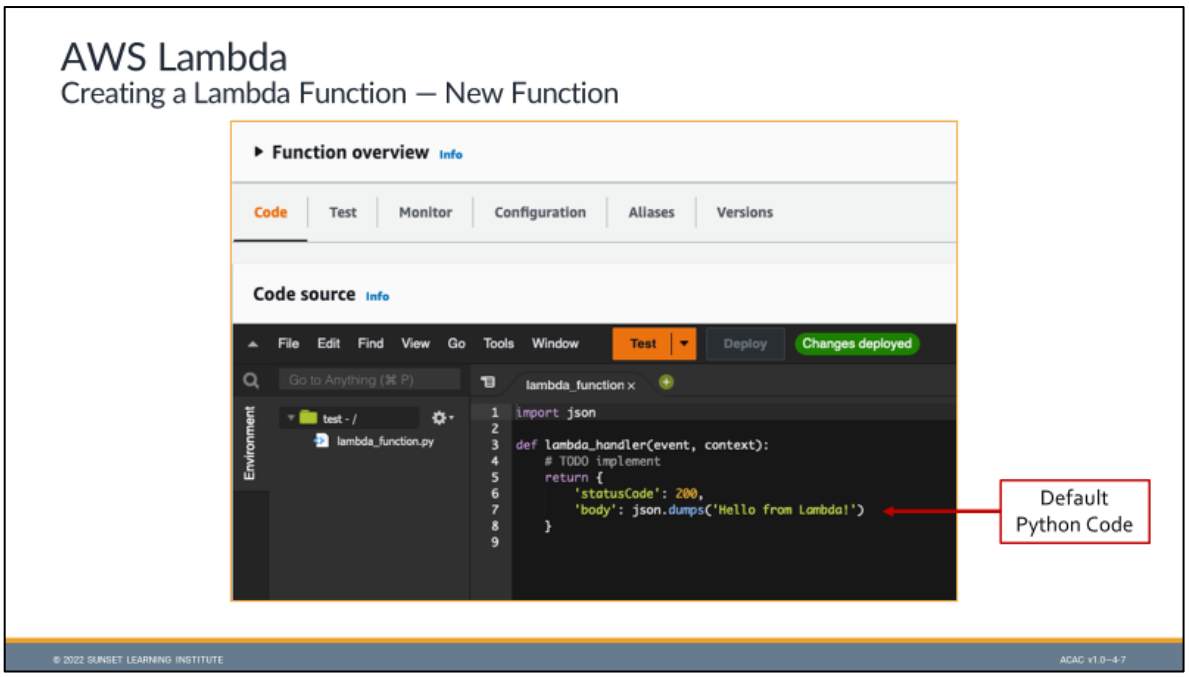
Step 3 Choose **Create** function.

Under Basic information, do the following:

Step 1 For Function name, enter **my-function**.

Step 2 For Runtime, confirm that **Node.js 18.x** is selected. Note that Lambda provides runtimes for .NET (PowerShell, C#), Go, Java, Node.js, Python, and Ruby. There are also different versions of those runtimes.

Step 3 Choose **Create function**.



The function above is the default Python code that comes with a new Python function.



Paste in the code for your function and click **Save**.

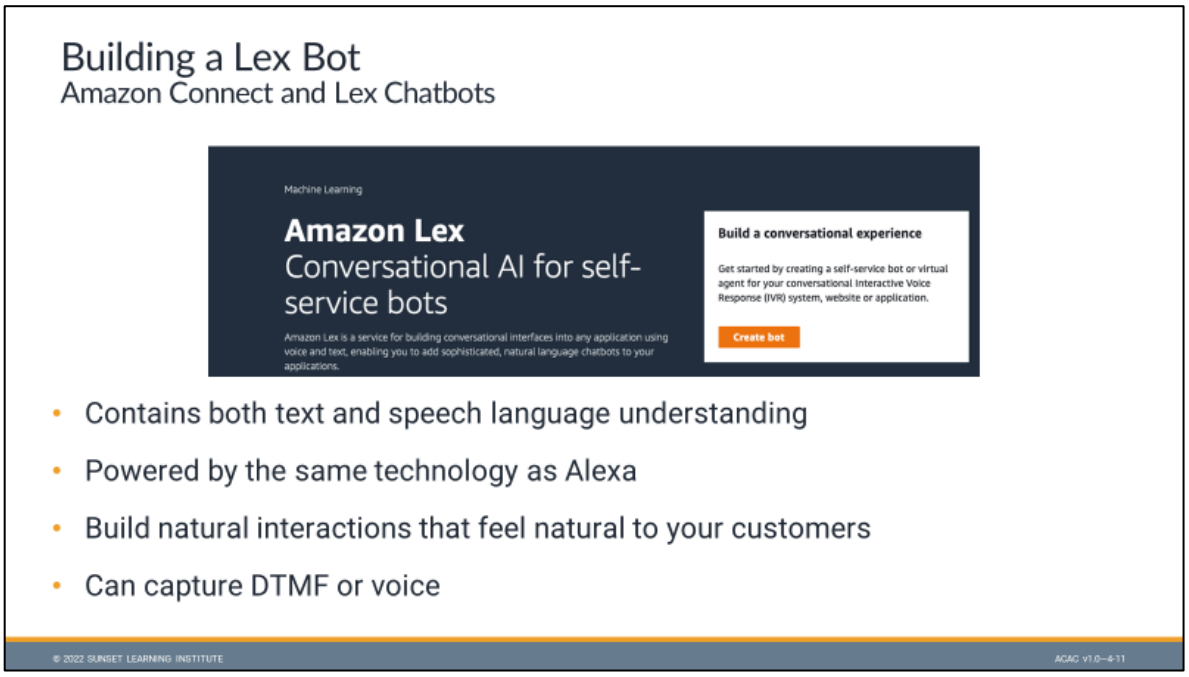
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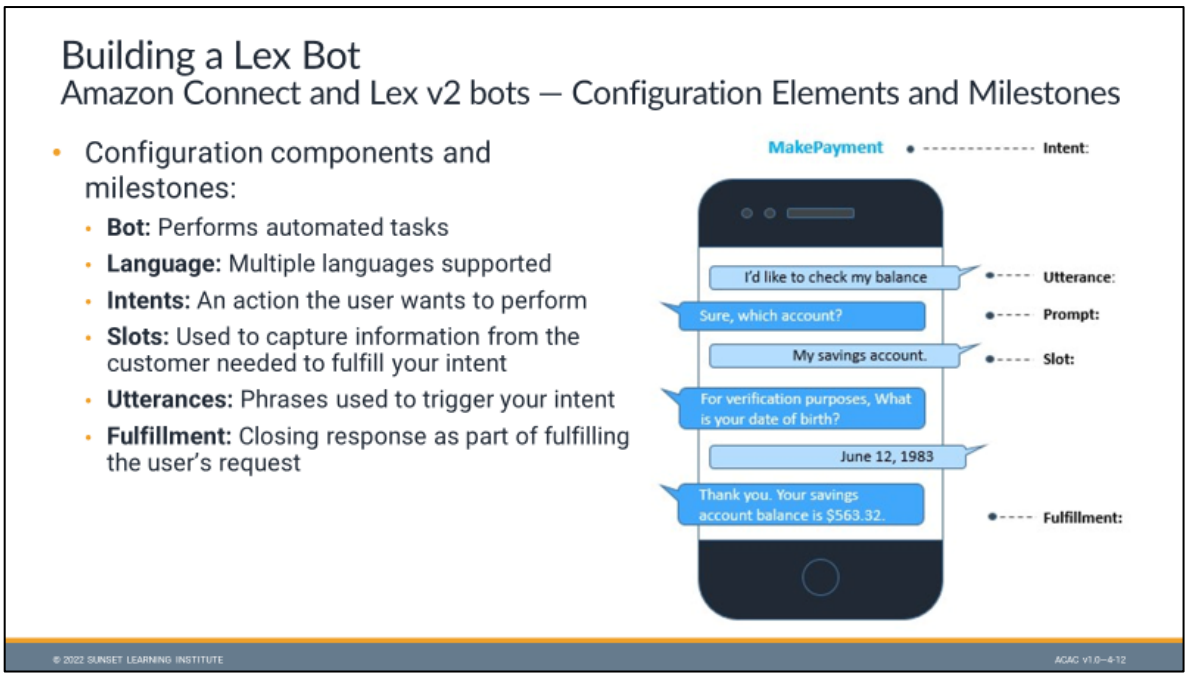
Building a Lex Bot

Lex is an AWS service for building conversational interfaces for voice and text applications. This section describes Amazon Lex and how to build a Lex Bot using Amazon Connect. To access Lex, use the URL https://console.aws.amazon.com/lex/.



With Amazon Lex, you can build conversational interactions (bots) that feel natural to your customers. Amazon Connect with Amazon Lex Bots can also capture customer input as digits that customers enter on their numeric keypad when used in an Amazon Connect contact flow. This way customers can choose how they want to enter sensitive information, such as account numbers.

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Bot

Start creating your bot by defining the name, description, and some basic information.

Languages

You add one or more languages and locales to your bot to enable it to communicate with users in their languages. You define the intents, slots, and slot types separately for each language so that the utterances, prompts, and slot values are specific to the language.

Intents

Intents are the goals that your users want to fulfill, such as ordering flowers or booking a hotel. Your bot needs to have at least one intent.

By default, all bots contain a single built-in intent: the fallback intent. This intent is used when Amazon Lex v2 does not recognize any other intent. For example, if a user says, "I want to order flowers" to a hotel booking intent, the fallback intent is triggered.

Slots

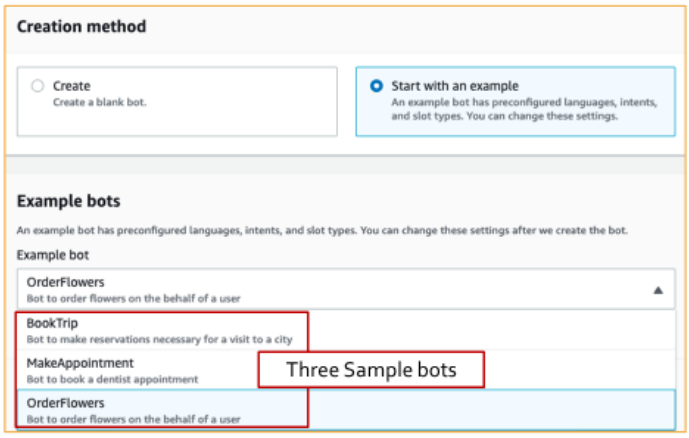
Slot types define the values that users can supply for your intent variables. You define slot types for each language so that the values are specific to that language. For example, for a slot type that lists paint colors, you could include the value *red* in English, *rouge* in French, and *rojo* in Spanish.

You can also use built-in slot types for standard values. For example, you can use the built-in slot type AMAZON.Country for a list of countries in the world.

Building a Lex Bot

Lex Bots — Creating a New Bot From a Sample

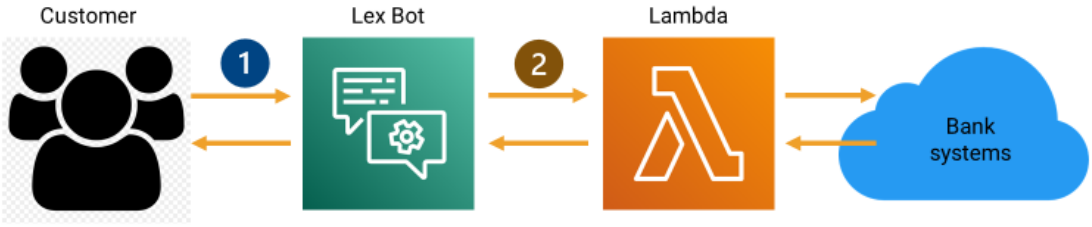
- Click **Create a blank bot**
- See three sample bots that may be customized or used for learning
- Has the option to create a new blank bot available



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Building a Lex Bot

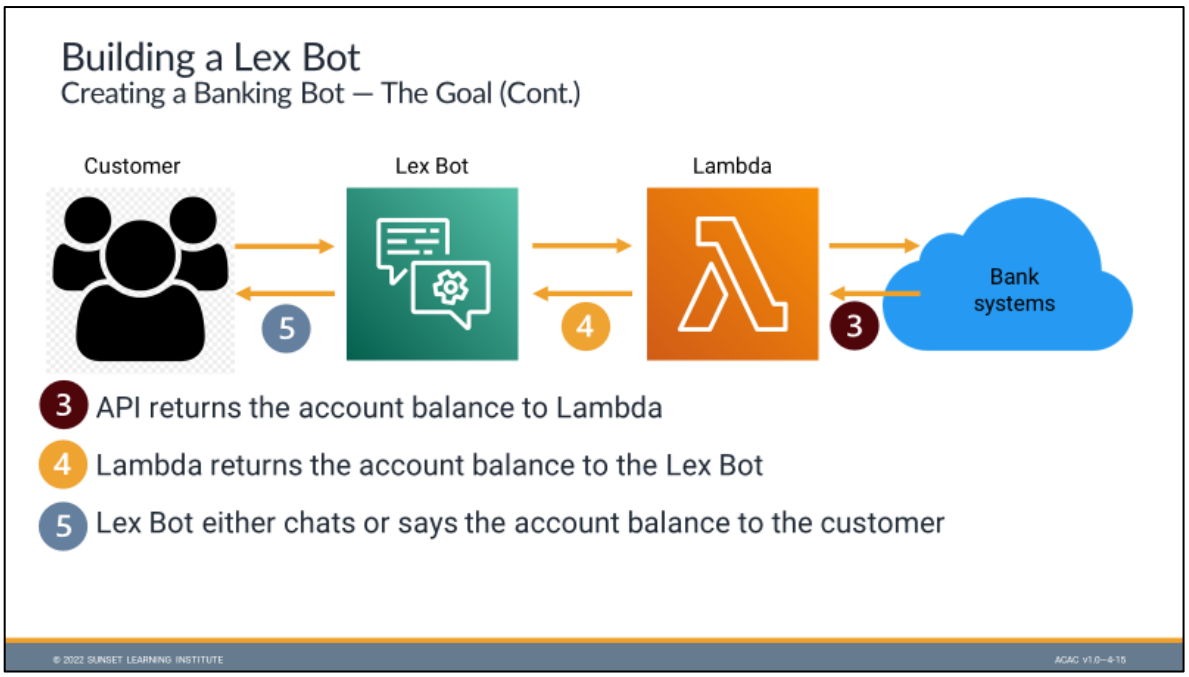
Creating a Banking Bot — The Goal

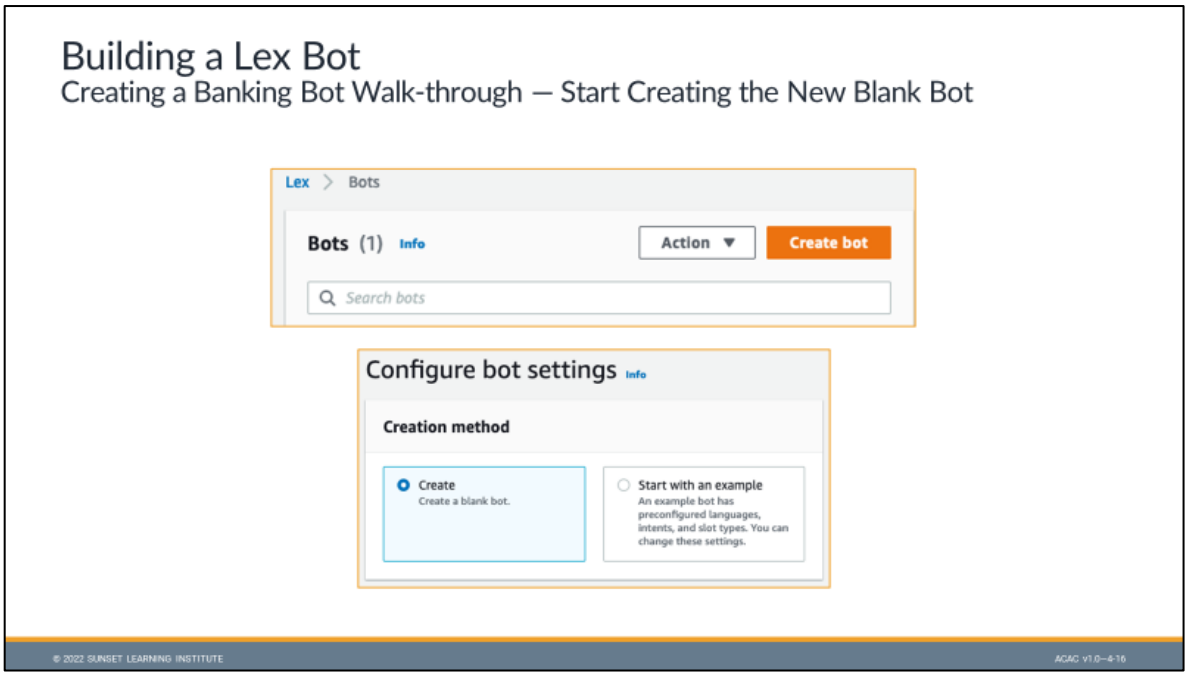


1. Customer contacts Amazon Connect which points to a Lex Bot to do banking
   - Lex Bot matches an utterance from the customer, "What is my account balance?"
   - This matches a configured intent
2. The intent invokes a Lambda function that issues an API call to backend banking systems to fulfill the request

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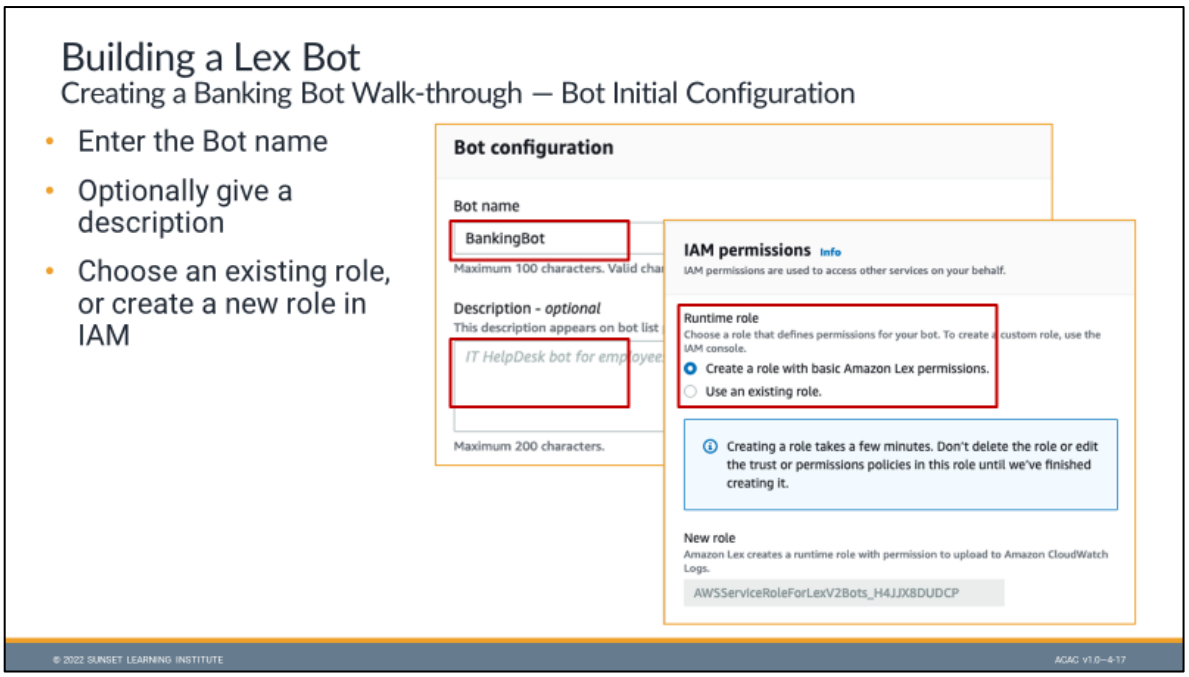




Step 1 On the **Amazon Lex V2 Console**, choose **Bots**. If you're in the Lex v1 console, click on **Switch to the new Lex V2 Console** located in the left-hand menu.

Step 2 Choose **Create bot**.

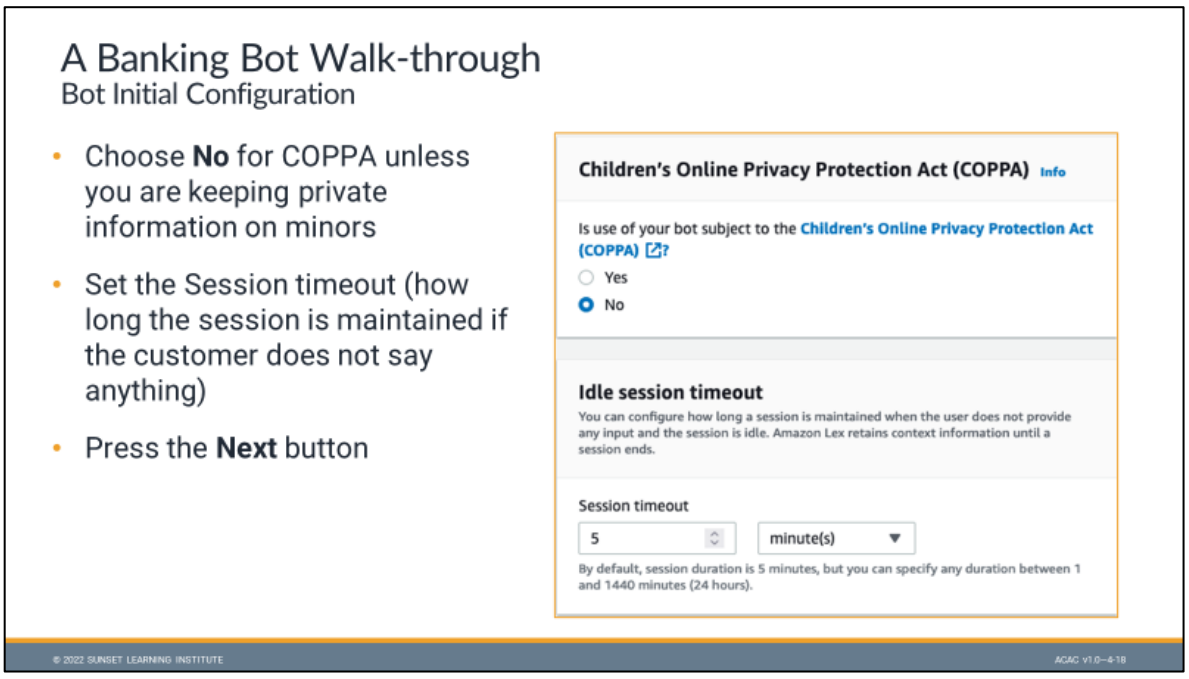
Step 3 For Creation method, select **Create**.



Step 1 For Bot name, enter **BankingBot**.

Step 2 Optionally, enter a description.

Step 3 For Runtime role, select **Create a role with basic Amazon Lex permissions**.

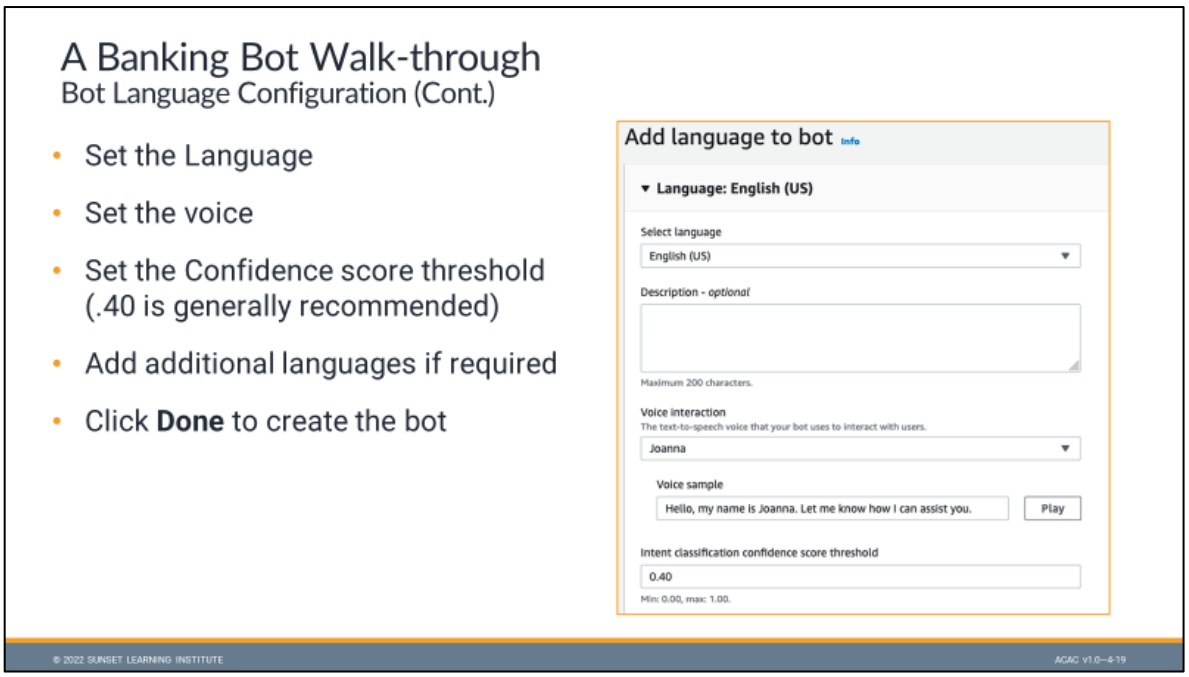


Step 1 Because this bot is only for demo purposes, it's not subject to COPPA. So, select **No**.

Step 2 Leave the Idle session timeout and Advanced settings at their default.

Step 3 Choose **Next**.

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If you're building a voice-based bot, Amazon Lex comes pre-integrated with the neural speech-to-text voices from **Amazon Polly**. Try them out and see what voice fits your bot.

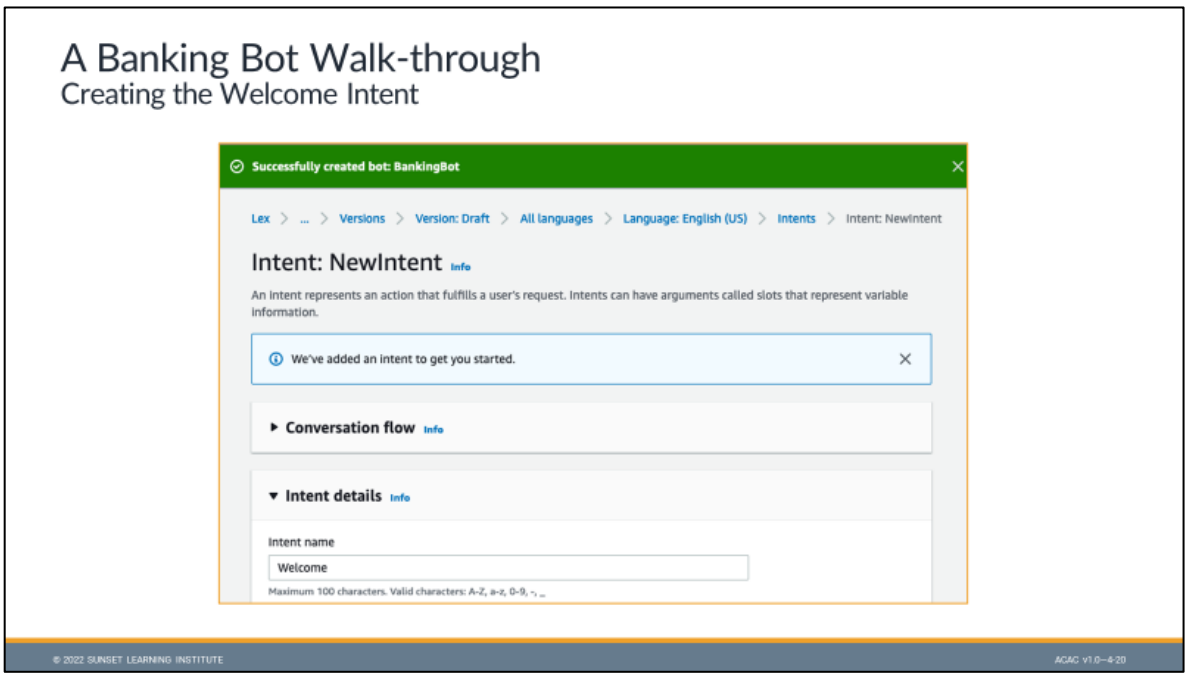
Step 1 For Select language, choose **English (US).**

Step 2 Add a description if desired.

Step 3 Choose the preferred voice if this will have voice interactions and not just chat.

Step 4 Set the **minimum confidence score.** (Higher numbers will require more exact matches to voice patterns.)

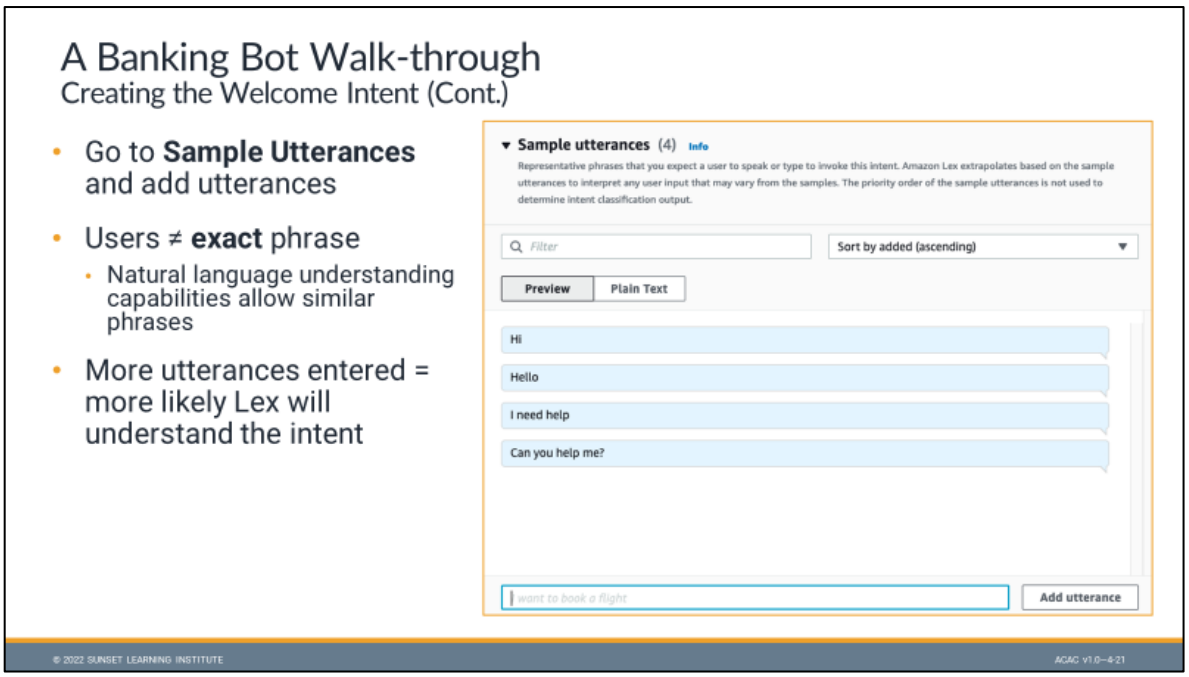
Congratulations! You have successfully created your BankingBot! Now, let's bring it to life.



At this point, the console automatically takes you into the Intent editor page, where a **NewIntent** is ready for you to configure. The BankingBot is a friendly bot. So, let's start by creating a simple Welcome intent to greet users.

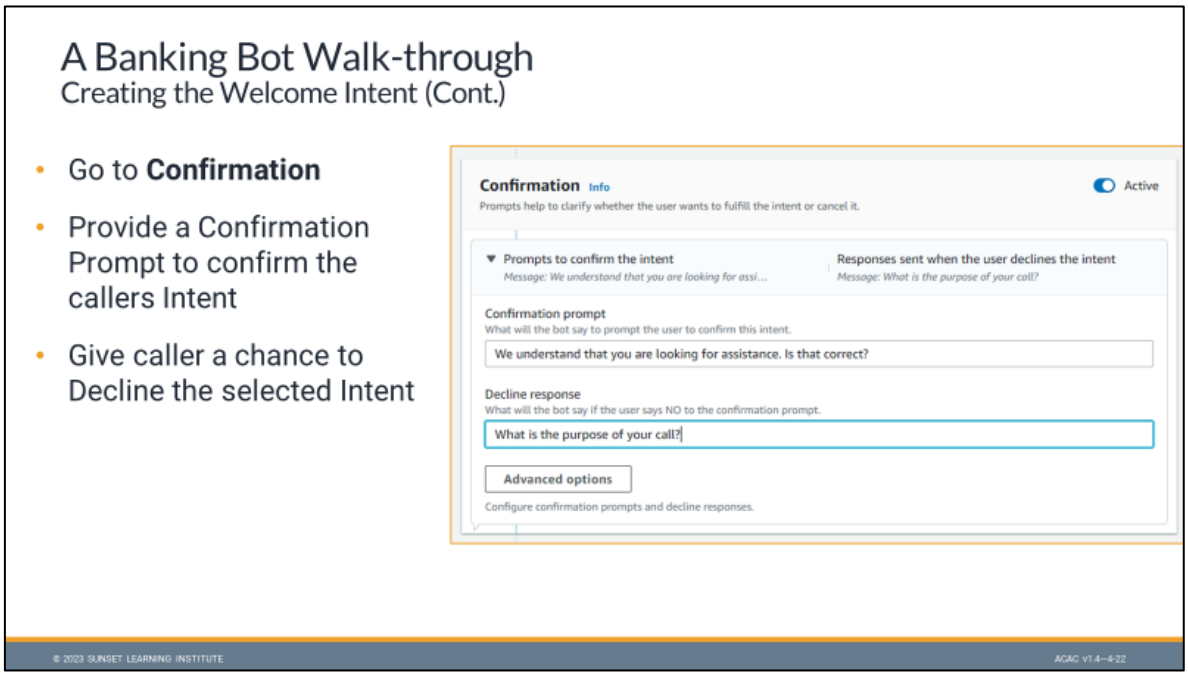
Step 1 Scroll to **Intent** details. For Intent name, replace NewIntent with **Welcome**.

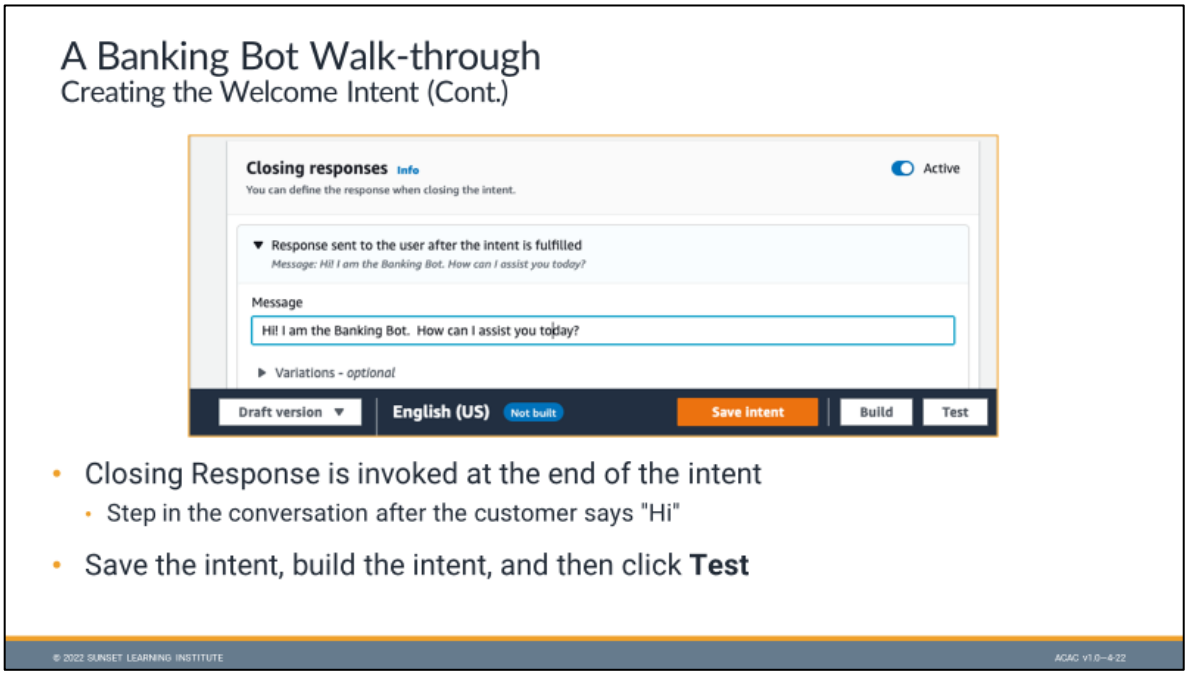
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Step 1 Under **Sample utterances**, choose the **Plain Text** tab and add the following:

- Hi.
- Hello.
- I need help.
- Can you help me?



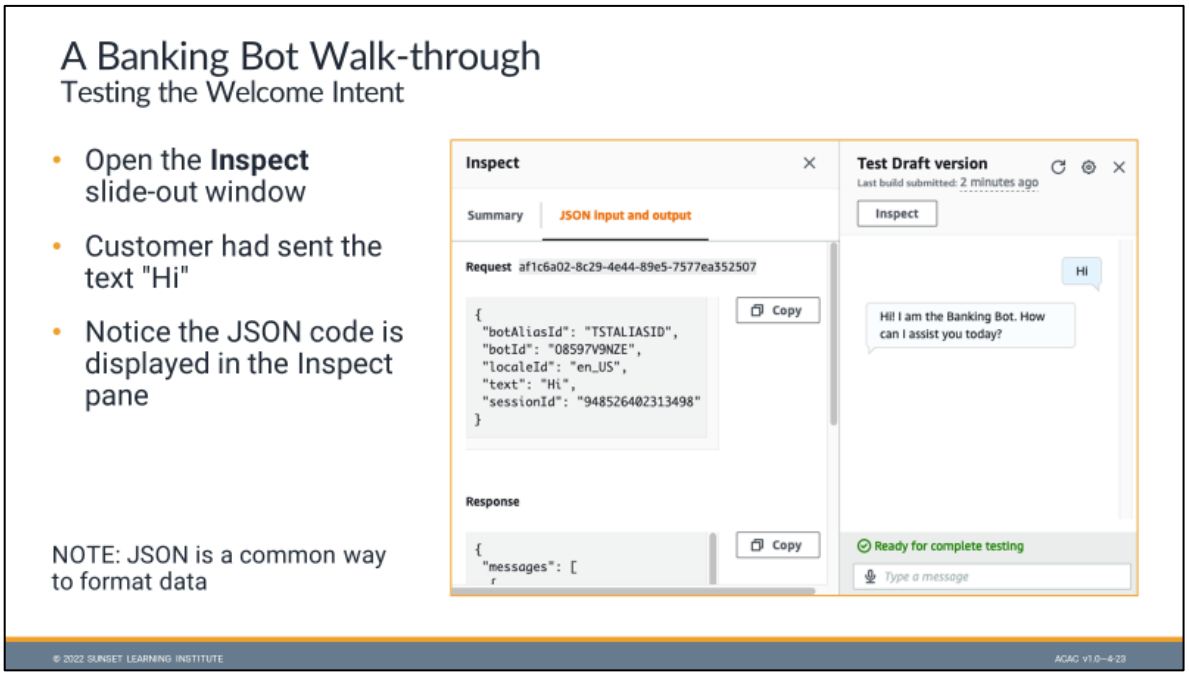


Step 1 Under Closing responses, for Message, enter: **Hi! I'm BB, the BankingBot. How can I help you today?**

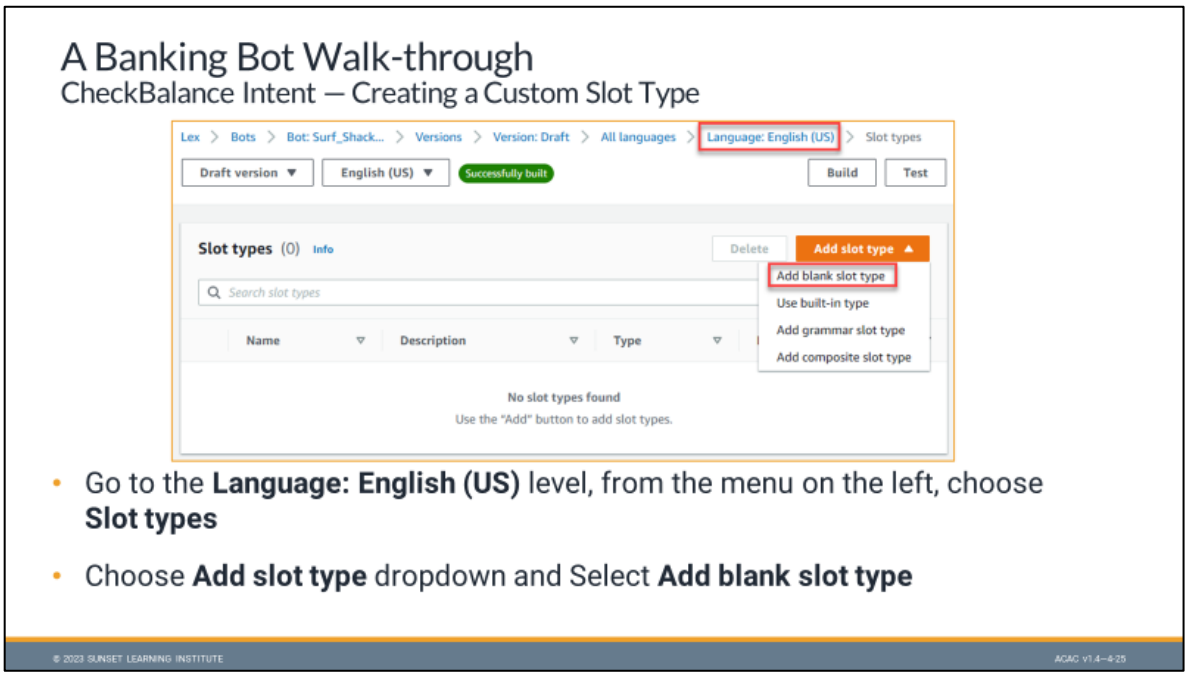
Step 2 Choose **Save intent**.

Step 3 After your intent is saved, choose **Build**.

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Now that you've successfully built your first intent, choose **Test** and give it a try. Open the **Inspect** pane to view more details of the JSON interactions taking place in the background.



Now let's get a bit fancier with a CheckBalance intent. This intent allows a user to check an account balance. The bot first validates the user by requesting the date of birth and then asks which account to check. This intent requires you to create a custom slot type, set up the intent, and finally first set up a Lambda function for fulfillment.

Custom Slots

For each intent, you can specify parameters that indicate the information that the intent needs to fulfill the user's request. These parameters, or slots, have a type. A *slot type* is a list of values that Amazon Lex uses to train the machine learning model to recognize values for a slot. For example, you can define a slot type called **Genres**. Each value in the slot type is the name of a genre: "comedy," "adventure," "documentary," etc. You can define a synonym for a slot type value. For example, you can define the synonyms "funny" and "humorous" for the value "comedy."

You can configure the slot type to restrict resolution to the slot values. The slot values will be used as an enumeration, and the value entered by the user will be resolved to the slot value only if it is the same as one of the slot values or a synonym. A synonym is resolved to the corresponding slot value. For example, if the user enters "funny," it will resolve to the slot value "comedy."

If you are using a Lambda function, the input event to the function includes a resolution list called **slotDetails**. The following example shows the slot and slot details section of the input to a Lambda function:

```
"slots": {
  "MovieGenre": "funny";
},
"slotDetails": {
  "Movie": {
    "resolutions": [
      "value": "comedy"
    ]
  }
}
```

Creating a custom slot

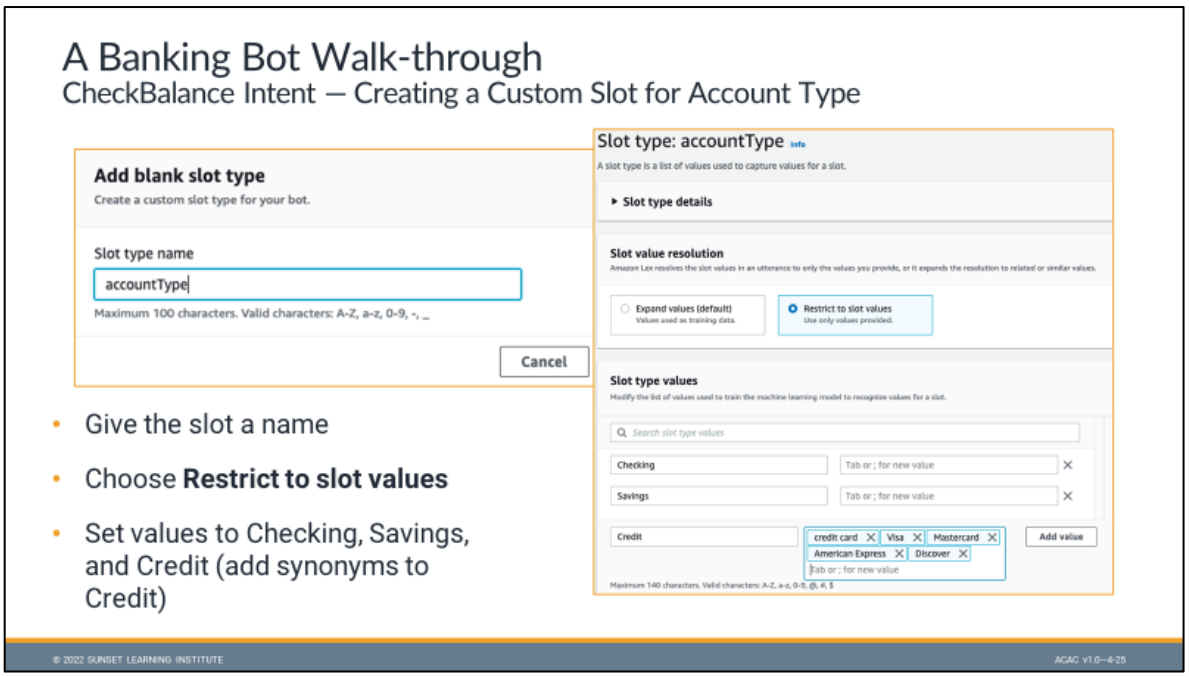
Now that you have added your Lambda function, you need to create a custom slot that can capture a user's account types with valid values, such as Checking, Savings, and Credit before creating the intents. To create a custom slot, follow these steps:

Step 1 In the Navigation pane, drill down to the English (US) version of your bot.

Step 2 Under English (US), choose **Slot types**.

Step 3 On the Add slot type menu, choose **Add blank slot type**.

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Step 1 For Slot type name, enter **accountType**.

Step 2 Choose **Add**.

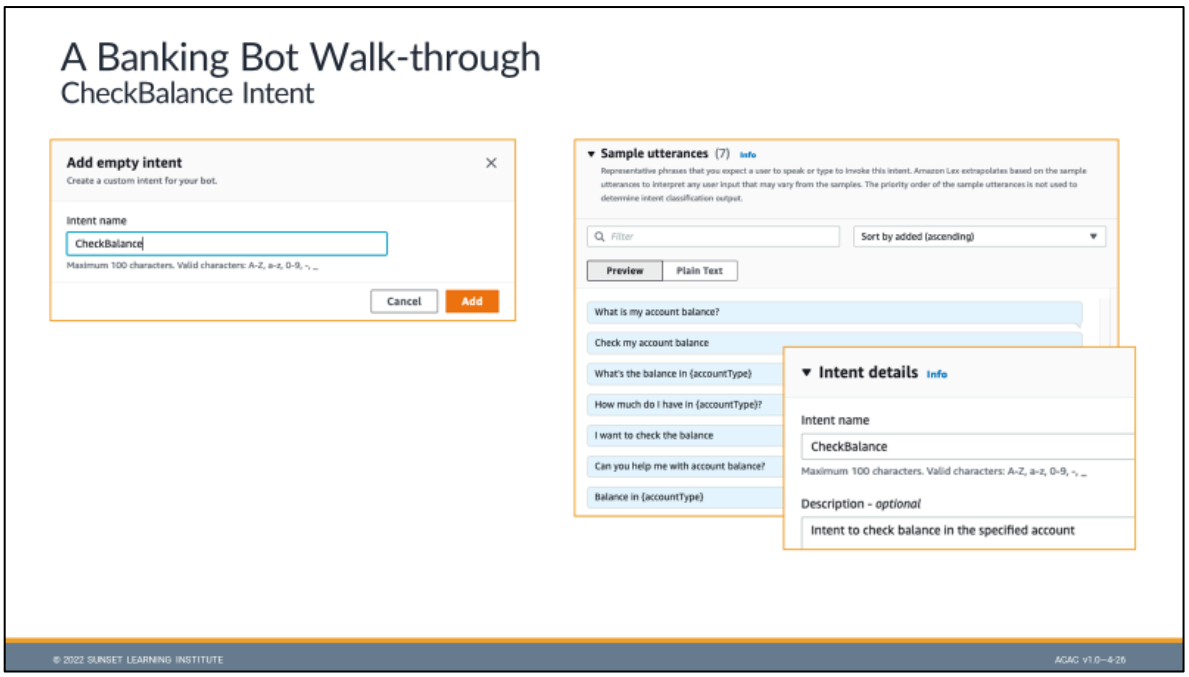
Step 3 For Slot value resolution, select **Restrict to slot values**.

Step 4 Under Slot type values, add values for **Checking**, **Savings**, and **Credit**. You've now created a custom slot type for **accountType**.

Step 5 You can also add synonyms in the second column to help the bot recognize additional references to the Credit slot, such as credit card, Visa, Discover, American Express, and Mastercard.

Step 6 Choose **Save slot type**.

Congratulations! You now have your first custom slot type.



Step 1 In the Navigation pane under English (US), choose **Intents**. There should be 2 currently.

Step 2 Choose **Add intent**.

Step 3 For Intent name, enter **CheckBalance**.

Step 4 Choose **Add**.

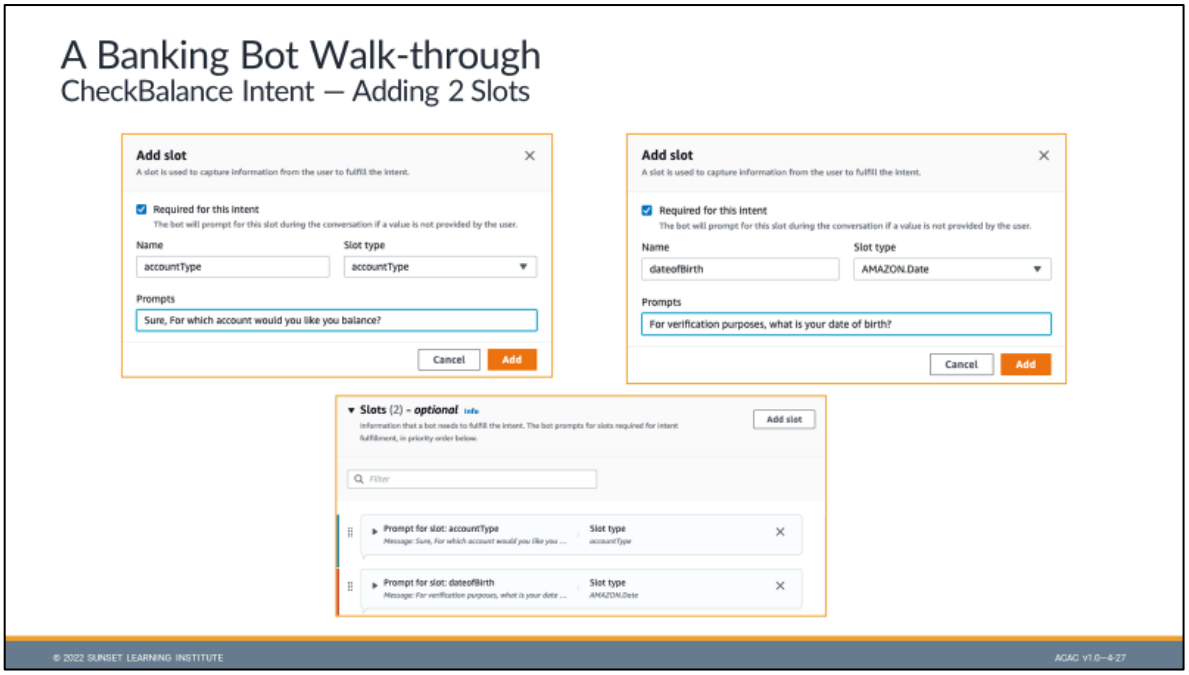
Step 5 Under Intent details for Description, add a description.

Step 6 Under Sample utterances, choose the **Plain Text** tab and enter the following utterances:

- What's the balance in my account?
- Check my account balance.
- What's the balance in {accountType}?
- How much do I have in {accountType}?
- I want to check the balance.
- Can you help me with account balance?
- Balance in {accountType}

Step 7 Choose **Save intent**.

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For the CheckBalance intent, we set up two slots: **account type** and **date of birth**.

Step 1 For Name, enter **accountType**.

Step 2 For Slot type, choose **accountType**.

Step 3 For Prompts, enter: **Sure. For which account would you like your balance?**

Step 4 Choose **Add**.

Step 5 Choose **Add slot**.

Step 6 For Name, enter **dateofBirth**.

Step 7 For Slot type, choose **AMAZON.Date**.

Step 8 For Prompts, enter: **For verification purposes, what is your date of birth?**

Step 9 Choose **Add**.

Step 10 **Save the Intent** when done.



Preparing for Intent 3: FollowupCheckBalance with context

Understanding the direction and context of an ever-evolving conversation is beneficial to building natural, human-like, conversational interfaces.

Being able to classify utterances as the conversation develops requires managing context across multiple turns. Consider when a user wants to follow up and check their account balance in a different account. You don't want the bot to ask the user for their date of birth again. You want the bot to understand the context of the question and carry over the date of birth slot value from this intent into the follow-up intent.

To prepare for the third BankingBot intent, FollowupCheckBalance, you need to preserve this CheckBalance context as an output for future use.

Step 1 Under Contexts, for Output contexts, choose **New Context tag**.

Step 2 For Context tag name, enter **contextCheckBalance**.

Step 3 Choose **Add**.

Now your context is stored for future use.

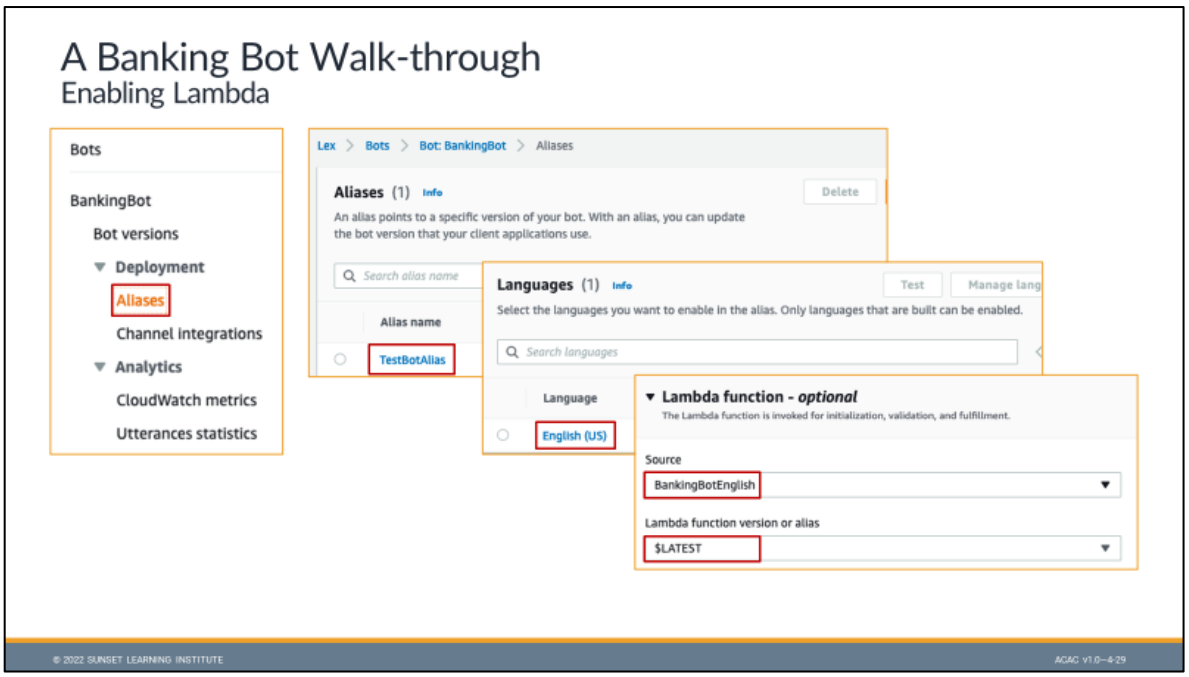
Step 4 Under Fulfillment, go to **Advanced**, and select **Use a Lambda function for fulfillment**.

Step 5 Choose **Save Intent**.

Step 6 Choose **Build**.

Step 7 After the bot-building process is complete, you can test the intent by choosing **Test**.

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Now you have set up your Lambda function. In **Amazon Lex V2 Console**, Lambda functions are defined at the bot alias level. Follow these steps to set up your bot to use a Lambda function:

Step 1 On the Amazon Lex V2 Console, in the Navigation pane, under your bot, choose **Aliases**.

Step 2 Choose **TestBotAlias**.

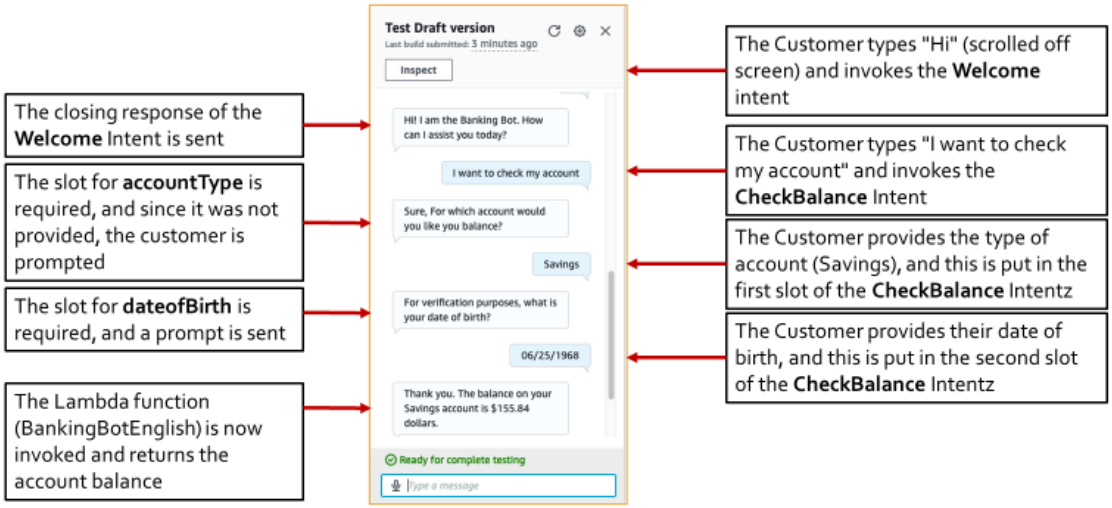
Step 3 For Languages, select **English (US)**.

Step 4 For Source, choose **BankingBotEnglish** as your source.

Step 5 For Lambda function version or alias, choose the function **BankingBotEnglish** and **$LATEST**.

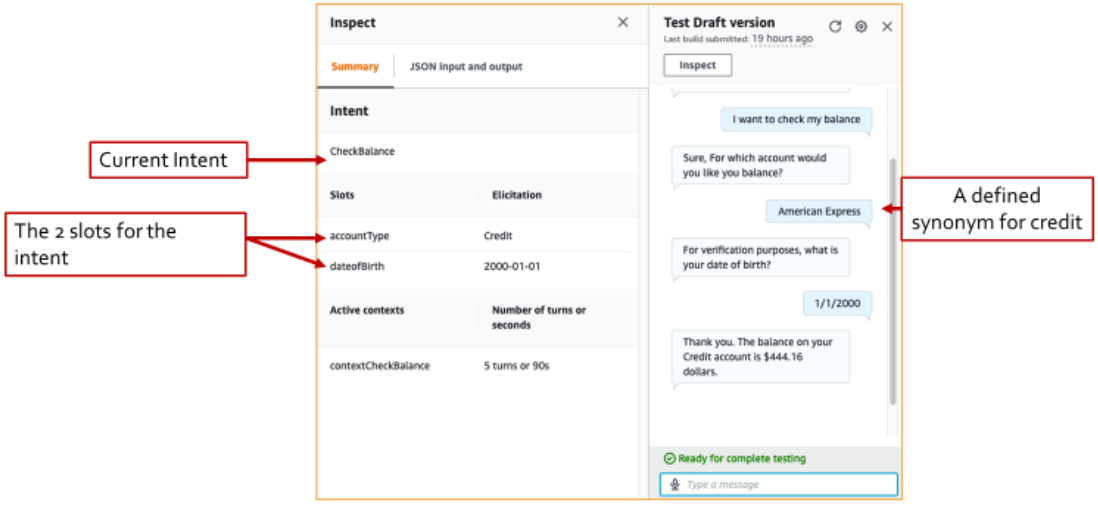
Step 6 Choose **Save**.

A Banking Bot Walk-through
Testing your Banking Bot



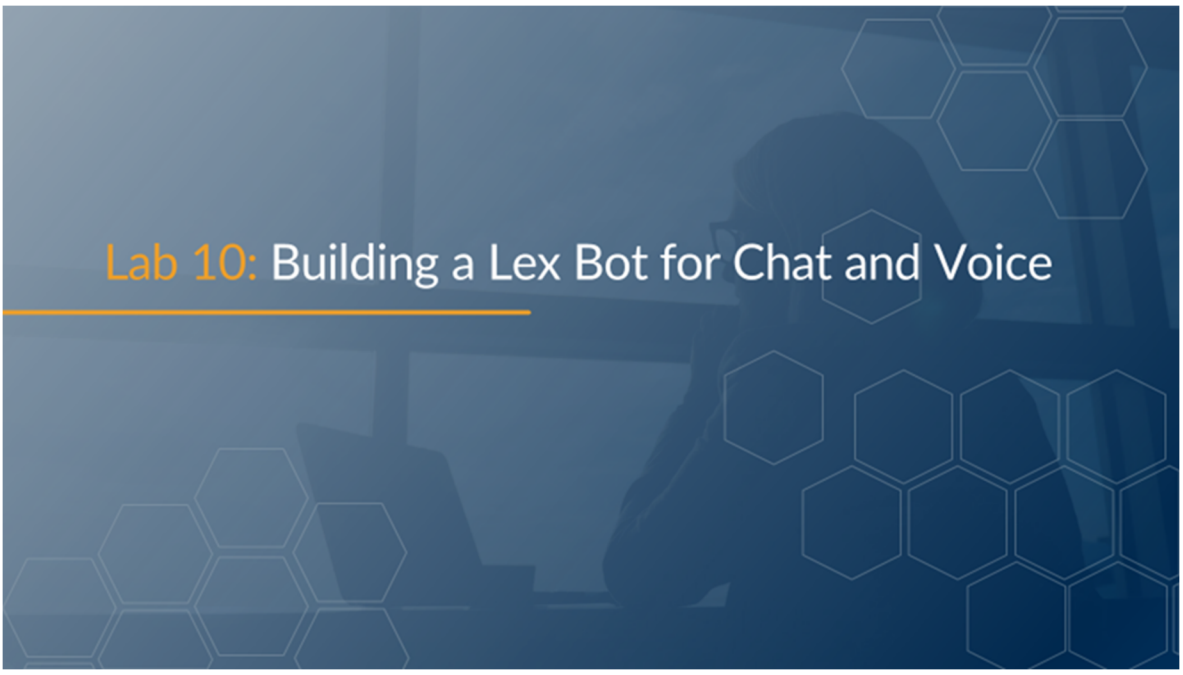
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A Banking Bot Walk-through
Testing your Banking Bot (Cont.)



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Module 5

Advanced Features

Overview

Amazon Connect offers many advanced features. Recognizing that customer records already have a place in your environment, Amazon Connect Customer Profiles has the ability to ingest your customer records for display directly into the Agent Workspace without custom development.

Other examples of advanced features include, the Amazon Connect Streams API and the Queued Callback feature make your business run more smoothly and with greater efficiency. The Amazon Connect Streams API (Streams) gives you the power to integrate your existing web applications with Amazon Connect. The Queued Callback feature improves the customer experience and eliminates excessive hold times while awaiting service by an Agent.

Objectives

Upon completing this module, you will be able to implement the advanced features of Amazon Connect. This ability includes being able to meet this objective:

- Connect to a DynamoDB to retrieve data about the caller, create the Lambda function, enable the Lambda function in your instance of Amazon Connect, create the contact flow.
- Enable Profiles and use Agent Workspace to present customer data
- Create a custom attribute mapping for an S3 data source
- Enable Cases for use within Agent Workspace
- Enable Views Guided Experience for use within Agent Workspace
- Implement a queued callback feature.
- Building a Screen Pop Using the Connect Streams API

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Options to present Customer data to Agents

Amazon Connect offers three primary ways for agents to interact with customer data, Contact Control Panel (CCP), Agent Workspace, and custom development using the Connect Streams API. Providing customer data to agents occurs as a result of displaying contact attributes that contain customer data. CCP provides minimal customer data. Using contact attributes, Connect Streams API allows developers to define the display according to the options of other applications, whether known CRM solutions or custom developed applications. Agent Workspace is a built-in interface that adjusts based on the features enabled such as Customer Profile, Cases, and Agent Guided Views.

Within these interfaces, the customer data that is displayed can either be defined during the contact, or accessed from other sources. For example, the Contact Flow can execute a Lambda function that can perform a database lookup with data stored in Connect as Contact Attributes for the call. Those attributes are then mapped to fields available for display via the Agent Workspace or passed via the Connect Streams API to an alternate agent application.

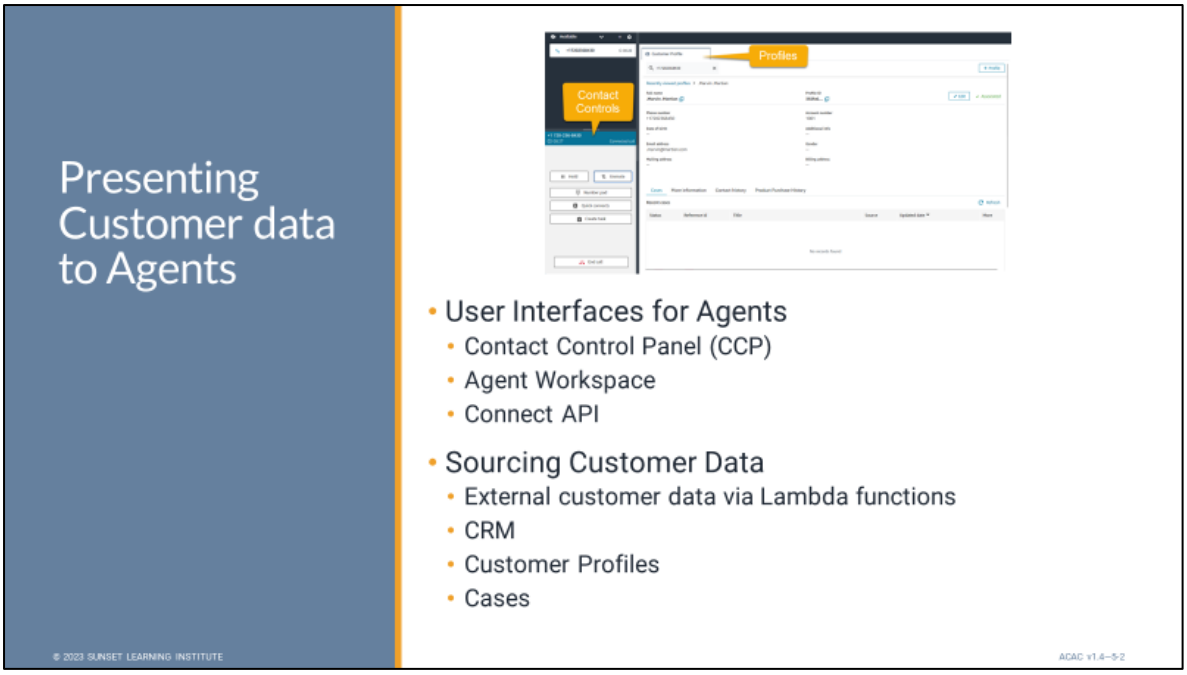
The key factor is where does the customer data exists. The example defined above accesses data contained in an external data source such as a DynamoDB or SQL Database. That requires detailed knowledge of the data source and the expertise required to develop the code necessary to access the data.

Many organizations leverage applications such as Customer Relationship Management (CRM) to manage and store customer data. CRM applications provide their own user interface to access customer data. Developers may be able to customize the display to include Amazon Connect Agent controls for calls and embed contact attributes into that interface.

On the other hand, Agent Workspaces natively provides support for Customer Profiles, Cases, and Views. Amazon Connect Customer Profiles offers the ability to ingest data from CRM solutions and provides pre-defined mappings of the CRM data to Profiles. These pre-defined mappings are available for common CRM applications such that detailed development is not required. Instead, from within Amazon Connect, administrators configure a connector to the CRM solution, then select a mapping for that CRM. At this point, customer records will then be automatically ingested into Customer Profiles. Once a Profile is available, a Flow may be configured to access the Profile. Or if a Profile does not previously exist, a Flow may create a new Profile for that customer.

Then, Agent Workspace has a native interface that displays the customer Profile based on the contact attributes that have already been defined.

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You can use contact attributes to capture information about the contact and then present it to the Agent. For example, you might want to do this to customize the Agent experience while integrated with a customer relationship management (CRM) application.

Our first example is to leverage a Lambda function to access data from an external source such as DynamoDB.

Acquiring Customer Data

One of the many benefits of Amazon Connect is the flexibility to customize the Agent interface as well as the possible customer data integration options. Customer data may reside in various data sources. It is important to have the tools necessary to access the data and display the data to agents for efficient customer service interactions. Two examples covered in this section will be accessing customer data using a Lambda function and integrating data from CRM solutions into Customer Profiles.

Also use them when integrating Amazon Connect with a custom application using the Amazon Connect Streams API or Amazon Connect API. You can use all user-defined attributes, in addition to the customer number and the dialed number.

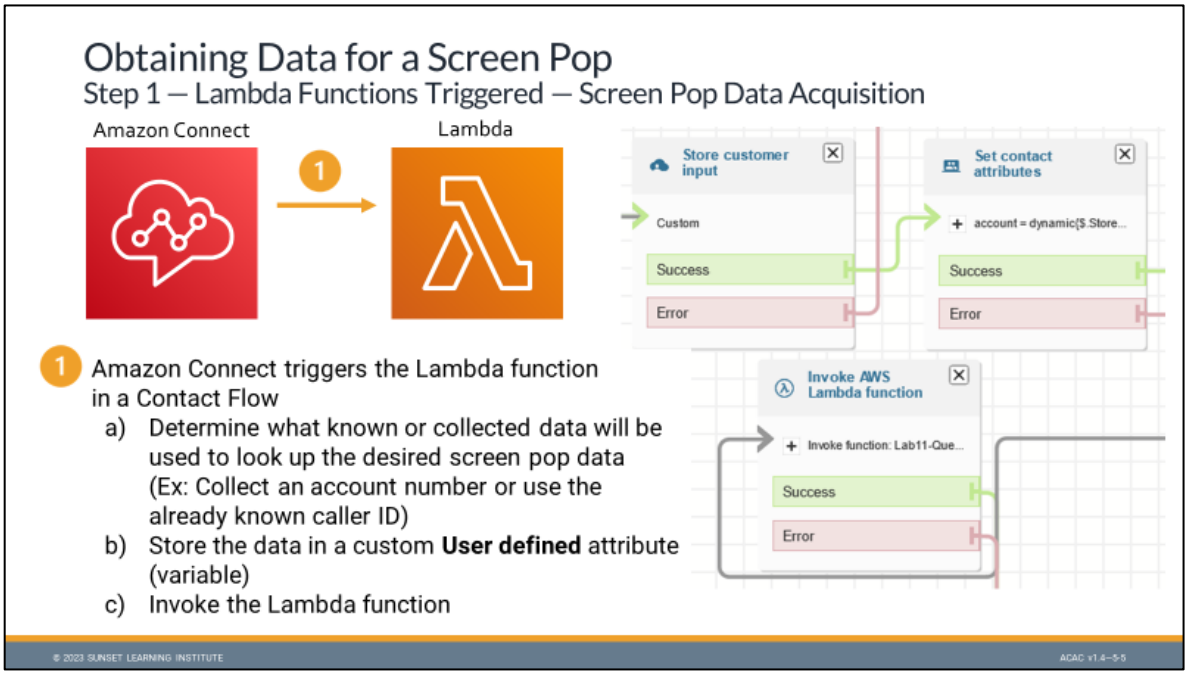
In addition to the CCP, there is another interface named Agent Workspace which already has the capability to display customer data ingested from CRM data.

This section describes how you obtain data and access the attribute directly from a Lambda function. Later, we will discuss methods to present that data to the agent.



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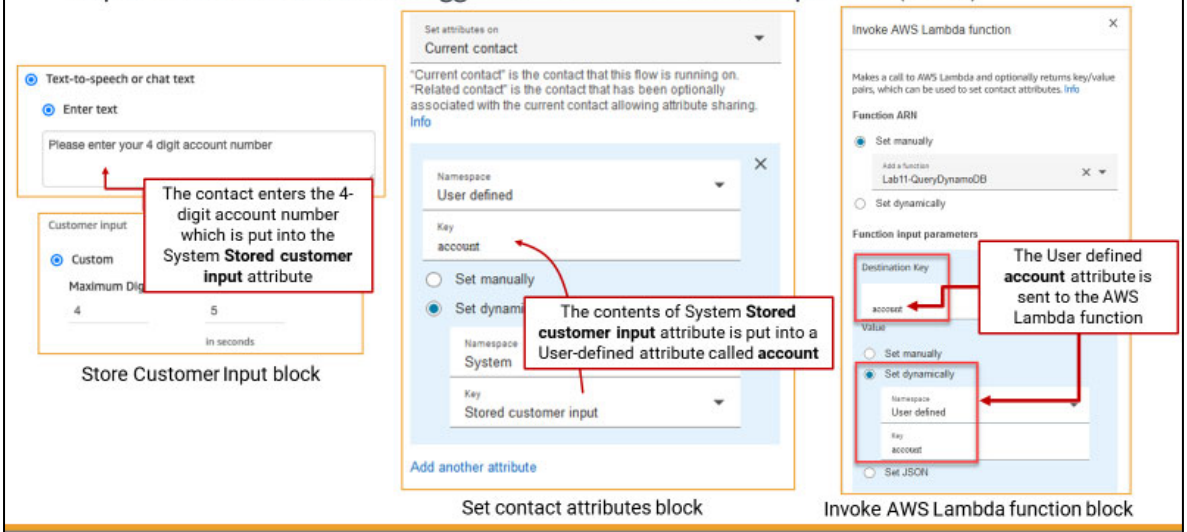
Using Lambda to access Customer data



To access the attribute directly from a Lambda function, use **$.External.AttributeName**. If the attribute is stored to a user-defined attribute from a **Set contact attributes** block, use **$.Attributes.AttributeName**.

Obtaining Data for a Screen Pop

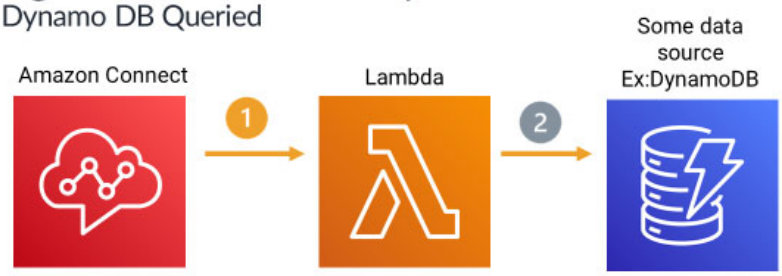
Step 1 — Lambda Functions Triggered — Customer Data Acquisition (Cont.)



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Obtaining Data for a Screen Pop

Step 2 — Dynamo DB Queried



1. Amazon Connect triggers the Lambda function in a Contact Flow
2. Lambda invokes a serverless application that queries a DynamoDB, or other data source, using call information or collected information

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Obtaining Data for a Screen Pop
Step 2 — Dynamo DB Queried (Cont.)

```
1  import json
2  import boto3
3  import pprint
4
5  client = boto3.client('dynamodb')
6
7
8  def DBread(event, context):
9    account = event["Details"]["ContactData"]["Attributes"]["account"]
10   data = client.get_item(TableName='ConnectDB' Key={'acctNumber':{'S': account}})
11
12   response = {'statusCode': 200,'body': json.dumps(data),'headers': {'Content-Type': 'application/json','Access-Control-Allow-Origin': '*'}}
```

In this example a Python script is executed as the Lambda function.

A python module called boto3 is used to query the DynamoDB and get an item using a key of **account** which was passed by Amazon Connect.

acctNum...	FirstName	LastName	ServiceLevel
1002	Mike	Keutzer	Gold
1001	Jed	Demeule	Silver
1003	Bruce	Wilkinson	Platinum

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Obtaining Data for a Screen Pop
Step 3 — Dynamo DB Results Returned



1. Amazon Connect triggers the Lambda function in a Contact Flow
2. Lambda invokes a serverless application that queries a DynamoDB, or other data source, using call information or collected information
3. DynamoDB query returns results to the Lambda function

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Obtaining Data for a Screen Pop
Step 3 – Dynamo DB Results Returned (Cont.)

The Dynamo DB returns the following JSON data to the AWS Lambda function

Amazon Connect will not be able to understand this nested JSON

A flat JSON file is required for Amazon Connect, no nested values allowed

Valid formatting Example:
{key1:value1 ,
key2:value2,
key3:value3
}


```
{
    'Item': {
        'acctNumber': {
            'S': '1003'
        },
        'ServiceLevel': {
            'S': 'Platinum'
        },
        'FirstName': {
            'S': 'Bruce'
        },
        'LastName': {
            'S': 'Wilkinson'
        }
    },
    'ResponseMetadata': {
        'RequestId': 'CGVC45TE8IRP6L995E8NHI6RJNVV4KQNSO5AEMVJF66Q9ASUAAJG',
        'HTTPStatusCode': 200,
        'HTTPHeaders': {
            'server': 'Server',
            'date': 'Mon, 06 Dec 2021 20:57:46 GMT',
            'content-type': 'application/x-amz-json-1.0',
            'content-length': '123',
            'connection': 'keep-alive',
            'x-amzn-requestid': 'CGVC45TE8IRP6L995E8NHI6RJNVV4KQNSO5AEMVJF66Q9ASUAAJG',
            'x-amz-crc32': '930411053'
        },
        'RetryAttempts': 0
    }
}
```


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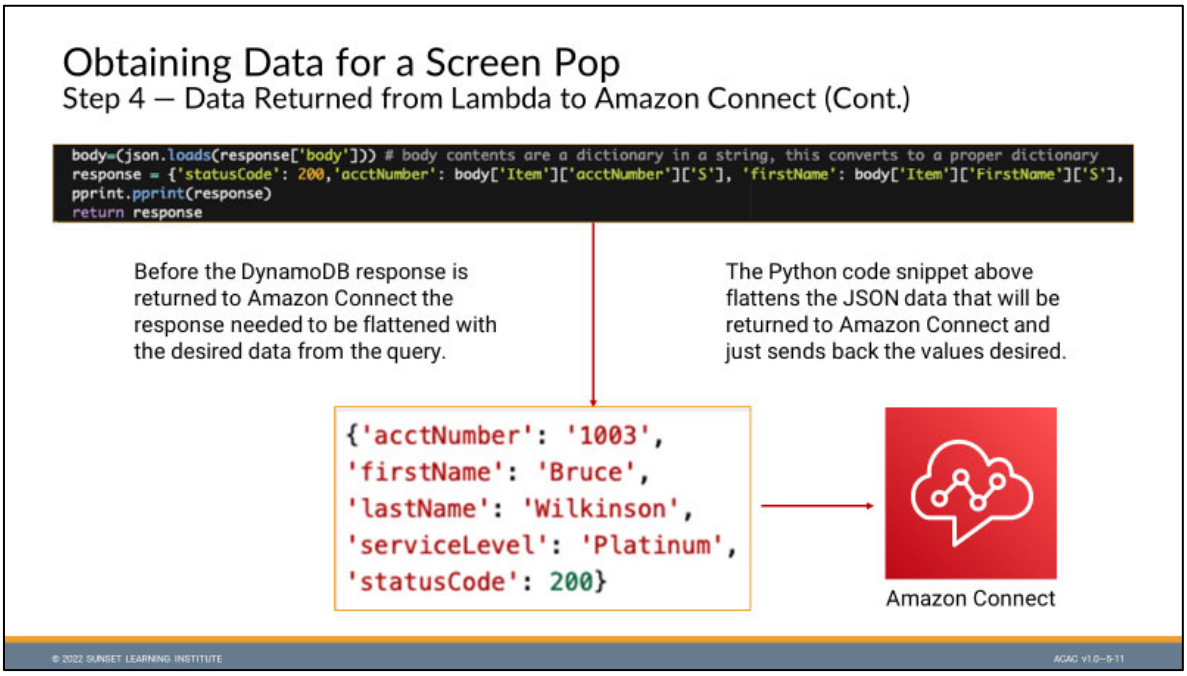
Obtaining Data for a Screen Pop
Step 4 – Data Returned from Lambda to Amazon Connect



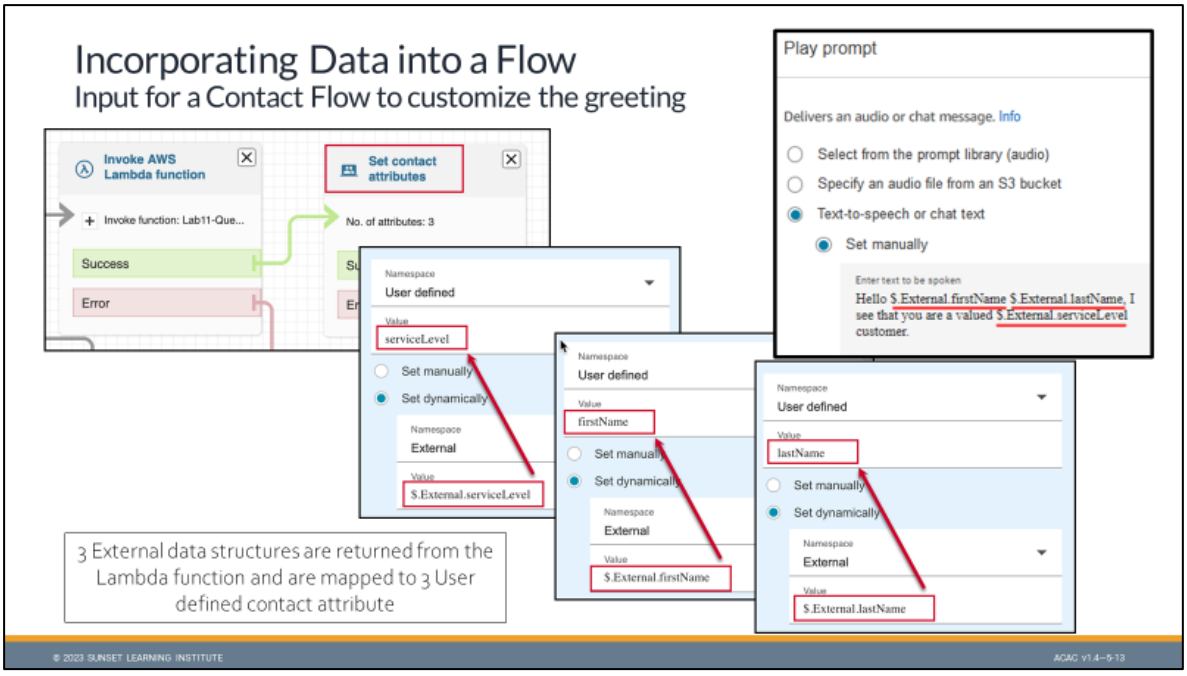
1. Amazon Connect triggers the Lambda function in a Contact Flow
2. Lambda invokes a serverless application that queries a DynamoDB, or other data source using call information or collected information
3. DynamoDB query returns results to the Lambda function
4. The Lambda function formats the data (flattens the JSON data) and returns it to Amazon Connect

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Once Lambda returns the customer record from the DynamoDB, the Flow defines how we will utilize the customer information to the Agent. In this first example, we use the Lambda values to customize the customer greeting.

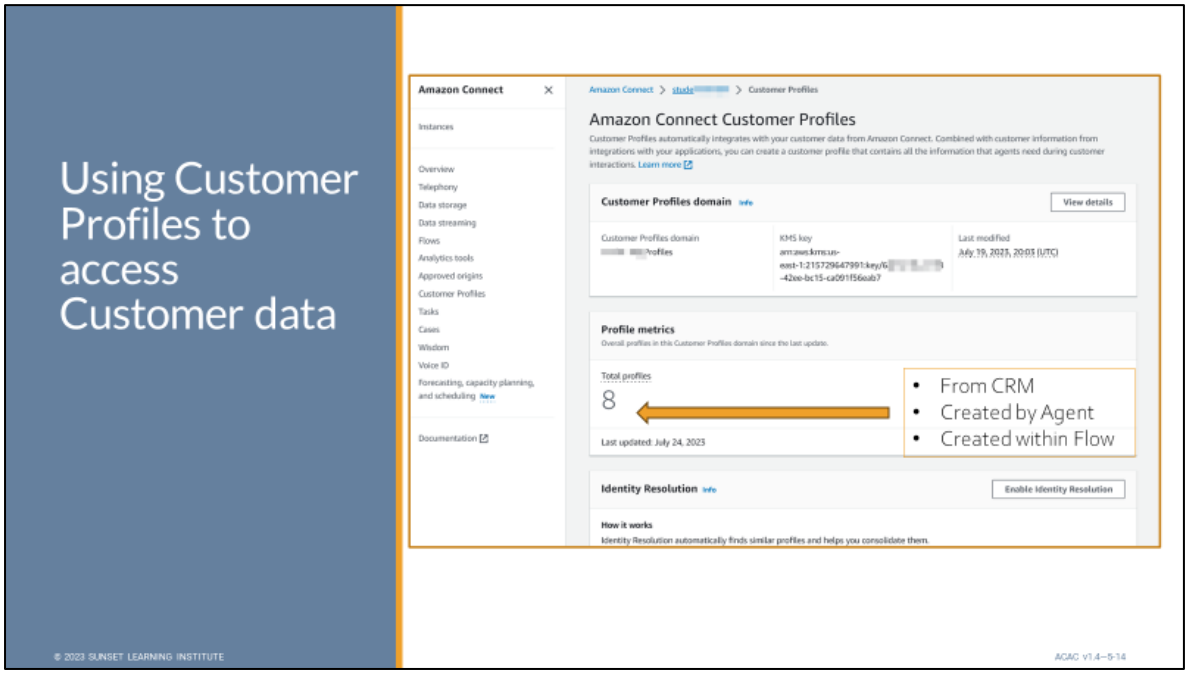


Using Customer Profiles to access Customer data

The industry has provided many Customer Relationship Management (CRM) applications for organizations to handle customer records and data in a manner that supports the needs of the organization.

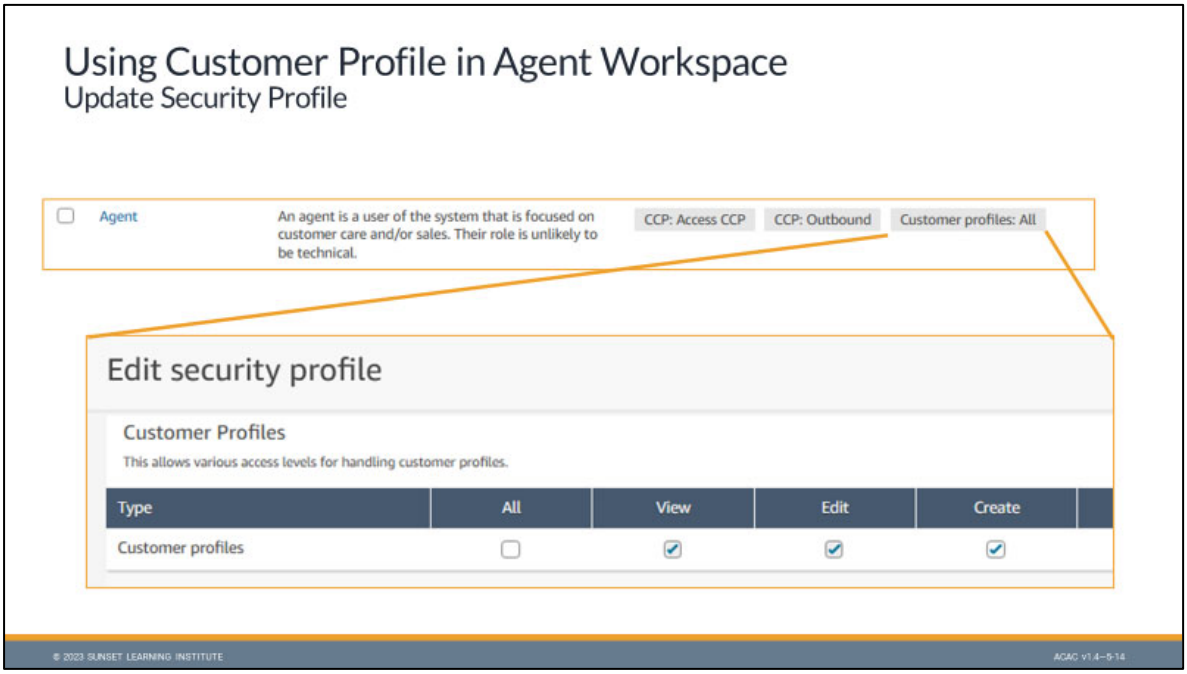
Rather than attempt to replace the CRM solution, Amazon Connect has managed to work with those solutions.

Amazon Connect Customer Profiles has a structure to support customer data in a format usable by the Flow designer.

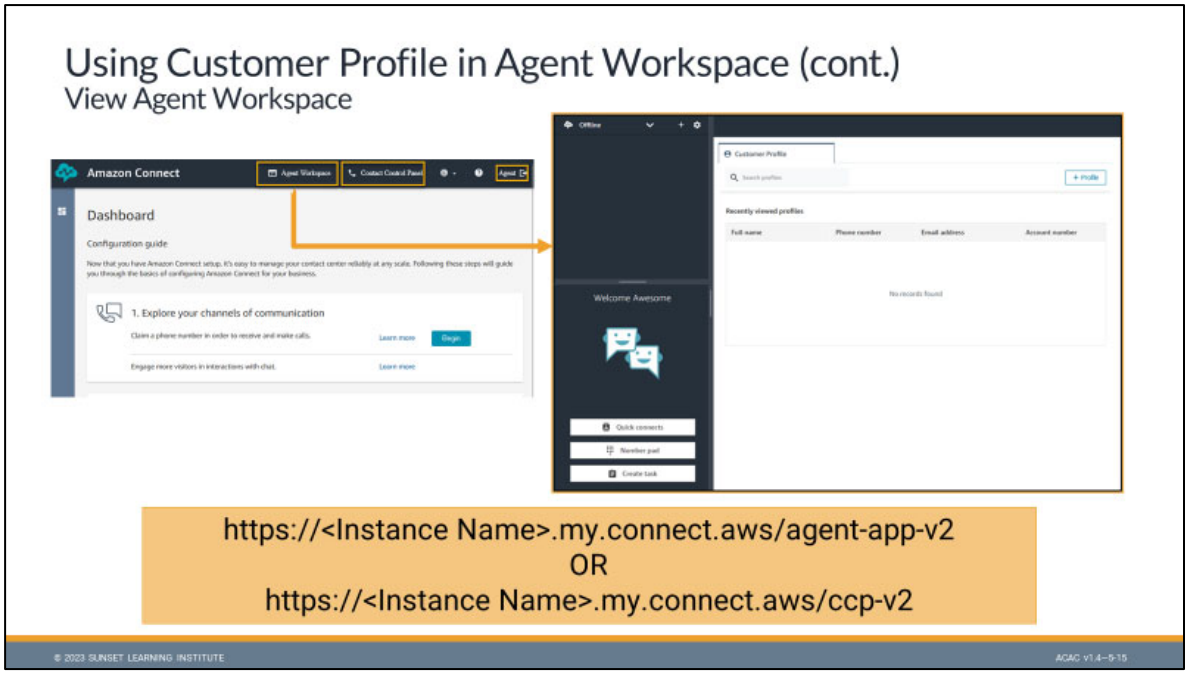


Amazon Connect Customer Profiles provides access to customer data records. Customer Profiles can be ingested from external CRM solutions, created by Agents from within the Agent Workspace tool, or created and updated from within Contact Flows.

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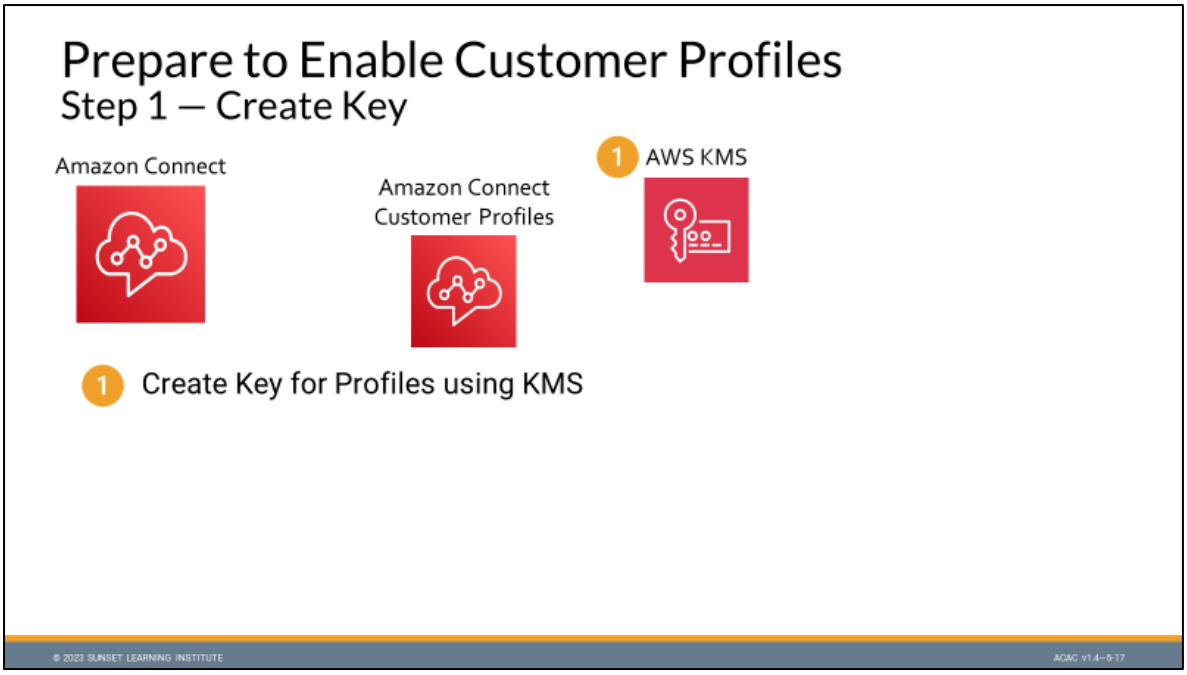
Customer Profiles need to be enabled for Agents via the Security Profile for Agent Workspace to display the Profile.



Once access to Customer Profiles has been enabled for Agents (or Supervisors for that matter), Customer Profiles are accessible using the Agent Workspace application. Users can access Agent Workspace using the URL: **https://<Instance Name>.my.connect.aws/agent-app-v2**. Alternately, Agent Workspace can be accessed from the Amazon Connect Dashboard.

Agents have the choice of using Agent Workspace or the CCP by accessing the URL: https://<Instance Name>.my.connect.aws/ccp-v2.

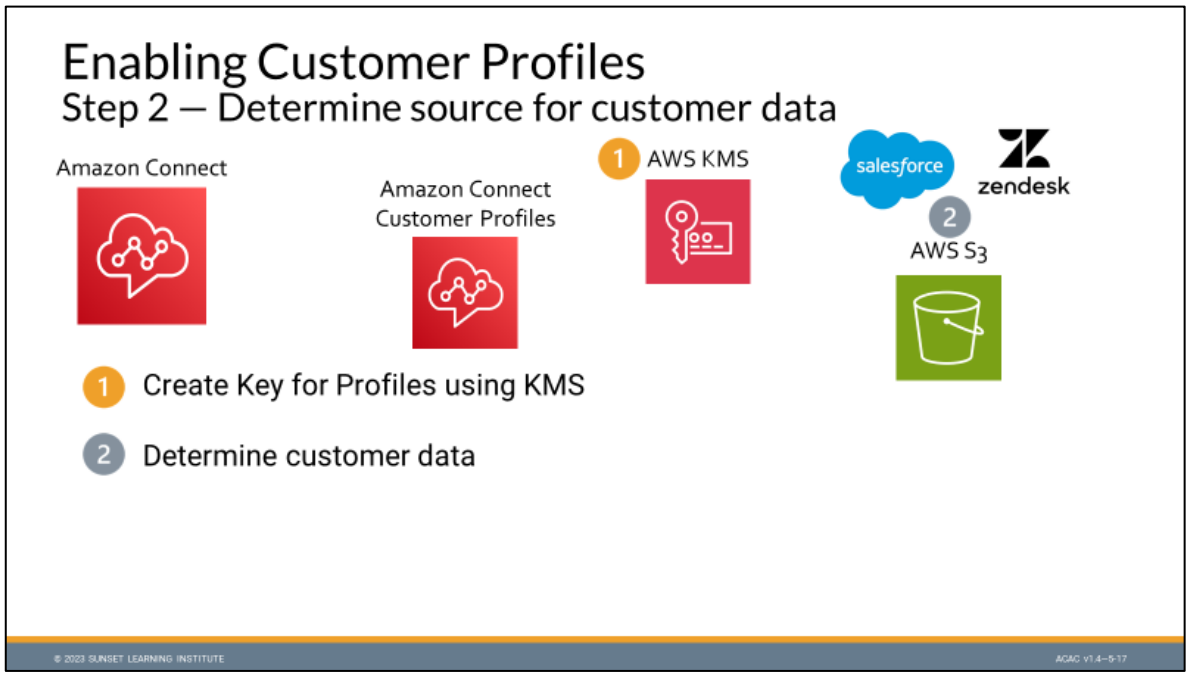
But first, we need to make sure that Customer Profiles have been enabled. To access the Profiles, a secure encryption key needs to be created and defined.



It is possible that a Customer Profile domain is created automatically within the Amazon Connect instance. However, if you need to create a new domain or choose to delete the previous domain and recreate a new Profile domain manually, you will first need an encryption Key in order to protect the data in the Profile.

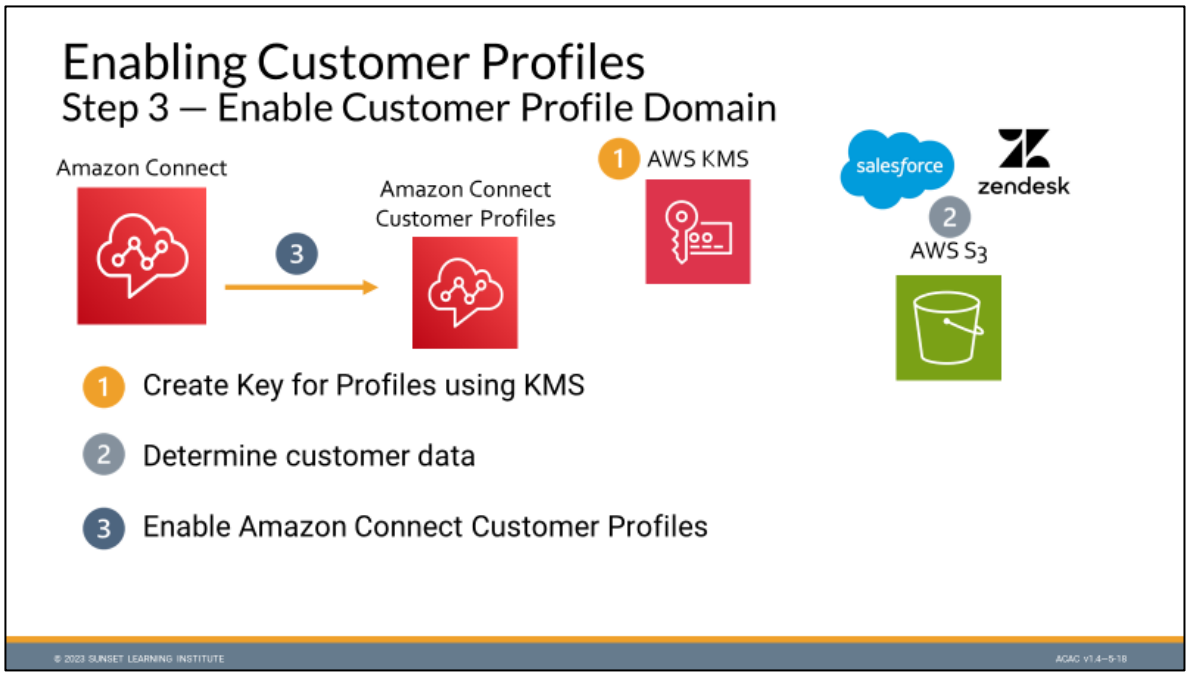
To create a key for the Profile domain, you can go directly to KMS, or from the Profile tool, click **Create Key** to redirect to KMS.

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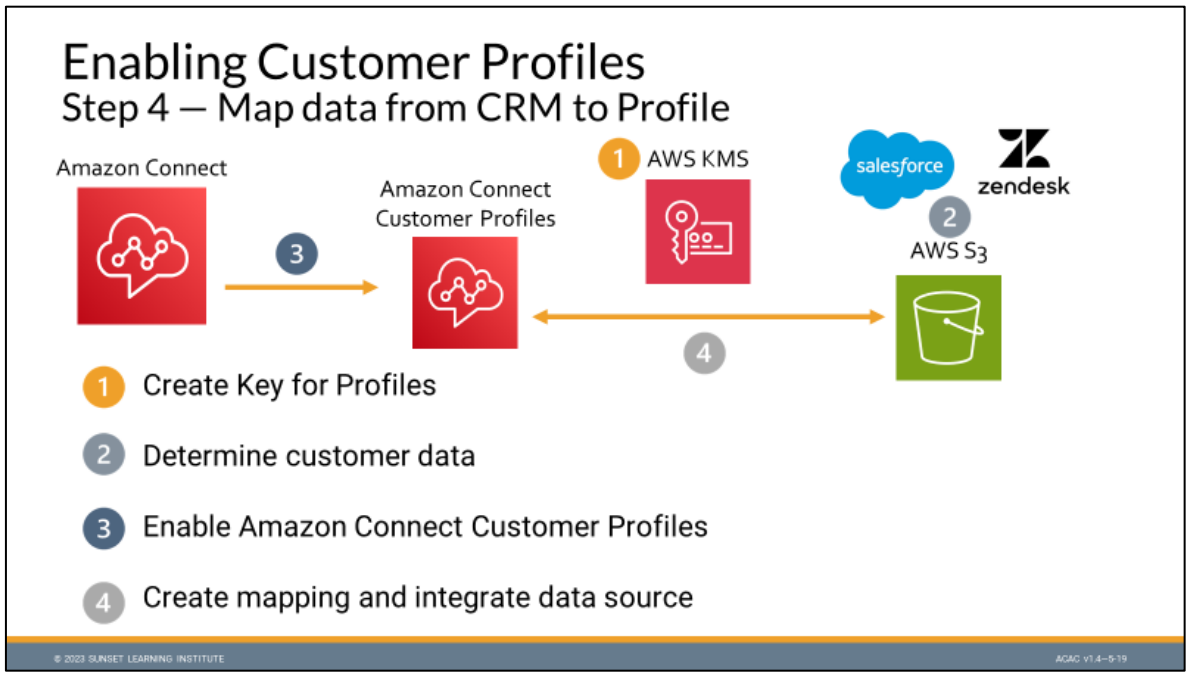
Customer Profiles may be populated in different ways. One method to populate Customer Profile is to ingest customer data from your CRM. Given that customer data may be drawn from multiple sources, determine what customer data is needed. Other methods include ingesting data from an S3 bucket. Additionally, new profiles may be created by Contact Flows when customers call for the first time, or agents may create profiles for customers.

Here, we will start with customer records provided by a file in an S3 bucket.



Now, enable a Profile domain for your Amazon Connect instance. You will name the domain and define which Key to protect your customer profile data. The Profile objects may not directly map to the data from CRM solutions. Next, we will define how to map the CRM data to Profile attributes.

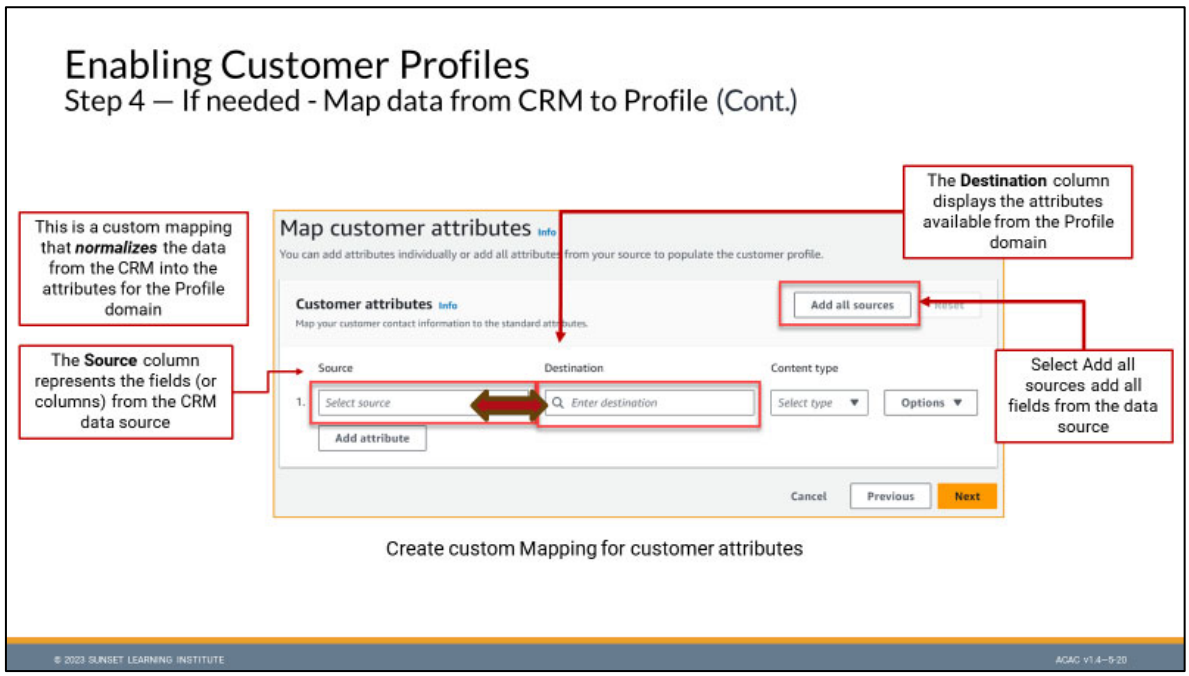
In order to enable a Profile domain, the AWS Administrator needs to have IAM Permissions specifically for Profiles and other AWS services. This is in addition to the permissions to create a Connect Instance.



Once your domain is created, you have the opportunity to create a data mapping to ingest the data from the CRM source into the Profile domain.

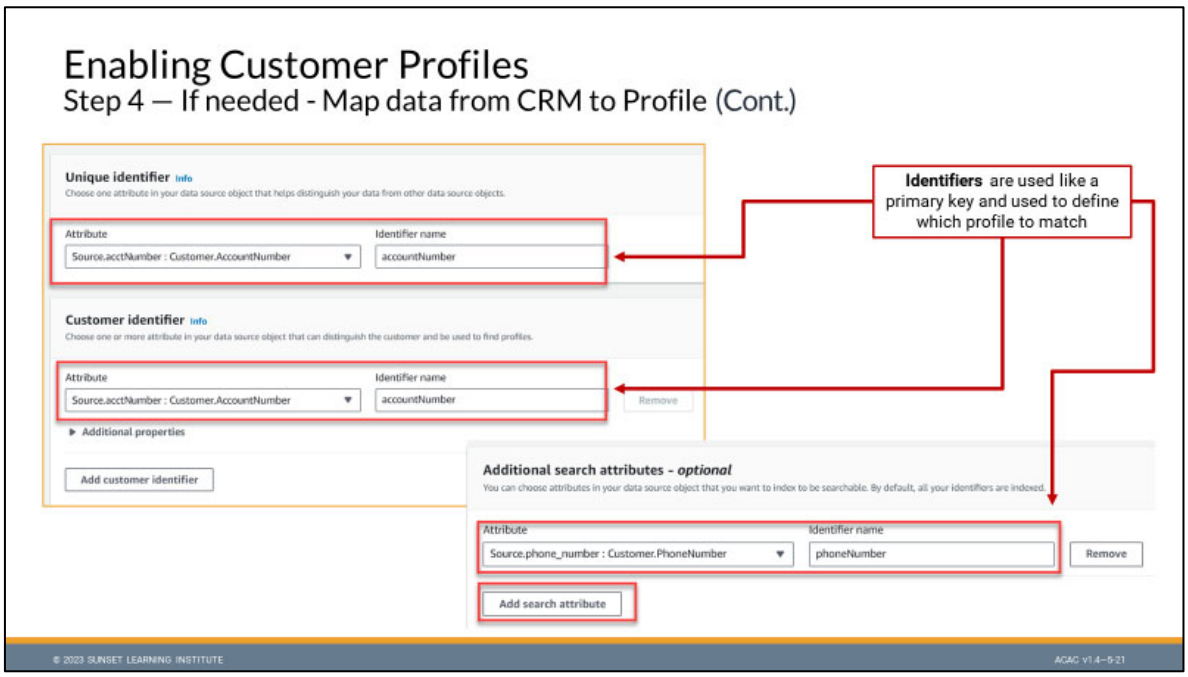
The last step is to integrate the source data to the Profile based on the mapping.

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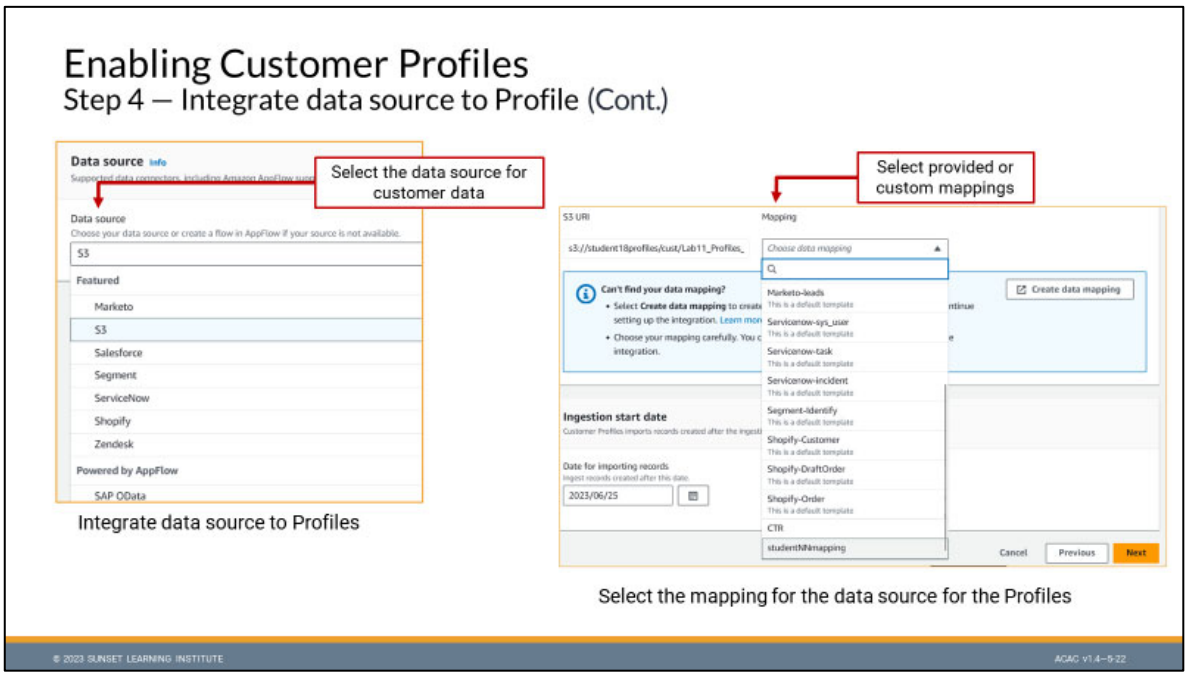


AWS provides multiple pre-built attribute mappings for popular CRM solutions such as Salesforce, Zendesk, and ServiceNow. In addition, certain data sources may need an attribute mapping that is not available by default. AWS provides an option to create a custom attribute mapping to normalize the fields from the data source based on the attributes for Customer Profiles.

For example, your customer data may be stored in a custom CSV file stored in a S3 bucket. Obviously, your custom CSV file has a custom definition which requires a custom mapping to ingest the data into the correct attribute of the Customer Profile.

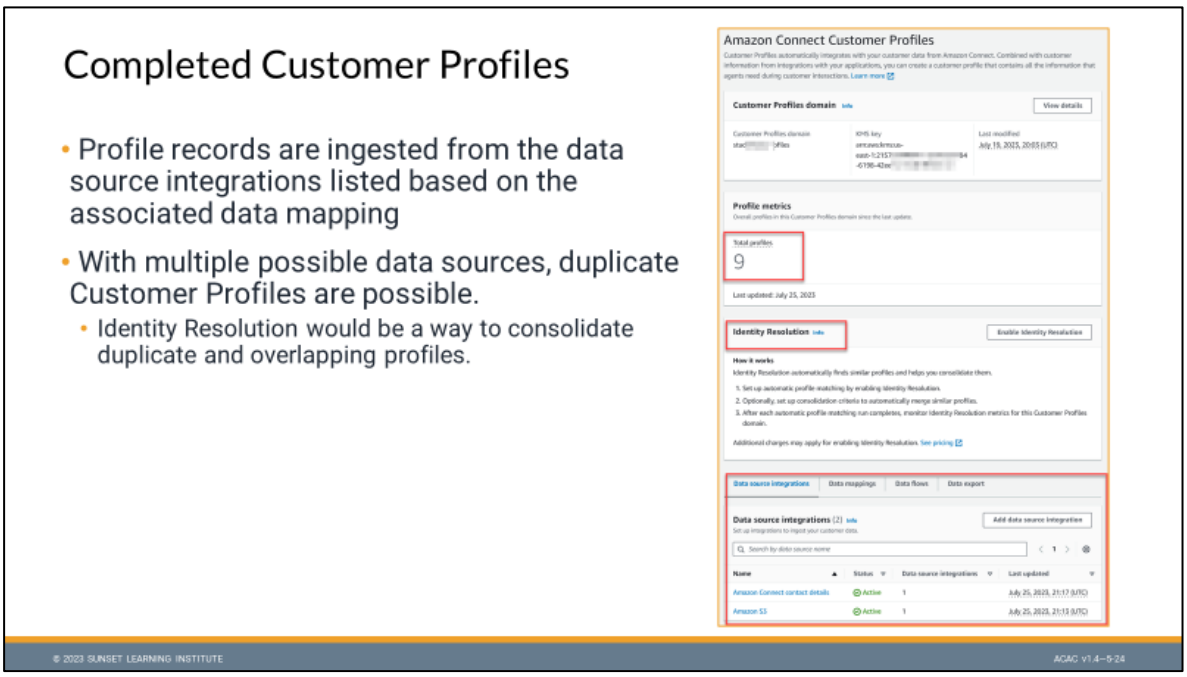


Within your custom mapping, it is important to determine which attribute will be used to match customers with a specific Customer Profile.



Whether you create a custom attribute mapping or not, you will need to create a data integration. The Customer Profile data integration defines the customer data source, such as S3, ServiceNow, Salesforce, etc., and the attribute mapping to be used to ingest the data and assign the appropriate Profile attribute as it is stored in the Profile domain.

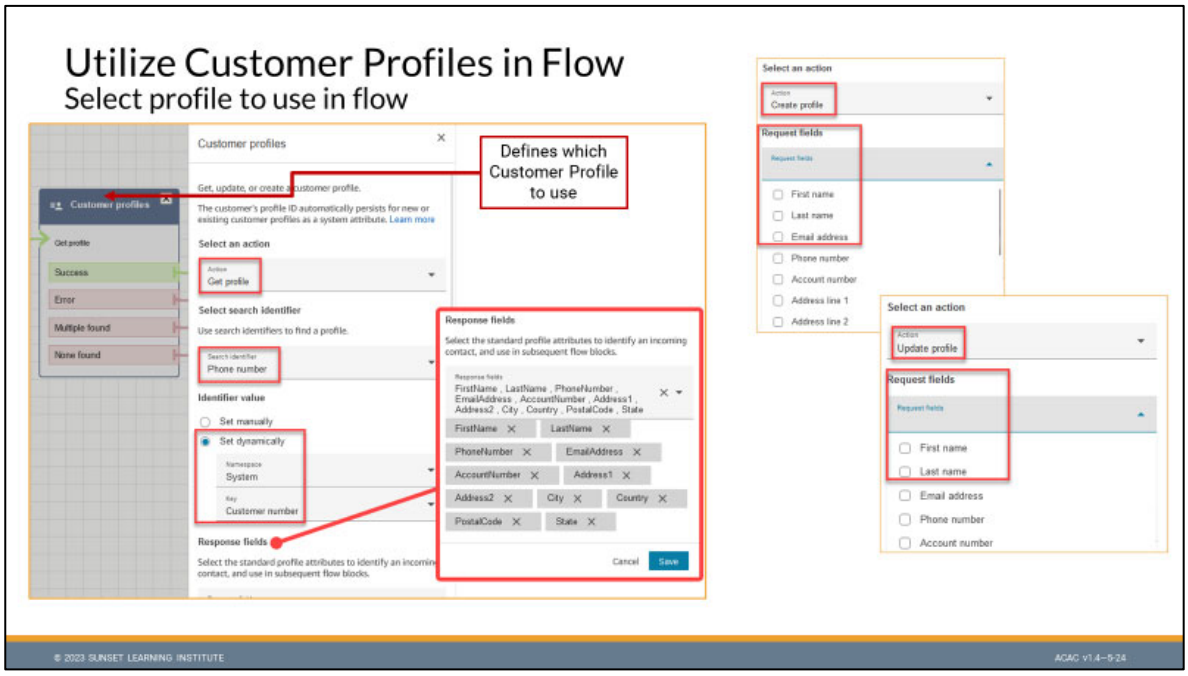
AWS has pre-built attribute mappings for popular CRM integrations.



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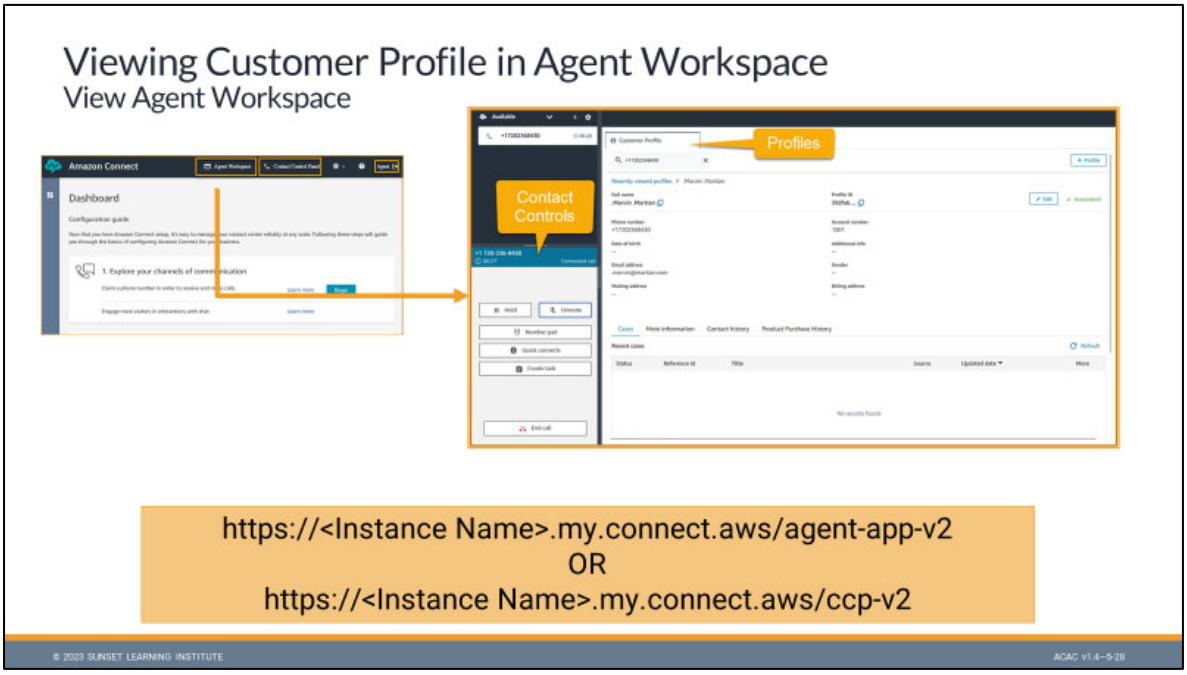
Once the data integration is complete, monitor the Customer Profile to confirm the ingestion of Profiles from the data sources. Notice that the Data source integrations are listed at the bottom of the screen. Once the integration is created, it initially displays a Status of Pending while it ingests the records. Once complete, the Status will change to Active.

Since there is the potential for ingesting data from multiple integrations, there is the potential for duplicate and overlapping profiles. Identity Resolution allows you to create criteria used to identify overlapping profiles that should be consolidated relative to unique profiles. Time stamps is one method that Identity Resolution may be used to resolve potential conflicts.



Keep in mind that Customer Profiles are a collection of customer attributes. However, when a customer call enters the contact center, the Contact Flow defines the logic used to handle the contact. Flow Designer has a block that is used for the Flow logic to integrate with the appropriate Customer Profile.

In the Contact Flow, the Customer Profiles block can be found in the Integrate folder and supports three Actions: Get Profile, Create Profile, and Update Profile.



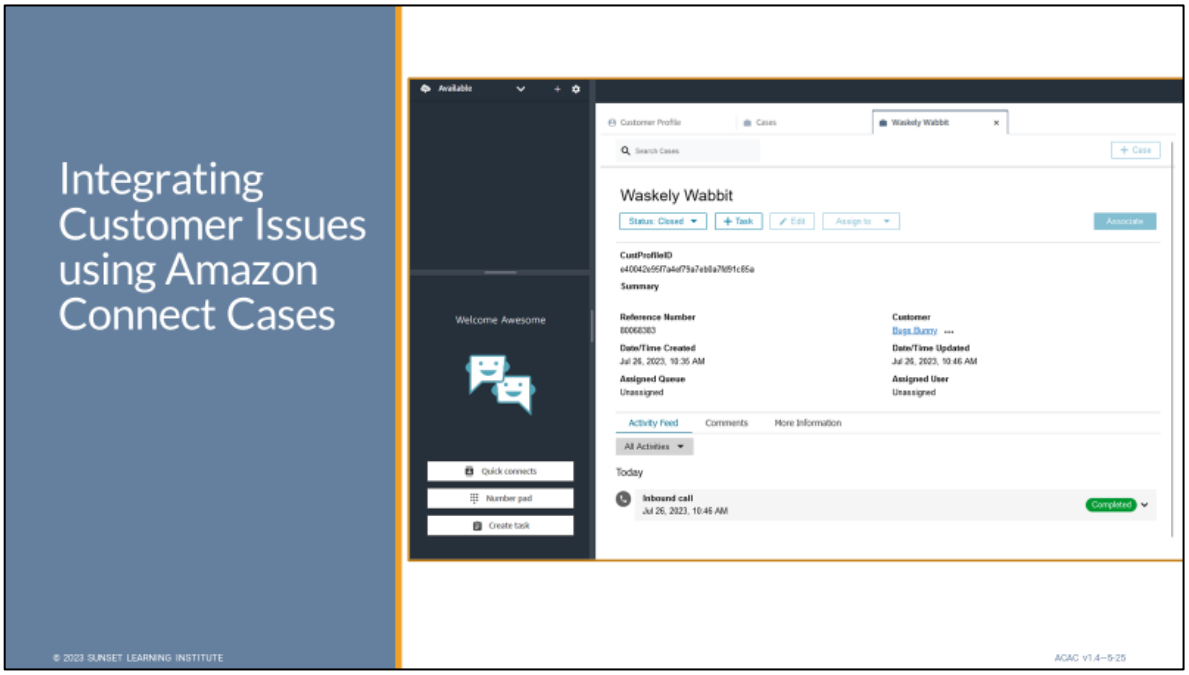
With an appropriate security profile combined enabled Profiles while using Agent Workspaces, Agents have access to view the Profile for a given caller.

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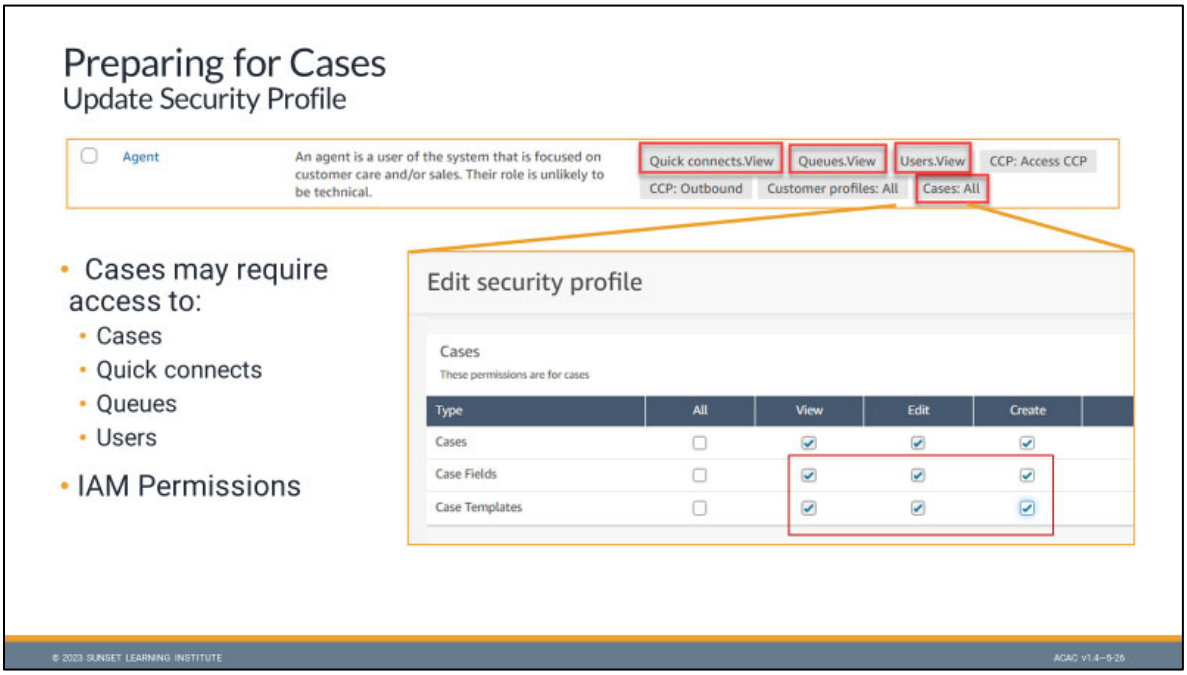
Integrating Customer Issues using Amazon Connect Cases

Typically, customers reach out to a Contact Center because they have an issue that needs to be resolved. In order to track and handle customer issues, Amazon Connect offers the Cases feature. Once a Customer Case has been created, Cases can track the steps and interactions to handle customer issues as well as the outcome. By managing the state of the Case, the Contact Center can make sure that the Case stays open until it is closed. The case can be assigned or re-assigned to specific queues or agents.

Before an Agent can create a new Case, the Agent needs to have an active customer contact that has been associated to a Customer Profile.



Amazon Connect Cases presents the Agent with options to track and resolve customer issues. Cases may be generated automatically as part of the Contact Flow or manually generated by Agents using the Agent Workspace interface.



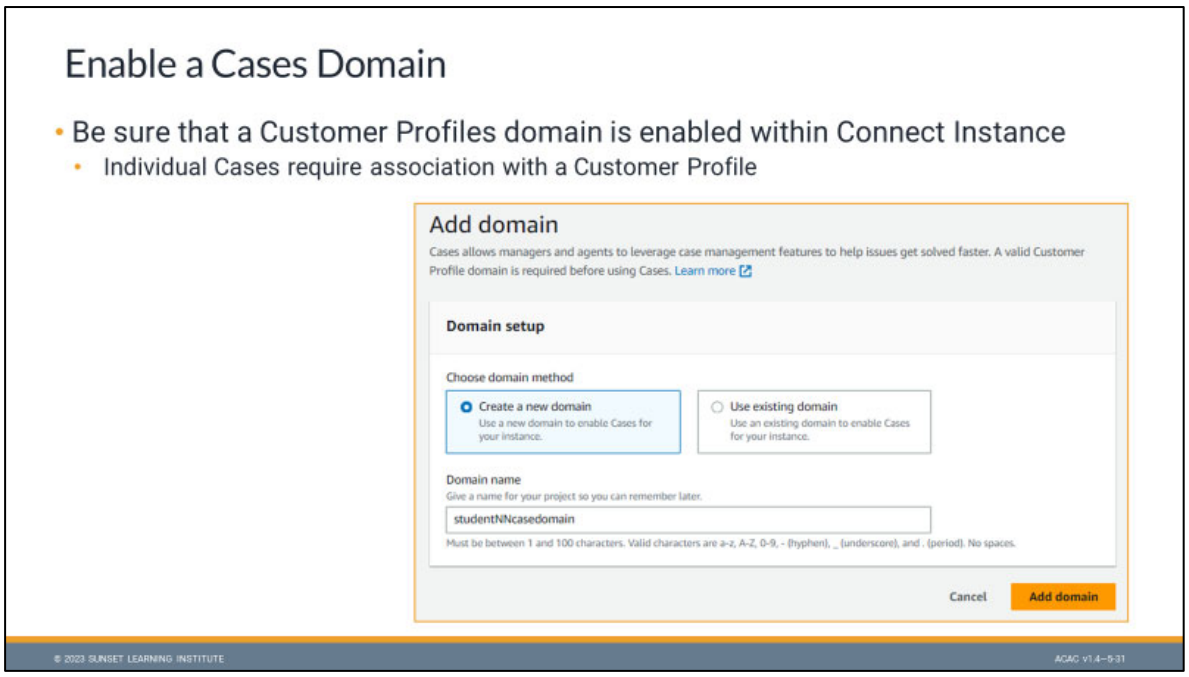
In order for Agents to access Cases, similar to Customer Profiles, permissions need to be enabled for Agents via the Security Profile. While updating the Case for a customer, the Agent may also need to assign Quick Connects, Queues, as well as Users to a case to ensure that case is routed to the appropriate resource to resolve the case. Therefore, View permissions for Quick Connects, Queues, and Users will also be required for Agents to make those assignments.

Amazon Connect administrators will need permissions to view, create, edit both Case Fields and Case Templates.

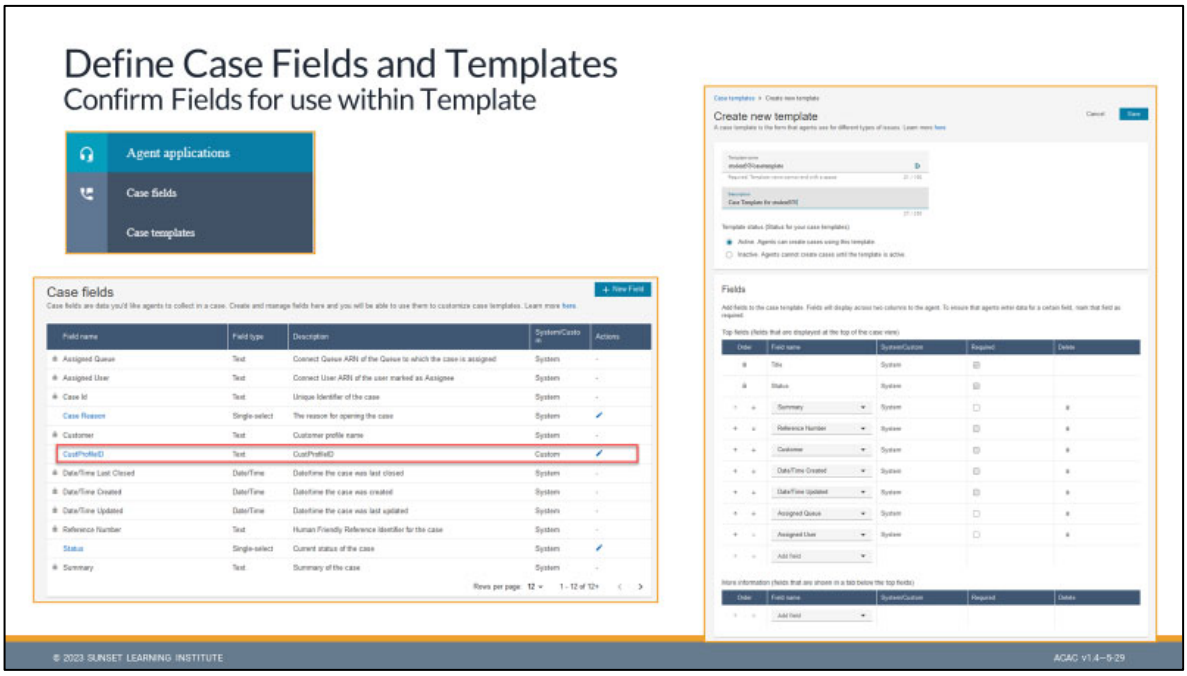
Just as Customer Profiles required IAM permissions, the administrator that creates the domain for Cases need to have the following:

- connect:ListInstances
- ds:DescribeDirectories
- connect:ListIntegrationAssociations
- cases:GetDomain
- cases:CreateDomain
- connect:CreateIntegrationAssociation
- connect:DescribeInstance
- iam:PutRolePolicy.

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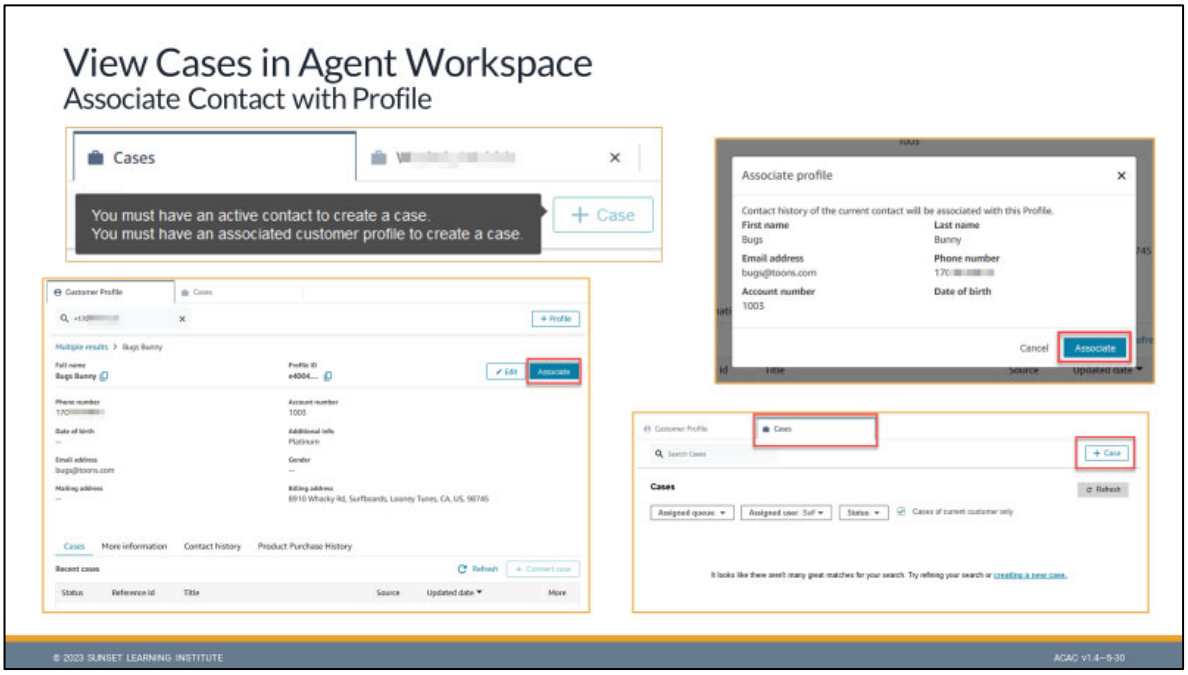


Cases require association with a Customer Profile, so make sure that the Customer Profile domain has already been enabled.



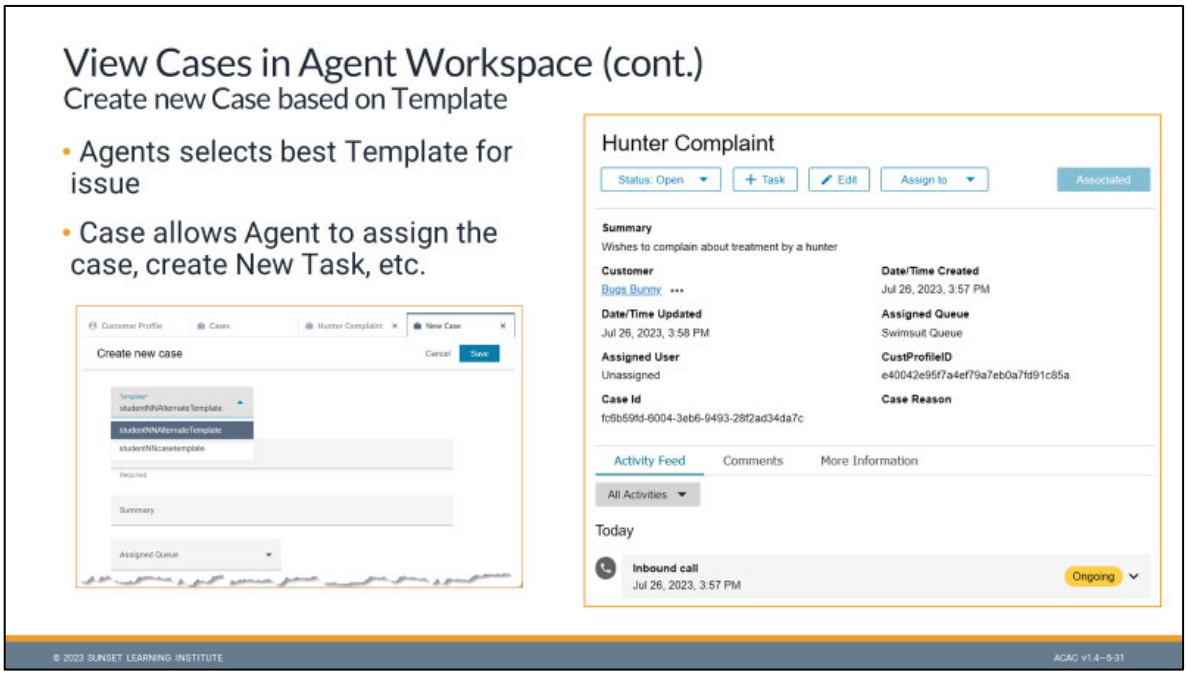
Case Templates are used to define which Case Fields are accessible to the Agent while editing a case. Several System Case fields are available. Additionally, you may create Custom Fields using the following data types:

- Number
- Single-select
- Text
- True/False
- URL

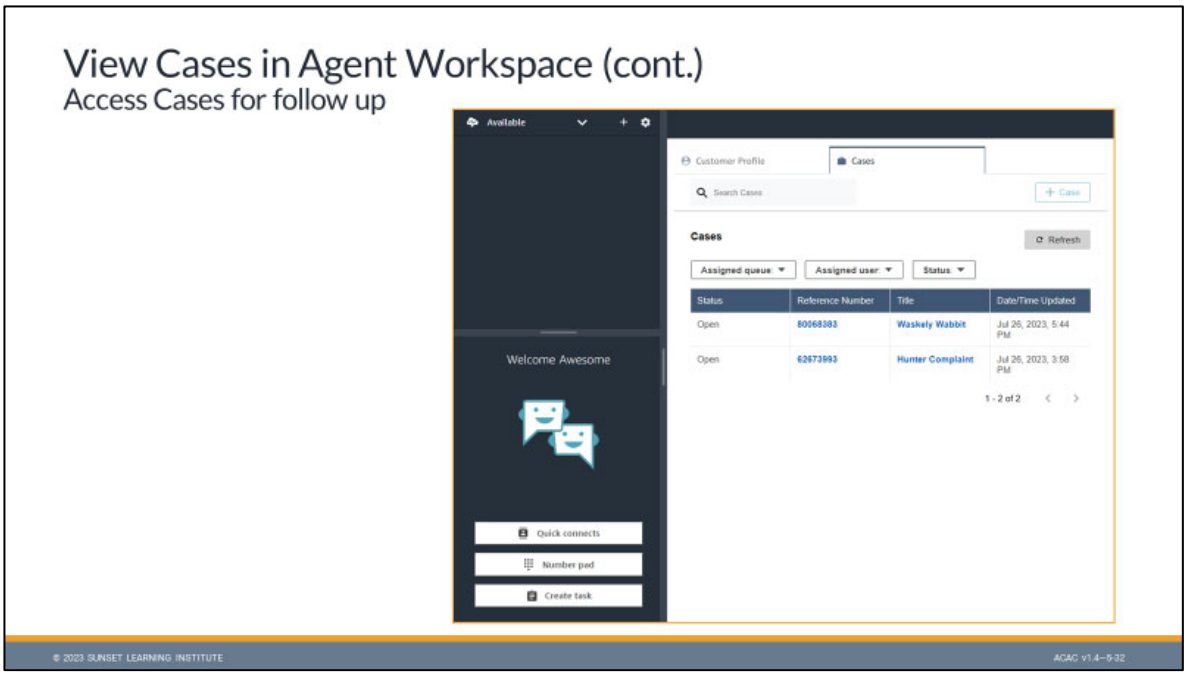


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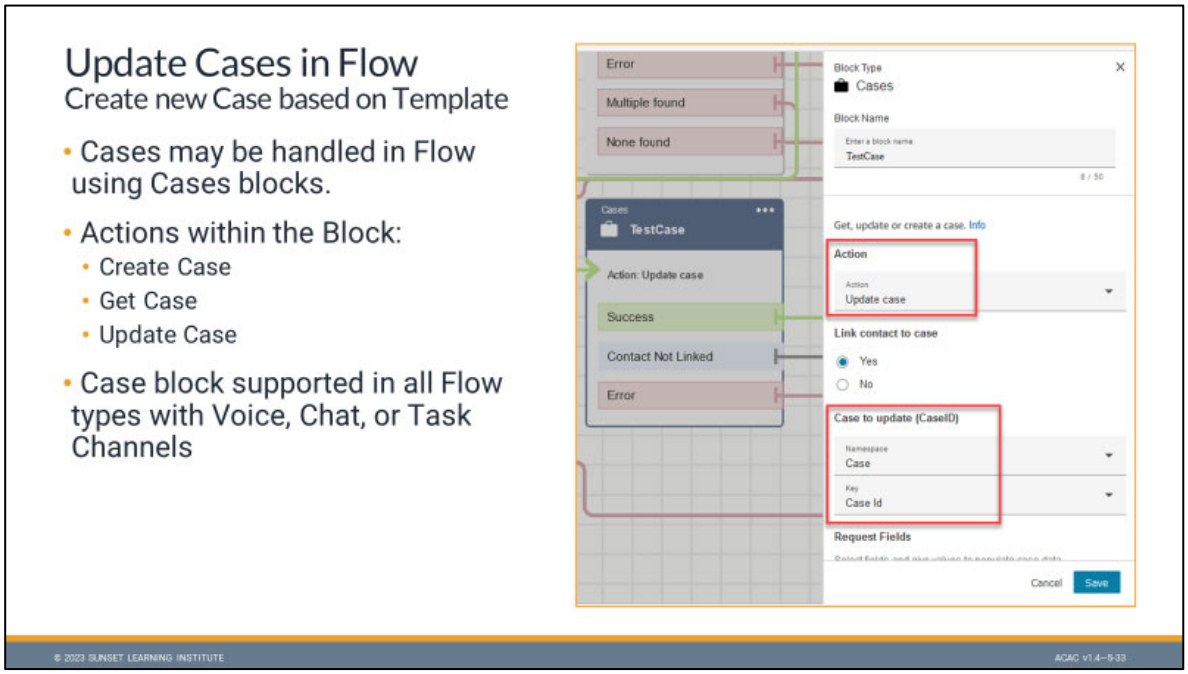
For Agents to create a new Case, they must be on an active contact that is associated with a customer profile. Once that is complete, the agent may create a new case.



When creating a new Case, agents choose the template for the case. Once the case is created, agents may update the values including assigning the case to a queue or agent or generate a new Task to be processed by a Flow.



Agent Workspace allows Users to search for Cases for follow-up.



With the ability to **Invoke Lambda functions** or access **Customer Profiles**, Flows are able to automatically manage **Cases** using the **Cases** block. Actions available in the **Cases** block include:

- **Create Case**
- **Get Case**
- **Update Case**

The Case block supports Voice, Chat, or Task Channels using any Flow types.

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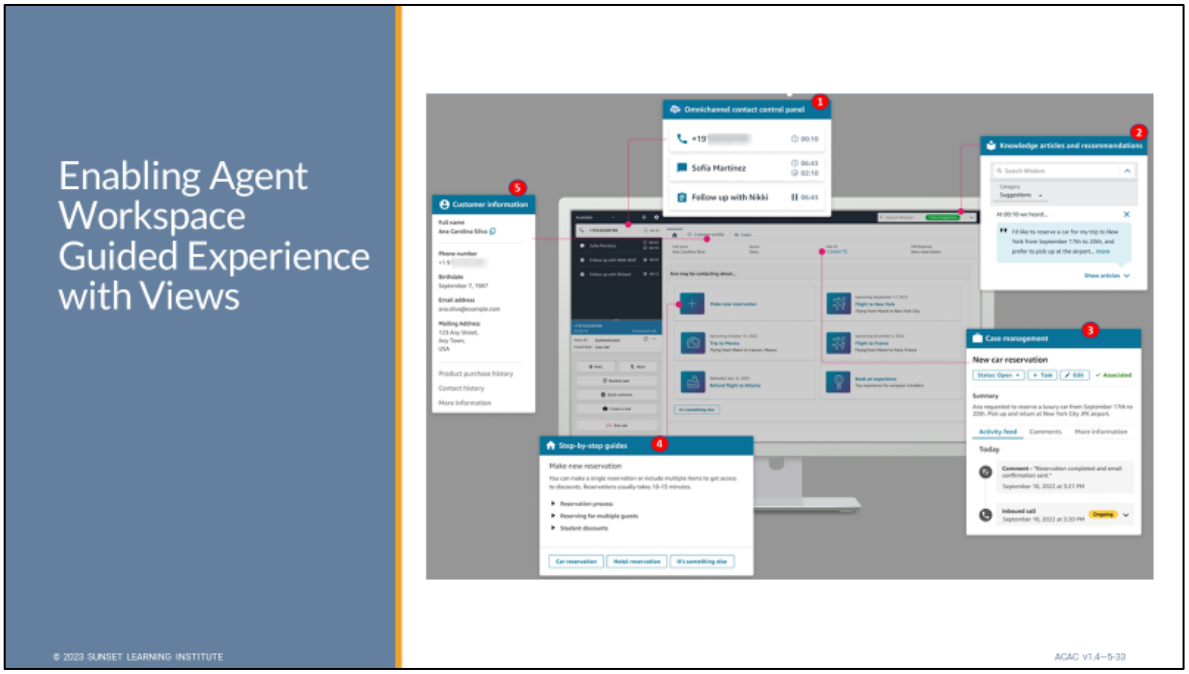
Enabling Agent Workspace Guided Experience with Views

Out-of-the-box, the Agent Workspace integrates all of your agent facing capabilities on one page. For example, when an agent accepts a call, chat, or task, they are given necessary information about the case and customer, plus real-time recommendations.

You can customize the agent workspace by enabling guided experiences, for example, and customizing the look and feel of the View resources in the agent workspace.

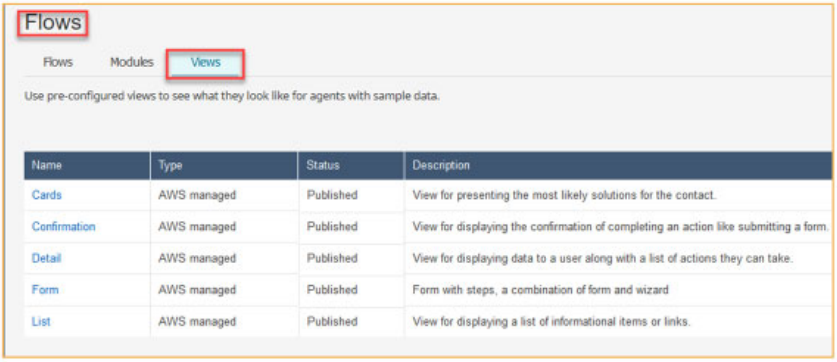
The following image shows the parts of the agent workspace.

1. The Contact Control Panel, which agents use to accept calls, chats, and tasks.
2. Real-time recommendations, powered by Amazon Connect Wisdom.
3. The case ID, and other info on the Cases tab, powered by Amazon Connect Cases.
4. Step-by-step guides.
5. Customer information on the Customer profile tab, powered by Amazon Connect Customer Profiles.



Agent Workspace Guided Experience

- Pre-built Views offer customized interfaces for Agents
- Available Views
  - Cards
  - Confirmation
  - Detail
  - Form
  - List

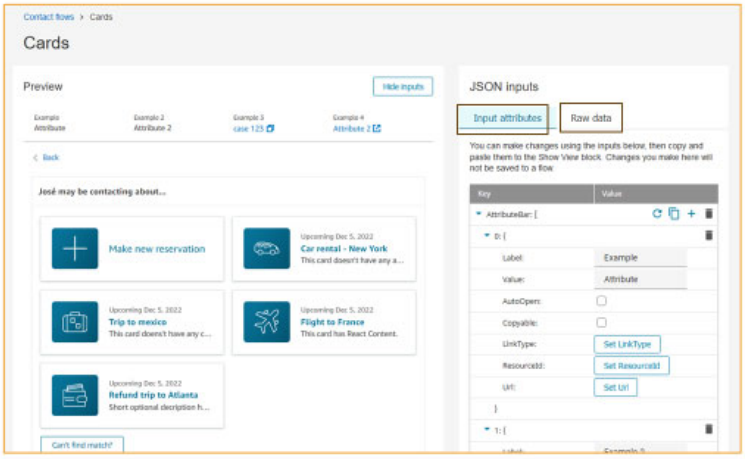


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Navigate to the Flow Designer and click on the Views tab. Currently, there are five system defined Views. As a part of the Contact Flow, you are able to define a Show View block indicating which View to display by setting values for fields defined within the View.

Agent Workspace Guided Experience

- Each View offers customizable attributes

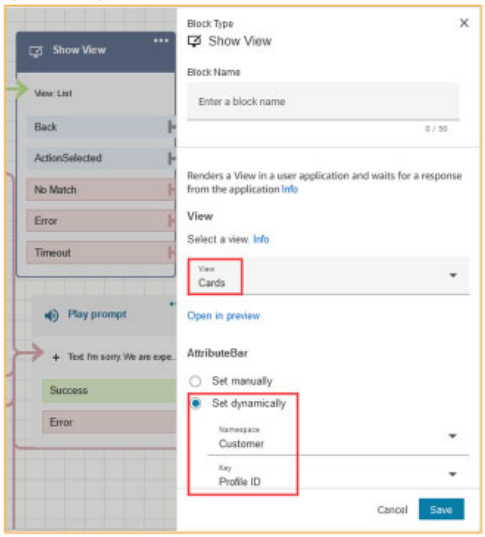


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Agent Workspace Guided Experience (cont.)

- Show View block allows choice of View
- Specify the value of the Views attributes using:
  - Set manually
  - Set dynamically
  - Set JSON

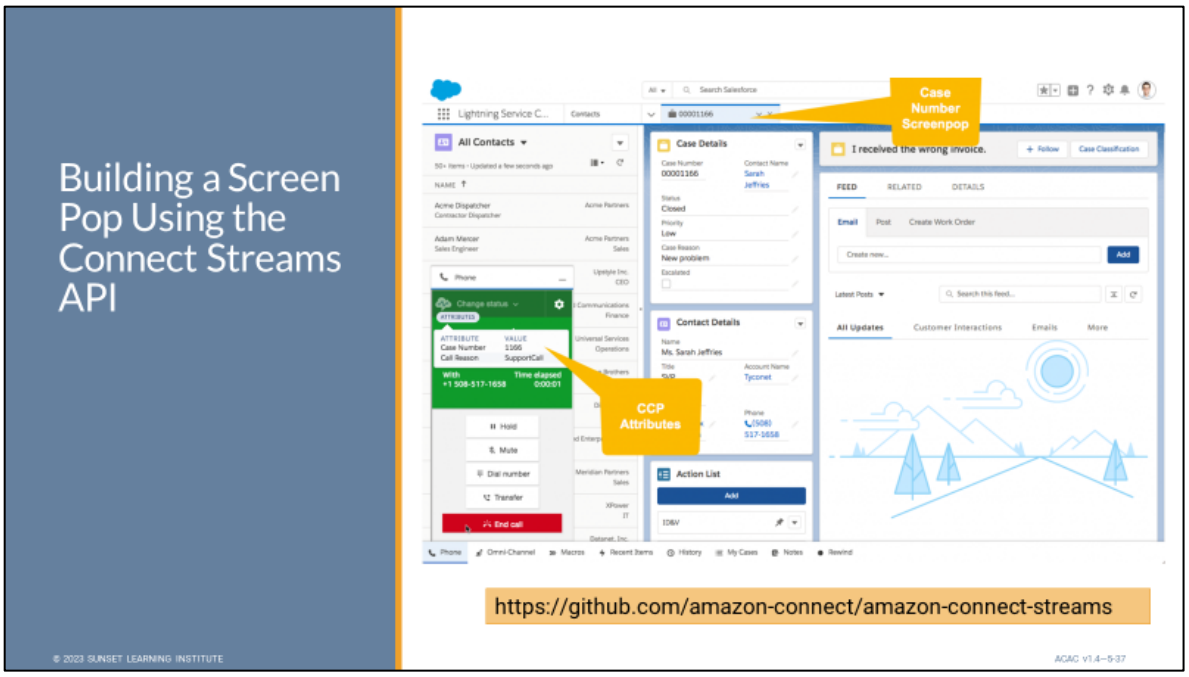


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Building a Screen Pop Using the Connect Streams API

The Amazon Connect Streams API (Streams) gives you the power to integrate your existing web applications with Amazon Connect. Streams let you embed the CCP and customer profiles app UI into your page. It also enables you to handle Agent and contact state events directly through an object-oriented event-driven interface. You can use the built-in interface or build your own from scratch: Streams gives you a choice. This section describes how you obtain data from a screen pop and access the attribute directly from a Lambda function.

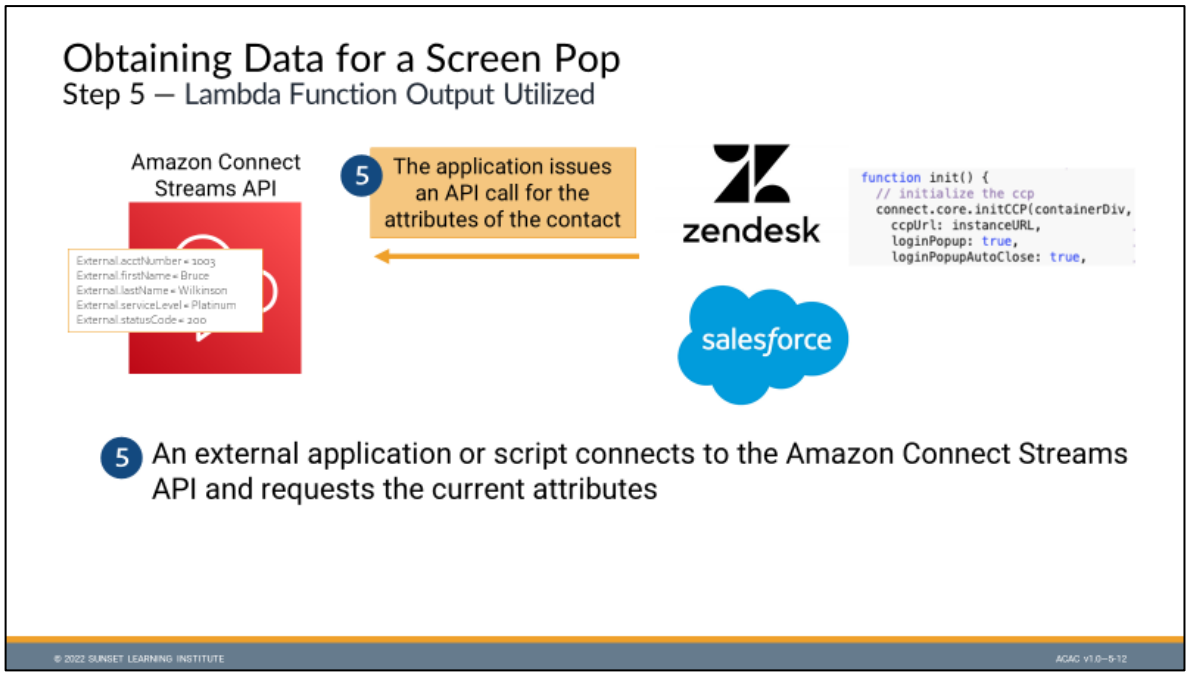


You can use contact attributes to capture information about the contact and then present it to the Agent through the Contact Control Panel (CCP). For example, you might want to do this to customize the Agent experience when using the CCP integrated with a customer relationship management (CRM) application.

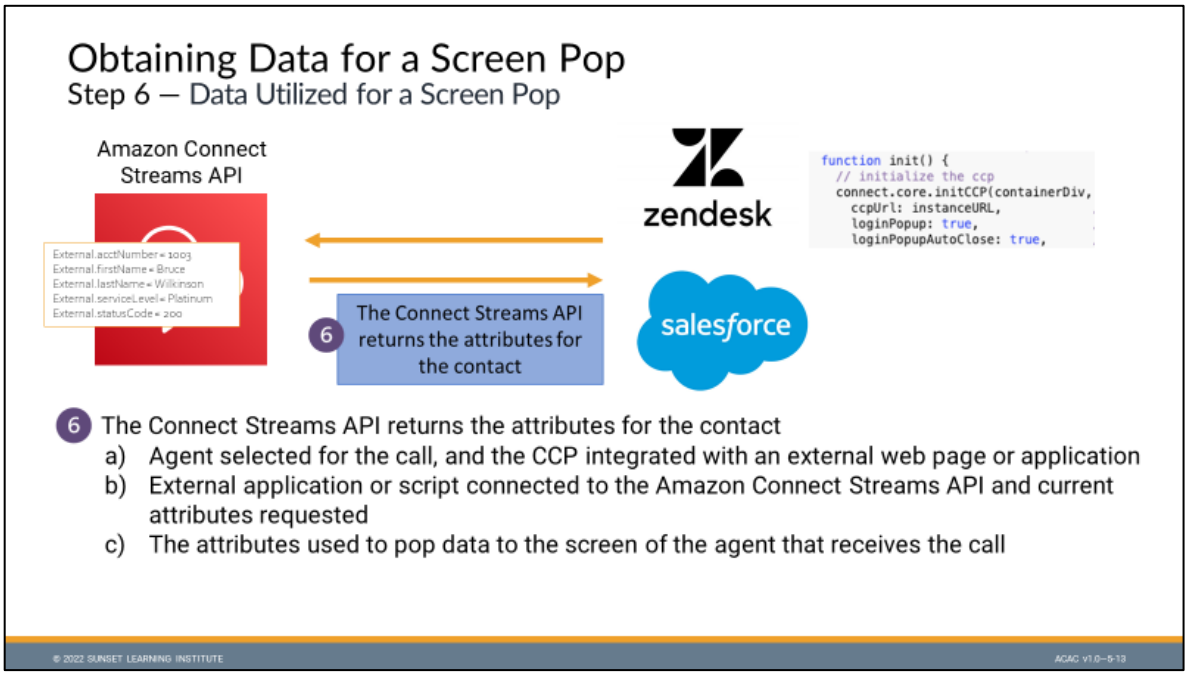
Also use them when integrating Amazon Connect with a custom application using the Amazon Connect Streams API or Amazon Connect API. You can use all user-defined attributes, in addition to the customer number and the dialed number, in the CCP using the Amazon Connect Streams JavaScript library.

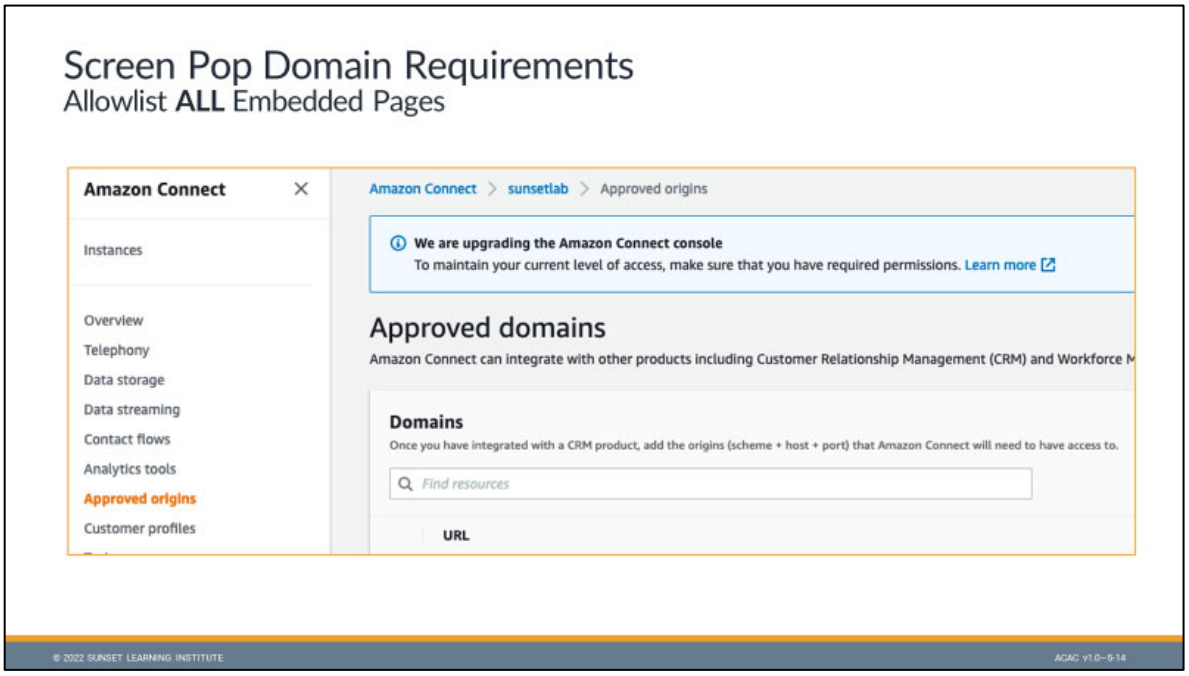
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Continuing where we left off with the Lambda example earlier.



The Amazon Connect Streams API (Streams) gives you the power to integrate your existing web applications with Amazon Connect. Streams lets you embed the Contact Control Panel (CCP) and Customer Profiles app UI into your page. It also enables you to handle agent and contact state events directly through an object-oriented event driven interface. You can use the built-in interface or build your own from scratch: Streams gives you the choice.





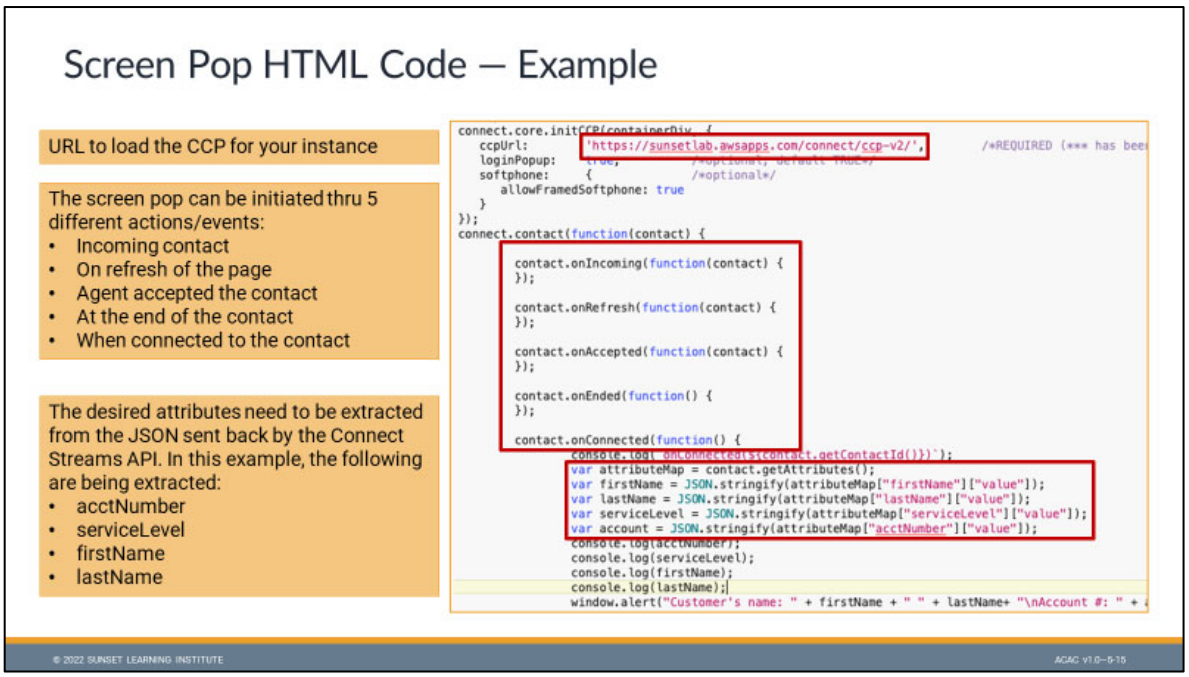
Allowlisting

The first step to using the Streams API is to allowlist the pages you wish to embed. For our customer's security, we require that all domains which embed the CCP for a particular instance are explicitly allowlisted. Each domain entry identifies the protocol scheme, host, and port. Any pages hosted behind the same protocol scheme, host, and port will be allowed to embed the CCP components which are required to use the Streams library.

A few things to note:

- Allowlisted domains must be HTTPS.
- All of the pages that attempt to initialize the Streams library must be hosted on domains that are allowlisted as per the previous steps.
- All open tabs that contain an initialized Streams library or any other CCP tabs opened will be synchronized. This means that state changes made in one open window will be communicated to all open windows.
- Using multiple browsers at the same time for the same connect instance is not supported and causes issues with the RTC communication.

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Initializing the Streams API is the first step to verify that you have everything set up correctly and that you are able to listen for events. The sequence starts by initializing the Connect Control Panel.

connect.core.initCCP()

```
<html>
<head>
 <script src="https://sunset-connect-screenpop-lab.s3.us-west-2.amazonaws.com/amazon-connect.min.js"></script>
</head>
<body>
<h1> Amazon Connect </h1>
<h2> Screenpop Example </h1>
<div id="containerDiv" ></div>
<script>
connect.core.initCCP(containerDiv, {
  ccpUrl:       'https://sunsetlab.awsapps.com/connect/ccp-v2/',       /*REQUIRED (*** has been replaced) */
  loginPopup:    true,         /*optional, default TRUE*/
  softphone:     {             /*optional*/
    allowFramedSoftphone: true
  }
});
connect.contact(function(contact) {
```

```
        contact.onIncoming(function(contact) {
        });

        contact.onRefresh(function(contact) {
        });

        contact.onAccepted(function(contact) {
        });

        contact.onEnded(function() {
        });

        contact.onConnected(function() {
            console.log(`onConnected(${contact.getContactId()})`);
            var attributeMap = contact.getAttributes();
            var firstName = JSON.stringify(attributeMap["firstName"]["value"]);
            var lastName = JSON.stringify(attributeMap["lastName"]["value"]);
            var serviceLevel = JSON.stringify(attributeMap["serviceLevel"]["value"]);
            var account = JSON.stringify(attributeMap["acctNumber"]["value"]);
            console.log(acctNumber);
            console.log(serviceLevel);
            console.log(firstName);
            console.log(lastName);
            window.alert("Customer's name: " + firstName + " " + lastName+ "\nAccount #: " + acctNumber
+ "\nCustomer's Level #: " + serviceLevel);

        });
});
</script>
</body>
</html>
```

This integrates with Connect by loading the pre-built CCP located at ccpUrl into an iFrame and placing it into the containerDiv provided. To provide enhanced functionality, API requests are funneled through this CCP and Agent, and contact updates are published through it and made available to your JS client code.

- **ccpUrl**: The URL of the CCP. This is the page you would normally navigate to in order to use the CCP in a standalone page; it is different for each instance.
- **region**: Amazon connect instance region. ex: us-west-2. This is only required for Chat Channel.

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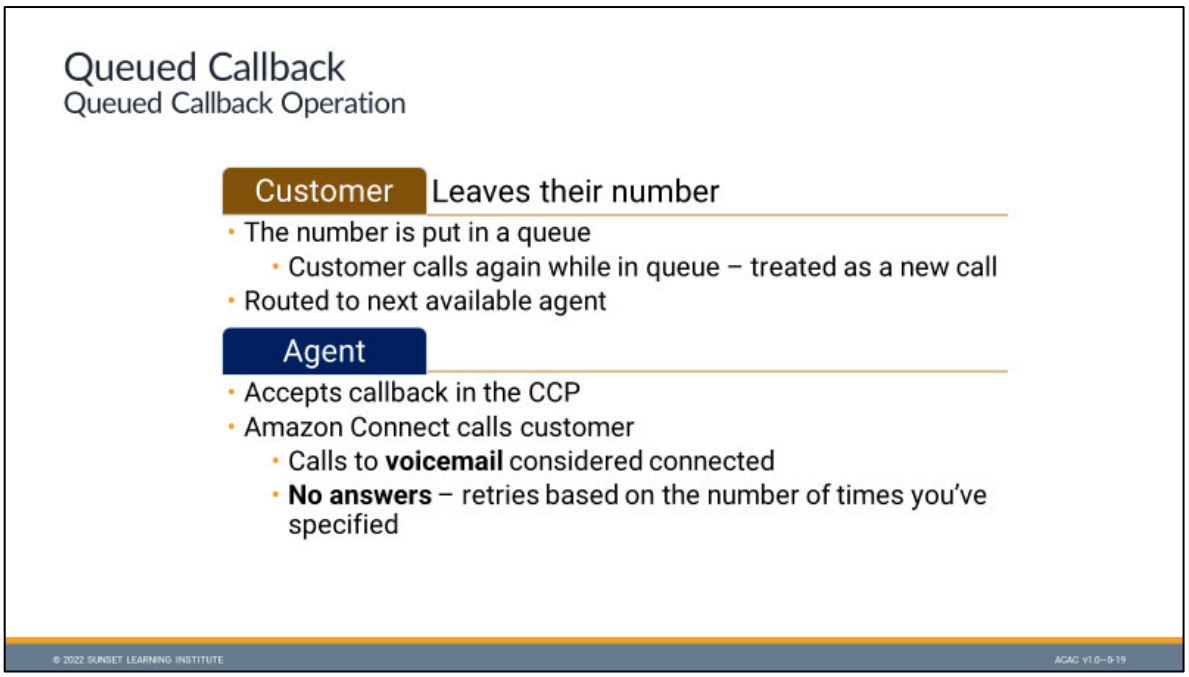
- **loginPopup**: Optional, defaults to true. Set to false to disable the login popup which is shown when the user's authentication expires.
- **loginOptions**: Optional, only valid when loginPopup is set to true. Provide an object with the following properties to open loginpopup in a new window instead of a new tab:
  - **autoClose**: Optional, defaults to false. Set to true to automatically close the login popup after the user logs in.
  - **height**: This allows you to define the height of the login pop-up window.
  - **width**: This allows you to define the width of the login pop-up window.
  - **top**: This allows you to define the top of the login pop-up window.
  - **left**: This allows you to define the left of the login pop-up window.
- **loginPopupAutoClose**: Optional, defaults to false. Set to true in conjunction with the loginPopup parameter to automatically close the login Popup window once the authentication step has completed. If the login page opened in a new tab, this parameter will also auto-close that tab. This can also be set in loginOptions if those options are used.
- **loginUrl**: Optional. Allows custom URL to be used to initiate the CCP, as in the case of SAML authentication.
- **softphone**: This object is optional and allows you to specify some settings surrounding the softphone feature of Connect. These default softphone features normally work well, but they can be overridden.
  - **allowFramedSoftphone**: Normally, the softphone microphone and speaker components are not allowed to be hosted in an iFrame. This is because the softphone must be hosted in a single window or tab. The window hosting the softphone session must not be closed during the course of a softphone call, or the call will be disconnected. If allowFramedSoftphone is true, the softphone components will be allowed to be hosted in this window or tab.
  - **disableRingtone**: This option allows you to completely disable the built-in ringtone audio that is played for an incoming call.
  - **ringtoneUrl**: If the ringtone is not disabled, this allows for overriding the ringtone with any browser-supported audio file accessible by the user.
- **pageOptions**: This object is optional and allows you to configure which configuration sections are displayed in the settings tab. Providing Agents with the ability to display and change Audio Device and Phone Type should be a consideration during the Agent Capabilities design phase.
  - **enableAudioDeviceSettings**: If true, the settings tab will display a section for configuring audio input and output devices for the Agent's local machine. If false, or if pageOptions is not provided, the Agent will not be able to change audio device settings from the settings tab, and it will not be displayed.
  - **enablePhoneTypeSettings**: If true, or if pageOptions is not provided, the settings tab will display a section for configuring the Agent's phone type and desk phone number. If false, the Agent will not be able to change the phone type or desk phone number from the settings tab.
- **ccpAckTimeout**: A timeout in milliseconds that indicates how long streams will wait for the iFramed CCP to respond to its SYNCHRONIZE event emissions. These happen continuously from the first time initCCP is called. They should only appear when there is a problem that requires a refresh or a re-login.
- **ccpSynTimeout**: A timeout in milliseconds that indicates how long streams will wait to send a new SYNCHRONIZE event to the iFramed CCP. These happens continuously from the first time initCCP is called.
- **ccpLoadTimeou**t: A timeout in milliseconds that indicates how long streams will wait for the initial ACKNOWLEDGE event from the shared worker while the CCP is still standing itself up.



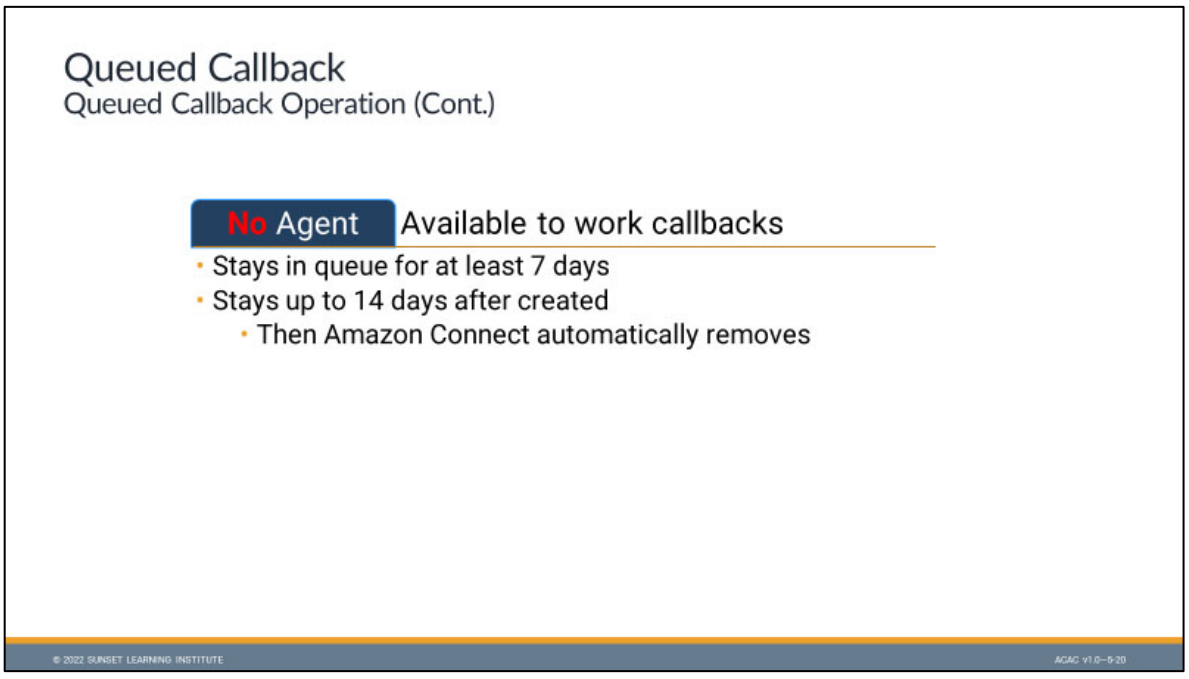
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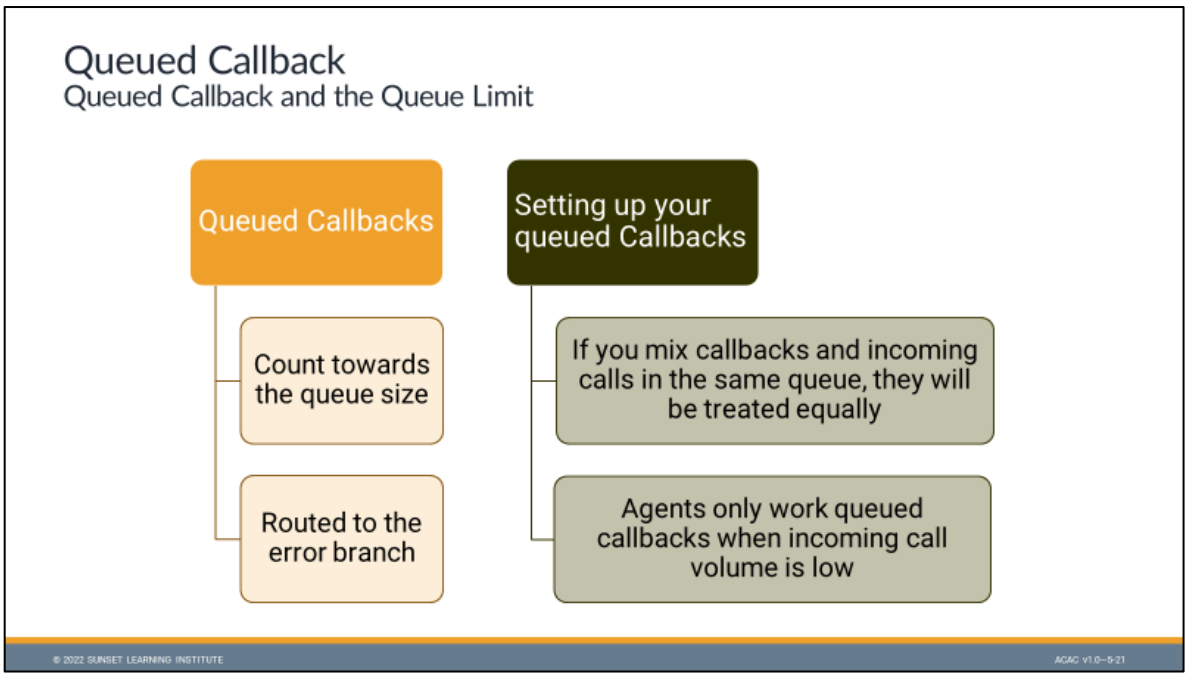
Queued Callback

What is a queued callback? A queued callback is a contact flow that allows customers to leave their phone number and get a callback from the Agent. This section describes the steps to set up a queued callback.



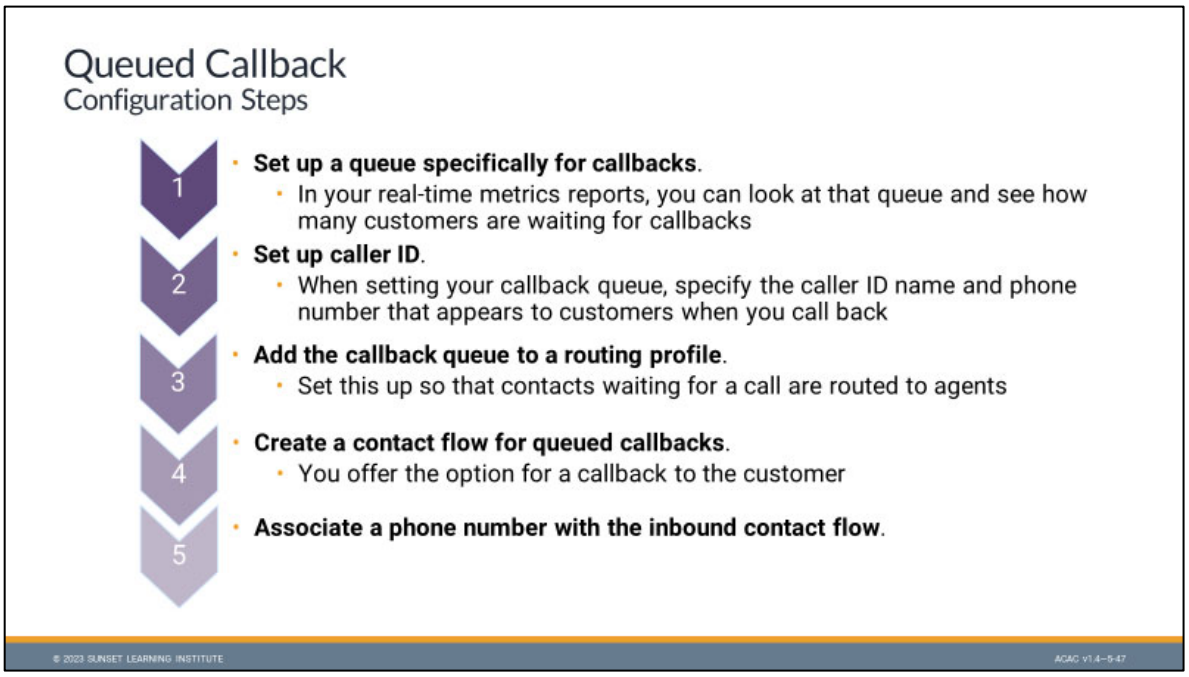
You can create contact flows that provide the ability for customers to leave their phone number and get a callback from an Agent.





If you have a queue that handles callbacks and incoming calls, and that queue reaches the size limit:

- The next callback is routed to the error branch.
- The next incoming call gets a reorder tone (also known as a fast busy tone), which indicates no transmission path to the called number is available.



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Queued Callback
Configuration Steps (Cont.)

Optional

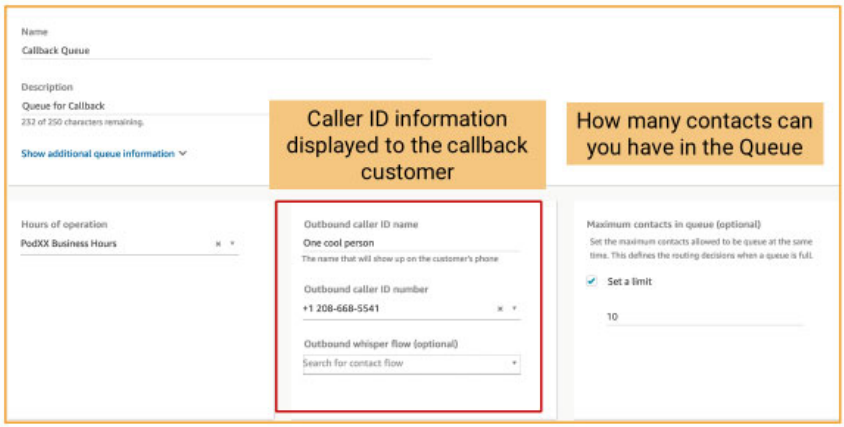
- **Create an outbound whisper flow.**
  - When a queued call is placed, the customer hears this message after pick up and before connected to the agent
  - For example, "Hello, this is your callback..."

Optional

- **Create an agent whisper flow.**
  - What the agent hears right after accepting the contact, before being joined to the customer
  - For example, "You're about to be connected to Customer John, who requested a refund for...

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Queued Callback Configuration
Create a Callback Queue



- Having a dedicated queue for callback allows you to report on how many calls are in queue without seeing other calls intermingled

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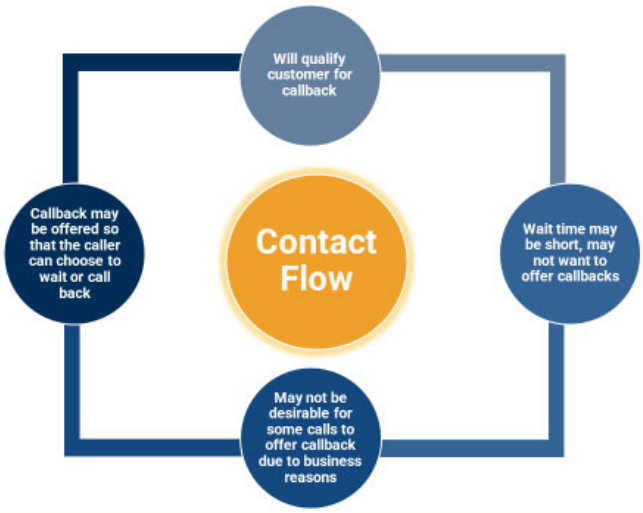
Queued Callback Configuration
Add the Callback Queue to the Routing Profile

Name	Channels	Priority	Delay (in seconds)
PodXX Swimsuit Queue	Voice Chat Task	1	0
PodXX Shark Repellant Queue	Voice Chat Task	1	0
PodXX Surfboard Queue	Voice Chat Task	1	0
Agent Quick Connect	Voice Chat Task	1	0
Callback Queue	Voice Chat Task	2	0

- Add **Callback queue** to the Routing profile
- Optionally may set the priority to **2** to give priority to inbound calls over callbacks

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Queued Callback Considerations
Prior to Creating the Contact Flow



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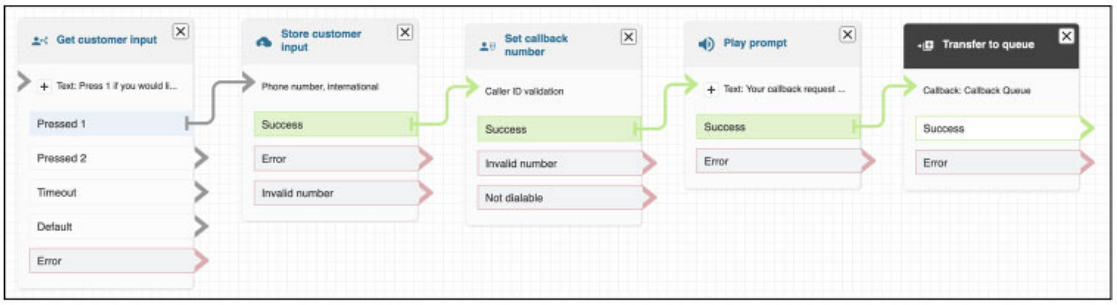
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Queued Callback Considerations
Prior to Creating the Contact Flow (Cont.)

- Request the callback number from the caller as the customer may wish to be called back at a different number than originally calling from
- Store the number as an attribute
- Reference the attribute in a **Set callback number** block in the contact flow
- Transfer to the callback queue and then disconnect the caller

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Queued Callback Example
Basic Contact Flow for Callback



Press 1

12065551234 > System Stored customer input

Choose Callback or wait

Collect the Callback Number and store in a System attribute

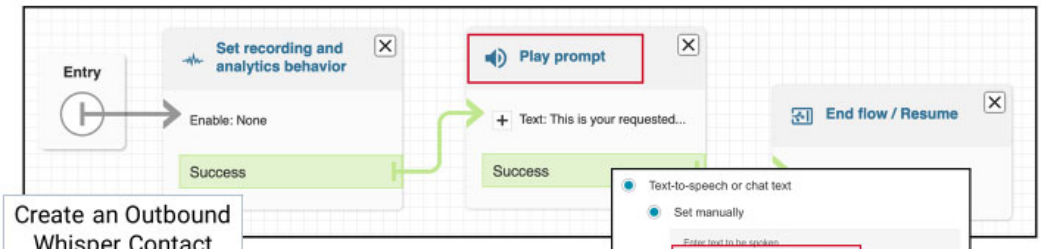
Set the Callback number from the **System.Stored customer input** attribute

Tell the caller that a callback is set and that the call will be disconnected

Queue the Callback and, upon success, disconnect

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Queued Callback Example
Create an Outbound Whisper *(Optional)*



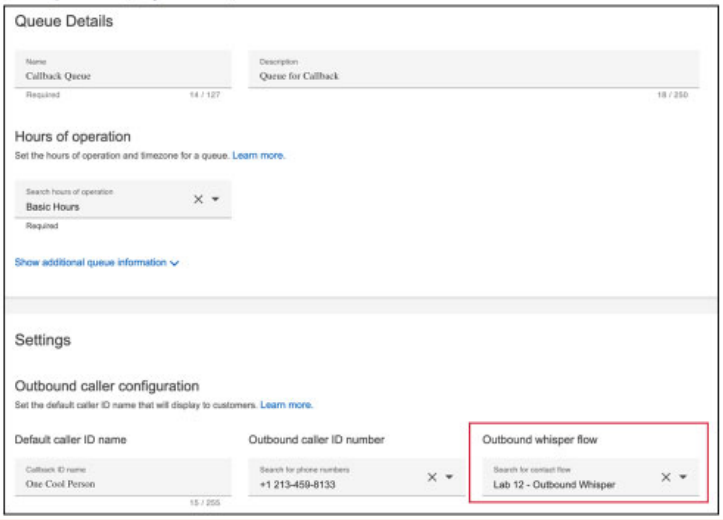
Create an Outbound Whisper Contact Flow

Enter the audio that you want played to the customer when called back, but before the agent is connected to the customer

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Queued Callback Example
Add Outbound Whisper to the Queue *(Optional)*

- Assign the Outbound Whisper Contact Flow to the Callback Queue



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Lab 12: Implementing Queued Callback

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Module 6

Advanced Agent Configurations

Overview

In this module, you will learn about Quick Connect, a method that allows you to connect calls to another agent, a queue, or a third party. Managers and agents-in-training can monitor live conversations between agents and customers.

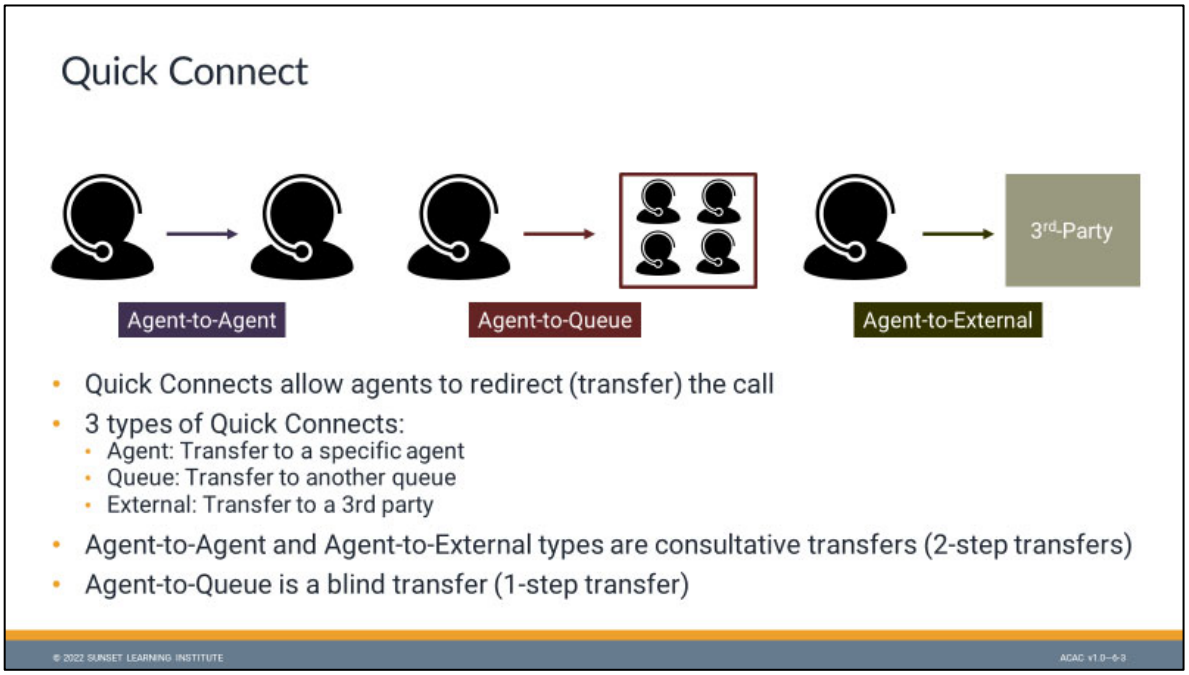
Objectives

Upon completing this module, you will be able to use Agents' settings to enhance Amazon Connect. This ability includes being able to meet these objectives:

- Configure Quick Connect
- Configure basic Agent settings for call monitoring, call recording, and listen to live calls and recordings, and view the chat transcripts

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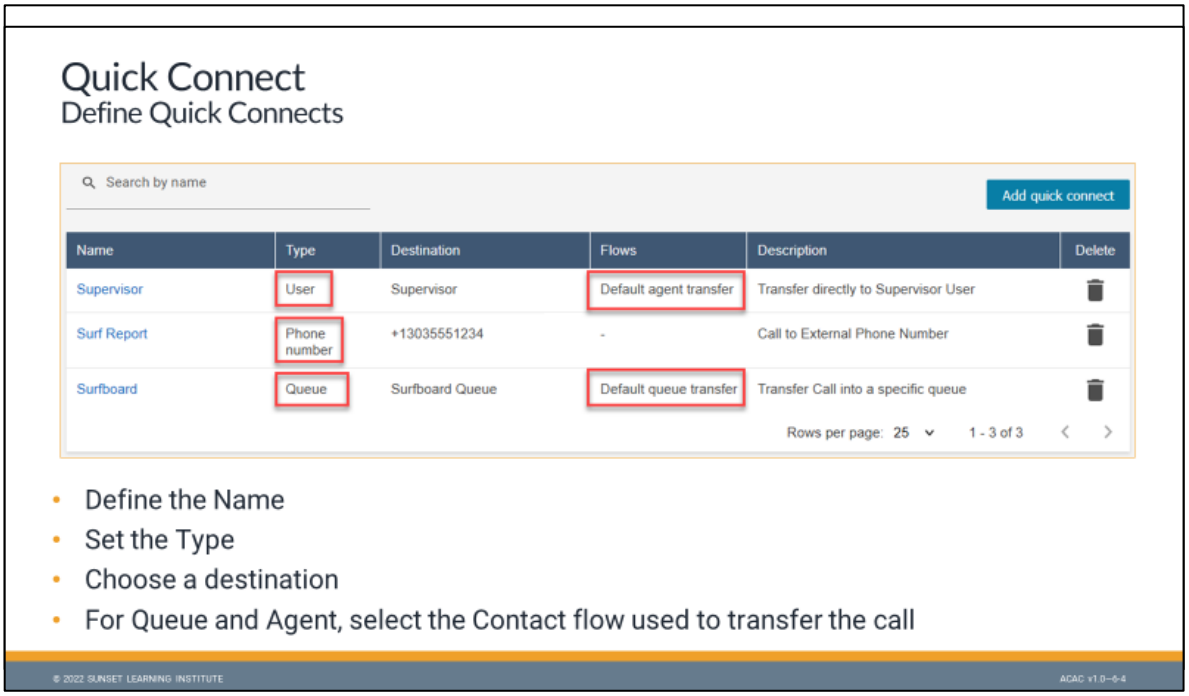
Quick Connect



Agents are responsible for setting their status in the Contact Control Panel (CCP). In fact, the only time an agent's status changes is when they manually change it in the CCP, or when their supervisor changes it in a real-time metrics report.

Amazon Connect provides two default status values:

- Available
- Offline



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Quick Connect

Agent-to-Agent Quick Connect (Cont.)

- Point to the agent in the Quick Connect
- Create a Queue for the Agent
- Include the Queue for the Routing profile and enable Task
- Only enabled when on a call
- Transfers to type Agent are successful only if the End Agent status is Available; otherwise, the transfer will fail
- The transferring agent must press the **Join** button when the target agent is on the line; then the transferring agent may choose to stay on the line (3-party conference) or leave the call

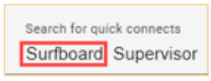
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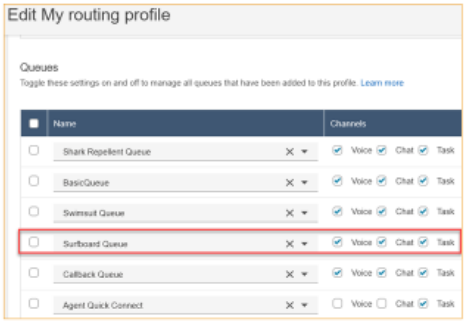
Quick Connect
Agent-to-Queue Quick Connects



Agent-to-Queue



Queue

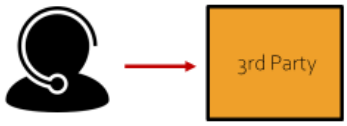


Routing Profile

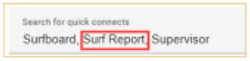
- Add the Quick Connect to one or more existing Target Queues
- Ensure the Routing Profile has the appropriate Queues
- The Quick Connects that show are determined by the Queue of the call at the agent
- Only visible when on an active call

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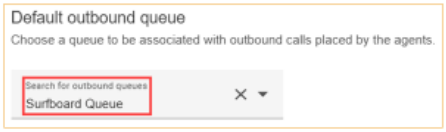
Quick Connect
Agent-to-3rd Party Quick Connects



Agent-to-3rd Party



Queue



Routing Profile

- Must be in E.164 format (Ex. +12066837777)
- Point to an External number
- Add the Quick Connect to the Default outbound queue
- Task for that Queue must be enabled
- Agent-to-3rd Party Quick Connects are visible based on the Default outbound queue settings

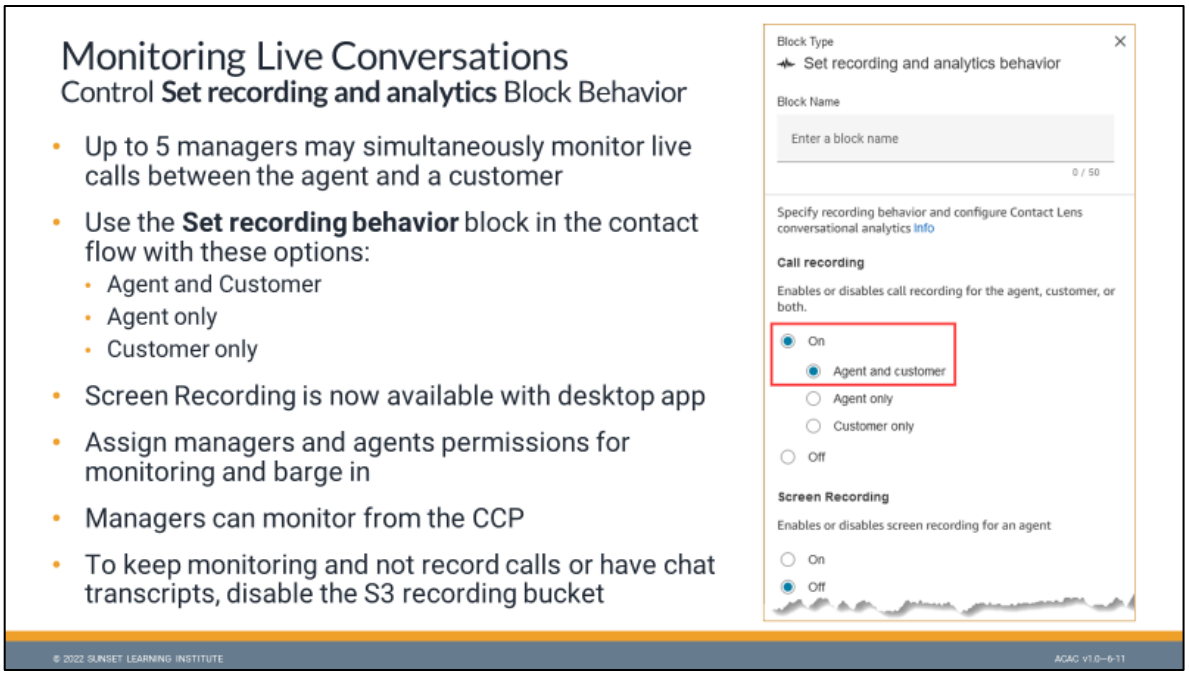
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Lab 13: Configuring Quick Connects

Monitoring and Recording Calls and Chats

Managers and agents-in-training can monitor live conversations between agents and customers.



Monitor live conversations

Managers and agents-in-training can monitor live conversations between agents and customers. To set this up, you need to add the **Set recording behavior** block to your voice/chat contact flow, assign managers and trainees the appropriate permissions, and then show them how to monitor the conversations.

Set up live monitoring for voice and/or chat

Add the **Set recording and analytics behavior** block to your contact flow. Do this to monitor calls, chats, or both.

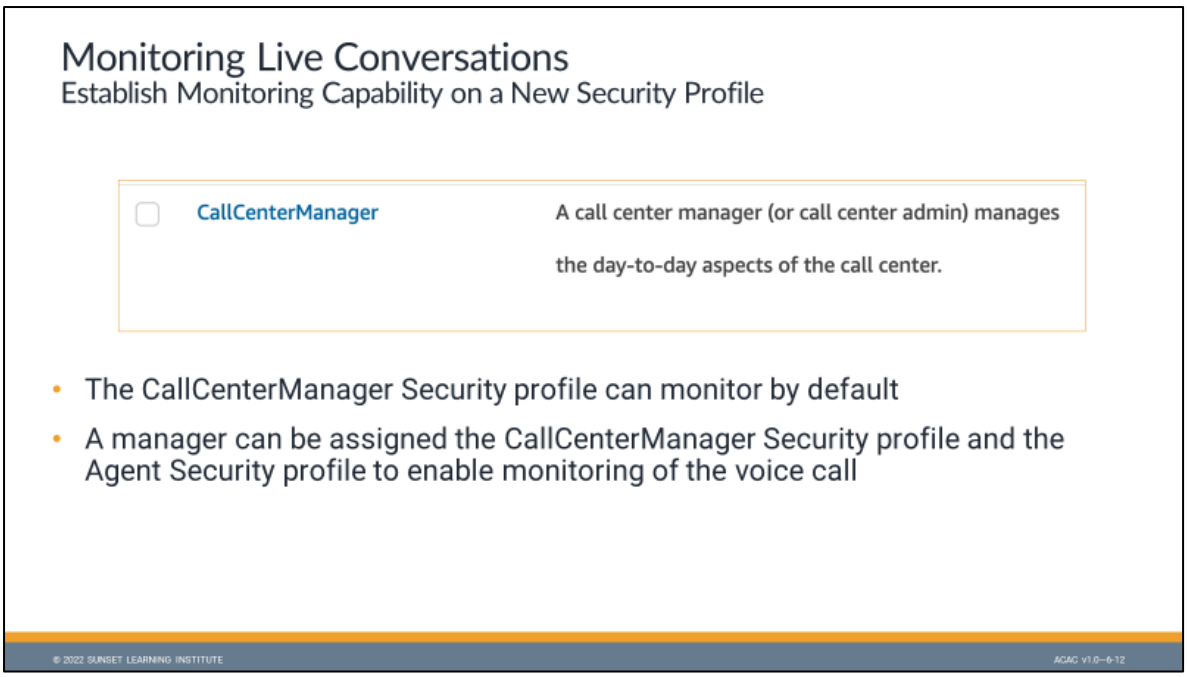
To enable monitoring of voice and/or chat conversations, in the block's properties choose **Agent and Customer**.

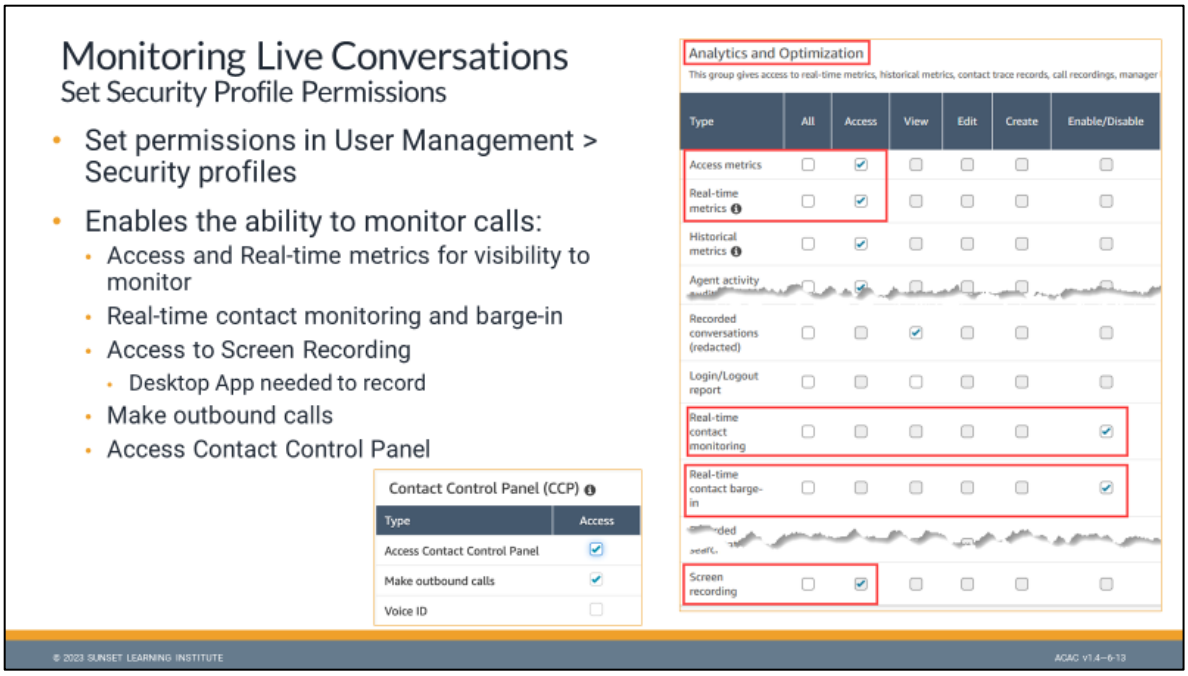
Choose whether to record the conversations you monitor.

Although you need to add the **Set recording behavior** block to your contact flow, you don't need to record voice and/or chat conversations for monitoring to work. By default, when you set up your instance, Amazon S3 buckets are created to store call recordings and chat transcripts. The existence of these buckets enables call recording and chat transcripts at the instance level.

To not record the calls or chats you're monitoring, disable the Amazon S3 buckets.

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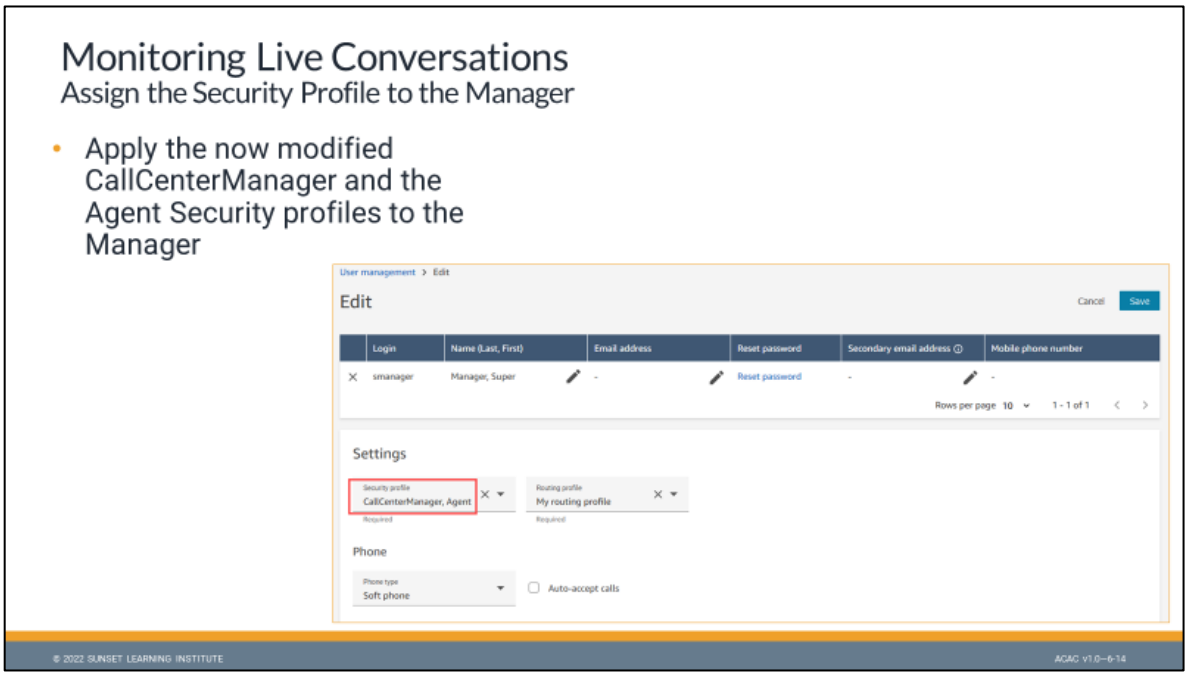
To access monitoring, Users require permissions for the monitoring applications. Navigate to User Management > Security profiles to Set permissions then enable the following permissions to give Users the ability to monitor calls:

- Access and Real-time metrics for visibility to monitor
- Real-time contact monitoring and barge-in
- Access to Screen Recording

- Desktop App needed for screen record
- Make outbound calls
- Access Contact Control Panel

Screen recording is a feature that was recently added. For more information: https://docs.aws.amazon.com/connect/latest/adminguide/enable-sr.html

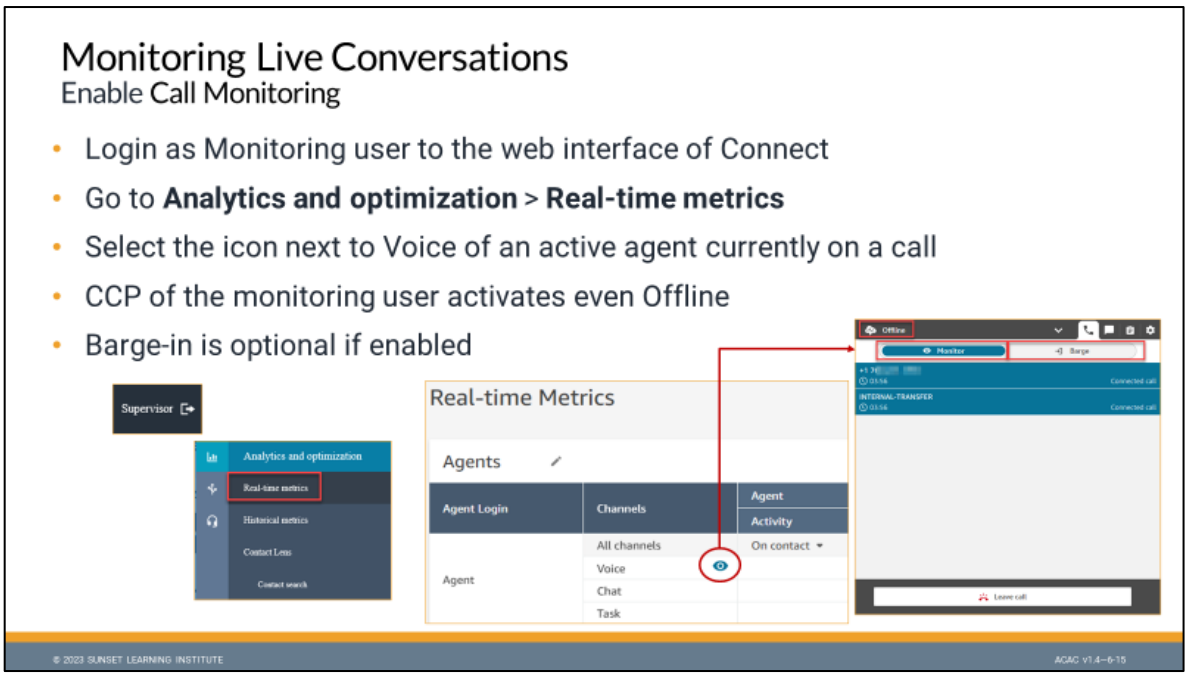
To download the recording application: https://docs.aws.amazon.com/connect/latest/adminguide/amazon-connect-client-app.html#client-app-download



Assign permissions to monitor live conversations

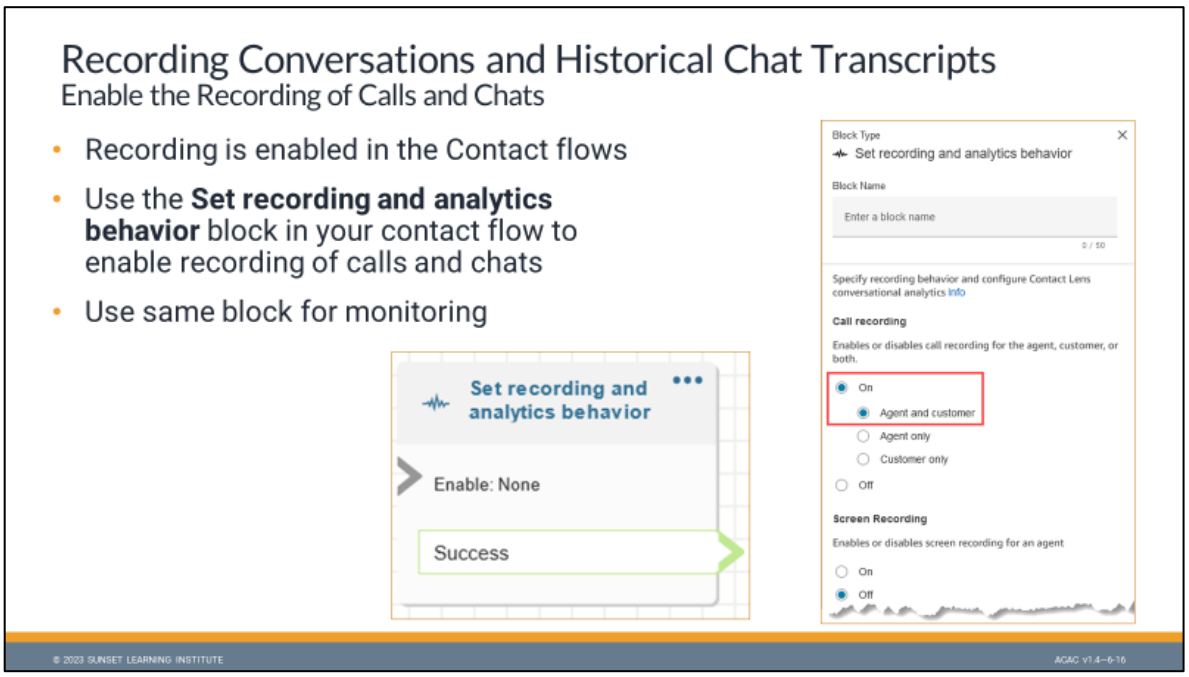
For managers to monitor live conversations, you assign them the **CallCenterManager** and **Agent** security profiles.

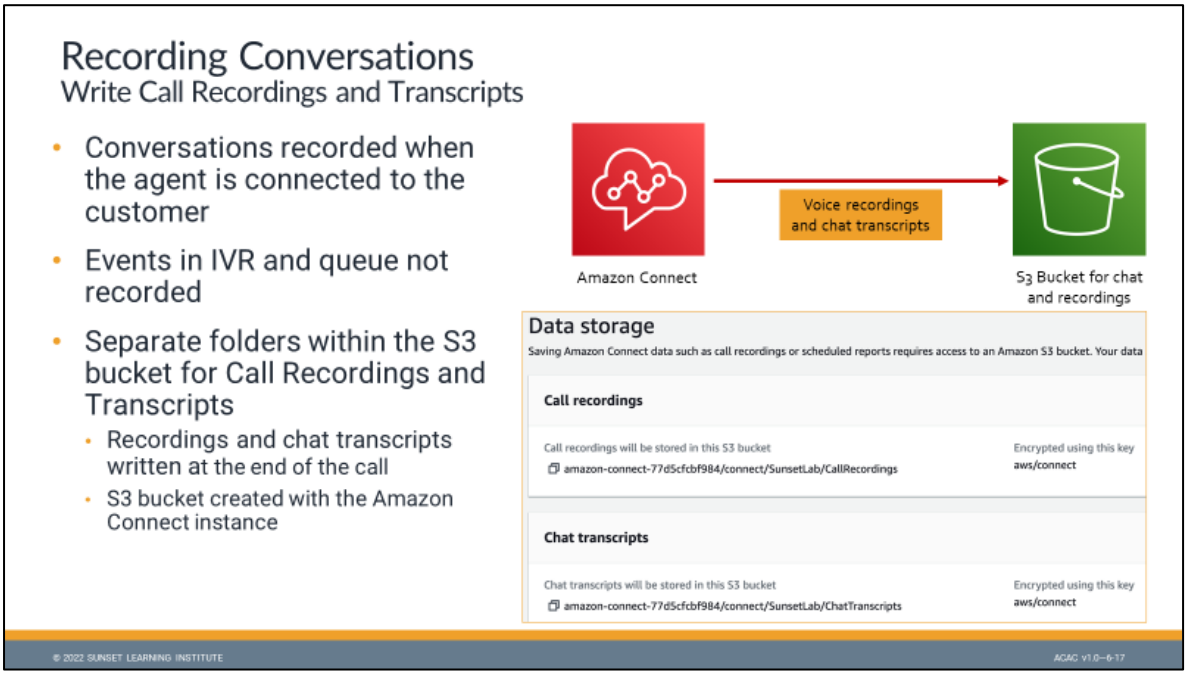
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To monitor a call, from the Amazon Connect Dashboard menu, navigate to **Analytics and optimization** -> **Real-time metrics**. Select **Agents Real-time Metrics**. Active contacts will display as the eye icon. Make sure that CCP is open and click on the eye icon. The monitoring user will need to **Accept Call** but does not require to be **Available**.

By default, the monitor will open as a listen only, but if the User has permissions, the **Barge** option will be available.



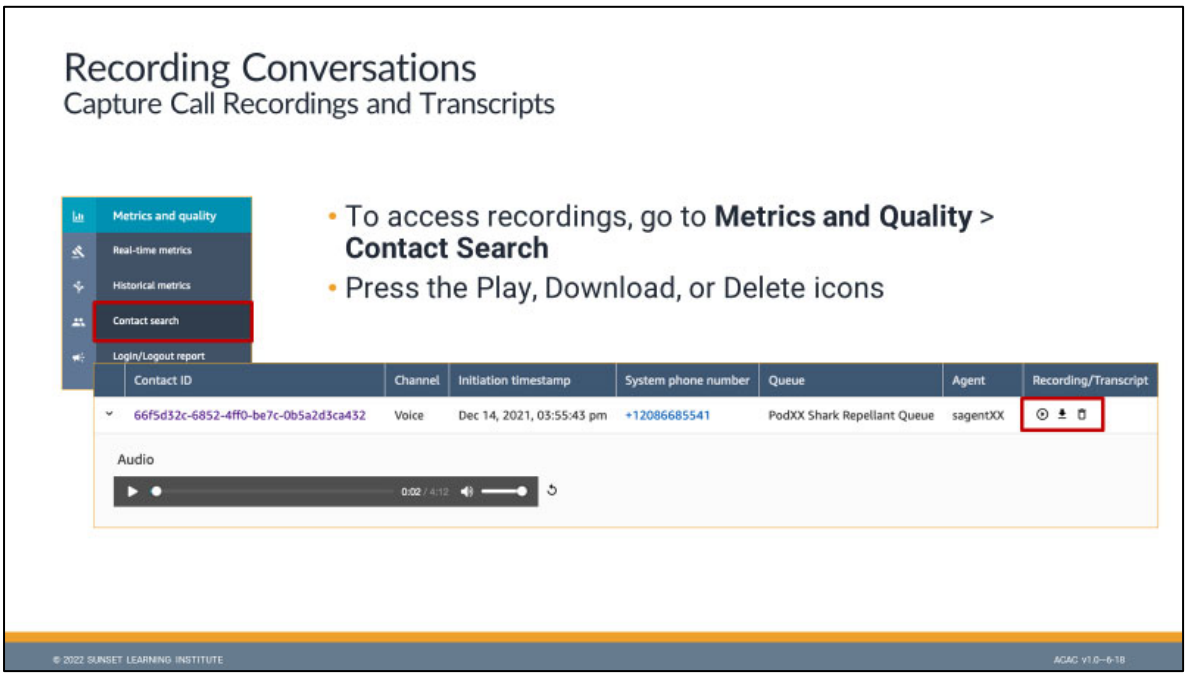


When is a conversation recorded? A conversation is recorded only when the contact is connected to an agent. The contact is not recorded before then when they are connected to the IVR. If the call is transferred externally, the call recording stops when the agent drops from the call.

When call recording is enabled, the recording is placed in your S3 bucket shortly after the contact is disconnected. Then the recording is available for you to review it using the steps in this article.

You can also access the recording from the customer's contact record. The recording is available in the contact record, however, only after the contact has left the After Contact Work (ACW) state.

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These are the steps that a manager does to review past recordings/transcripts of conversations.

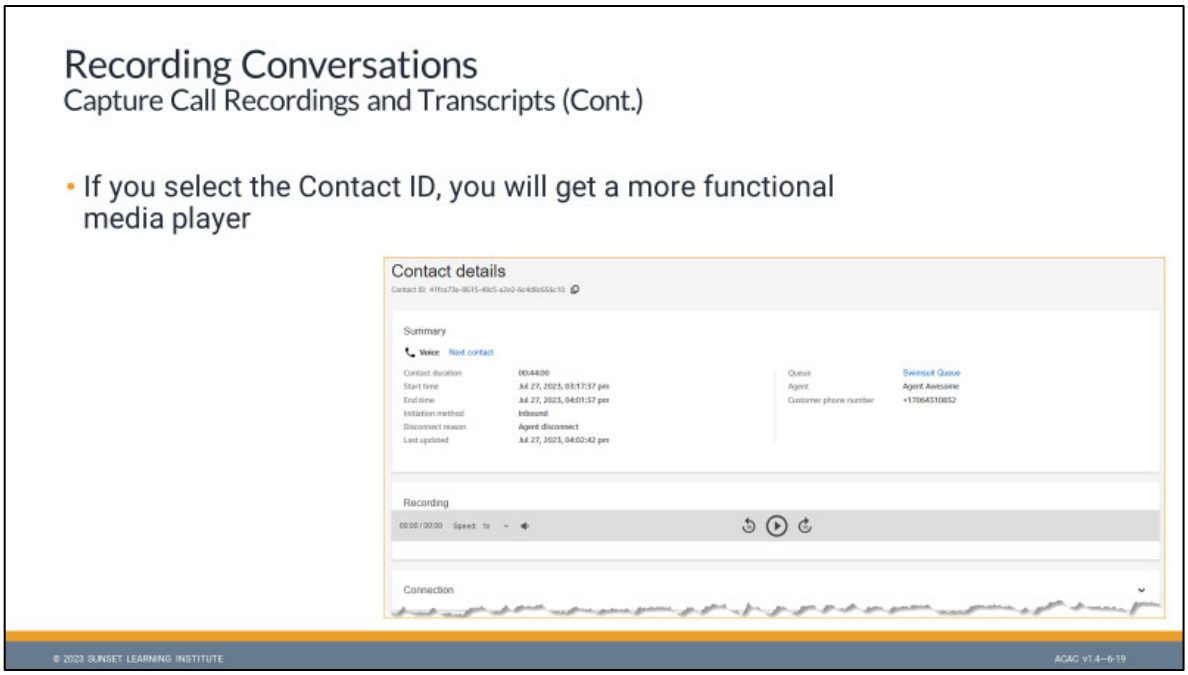
Step 5 Log in to Amazon Connect with a user account that has permissions to access recordings.

Step 6 In Amazon Connect, choose **Analytics and optimization > Contact Lens > Contact Search**.

Step 7 Filter the list of contacts by date, agent login, phone number, or other criteria. Choose **Search**.

Step 8 Conversations that were recorded have icons in the Recording/Transcript column. If you don't have the appropriate permissions, you won't see these icons.

Step 9 To listen to a recording of a voice conversation or read the transcript of a chat, choose the **Play** icon.



To get a media player with pause, rewind, and fast forward, do the following:

- Instead of choosing the **Play** icon, choose the contact ID to open the contact record.
- On the **Contact record** page, there are more controls to navigate the recording.
  - Click or tap to the time you want to investigate.
  - Adjust the playing speed.
  - Play, pause, or skip backwards or forwards in 10 second increments.

Note If you are unable to pause, rewind, or fast-forward recordings on the **Contact search** page, one possible reason could be that your network is blocking HTTP range requests.



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Module 7

Contact Lens

Overview

This module provides an overview of the process and key tasks for successfully creating, configuring, and enabling Contact Lens within a contact flow.

Objectives

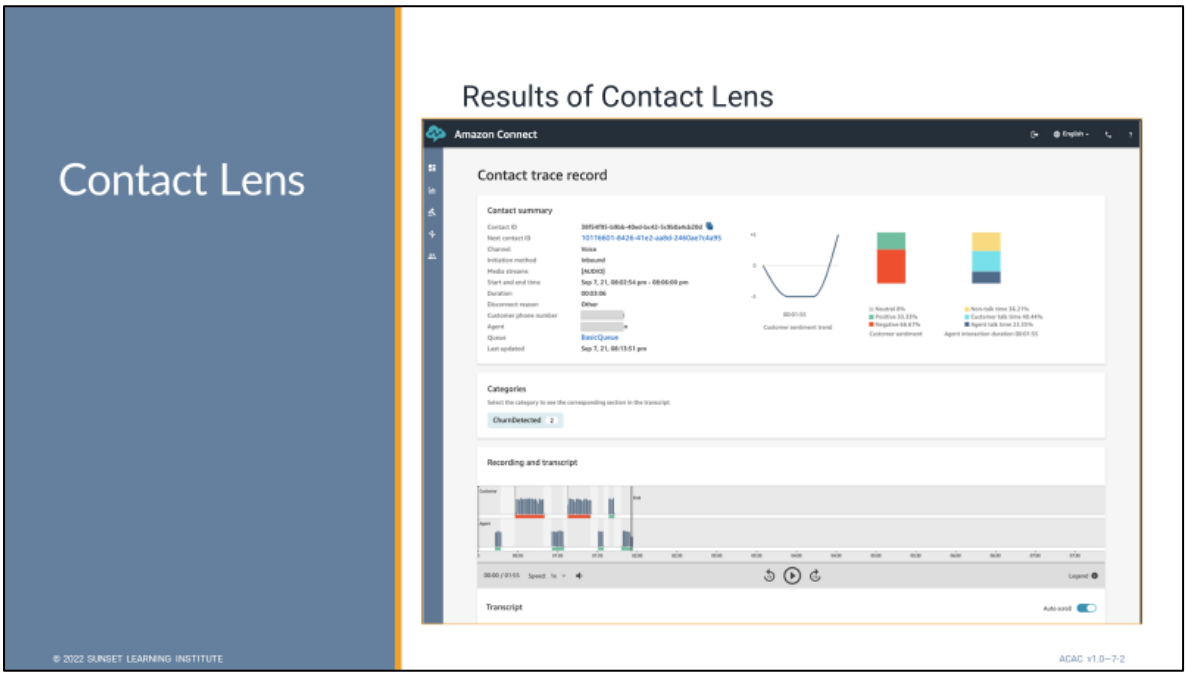
Upon completing this module, you will show an understanding of the basic premise of Contact Lens. This ability includes being able to meet these objectives:

- Configure Amazon Connect to support Contact Lens
- Enable Contacts Lens
- Configure a contact flow to enable to Contact Lens
- Configure a Contact Lens Rule to send an email to the Supervisor's email
- Configure a Contact Lens Rule to create task requiring follow up

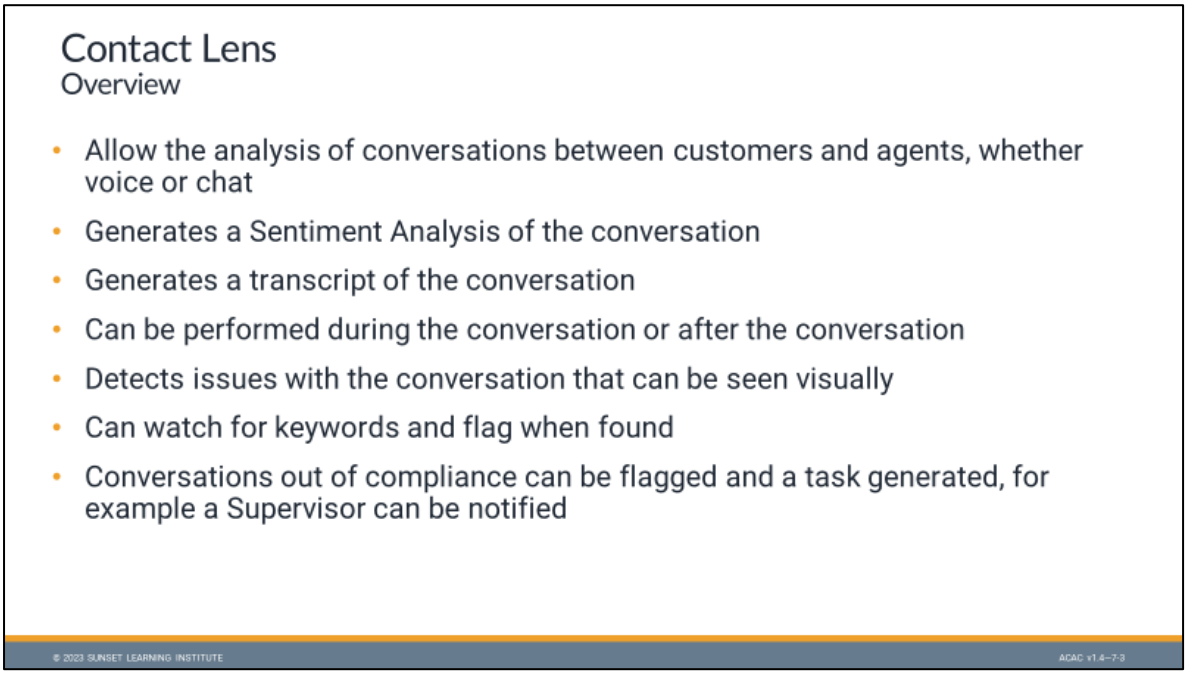
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Contact Lens Overview

This module provides an overview of the Contact Lens feature that enables you to better understand the sentiment and trends of customer conversations to identify crucial company and product feedback. You can also track the compliance of customer conversations in your Contact Center to ensure standard greetings and signoffs are used, help train agents, and replicate successful interactions. Supervisors can conduct fast full-text search on all transcripts to quickly troubleshoot customer issues. In addition, with real-time capabilities, you can get alerted to issues during live customer calls and can deliver proactive assistance to agents while calls are in progress, improving customer satisfaction.



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Contact Lens for Amazon Connect, a feature of Amazon Connect, helps you follow the sentiment and trends of customer conversations in real time to identify crucial company and product feedback. You can also track the agent compliance of customer conversations in your Contact Center to ensure standard greetings and sign-offs are used, help train agents, and replicate successful interactions. Supervisors can conduct fast full-text search on all transcripts to quickly troubleshoot customer issues. Using real-time analytics powered by machine learning (ML), you can also get alerted to issues during live customer calls and deliver coaching to agents while calls are in progress, improving customer satisfaction.

Contact Lens is supported by both voice and chat channels.

Capabilities

- **Automated call categorization:** Track all customer conversations for compliance with company policies or regulatory requirements. Define and manage categories based on your specific criteria (for example, words, phrases, conversation characteristics like sentiment, interruptions, and non-talk time) directly within Amazon Connect—using an ML-powered categorization engine that is trained to understand spoken phrases, their intent, and context.
- **Advanced conversational search:** Conduct full-text search while on calls. Search by keywords, sentiment score, categories, “non-talk” time, and specific customers and agents to identify relevant calls. This helps you understand customer trends and insights and how to improve customer satisfaction.
- **Call summarization:** Using ML, call summarization automatically identifies key parts of the customer conversation, assigns tags (for example, issue, outcome, or action item), and displays a summary that can be expanded to view the full transcript of the call.
- **Analytics dashboard:** View all call analytics in the Contact Detail page, including call transcripts, sentiment, and conversation characteristics to detect issues and customer trends.
- **Real-time alerts:** Easily create rules to flag any customer experience issue in real time, with categories based on keywords, sentiment, and phrase matching. This automatically alerts your supervisors in real time when they need to assist an agent on live calls, so they can provide guidance via chat or have the agent transfer the call.

- **Sentiment analysis:** ML-powered natural language processing (NLP) captures and analyzes the sentiment of words being spoken by the customer, generating a score between -5 (most negative) to +5 (most positive). You can then use the Contact Search page to search calls based on these scores to identify customer experience issues.
- **Sensitive data redaction:** Automatically redact sensitive data, such as names, addresses, credit card details, and social security numbers. Data redaction removes sensitive information from both the call transcript and audio recording.

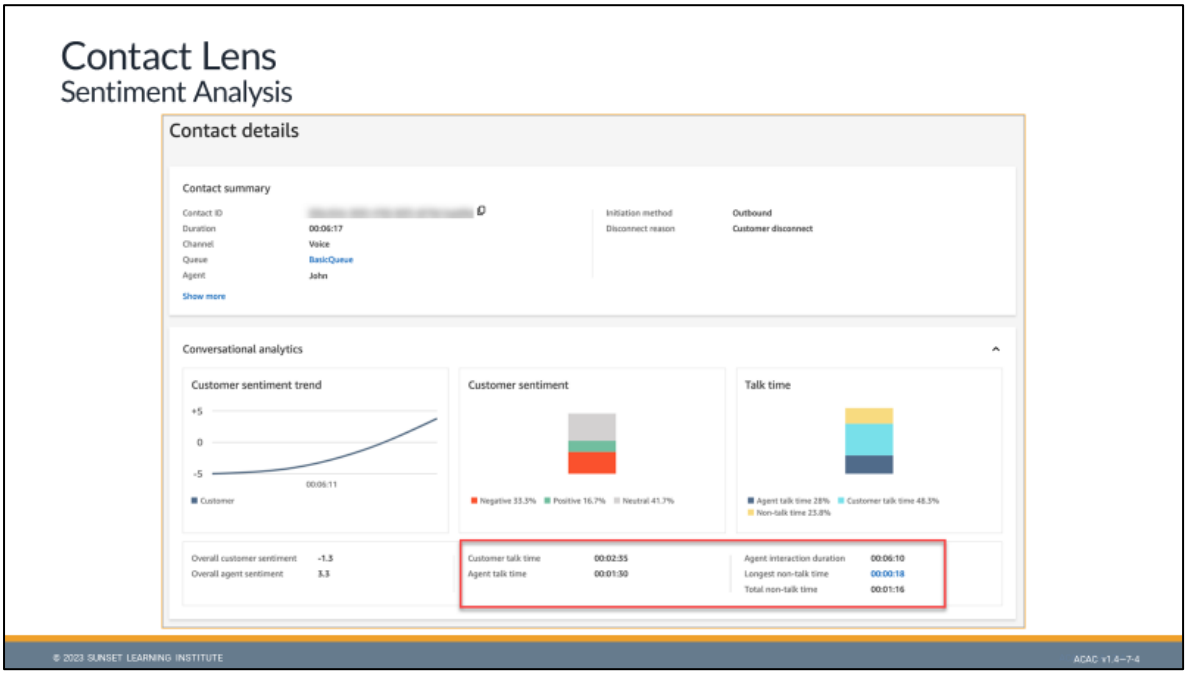
Speech analytics support

- **Real-time call analytics:** Use to detect and resolve customer issues more proactively while the call is in progress. For example, it can analyze and alert you when a customer is getting frustrated because the agent is unable to resolve a complicated problem. This allows you to provide more immediate assistance.
- **Post-call analytics:** Use to understand trends of customer conversations and agent compliance. This helps you identify opportunities to coach an agent after the call.

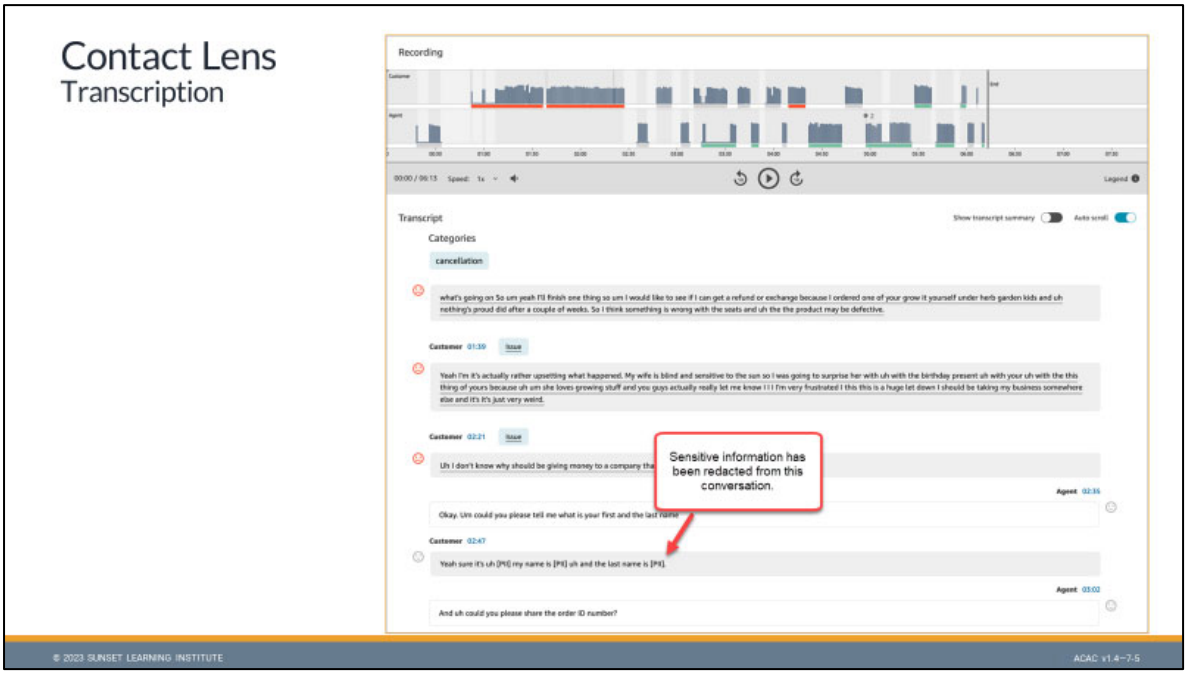
Chat analytics support

- **Post-chat analytics:** Use to understand trends of customer conversations with both bots and agents. It provides information specific to a chat interaction, such as the agent greeting time, and agent and customer response times. The response times and sentiments help you investigate the customer's experience with the bot versus the agent and identify areas for improvement.

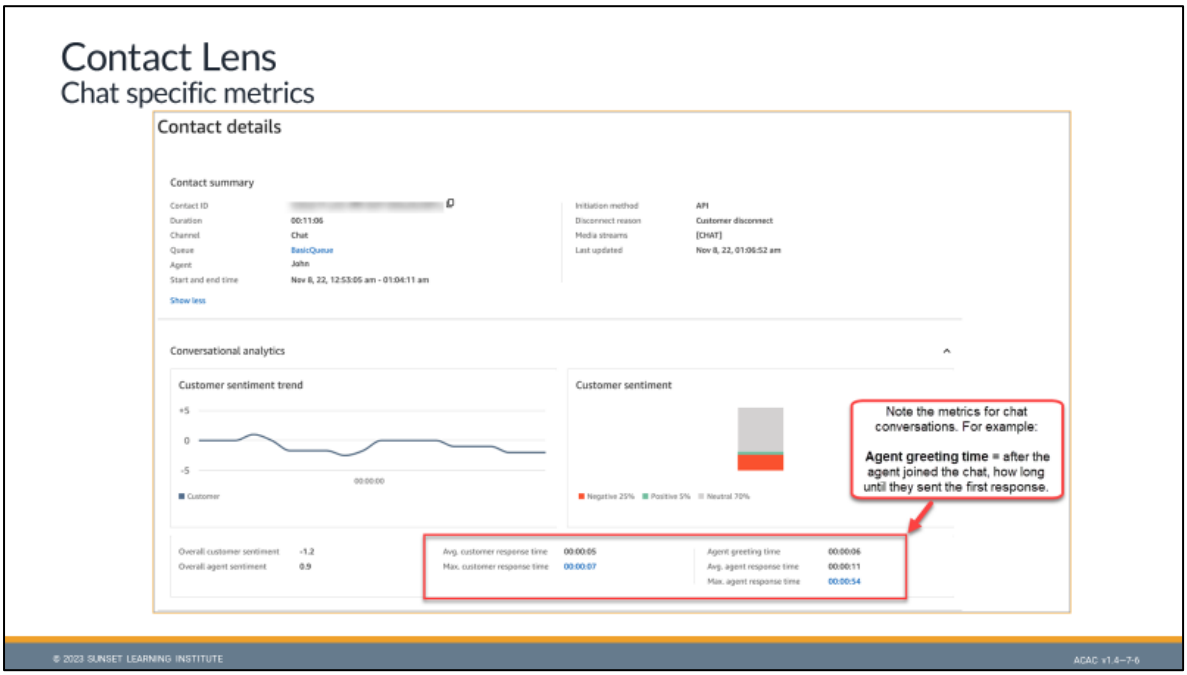
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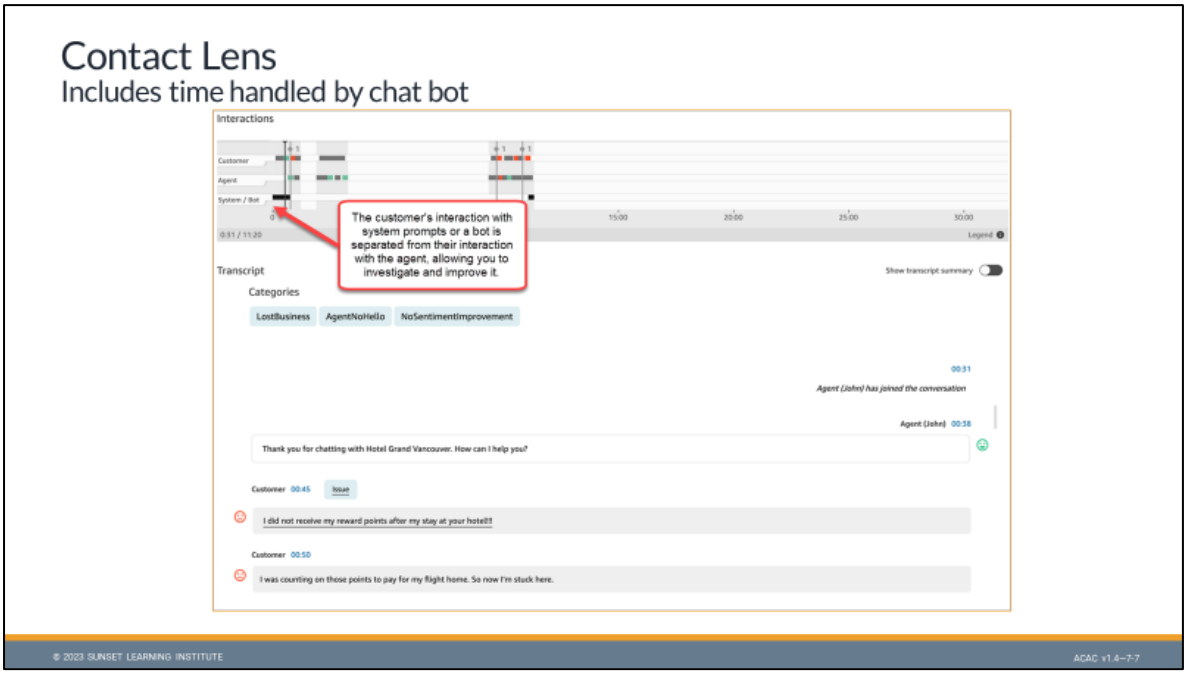
Easily understand sentiment, conversation characteristics, and agent compliance risks in call transcripts using natural language processing (NLP) and ML-powered speech-to-text-analytics to discover customer insights and identify agent coaching opportunities. Set real-time alerts to flag customer issues for detailed analysis in the analytics dashboard.



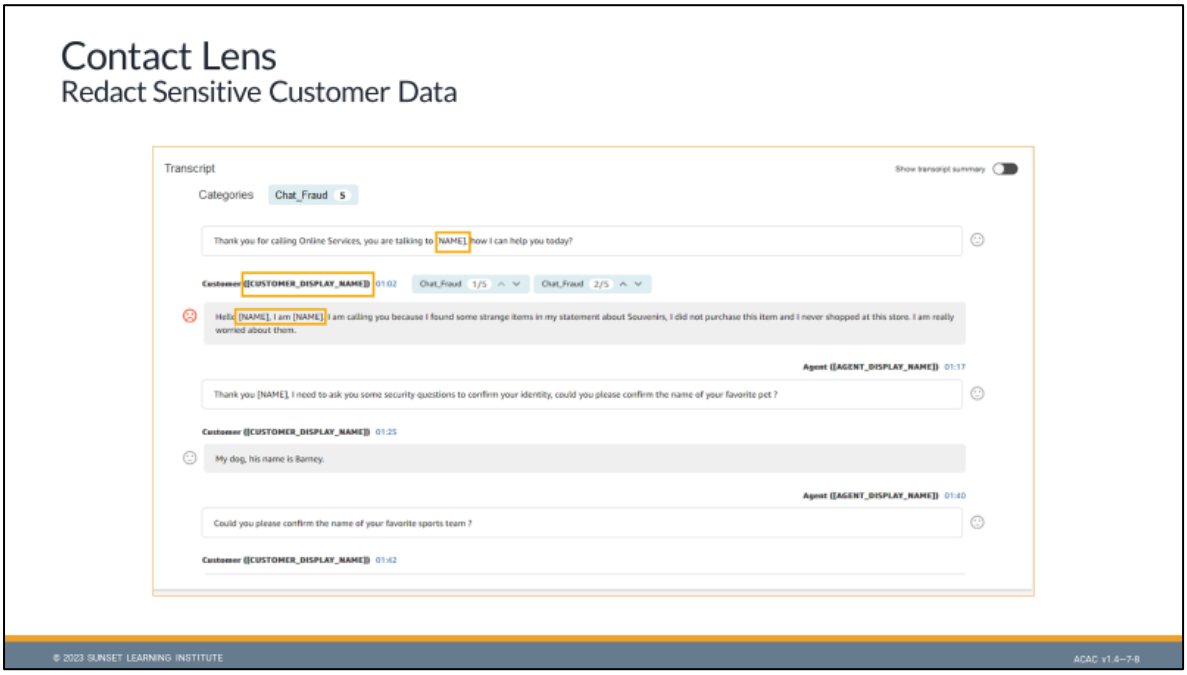
Save your agents' valuable time on every call by automatically recording and transcribing key parts of customer interactions without the need to take detailed notes. Managers can view call summaries alongside the call recording and contact details in Amazon Connect to quickly understand the context of an interaction, help with follow-up items such as issuing refunds, and provide precise feedback to agents.



Recognizing that chat may require metrics that differ from voice, Contact Lens is able to reflect metrics such as the time it takes for the agent to send the first message or time spent interacting with bots before the chat is assigned to the agent.



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Automatically detect and redact sensitive customer data such as name, address, and social security number from call recordings and transcripts. Improve agent compliance with company policies or regulatory requirements by tracking all customer conversations with categories that are understood by an ML-powered categorization engine and based on criteria you determine—for example, words or phrases used in standard greetings and sign-offs.

Configuring Contact Lens

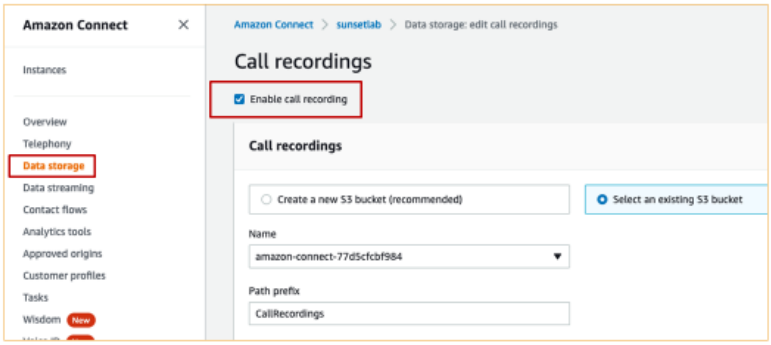
Contact Lens
Configuration Tasks Overview

1. Configure Amazon Connect for Recording of Calls and Chat Transcripts (enabled by default)
2. Setting permissions in a Security Profile for the Supervisor to search for and access the Contact Lens results
3. Enable Contact Lens on your Amazon Connect instance
4. Configure Contact Lens settings in your Inbound Contact Flow
   - Enable Contact Lens (Analytics)
     - Post-call analytics
     - Real-time and Post-call analytics
     - Chat analytics
   - Language
   - Redact sensitive data
5. Configure Contact Lens rules
   - Contact Lens post-call
   - Contact Lens post-chat
   - Contact Lens real-time

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Contact Lens – Configuration Steps
Task 1 – Configure Data Storage on the Amazon Connect Instance

- The ability to record conversations is enabled on the Amazon Connect instance with default settings
- If disabled enable on the the instance



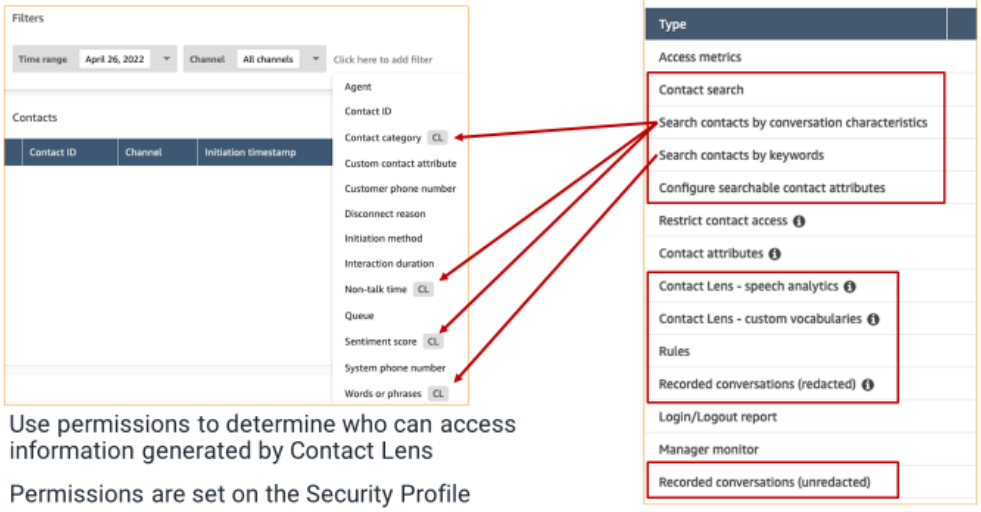
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In order to provide analytics including call recordings and transcripts, call recording needs to be enabled. This also applies to storing chat transcripts.

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Contact Lens – Configuration Steps

Task 2 - Assign Permissions



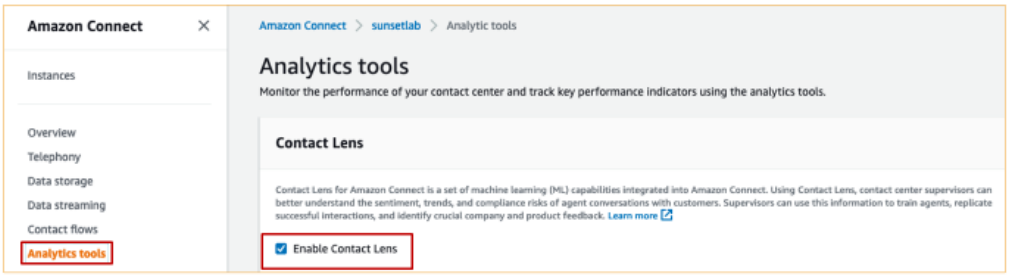
- Use permissions to determine who can access information generated by Contact Lens
- Permissions are set on the Security Profile

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Contact Lens – Configuration Steps

Task 3 – Enable Contact Lens on the Amazon Connect Instance

- Access the Amazon Connect instance in the AWS Console, then go to the Analytics page
- Enable Contact Lens for the the instance (this does not by itself analyze conversations)
- The Contact flow will additionally need to enabled for Contact Lens conversations to be analyzed

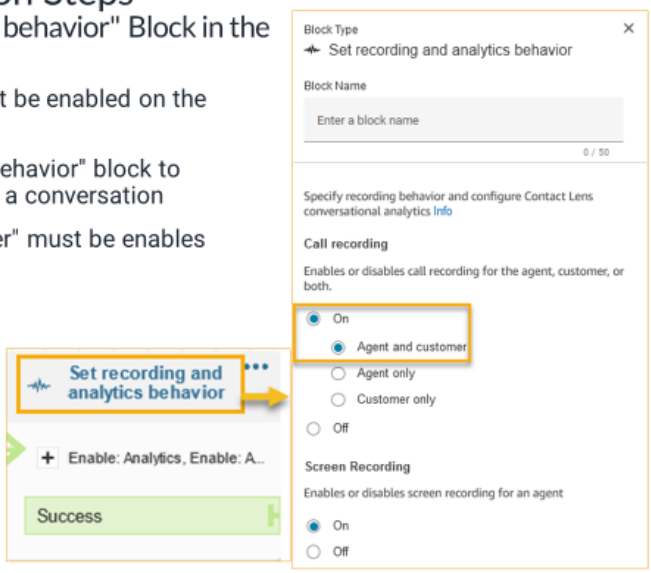


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Contact Lens – Configuration Steps

Task 4 – "Set recording and analytics behavior" Block in the Contact Flow

- For Contact Lens to be invoked it must be enabled on the inbound Contact Flow
- Use the "Set recording and analytics behavior" block to enable Contact Lens when desired for a conversation
- Call recording for "Agent and Customer" must be enables for Contact Lens to be available

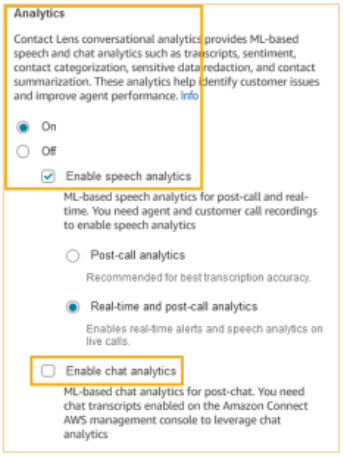


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Contact Lens – Configuration Steps

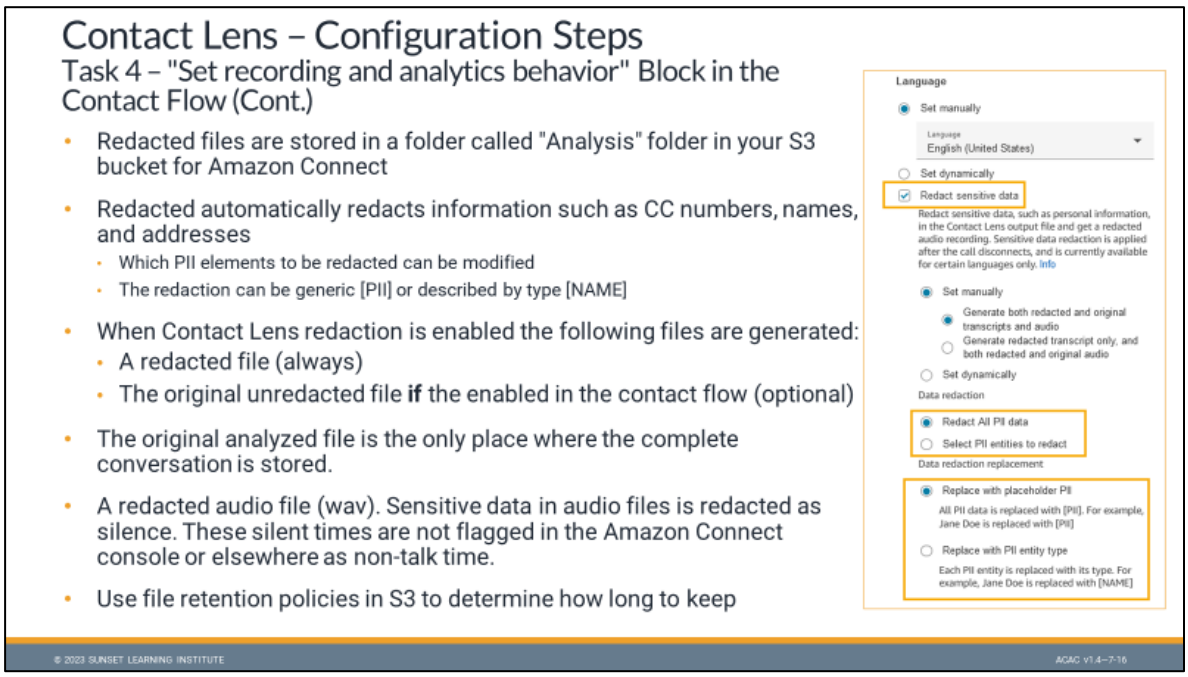
Task 4 – "Set recording and analytics behavior" Block in the Contact Flow (Cont.)

- Enabling Contact Lens (Analytics) will bring up other configuration options
- Post-call analytics occur when the conversation is over, and this has the best transcription accuracy
- Real-time and post-call analytics enables real-time alerts on and analytics and useful when a supervisor should be notified to join a live conversation
- Post-chat analytics provides analytics once the chat is complete.
- The expected language should be set for this conversation, many languages are supported.



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To help you protect your customer's privacy, Contact Lens lets you automatically redact sensitive data from conversation transcripts and audio files. It redacts sensitive data, such as name, address, and credit card information using Natural Language Understanding.

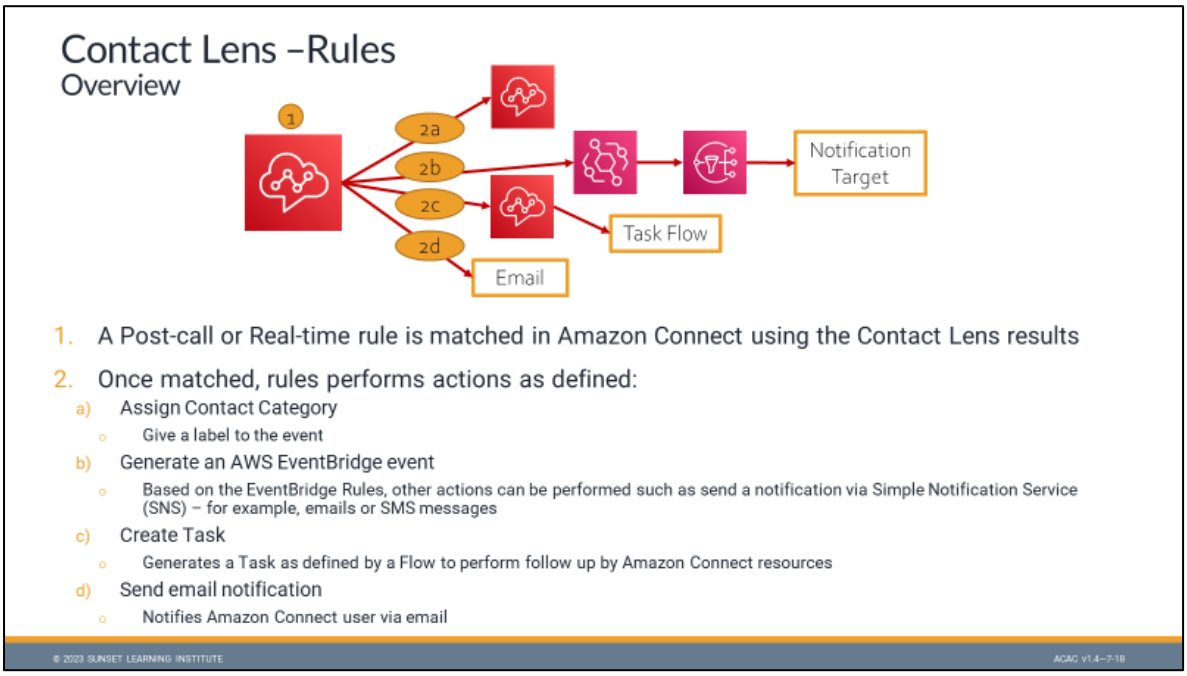
No configuration is required since Contact Lens is able to determine what data can be redacted. However, contact centers may select specific fields to not redact based on the type of PII.

Additionally, choose whether the PII is replaced with a generic label [PII] or the name of the PII type [NAME].

Sensitive data redaction is applied after a call disconnects.

Configuring Contact Lens Rules

Contact Lens for Amazon Connect enables you to analyze conversations between customer and agents, by using rules to analyze speech transcription, natural language processing, and intelligent search capabilities.



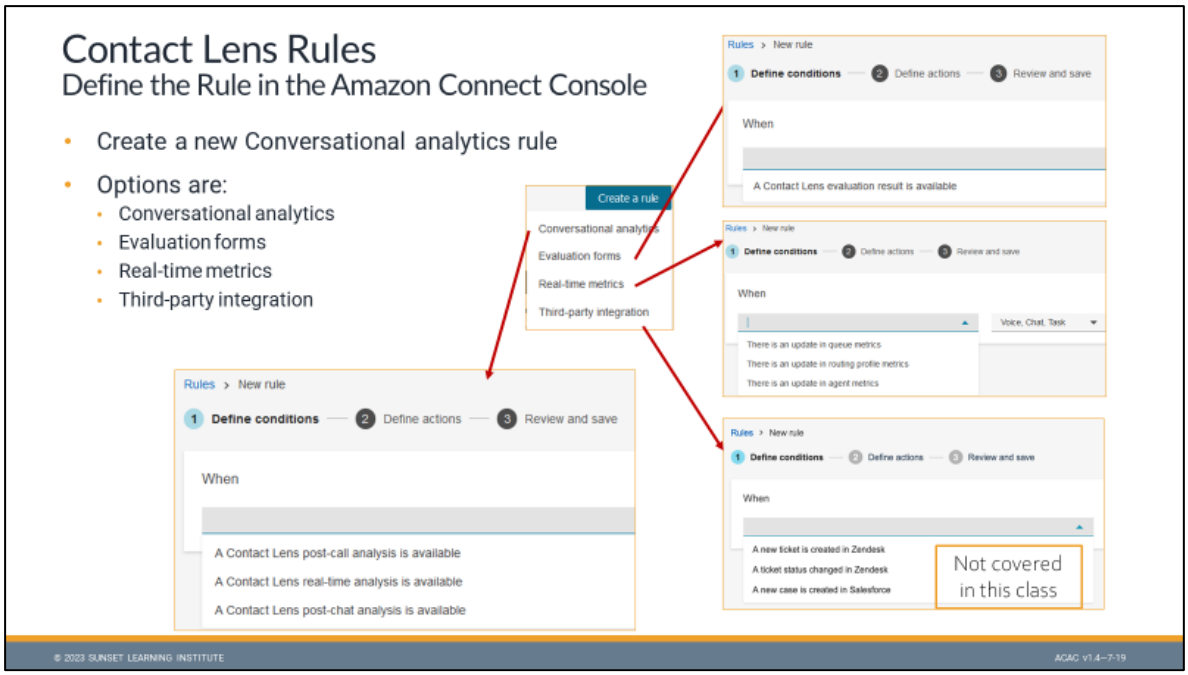
A rule is an action that Amazon Connect automatically performs, based on conditions you specify. Contact center managers, supervisors, and QA analysts can quickly create rules from the Amazon Connect console. No coding is required. Contact Lens rules allow you to automatically categorize contacts, receive alerts, or generate tasks based on uttered keywords, sentiment scores, customer attributes, and other criteria.

You can set up Contact Lens to track issues that you know exist in your Contact Center ("known knowns") and monitor any changes over time.

You can label your contacts with predefined criteria you set up, that is, keywords and phrases you want to detect. Through categorization, each contact is analyzed for these criteria and labeled.

This is useful to do when, for example, you want to ensure that agents are speaking certain words or phrases for compliance reasons. Or, for example, you want to investigate when customers use certain words and have a negative sentiment.

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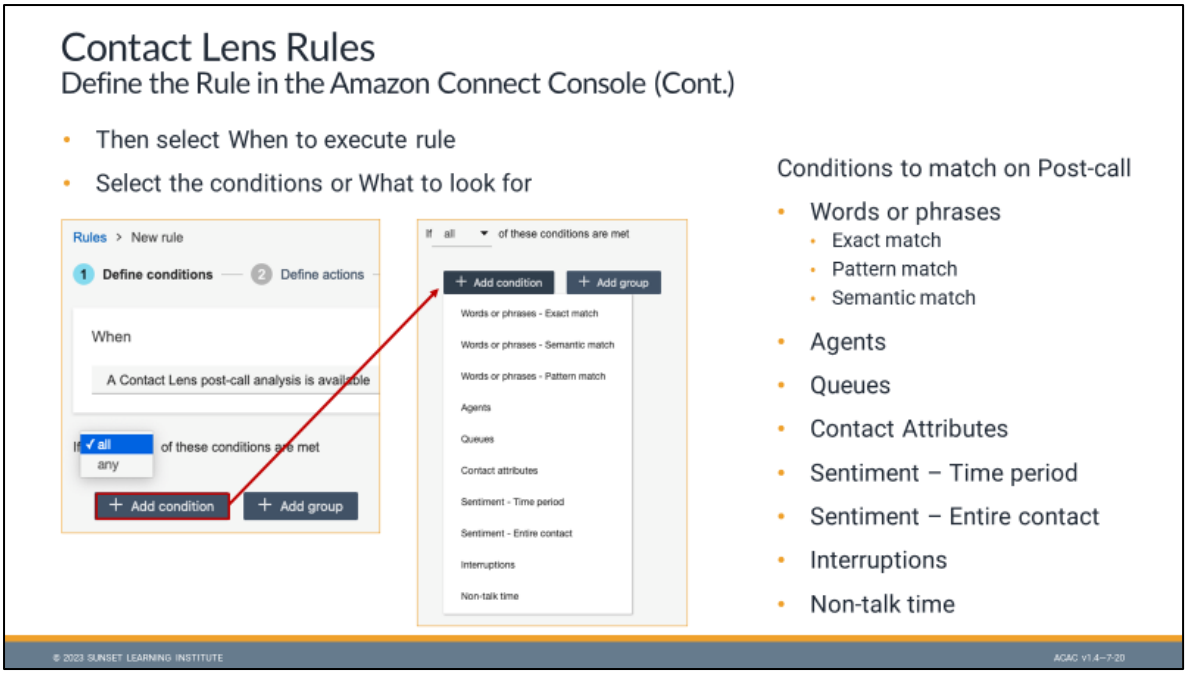
There are four types of rules available in Amazon Connect:

- Conversational analytics
- Evaluation forms
- Real-time metrics
- Third-party integration

Conversational analytics rules are primarily defined for Contact Lens. That said, rules to evaluate agent performance can be used to automatically complete agent evaluation forms that have already been created.

Rules for real-time metrics can send notifications via email or EventBridge events as well as create tasks.

Third-party integration rules identify when certain events occur in Salesforce or ZenDesk.



Once the type of rule has been selected, choose the WHEN the rule will execute followed by the What to look for.

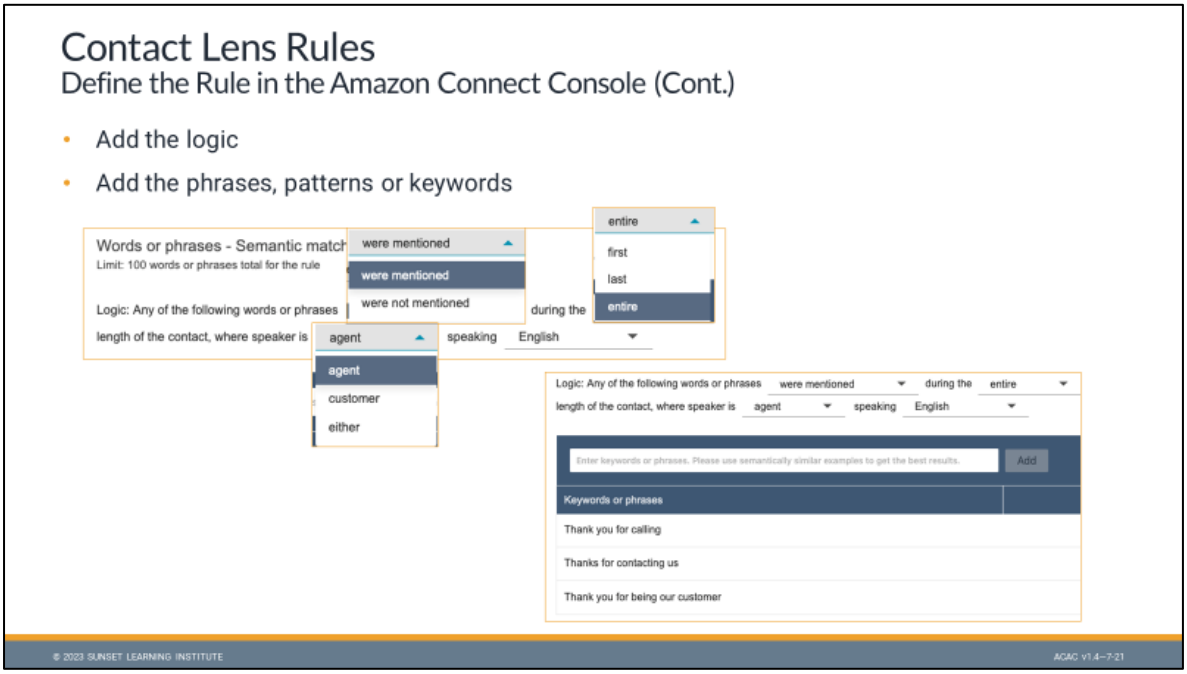
You can combine criteria from a large set of conditions to build very specific Contact Lens rules. Following are the available conditions:

- **Words or phrases:** Choose from Exact match, Pattern match or Semantic match to trigger an alert or task when keywords are uttered.
    - **Exact Match:** An exact word match, singular or plural.
    - **Pattern Match:** If you want to match related words, append an asterisk (*) to the criteria. For example, if you want to match on all variations of "neighbor" (neighbors, neighborhood) you would type neighbo*.
    - **Semantic Match:** Supported only for post-call analysis,
- **Agent:** Build rules that run on a subset of agents. For example, create a rule to ensure newly hired agents comply with company standards. To see agent names so you can add them to rules, you need Users – View permissions in your security profile.
- **Queues:** Build rules that run on a subset of queues. Often organizations use queues to indicate a line of business, topic, or domain. For example, you could build rules specifically for your sales queues, tracking the impact of a recent marketing campaign or alternatively rules for your customer support queues, tracking overall sentiment. To see the queue names so you can add them to rules, you need Queues – View permissions in your security profile.
- **Contact attributes:** Build rules that run on the values of custom contact attributes. For example, you can build rules specifically for a particular line of business or for specific customers, such as based on their membership level, their current country of residence, or if they have an outstanding order. You can add up to five contact attributes to a rule.
- **Sentiment – Time period:** Build rules that run on the sentiment analysis results (positive, negative, or neutral) over a trailing window of time. For example, you can build a rule for when customer sentiment has remained negative for a set period of time.
- **Sentiment – Entire contact:** Build rules that run on the value of sentiment scores over an entire contact. For example, you can build a rule when customer sentiment has remained low for the

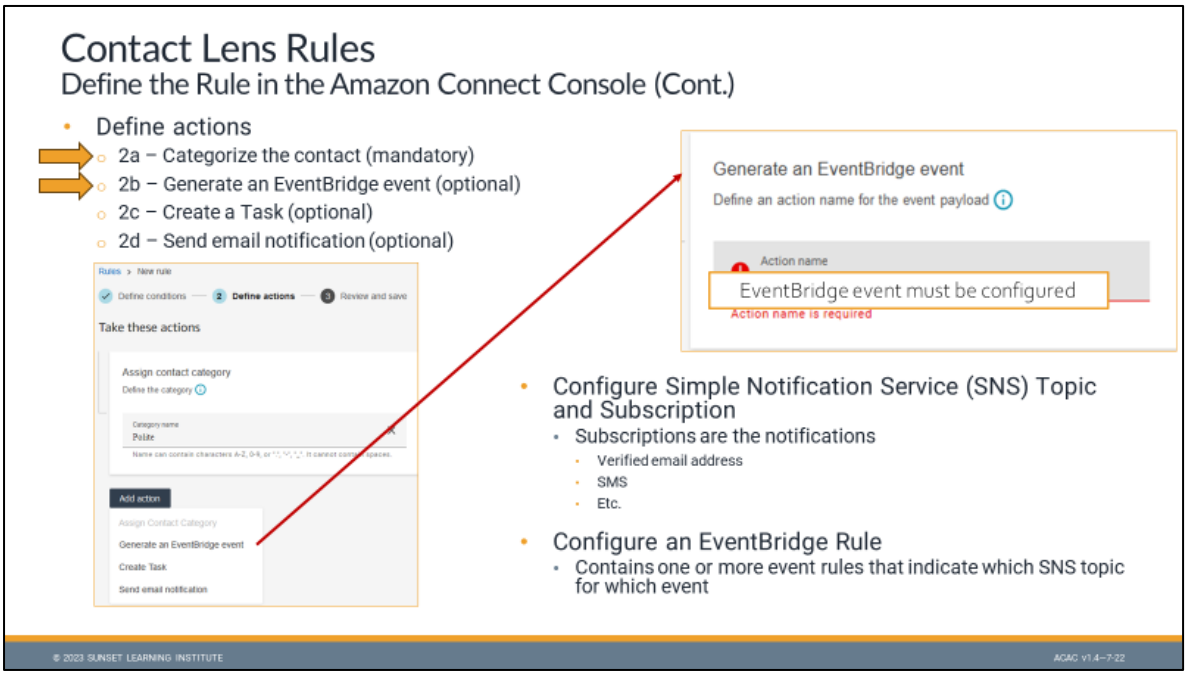
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entire contact, you can create a task for a customer experience analyst to review the call transcript and follow-up.

- **Interruptions:** Build rules that detect when the agent has interrupted the customer for more than X times.
- **Non-talk time:** Build rules that run when periods of no talk time are detected. For example, when a customer and agent have not spoken for over 30 seconds which may indicate unnecessary customer wait time or highlight a customer services process that would benefit from optimization.



Rules logic define when, what, and who. The who could be the agent, the customer, or either.

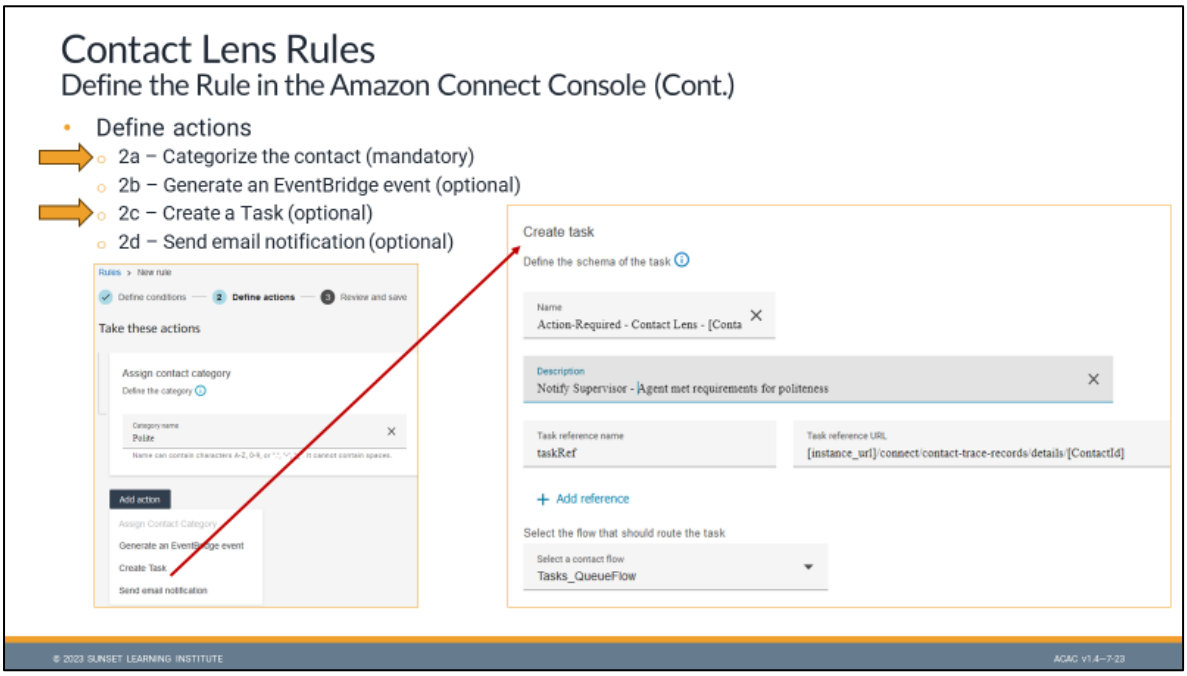


Once the condition is matched, Contact Lens rules allows the administrator to define actions to take. At a minimum, the category for the contact will be defined. There are three additional actions that could be selected, generate an EventBridge event, create a Task, or send an email.

Amazon EventBridge is a serverless event bus that makes it easier to build event-driven applications at scale using events generated from your applications, integrated Software-as-a-Service (SaaS) applications, and AWS services. EventBridge events are able to invoke other actions such as SNS notifications.

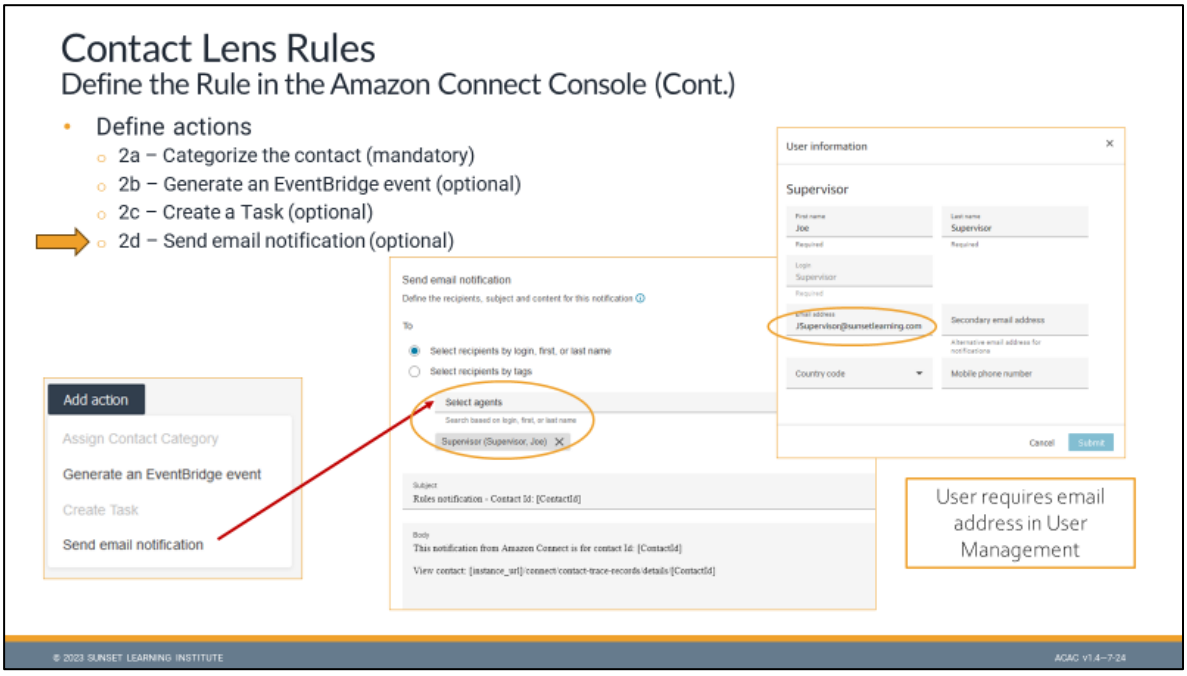
Due to the scope of this class, we will not cover the configuration of EventBridge and SNS further. The appendix of the Lab Guide contains instructions for configuring these services since you may run across these in other context.

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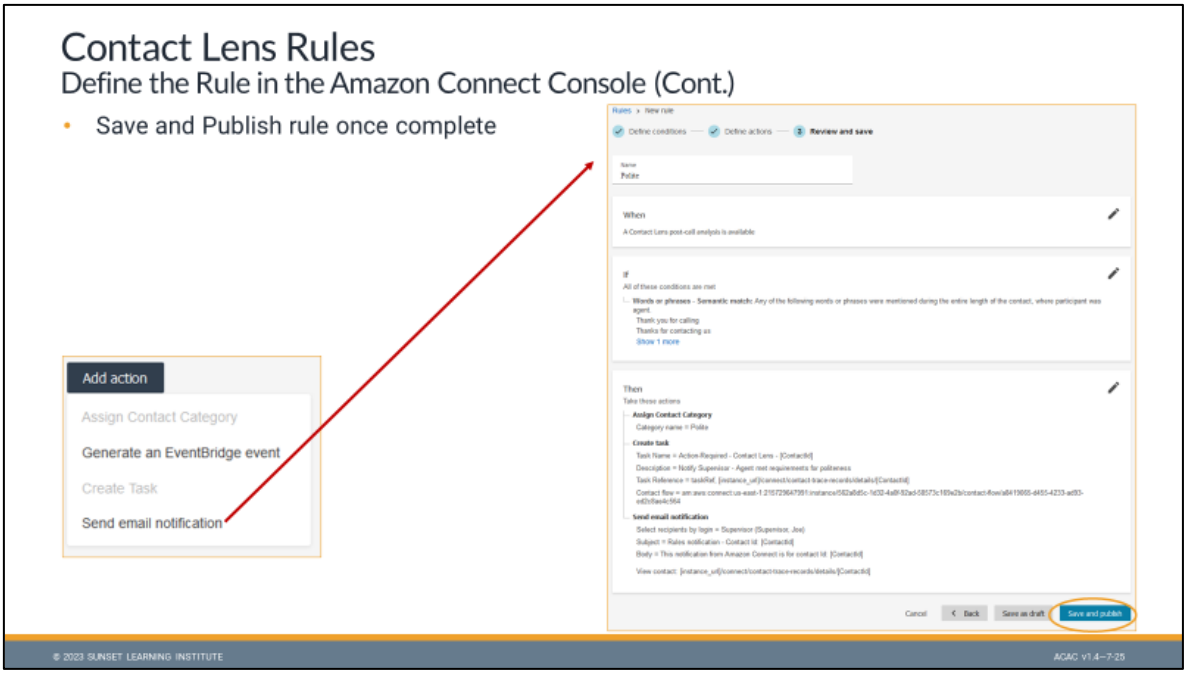
In addition to naming the category for the contact, a task can be created. Recall that Amazon Connect had three channels or media types, Voice, Chat, or Tasks.

Tasks are contact events that Connect can be created which will be processed by specific Contact Flows and can be assigned to specific queues to ensure follow up by Connect users.



Or an email can be sent by the rule. The email recipients can be defined either explicitly as Connect Users, or by selection of tags. Keep in mind that the Connect users require valid email addresses to be configured.

The email template can be modified to notify the email recipient of the situation that matched the rule.

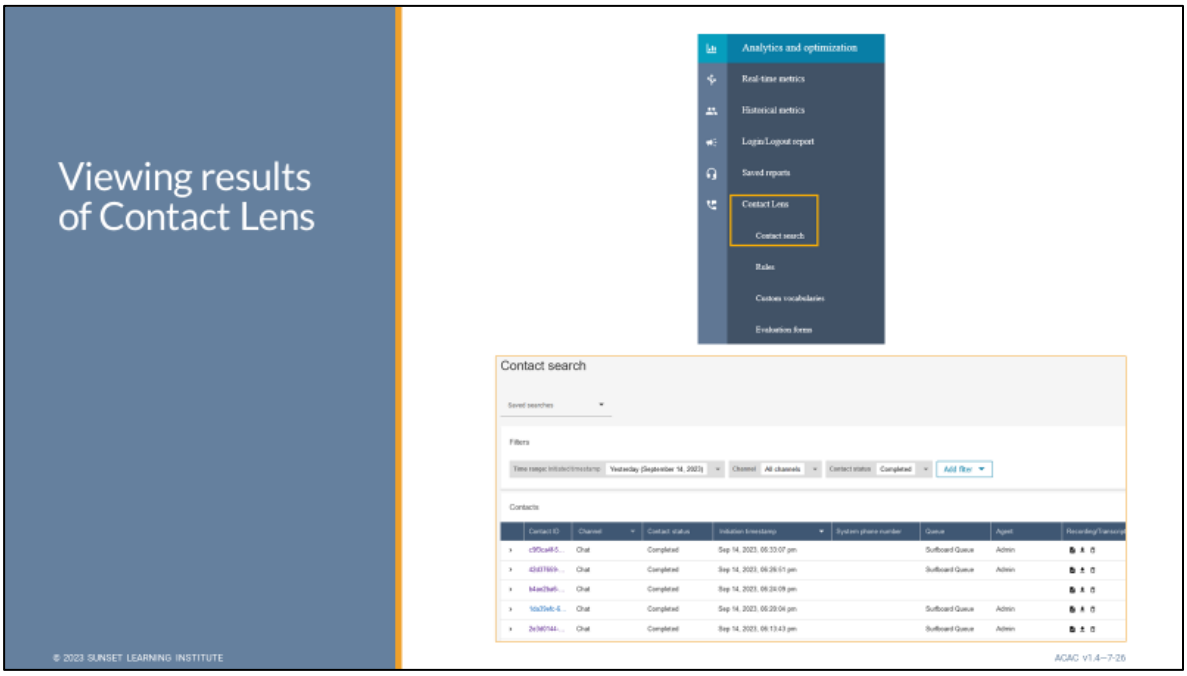


Once the rule has been configured, you have the opportunity to confirm the rule or reconfigure. You have the option to save this rule as a draft or Save and publish.

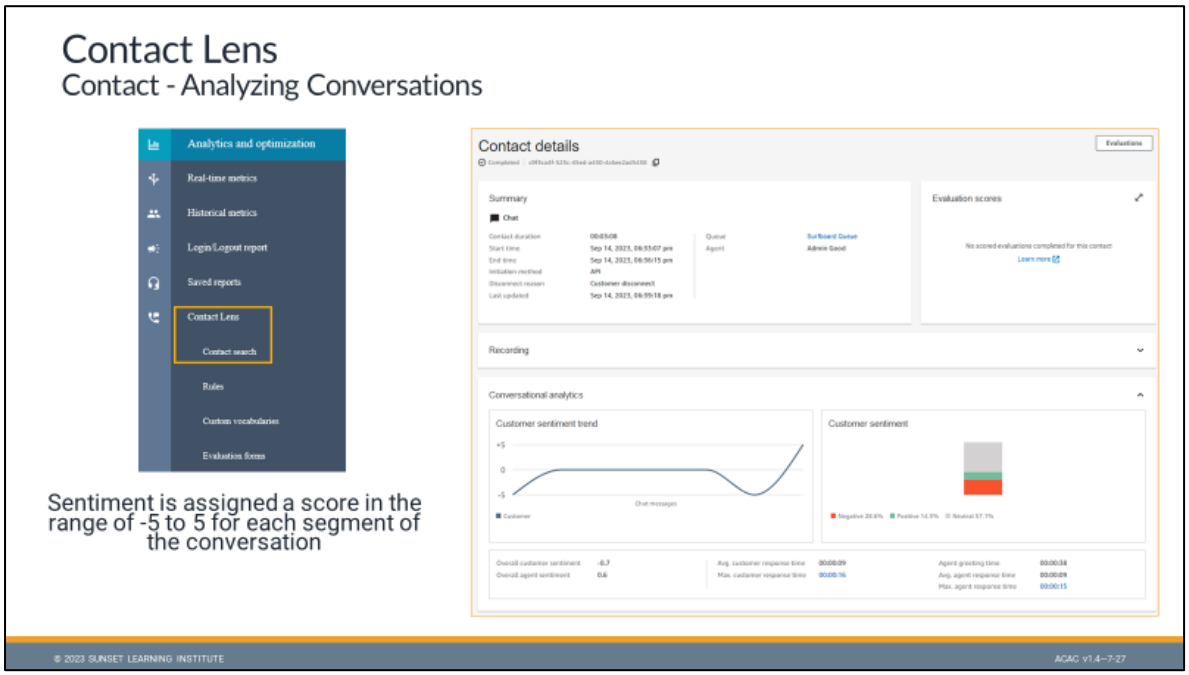
Don’t forget, creating the rule defines what happens when a contact has been assigned to Contact Lens. The Set recording and analytics behavior block in the Contact Flow defines when Contact Lens will be engaged.

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Viewing Results of Contact Lens



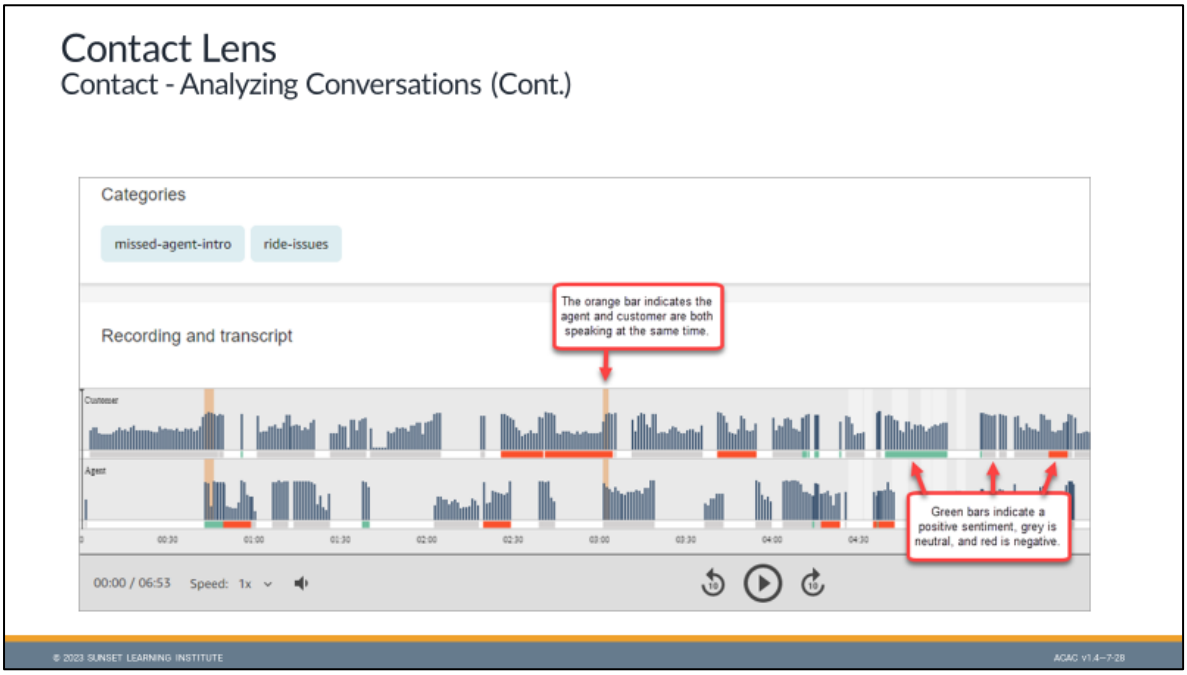
Contact Lens for Amazon Connect enables you to analyze conversations between the customer and agent, by using speech transcription, natural language processing, and intelligent search capabilities.

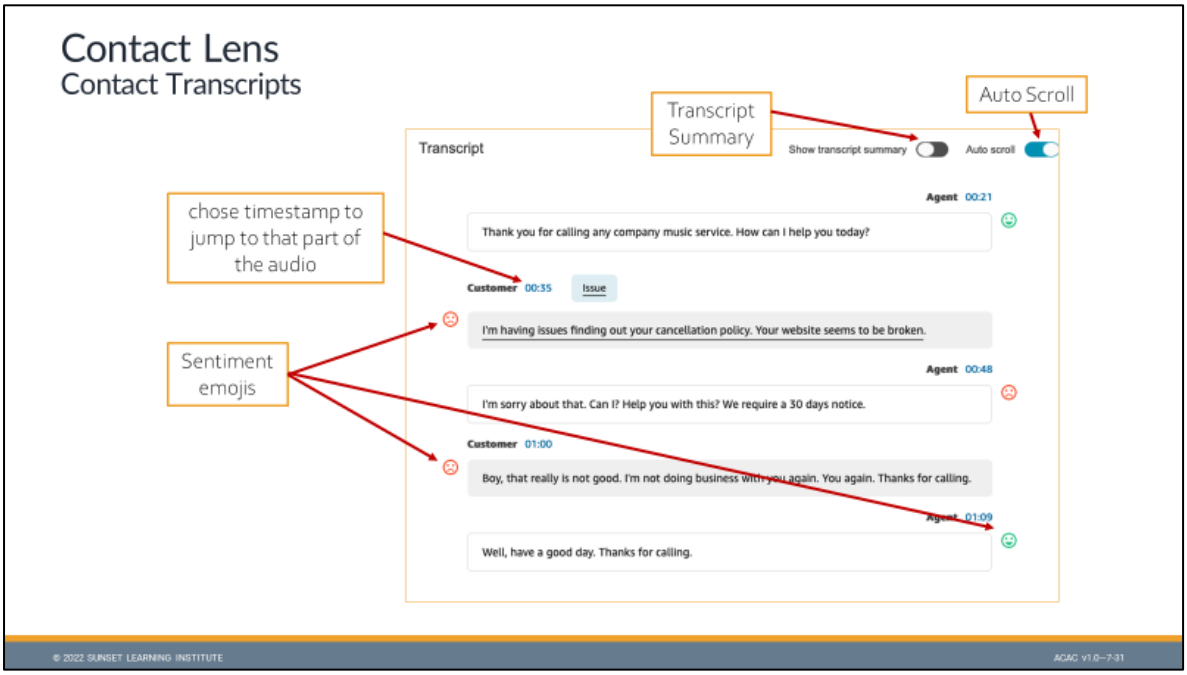


How sentiment scores are determined

To determine the sentiment score, Contact Lens for Amazon Connect analyzes the sentiment for every speaker turn during the conversation. It uses the frequency and proximity of the resulting sentiment for each speaker in turn to assign a score that ranges from -5 to +5 for each portion of the call.

The final sentiment score for the entire conversation is an average of the scores assigned during the call.





Supervisors are often required to review many agent's calls for quality assurance purposes. The turn-by-turn transcript and sentiment data helps you quickly identify and navigate to the portion of the recording that is of interest to you.

The following image shows features that enable you to quickly navigate transcripts and audio to find areas that need your attention.

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It can be time-consuming to review contact transcripts that are hundreds of lines long. To make this process faster and more efficient, Contact Lens provides the option for you to view a transcript summary. The summary shows only those lines where Contact Lens has identified an issue, outcome, or action item in the transcript.

Turn on Auto scroll to synchronize the transcript and audio

Auto scroll enables you to jump around the audio or transcript, and the two always stay in sync. For example:

- When you listen to a conversation, the transcript moves along with it, showing you sentiment emojis and any detected issue.

You can scroll through the transcript and choose the timestamp for the turn to listen to that specific point in the recording.

Because the audio and transcript are aligned, the transcript can help you understand what the agent and customer are saying. This is especially useful when:

- The audio is bad, maybe due to a connection issue. The transcript can help you understand what's being said.

There's a dialect or language variant. Our models are trained on different accents, so the transcript can help you understand what's being said.



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Module 8

Monitoring Metrics, Reports, and Dashboards in Amazon Connect

Overview

This module provides an overview of the process and key tasks for viewing, customizing, and saving Real-time metrics, Dashboards, Historical Metrics, and Login/Logout reports.

Objectives

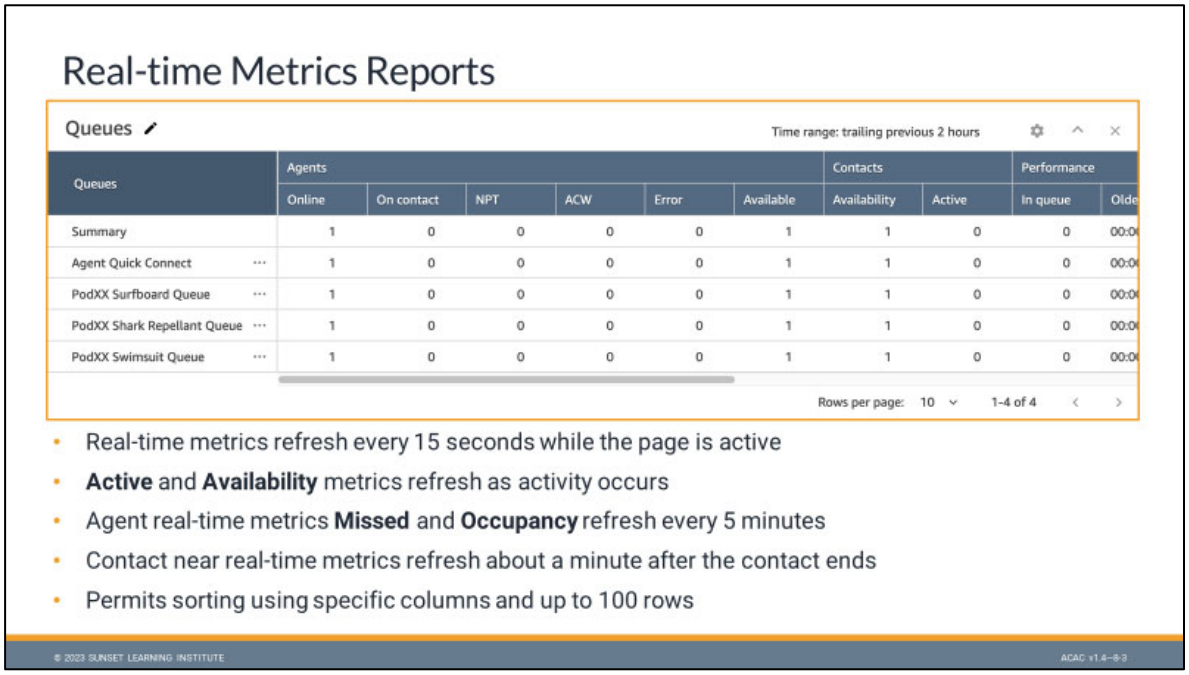
Upon completing this module, you will show an understanding of the basic premise of metrics. This ability includes being able to meet this objective:

* Identify methods of monitoring metrics, reports, and dashboards in Amazon Connect

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Real-time Metrics and Dashboards

Real-time metrics allows you to group your data by queues, agents, and routing profiles.

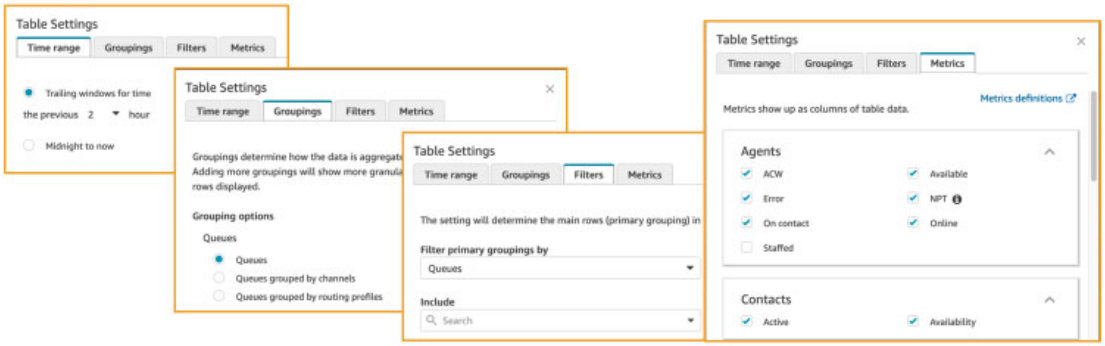


Data in real-time metrics reports is refreshed as follows:

- The **Real-time metrics** page refreshes every 15 seconds, as long as the page is active. For example, if you have multiple tabs open in your browser and navigate to a different tab, the real-time metric page won't be updated until you return to it.
- Metrics, such as **Active** and **Availability,** refresh as activity occurs with a small system delay for processing the activity.
- Agent near real-time metrics, such as **Missed** and **Occupancy**, refresh every 5 minutes.
- Contact near real-time metrics refresh about a minute after a contact ends.
- Permits sorting using specific columns and up to 100 rows.

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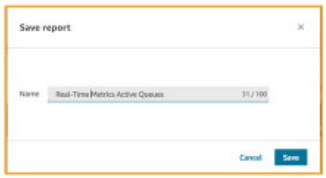
Real-time Metrics Reports (Cont.)

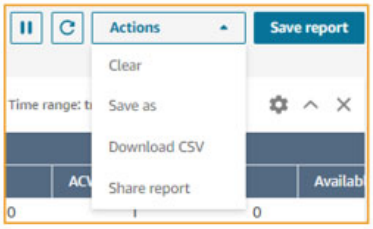


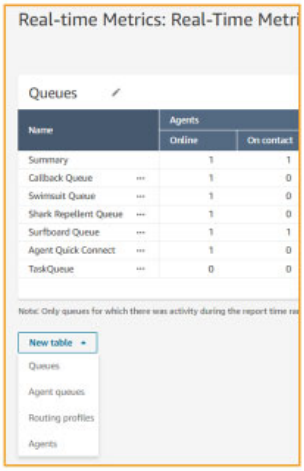
- Settings can be accessed by selecting the icon of a specific metric
- Additional metrics can be added to the page
- Page can be saved and viewed later

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Real-time Metrics Reports (Cont.)



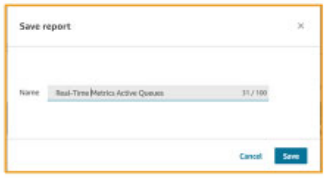


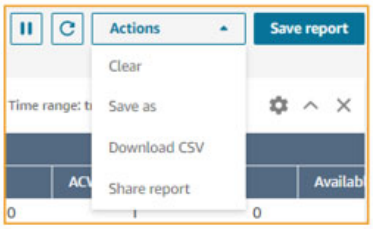


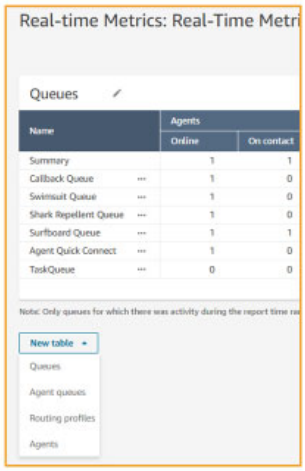
- Once custom settings have been defined, specific settings can be saved with a unique name
- Real-time Metrics reports may contain multiple tables
- Other Actions are available, such as:
  - Downloading current results as a CSV file
  - Share the report – creates a link of the saved report that can be shared with other Connect users

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Real-time Metrics Reports (Cont.)



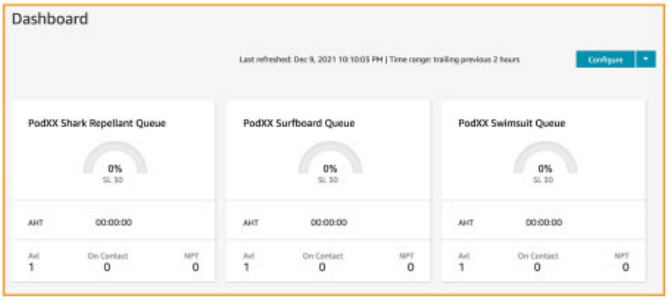




- Once custom settings have been defined, specific settings can be saved with a unique name
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- Other Actions are available, such as:
  - Downloading current results as a CSV file
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Dashboards



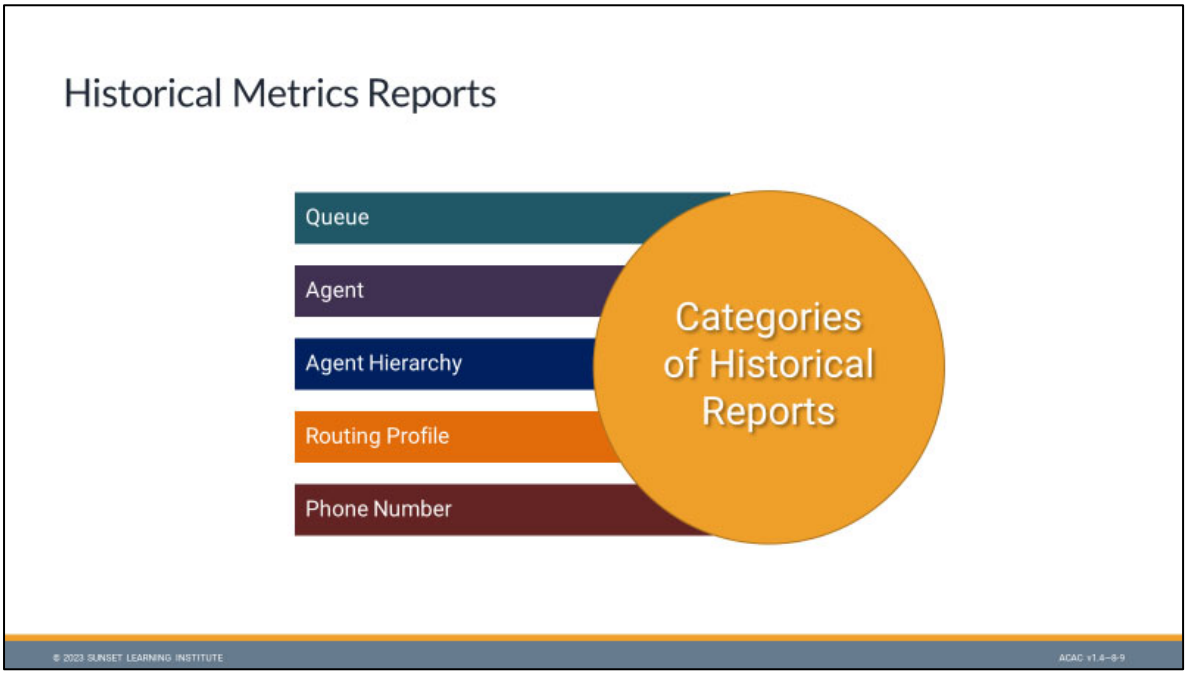
- Displayed in Dashboards under the step-by-step guide
- Dashboards can be customized and saved
- Widgets can be customized with these settings:
  - Time Range
  - Queues to view
  - Color coding of performance metrics

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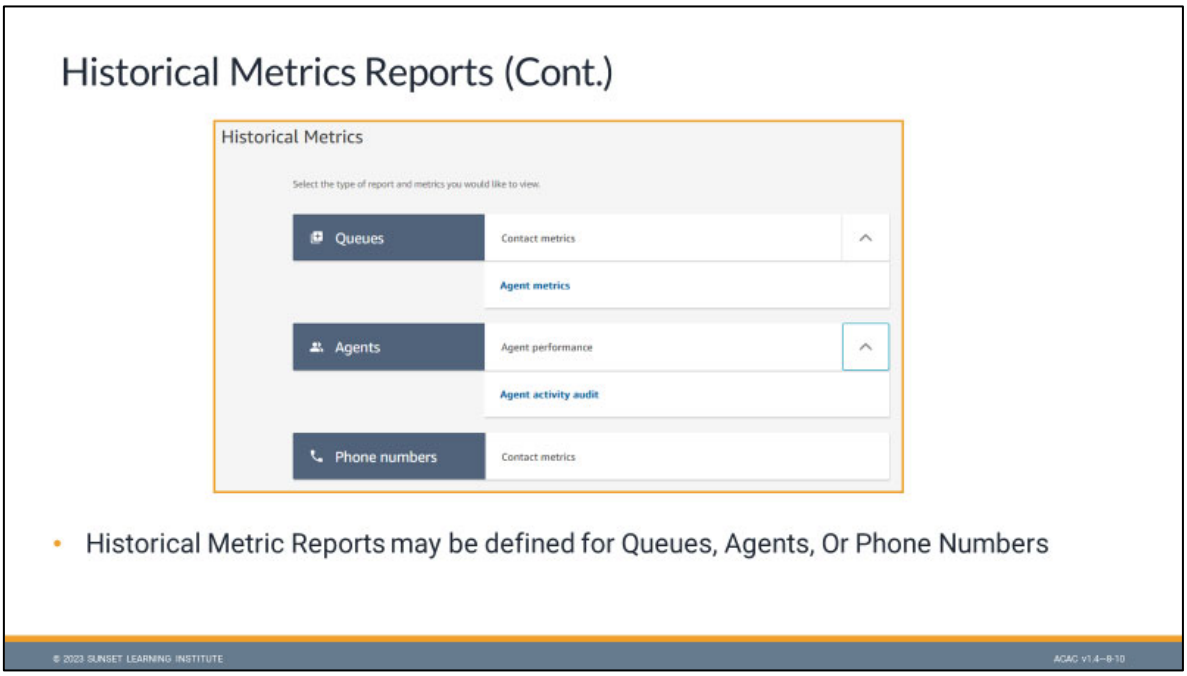
Historical Metrics

Amazon Connect includes built-in historical reports; however, you can create your own custom reports, so you look at only the data that's of interest to your organization.

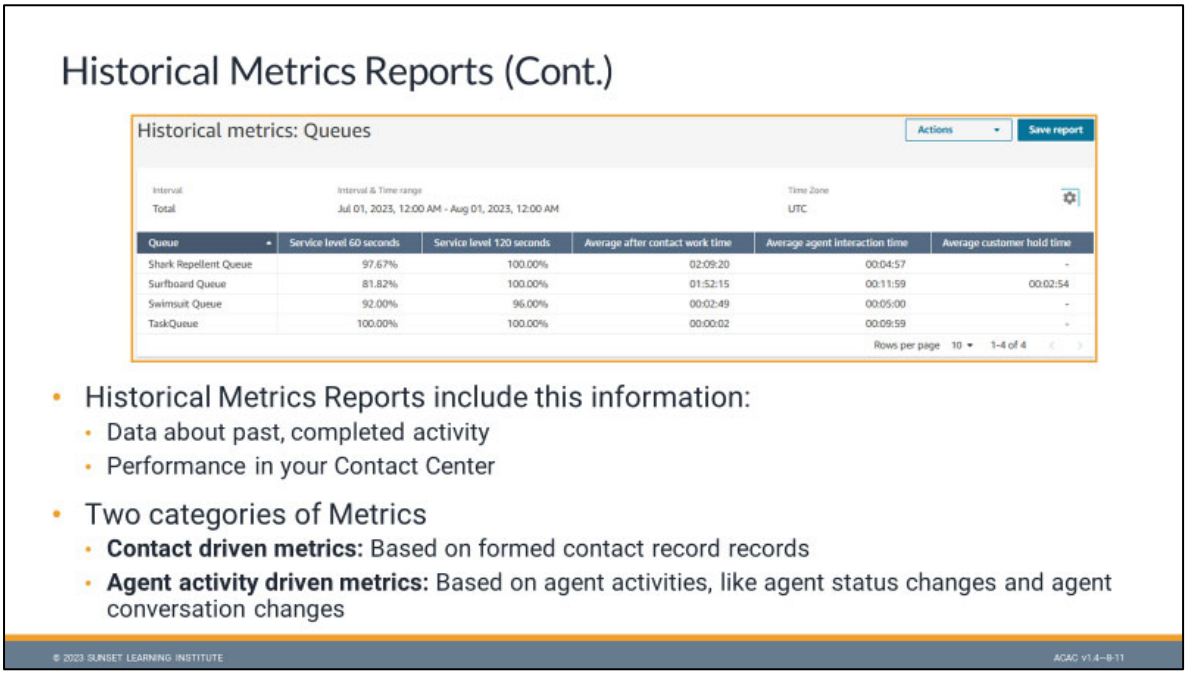


You can group reports by the following:

- Queue
- Agent
- Agent hierarchy
- Routing profile
- Phone number



Amazon Connect includes built-in historical reports; however, you can create your own custom reports, so you look at only the data that's of interest to your organization.



Although Amazon Connect includes built-in historical reports, you can create your own custom reports, so you look at only the data that's of interest to your organization.

Requirement

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- You must have permission to access metric data. The following security profiles include this permission: **CallCenterManager** and **QualityAnalyst**.

Grouping options

You can group the metrics included in your reports in different ways to provide greater insight into how your Contact Center is performing.

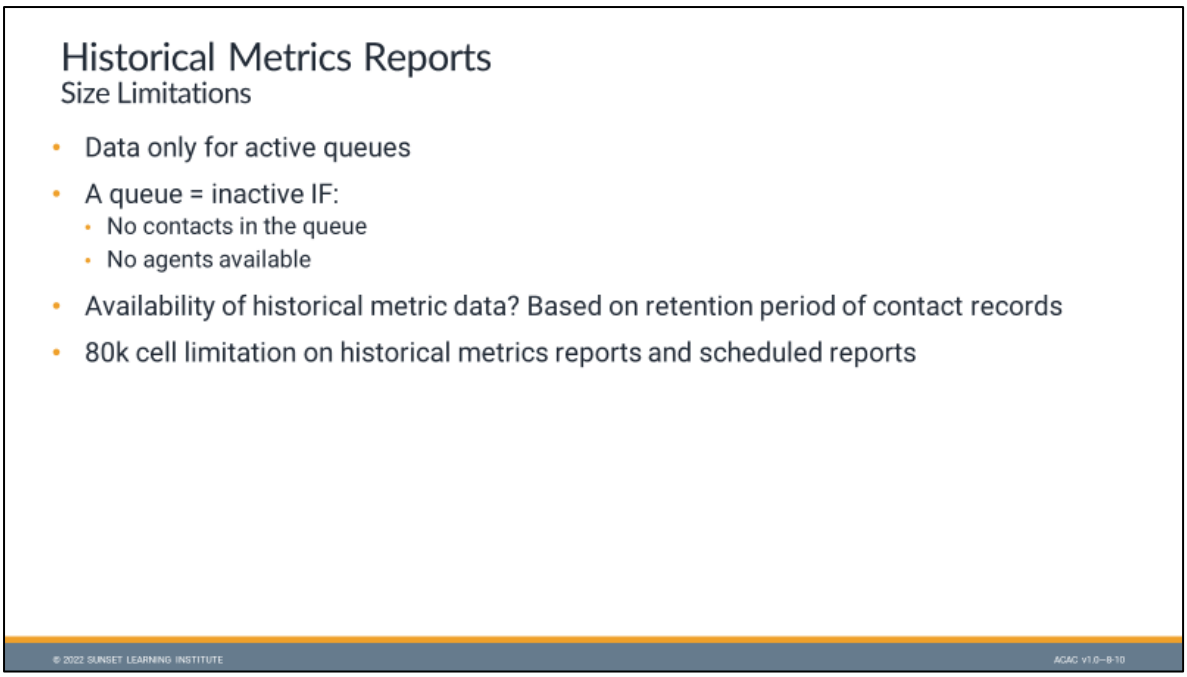
You can group reports by queue, agent, agent hierarchy, routing profile, or phone number. The metric calculations, and therefore metrics values displayed in the report, are different when reports are grouped differently. For example, if you group a report by queue, the value of a metric includes all contacts associated with the queue. If you group a report by agent, the values for the metrics associated with queues might not provide much insight.

When you create a report, the values for calculated metrics are displayed as rows in the report. The rows in the report are grouped by the grouping options you select. Grouping the data enables you to generate global data for your Contact Center, or more specific data for queues, agents, routing profiles, or agent hierarchy defined in your Contact Center.

For example, consider **the Contacts handled** metric. This metric is a count of the contacts handled during the time range defined for the report. Here are the results based on the grouping:

- **Queue**: The metric is the total number of contacts handled during the time range from that queue by all agents in your Contact Center.
- **Agent**: The metric is the total number of contacts handled by that agent during the time range across all queues and routing profiles.
- **Routing Profile**: The metric is the total number of contacts handled during the time range by agents assigned that routing profile.
- **Queue**, then **Agent**, then **Routing Profile**: The metric is the total number of contacts that agent assigned that routing profile handled from that queue.

Agent activity can be included in one routing profile at a time, but agents can switch between routing profiles over the reporting time interval. If agents are assigned multiple routing profiles and handle contacts from multiple queues, there are multiple rows in the report for each routing profile assigned to the agent and the queue that the agent handled contacts from.



Historical metrics reports have the following limits:

Data only for active queues

You can get data only for active queues. A queue is inactive if there are no contacts in the queue and no agents available.

Query data for three days at a time, for the past 35 days

When you create a report that uses intervals (such as 30-minute intervals), you can return data for three days at a time, for the past 35 days.

The availability of historical metric data is based on the retention period of contact records

Historical metrics are based on contact records for the current retention period for contact records.

80k cell limit

There is currently an 80k cell limitation on historical metrics reports and scheduled reports. This applies to the total number of cells (columns * rows), accounting for grouping and filtering.

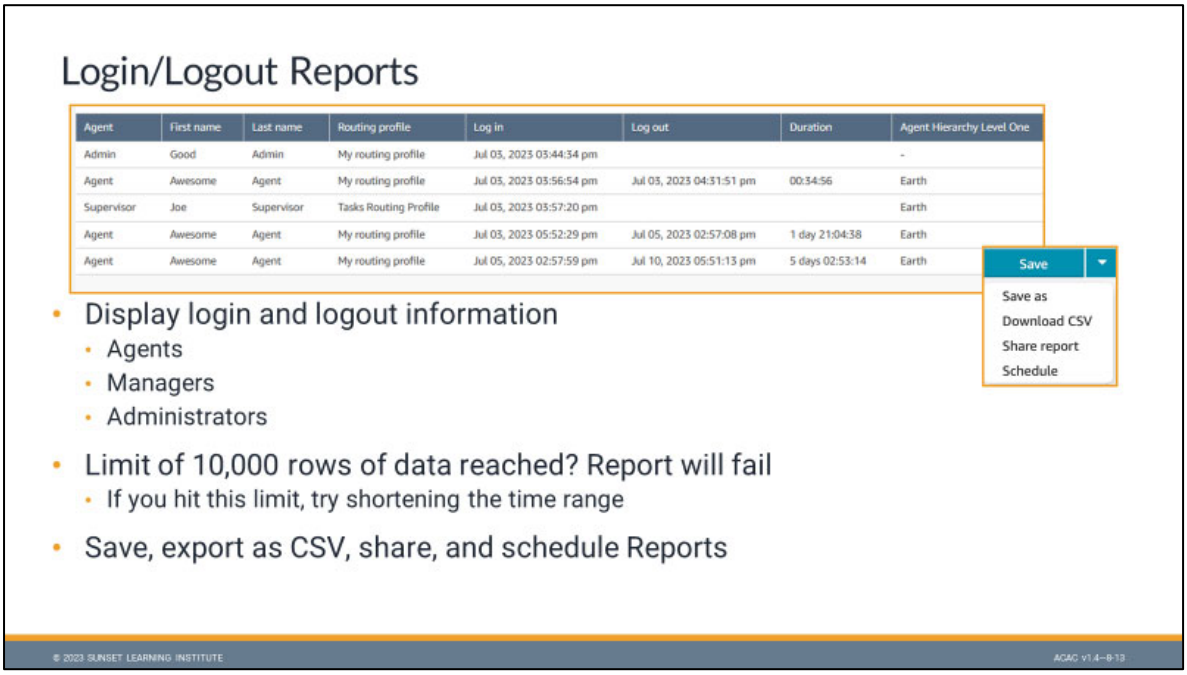
For example, let's say you create a historical metrics report with these criteria:

- Grouped by agents
- With an interval of 30 minutes
- For the last 24 hours
- Configured to include only 5 metrics
- Filtered to show only contacts handled in BasicQueue

If only 10 agents handled contacts in BasicQueue during this time, then you would expect to see (24*2)*5*10 = 2400 cells that count towards the 80k limit.

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A message informs you if you reach the limit.



The Login/Logout report displays the login and logout information for the users in your Contact Center (for example, agents, managers, and administrators). For each user session, the login and logout times are displayed as a row in the report. You can use the report to determine the time users were logged in to Amazon Connect. The report also displays the amount of time for each session that user was logged in to Amazon Connect.

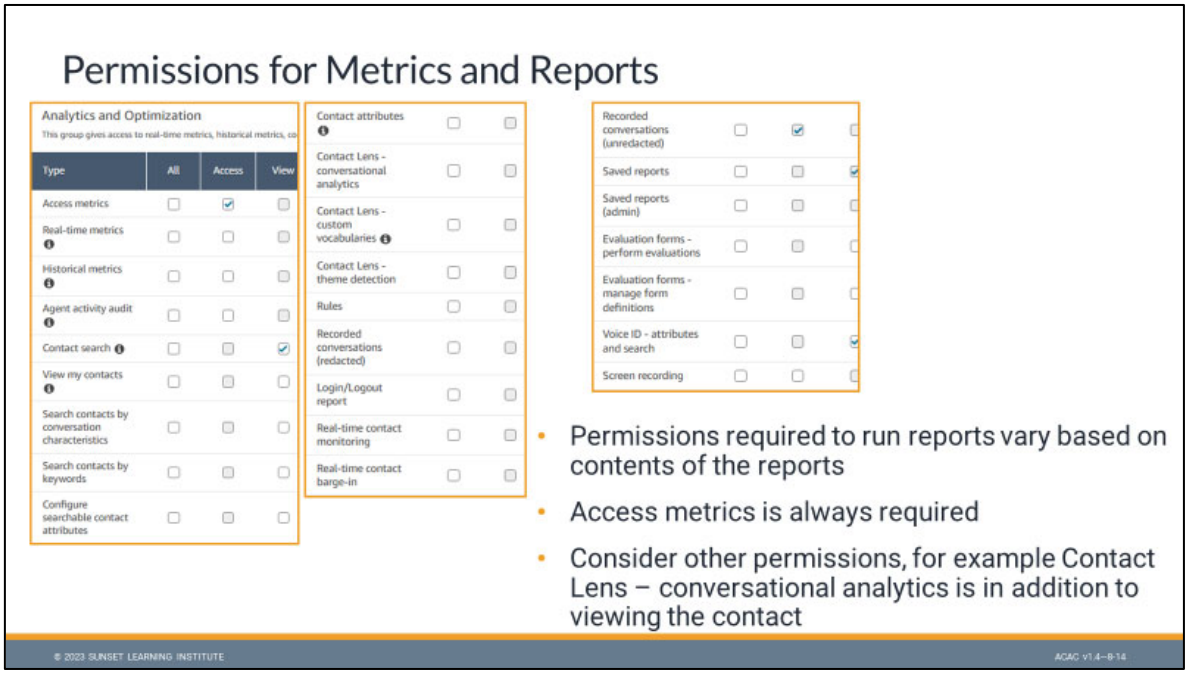
By default, when an agent closes their CCP window, they are not logged out. Unless you have customized your CCP for automatic logout, agents must choose the **Logout** button. Until they choose the **Logout** button, the Login/Logout report shows them as logged in.

Login/Logout report limit: 10,000 rows

- If you try to generate a Login/Logout report that has more than 10,000 rows, it won't complete.
- The Login/Logout report page displays only 10,000.
- If you schedule a Login/Logout report that contains more than 10,000 rows, the report will fail. In addition, no report output will be saved to your S3 bucket, and you cannot view the report.
- If you have a Contact Center with a lot of users, and your reports fail to complete, you can specify a shorter time range to reduce the size of the report generated, or apply filters to the report, such as routing profile and agent hierarchy. You can then use other filters to capture all of the login/logout data for your instance.

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Permissions for Metrics and Reports



To view historical metrics reports, you need to be assigned to a security profile that has Access metrics permission.

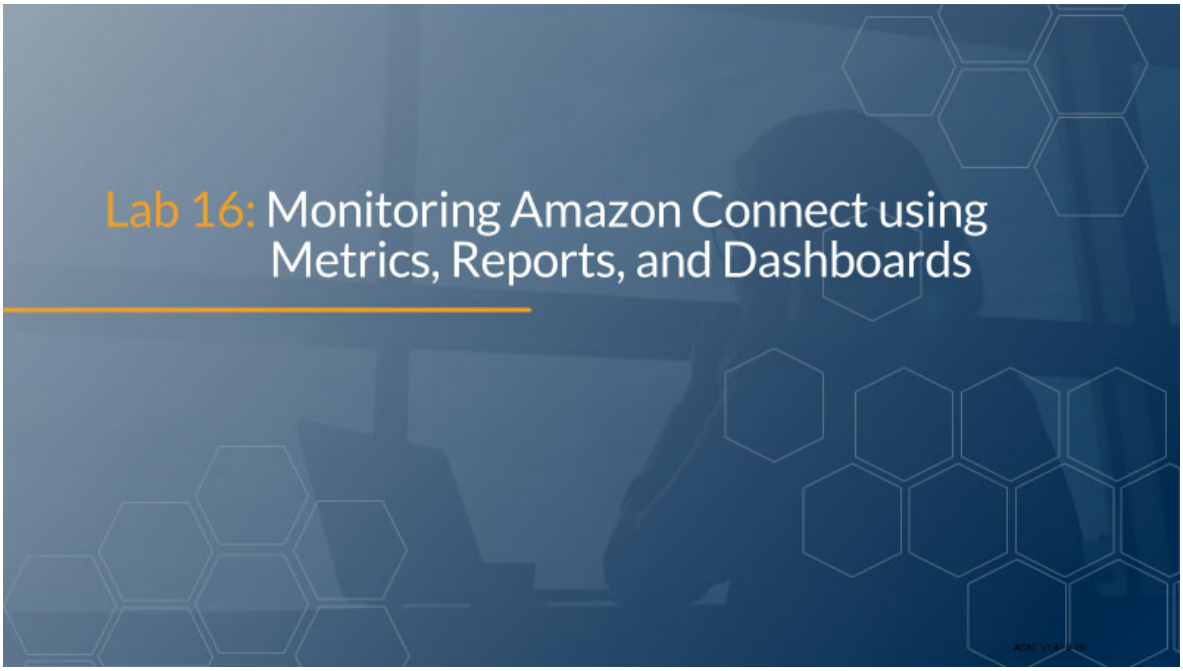
To create, share, and publish saved reports, you need the Saved reports, Create permission.

To view the Agent reports, for example - activity audit report, you need View permissions in the Users section.

Consider other permissions, for example Contact Lens – conversational analytics is in addition to viewing the contact.

Saved report permissions are required to create share and publish reports.

As new features become available, permissions may need to be reviewed.



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